Logic answer key (version 1.0.0)

Private grading material. Each entry gives the question ID, its page in the question collection, the correct response, the full reasoning, why each wrong choice fails, accepted alternatives, the grading rule and how the key was verified.

S1-M-Q1 · question page 2

LOGIC.SPATIAL.01 · Decide what a mirror reflection must show

The shaded shape is reflected in the dashed mirror line on its right. Which figure shows its mirror image? (Only the shape matters, not where it sits in its box.)

Answer: C. Figure C

Reasoning: A reflection in a vertical line keeps every cell in its row and swaps left and right. Figure C is the only panel whose rows match the shape row by row with left and right swapped.

- ✗ Figure A: a half-turn of the shape, not a reflection
- ✗ Figure B: the reflection in a horizontal line (top and bottom swapped)
- ✗ Figure D: a quarter-turn of the shape

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the panel check accepts only the keyed panel.

S1-M-Q2 · question page 3

LOGIC.SPATIAL.01 · Decide what a mirror reflection must show

Each figure shows a shape above a dashed water line and, below the line, what is supposed to be its reflection in the water. In which figure can the lower shape NOT be the reflection of the upper one?

Answer: D. Figure D

Reasoning: A reflection in a horizontal line keeps every cell in its column and reverses the order of the rows, measured from the line. In D the lower

shape is the upper shape moved down without flipping, so the row nearest the line is wrong.

- ✗ Figure A: the lower shape is the upper shape flipped top to bottom across the line, so it is a real reflection
- ✗ Figure B: the lower shape is the upper shape flipped top to bottom across the line, so it is a real reflection
- ✗ Figure C: the lower shape is the upper shape flipped top to bottom across the line, so it is a real reflection

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the panel check accepts only the keyed panel.

S1-M-Q3 · question page 4

LOGIC.SPATIAL.01 · Decide what a mirror reflection must show

Words are written left to right in block capitals and seen in a left-right mirror (a vertical mirror line). In the block capitals used here, exactly these letters look the same in a left-right mirror: A H I M O T U V W X Y. Which word looks exactly the same in the mirror?

Answer: A. TOOT

Reasoning: A left-right mirror reverses the order of the letters and mirrors each letter. The word looks the same only if it reads the same backward AND every letter is a listed symmetric letter. TOOT is the only choice that does both.

- ✗ NOON: N is not one of the listed letters, so each N appears reversed in the mirror
- ✗ ZOOM: Z is not listed, and the mirror reverses the letter order, giving MOOZ
- ✗ BIB: B is not listed, so each B appears reversed even though BIB reads the same backward

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the

check function accepts only the keyed choice.

S1-M-Q4 · question page 5

LOGIC.SPATIAL.01 · Decide what a mirror reflection must show

A word is written in block capitals and seen reflected in calm water below it (a horizontal mirror line). In the block capitals used here, exactly these letters look the same in a top-bottom mirror: B C D E H I K O X. The letter order stays the same in this reflection. Which word looks exactly the same in the water?

Answer: B. CODE

Reasoning: A top-bottom reflection keeps the order of the letters and flips each letter. The word looks the same when every letter is a listed letter; only CODE qualifies.

- ✗ CORE: R is not one of the listed letters, so it appears upside down
- ✗ BEAD: A is not listed, so it appears upside down
- ✗ BOXES: S is not listed, so it appears changed

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S1-M-Q5 · question page 6

LOGIC.SPATIAL.01 · Decide what a mirror reflection must show

The grid has 4 rows and 8 columns. The dashed mirror line runs between columns 4 and 5. The shaded cells are in row 1 column 2, row 2 columns 2, 3 and 4, and row 3 column 4. Mark the cells of their mirror image. (Cells are named row,column; row 1 is the top row and column 1 the left column.)

Answer: 1,7, 2,5, 2,6, 2,7, 3,5

Reasoning: Each cell keeps its row, and a cell in column c lands in column 9 - c (the same distance on the other side of the line between columns 4 and 5): (1,7), (2,7), (2,6), (2,5), (3,5).

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: the key is computed as column 9 - c for every shaded cell.

S1-B1-Q1 · question page 7

LOGIC.SPATIAL.01 · Strategy: reflect one cell at a time

A grid has 10 columns. A vertical mirror line lies between columns 5 and 6. A shaded cell is in column 2. In which column is its mirror image?

Answer: 9

Reasoning: Column 2 is 4 columns left of the line (columns 2, 3, 4, 5), so its image is 4 columns right of it: column 9. Check: 2 + 9 = 11 = 5 + 6.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S1-B1-Q2 · question page 8

LOGIC.SPATIAL.01 · Strategy: reflect one cell at a time

A grid has 8 rows. A horizontal mirror line lies between rows 4 and 5. A shaded cell is in row 1. In which row is its mirror image?

Answer: 8

Reasoning: Row 1 is 4 rows above the line, so the image is 4 rows below it: row 8.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S1-B1-Q3 · question page 9

LOGIC.SPATIAL.01 · Strategy: reflect one cell at a time

A reflection in a vertical mirror line keeps every cell in its own row.

Answer: True

Reasoning: Only the column changes; the row stays the same.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S1-B1-Q4 · question page 10

LOGIC.SPATIAL.01 · Strategy: reflect one cell at a time

A cell is 3 columns to the left of a vertical mirror line. Where is its mirror image?

Answer: D. 3 columns to the right of the line

Reasoning: A reflection keeps the distance to the line and changes the side.

- ✗ 3 columns to the left of the line: that is the original cell
- ✗ 6 columns to the right of the line: the distance from the line is kept, not doubled
- ✗ on the mirror line itself: only points on the line stay on it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S1-B1-Q5 · question page 11

LOGIC.SPATIAL.01 · Strategy: reflect one cell at a time

A row of 6 cells has a mirror line between cells 3 and 4. Cell 1 is shaded. Mark the cell of its mirror image. (The grid is 1 row by 6 columns; cells are named row,column.)

Answer: 1,6

Reasoning: Cell 1 is 3 cells left of the line, so its image is 3 cells right: cell 6.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: image column = 7 - 1.

S1-F1-Q1 · question page 12

LOGIC.SPATIAL.01 · Combine reflections and read mirrored clocks and words

A clock face with no numbers shows 2:20. You see it in a left-right mirror. What time does the mirrored clock appear to show?

Answer: C. 9:40

Reasoning: A left-right mirror sends a time t to 12:00 - t. 12:00 - 2:20 = 9:40. Check: the minute hand at 4 (20 minutes) appears at 8 (40 minutes), and the hour hand just past 2 appears just before 10.

- ✗ 10:20: only the hour hand was mirrored; the minute hand was left at 20 minutes
- ✗ 2:40: only the minute hand was mirrored
- ✗ 3:40: the hands were swapped instead of mirrored

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S1-F1-Q2 · question page 13

LOGIC.SPATIAL.01 · Combine reflections and read mirrored clocks and words

The shape is reflected in a vertical mirror line and then the result is reflected in a horizontal mirror line. Which figure shows the final result? (Only the shape matters, not where it sits.)

Answer: B. Figure B

Reasoning: A vertical reflection then a horizontal one sends (x, y) to (-x, -y): a half-turn. Figure B is the shape turned halfway around.

- ✗ Figure A: only the first (vertical) reflection
- ✗ Figure C: the starting shape: the two reflections do not cancel
- ✗ Figure D: only the second (horizontal) reflection

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the panel check accepts only the keyed panel.

S1-F1-Q3 · question page 14

LOGIC.SPATIAL.01 · Combine reflections and read mirrored clocks and words

A word is written DOWN the page, one block capital under another, and seen in a left-right mirror. In the block capitals used here, exactly these letters look the same in a left-right mirror: A H I M O T U V W X Y. Which word looks exactly the same?

Answer: A. MATH

Reasoning: Written downward, a left-right mirror keeps the order (top to bottom) and mirrors each letter, so the word looks the same when every letter is listed. Only MATH qualifies.

- ✗ MAZE: Z and E are not listed, so they appear reversed
- ✗ BATH: B is not listed
- ✗ MASK: S and K are not listed

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S1-F1-Q4 · question page 15

LOGIC.SPATIAL.01 · Combine reflections and read mirrored clocks and words

A 6 by 6 grid has two dashed mirror lines: a vertical one between columns 3 and 4 and a horizontal one between rows 3 and 4. The figure must look the same in BOTH mirrors. Four cells are already shaded in the top-left quarter: (1,1), (1,2), (2,2) and (3,3). Mark every shaded cell of the completed figure, including the four given cells.

Answer: 1,1, 1,2, 1,5, 1,6, 2,2, 2,5, 3,3, 3,4, 4,3, 4,4, 5,2, 5,5, 6,1, 6,2, 6,5, 6,6

Reasoning: Each given cell (r, c) also needs its images (r, 7 - c), (7 - r, c) and (7 - r, 7 - c). That gives 4 x 4 = 16 cells, one copy of the pattern in

each quarter.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: the key is the union of the four images of each given cell.

S1-F1-Q5 · question page 16

LOGIC.SPATIAL.01 · Combine reflections and read mirrored clocks and words

A figure looks the same after a reflection in a vertical line through its center and after a reflection in a horizontal line through its center. Then it also looks the same after a half-turn about its center.

Answer: True

Reasoning: Reflecting in the vertical line and then in the horizontal line sends (x, y) to (-x, y) and then to (-x, -y), which is a half-turn. Each reflection leaves the figure unchanged, so the half-turn does too.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S1-F2-Q1 · question page 17

LOGIC.SPATIAL.01 · Construct and count symmetric figures

The figure is on a 4 by 4 grid with a dashed vertical line between columns 2 and 3. Shaded cells can be added but not removed. What is the least number of cells you must add so that the figure looks the same in a mirror along the dashed line?

Answer: 5

Reasoning: For symmetry every shaded cell needs its image across the line (column c pairs with column 5 - c). Row 1: columns 1 and 2 need columns 4 and 3. Row 2: column 1 needs column 4; columns 2 and 3 are already each other's images. Row 3: column 1 needs column 4. Row 4: column 2 needs column 3. New cells: (1,3), (1,4), (2,4), (3,4), (4,3), so 5. Adding fewer leaves some shaded cell without its image.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S1-F2-Q2 · question page 18

LOGIC.SPATIAL.01 · Construct and count symmetric figures

Strings of 3 letters are made from the letters A, H, M, O and T (letters may repeat). Each is written left to right in block capitals. In the block capitals used here, exactly these letters look the same in a left-right mirror: A H I M O T U V W X Y. How many of these strings look exactly the same in a left-right mirror?

Answer: 25

Reasoning: All five letters are mirror-symmetric, so a string looks the same exactly when it reads the same backward: the first and last letters match. Choose the first letter (5 ways) and the middle letter (5 ways): 25.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S1-F2-Q3 · question page 19

LOGIC.SPATIAL.01 · Construct and count symmetric figures

Two vertical mirror lines are 3 cells apart: line 1 on the left, line 2 on the right. A shape is reflected in line 1 and the image is then reflected in line 2. Where does the final image end up?

Answer: D. the original shape moved 6 cells to the right, not flipped

Reasoning: Put line 1 at x = a and line 2 at x = b = a + 3. The first reflection sends x to 2a - x; the second sends that to 2b - (2a - x) = x + 2(b - a) = x + 6. Every point moves 6 to the right and nothing is flipped.

✗ the original shape moved 3 cells to the right, not flipped: two reflections move the shape twice the distance between the lines

✗ a mirror image of the shape, 6 cells to the right: two reflections undo each other's flip

✗ back exactly where the shape started: that only happens when both lines are the same line

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S1-F2-Q4 · question page 20

LOGIC.SPATIAL.01 · Construct and count symmetric figures

A row of 12 cells is numbered 1 to 12. A mirror line lies between cells 6 and 7. Give two cell numbers a and b, with a less than b, that are mirror images of each other.

Answer: any values that satisfy: a + b == 13 and a < b and a >= 1 and b <= 12 and isInteger(a) and isInteger(b) (example: a = 4, b = 9)

Reasoning: Cells that mirror each other are the same distance from the line, so their numbers add to 6 + 7 = 13: (1, 12), (2, 11), (3, 10), (4, 9), (5, 8) or (6, 7). Any one pair is correct.

Tempting answers the rule rejects: a = 4, b = 10; a = 6, b = 6; a = 9, b = 4

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 6 valid answers.

S1-F2-Q5 · question page 21

LOGIC.SPATIAL.01 · Construct and count symmetric figures

Count only mirror lines that run vertically, horizontally or diagonally through the center of the shape (the middle of the smallest rectangle around it). Which figure has exactly ONE mirror line?

Answer: A. Figure A

Reasoning: Testing each kind of line: A (a T shape) looks the same only in the vertical line. B looks the same after a half-turn but in no mirror; C and D each have four mirror lines.

- ✗ Figure B: no mirror line: it only looks the same after a half-turn
- ✗ Figure C: four mirror lines (vertical, horizontal and both diagonals)
- ✗ Figure D: four mirror lines (vertical, horizontal and both diagonals)

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the panel check accepts only the keyed panel.

S2-M-Q1 · question page 23

LOGIC.READING.01 · Read negations, units and relationships exactly as stated

Three students each make one statement about themself, and every statement is true. Dee: "It is not true that I finished the race." Eli: "I did not fail to finish the race." Fay: "I never said that I finished the race." Who definitely finished the race?

Answer: B. Eli only

Reasoning: Eli: "did not fail to finish" is a double negative, so Eli finished. Dee denies finishing. Fay's statement is about what she said, not about the race, so she may or may not have finished. Only Eli definitely did.

- ✗ Dee only: "It is not true that I finished" means Dee did not finish
- ✗ Eli and Fay: Fay only says what she never SAID; that tells us nothing about whether she finished
- ✗ Dee and Fay: Dee did not finish, and nothing is known about Fay

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S2-M-Q2 · question page 24

LOGIC.READING.01 · Read negations, units and relationships exactly as stated

Shoelaces are sold only in packs of 2 laces for $3 a pack. Each sneaker needs exactly one

lace. A club needs laces for 12 sneakers and also buys 4 whistles at $5 each. What is the least the club can spend, in dollars?

Answer: 38

Reasoning: 12 sneakers need 12 laces = 6 packs, costing 6 x $3 = $18. Whistles cost 4 x $5 = $20. Total $38. A common slip is to pay $3 per lace (12 x 3 = $36, total $56).

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S2-M-Q3 · question page 25

LOGIC.READING.01 · Read negations, units and relationships exactly as stated

In the Ortiz family, each son has exactly one brother, and each daughter has exactly two brothers and three sisters. How many children are in the family?

Answer: 6

Reasoning: A son's brothers are the other sons, so one brother means 2 sons. A daughter's sisters are the other daughters, so three sisters means 4 daughters, and each daughter's two brothers are the 2 sons. 2 + 4 = 6.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S2-M-Q4 · question page 26

LOGIC.READING.01 · Read negations, units and relationships exactly as stated

A box holds only red pens and blue pens: at least 3 red pens, at most 2 blue pens, and exactly 6 pens in all. Which statement MUST be true?

Answer: C. There are at least 4 red pens.

Reasoning: Blue is 0, 1 or 2, so red is 6, 5 or 4: always at least 4. Each

other statement fails in at least one of these three possible boxes.

- ✗ There are exactly 3 red pens.: then there would be 3 blue pens, but there are at most 2
- ✗ There are exactly 2 blue pens.: 1 or 0 blue pens also fit every condition
- ✗ There are no blue pens.: 1 or 2 blue pens also fit

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: all (red, blue) pairs enumerated: (4, 2), (5, 1), (6, 0). Automatic: the check function accepts only the keyed choice.

S2-M-Q5 · question page 27

LOGIC.READING.01 · Read negations, units and relationships exactly as stated

On this team, April, June and May are the names of three players (not months). A coach writes: "June scored after April and before May." Put the players in the order in which they scored, first to last.

Answer: April → June → May

Reasoning: The words are names here, so read the sentence literally: April, then June, then May.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 6 orderings of 3 items enumerated; 1 valid.

S2-B1-Q1 · question page 28

LOGIC.READING.01 · Strategy: rewrite each statement in plain form

Treat "unhappy" as meaning exactly "not happy". What does "Rui is not unhappy" say?

Answer: D. Rui is happy.

Reasoning: "Not unhappy" = not (not happy) = happy, under the stated meaning.

✗ Rui is unhappy.: the "not" cancels the "un-"

✗ Rui is sad and tired.: nothing is said about being tired

✗ Nothing at all about Rui.: it is a definite statement about Rui

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S2-B1-Q2 · question page 29

LOGIC.READING.01 · Strategy: rewrite each statement in plain form

A drawer holds 18 single socks that make complete matching pairs. How many pairs is that?

Answer: 9

Reasoning: A pair is 2 socks: 18 / 2 = 9.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S2-B1-Q3 · question page 30

LOGIC.READING.01 · Strategy: rewrite each statement in plain form

"At most 3 people may enter" allows exactly 3 people to enter.

Answer: True

Reasoning: "At most 3" means 3 or fewer, so 3 is allowed.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S2-B1-Q4 · question page 31

LOGIC.READING.01 · Strategy: rewrite each statement in plain form

"Not all of the doors are locked." Which statement means the same?

Answer: A. At least one door is unlocked.

Reasoning: "Not all are locked" is true exactly when at least one door is not locked.

- ✗ No door is locked.: that says more: some doors may still be locked
- ✗ Every door is unlocked.: that says more than the sentence
- ✗ Exactly one door is unlocked.: the sentence does not say how many

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S2-B1-Q5 · question page 32

LOGIC.READING.01 · Strategy: rewrite each statement in plain form

Lin has 3 brothers. Each of her brothers has exactly one sister. How many children are in the family?

Answer: 4

Reasoning: The brothers' one sister is Lin herself: 3 brothers + Lin = 4.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S2-F1-Q1 · question page 33

LOGIC.READING.01 · Read layered negations, units and family clues

Every statement below is true. Which person is going to the game? Gus: "It is not true that I am not going." Hal: "I am not not staying home." Ivy: "It is false that I won't go."

Answer: B. Gus and Ivy

Reasoning: Gus: not (not going) = going. Hal: not (not staying) = staying home. Ivy: false that (won't go) = will go.

✗ Gus only: Ivy's statement also cancels to "I will go"

✗ Hal and Ivy: Hal's "not not staying" means Hal is staying home

✗ all three: Hal is staying home

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S2-F1-Q2 · question page 34

LOGIC.READING.01 · Read layered negations, units and family clues

Eggs are sold in half-dozen boxes (6 eggs) for $2 a box. A recipe uses 4 eggs per cake, and a baker makes 9 cakes. What is the least she can spend on eggs, in dollars?

Answer: 12

Reasoning: 9 cakes x 4 eggs = 36 eggs = 6 boxes. 6 x $2 = $12.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S2-F1-Q3 · question page 35

LOGIC.READING.01 · Read layered negations, units and family clues

In the Novak family, each son has exactly 3 sisters and 2 brothers. Each daughter has exactly 2 sisters. How many children are in the family?

Answer: 6

Reasoning: A son has 2 brothers, so there are 3 sons. A son has 3 sisters, so there are 3 daughters; then each daughter has 2 sisters, as stated. 3 + 3 = 6.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S2-F1-Q4 · question page 36

LOGIC.READING.01 · Read layered negations, units and family clues

A bag holds only red and blue marbles: at least twice as many red as blue, at least 3 blue, and at most 10 marbles in all. Which statement MUST be true?

Answer: C. There are exactly 3 blue marbles.

Reasoning: If blue were 4, red would be at least 8, giving at least 12 marbles: too many. So blue is exactly 3, and red is 6 or 7.

- ✗ There are exactly 6 red marbles.: 7 red and 3 blue also fit
- ✗ There are exactly 10 marbles.: 6 red and 3 blue (9 marbles) also fit
- ✗ There are more blue marbles than red ones.: red is at least twice blue

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: every (red, blue) pair up to 10 enumerated: (6, 3), (7, 3). Automatic: the check function accepts only the keyed choice.

S2-F1-Q5 · question page 37

LOGIC.READING.01 · Read layered negations, units and family clues

"Only members may vote" means that everyone who votes is a member.

Answer: True

Reasoning: "Only members may vote": if someone votes, that person is a member. It does not say every member votes.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S2-F2-Q1 · question page 38

LOGIC.READING.01 · Precise reading in multi-step puzzles

Four statements: (1) "It is not the case that 7 is not odd." (2) "It is false that 10 is not even."

(3) "It is not true that 9 is not a multiple of 4." (4) "5 is not unequal to 5." How many of the four statements are true?

Answer: 3

Reasoning: (1) 7 is odd: true. (2) 10 is even: true. (3) says 9 IS a multiple of 4: false. (4) "not unequal" means equal: 5 = 5, true. Three are true.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S2-F2-Q2 · question page 39

LOGIC.READING.01 · Precise reading in multi-step puzzles

On a quiz show, Monday, Tuesday, Wednesday and Friday are the first names of four contestants. The host says: "Wednesday answered before Monday. Friday answered right after Tuesday. Monday did not answer last." Put the contestants in answering order, first to last.

Answer: Wednesday → Monday → Tuesday → Friday

Reasoning: The words are names. Tuesday and Friday form a block (Tuesday, then Friday). With Wednesday before Monday, the possible orders are: block, Wednesday, Monday; Wednesday, block, Monday; Wednesday, Monday, block. Monday is not last, so only Wednesday, Monday, Tuesday, Friday remains.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S2-F2-Q3 · question page 40

LOGIC.READING.01 · Precise reading in multi-step puzzles

A bag holds only red and blue marbles: the number of red marbles is at least twice the number of blue, there are at least 3 blue marbles, and there are at most 10 marbles in all. Give a possible number of red marbles r and blue marbles b.

Answer: any values that satisfy: r >= 2*b and b >= 3 and r + b <= 10 and isInteger(r) and isInteger(b) (example: r = 6, b = 3)

Reasoning: Blue must be 3 (with 4 blue, red is at least 8, too many in all). Red can be 6 or 7. Either (6, 3) or (7, 3) is correct.

Tempting answers the rule rejects: r = 5, b = 3; r = 8, b = 4; r = 8, b = 2

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 2 valid answers.

S2-F2-Q4 · question page 41

LOGIC.READING.01 · Precise reading in multi-step puzzles

In a group of brothers and sisters, every girl has exactly as many brothers as sisters, and there are 3 boys. How many girls are there?

Answer: 4

Reasoning: A girl's brothers are the 3 boys; her sisters are the other girls. So girls - 1 = 3, and there are 4 girls.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S2-F2-Q5 · question page 42

LOGIC.READING.01 · Precise reading in multi-step puzzles

"No student who did not study passed the test." Which statement follows?

Answer: A. Every student who passed studied.

Reasoning: The sentence says: if a student did not study, the student did not pass. Equivalently, if a student passed, the student studied (the

contrapositive).

- ✗ Every student who studied passed.: that reverses the condition
- ✗ Some students who studied failed.: nothing is said about students who studied
- ✗ No student passed.: students who studied may have passed

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-M-Q1 · question page 44

LOGIC.ARGUMENT.01 · Identify statements, premises, conclusions and assumptions

"The trail is closed whenever it rains. It is raining now. So the trail is closed." Which sentence is the conclusion?

Answer: D. The trail is closed.

Reasoning: "So" introduces the conclusion: the trail is closed. The other two sentences are the reasons (premises).

- ✗ It is raining now.: this is given as a reason (a premise)
- ✗ The trail is closed whenever it rains.: this is a general premise
- ✗ Rain is likely tomorrow.: this sentence is not in the argument

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-M-Q2 · question page 45

LOGIC.ARGUMENT.01 · Identify statements, premises, conclusions and assumptions

Which of these is a statement (a sentence that is either true or false)?

Answer: B. Every square has four sides.

Reasoning: A statement makes a claim that is true or false. "Every

square has four sides" is a (true) statement.

- ✗ Please close the window.: a request is neither true nor false
- ✗ Is 15 a prime number?: a question is neither true nor false
- ✗ What a great game!: an exclamation of feeling is not a claim that is true or false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-M-Q3 · question page 46

LOGIC.ARGUMENT.01 · Identify statements, premises, conclusions and assumptions

"Since every ticket holder may enter, and Zoe holds a ticket, Zoe may enter." Which sentence is a premise?

Answer: C. Zoe holds a ticket.

Reasoning: "Since" introduces the reasons: every ticket holder may enter, and Zoe holds a ticket. The conclusion is that Zoe may enter.

- ✗ Zoe may enter.: this is the conclusion the argument reaches
- ✗ Zoe bought the ticket online.: the argument does not say this
- ✗ Only Zoe may enter.: the argument does not say this

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-M-Q4 · question page 47

LOGIC.ARGUMENT.01 · Identify statements, premises, conclusions and assumptions

"Ravi will be on time, because the bus left on schedule." Which unstated assumption does this argument need?

Answer: D. Ravi is on that bus.

Reasoning: The reason (the bus left on schedule) supports the

conclusion only if Ravi is riding it. That missing link is the assumption.

- ✗ The bus is red.: the color is irrelevant to the conclusion
- ✗ Ravi was late yesterday.: that would not help the conclusion
- ✗ Some buses are late.: that does not connect the bus to Ravi

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-M-Q5 · question page 48

LOGIC.ARGUMENT.01 · Identify statements, premises, conclusions and assumptions

"All of our club's robots passed inspection. Unit 7 is one of our club's robots. The room is cold. Therefore Unit 7 passed inspection." Which sentence plays no part in the reasoning?

Answer: A. The room is cold.

Reasoning: The conclusion follows from the first two sentences. The temperature of the room is irrelevant.

- ✗ Unit 7 is one of our club's robots.: this premise connects Unit 7 to the group
- ✗ All of our club's robots passed inspection.: this premise gives the general rule
- ✗ Therefore Unit 7 passed inspection.: this is the conclusion

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-B1-Q1 · question page 49

LOGIC.ARGUMENT.01 · Which sentences are statements?

"Close the door." is a statement.

Answer: False

Reasoning: It is a command; it is neither true nor false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S3-B1-Q2 · question page 50

LOGIC.ARGUMENT.01 · Which sentences are statements?

"7 is greater than 9." is a statement.

Answer: True

Reasoning: It is false, but it is still a statement: it makes a claim that is true or false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S3-B1-Q3 · question page 51

LOGIC.ARGUMENT.01 · Which sentences are statements?

Which is NOT a statement?

Answer: B. Who won the game?

Reasoning: A question asks; it does not claim.

- ✗ The game ended in a tie.: a claim that is true or false
- ✗ Nobody scored.: a claim that is true or false
- ✗ Our team won.: a claim that is true or false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-B1-Q4 · question page 52

LOGIC.ARGUMENT.01 · Which sentences are statements?

"x + 2 = 5" with no value of x given is a statement.

Answer: False

Reasoning: Its truth depends on x, which is not given; it becomes a statement once x is fixed (an open sentence).

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S3-B1-Q5 · question page 53

LOGIC.ARGUMENT.01 · Which sentences are statements?

Which sentence is a statement?

Answer: C. Paris is in Spain.

Reasoning: It is a false claim, but a claim: a statement.

- ✗ Let's go to Paris!: a suggestion
- ✗ Is Paris far?: a question
- ✗ Visit Paris.: a command

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-B2-Q1 · question page 54

LOGIC.ARGUMENT.01 · Find the conclusion by its signal words

Which word usually introduces a conclusion?

Answer: B. therefore

Reasoning: "Therefore" signals the conclusion.

- ✗ because: introduces a reason
- ✗ since: introduces a reason

✗ given that: introduces a reason

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-B2-Q2 · question page 55

LOGIC.ARGUMENT.01 · Find the conclusion by its signal words

Which word usually introduces a reason (a premise)?

Answer: C. because

Reasoning: "Because" signals a reason.

✗ so: introduces a conclusion
✗ thus: introduces a conclusion
✗ consequently: introduces a conclusion

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-B2-Q3 · question page 56

LOGIC.ARGUMENT.01 · Find the conclusion by its signal words

"The lake is frozen, so we can skate." What is the conclusion?

Answer: D. We can skate.

Reasoning: "So" introduces the conclusion.

✗ The lake is frozen.: the reason
✗ It is winter.: not stated
✗ Skating is fun.: not stated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S3-B2-Q4 · question page 57

LOGIC.ARGUMENT.01 · Find the conclusion by its signal words

"We should leave now, since the show starts at 7." What is the conclusion?

Answer: A. We should leave now.

Reasoning: The claim supported by the "since" reason is the conclusion.

- ✗ The show starts at 7.: the reason, after "since"
- ✗ The show is long.: not stated
- ✗ We are late.: not stated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S3-B2-Q5 · question page 58

LOGIC.ARGUMENT.01 · Find the conclusion by its signal words

In "Because the battery is dead, the car will not start," the conclusion is "the battery is dead."

Answer: False

Reasoning: "Because" marks the reason; the conclusion is that the car will not start.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S3-F1-Q1 · question page 59

LOGIC.ARGUMENT.01 · Arguments with several premises and hidden structure

"The museum must be open today. After all, it is open every weekday except Monday, and today is Thursday." Which sentence is the conclusion?

Answer: A. The museum must be open today.

Reasoning: The first sentence is supported by the other two ("after all" introduces reasons).

- ✗ Today is Thursday.: a premise
- ✗ It is open every weekday except Monday.: a premise
- ✗ The museum is closed on Monday.: part of a premise, not the claim being supported

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-F1-Q2 · question page 60

LOGIC.ARGUMENT.01 · Arguments with several premises and hidden structure

Put these sentences in the order of a clear argument: the general rule first, then the fact, then the conclusion.

Answer: Any whole number greater than 1 with a factor other than 1 and itself is not prime. → 91 = 7 x 13, so 91 has the factor 7. → Therefore, 91 is not prime.

Reasoning: Rule (B), fact about 91 (C), then the conclusion (A).

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 6 orderings of 3 items enumerated; 1 valid.

S3-F1-Q3 · question page 61

LOGIC.ARGUMENT.01 · Arguments with several premises and hidden structure

"Coach Lee will pick Maya, because Coach Lee always picks the fastest runner." Which assumption is needed?

Answer: D. Maya is the fastest runner.

Reasoning: The rule talks about the fastest runner; the conclusion needs Maya to be that runner.

✗ Maya wants to be picked.: not needed for the conclusion
✗ Coach Lee likes Maya.: not what the argument relies on
✗ Maya ran yesterday.: irrelevant to being the fastest

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-F1-Q4 · question page 62

LOGIC.ARGUMENT.01 · Arguments with several premises and hidden structure

An argument can have a true conclusion even when one of its premises is false.

Answer: True

Reasoning: For example: "All fish can fly; a pigeon is a fish; so a pigeon can fly." The premises are false, the conclusion is true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S3-F1-Q5 · question page 63

LOGIC.ARGUMENT.01 · Arguments with several premises and hidden structure

"If the alarm is set, it rings at 6. It did not ring at 6. Hence it was not set." How many premises does this argument have?

Answer: B. 2

Reasoning: Two premises (the conditional and the fact that it did not ring), and one conclusion after "hence".

✗ 1: there are two reasons before "hence"
✗ 3: the third sentence is the conclusion
✗ 0: the first two sentences are reasons

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-F2-Q1 · question page 64

LOGIC.ARGUMENT.01 · Evaluate structure and missing links

"All sprinters on the team train daily. So Kim trains daily." Which added premise makes the conclusion follow for certain?

Answer: C. Kim is a sprinter on the team.

Reasoning: The general premise is about sprinters on the team; Kim must be one of them.

- ✗ Kim trains often.: does not make Kim a sprinter on the team
- ✗ Some sprinters train daily.: the conclusion is about Kim, who is still not linked to the group
- ✗ Kim is on the team.: Kim might be on the team but not a sprinter

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-F2-Q2 · question page 65

LOGIC.ARGUMENT.01 · Evaluate structure and missing links

"Every number divisible by 6 is even. So 84 is even." Which added premise makes this valid?

Answer: D. 84 is divisible by 6.

Reasoning: The rule applies to numbers divisible by 6, so we need 84 to be one: 84 = 6 x 14.

- ✗ 84 is divisible by 4.: true, but the premise is about divisibility by 6
- ✗ 84 is greater than 6.: size does not matter
- ✗ Some even numbers are divisible by 6.: still says nothing about 84

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S3-F2-Q3 · question page 66

LOGIC.ARGUMENT.01 · Evaluate structure and missing links

In a valid argument, adding an extra true premise can make the conclusion no longer follow.

Answer: False

Reasoning: If the conclusion follows from some premises, it still follows when more premises are added.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S3-F2-Q4 · question page 67

LOGIC.ARGUMENT.01 · Evaluate structure and missing links

Arrange these sentences into an argument that ends with its conclusion: chain the two rules.

Answer: The dam opens. → If the dam opens, the river rises. → If the river rises, the ferry stops. → So the ferry stops.

Also accepted: If the dam opens, the river rises. → If the river rises, the ferry stops. → The dam opens. → So the ferry stops. | If the dam opens, the river rises. → The dam opens. → If the river rises, the ferry stops. → So the ferry stops.

Reasoning: State the first rule and the fact, then the second rule, then the conclusion. Orders that state both rules first are also accepted.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 3 valid.

S3-F2-Q5 · question page 68

LOGIC.ARGUMENT.01 · Evaluate structure and missing links

"Our new ads worked: sales rose this month." Which assumption is the argument making?

Answer: B. Nothing other than the ads caused most of the rise in sales.

Reasoning: Claiming the ads "worked" assumes the rise was due to the ads and not to something else.

- ✗ Sales rose this month.: that is the stated premise, not an assumption
- ✗ The ads were expensive.: cost is irrelevant to whether they worked
- ✗ Sales will rise next month.: the argument is about this month

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S4-M-Q1 · question page 70

LOGIC.CONSTRAINT.01 · Find the only candidate left after every clue

Which month is it? Its name has fewer than 8 letters. Its name contains the letter R. It is the 1st, 3rd, 5th, 7th, 9th or 11th month. Its name does not start with J.

Answer: A. March

Reasoning: Fewer than 8 letters and an R: January, March, April, October. Odd-numbered month (1st, 3rd, ...): January, March. Not starting with J: March.

- ✗ January: starts with J
- ✗ May: has no R
- ✗ September: has 9 letters

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the four clues applied to all 12 months leave only March. Automatic: the check function accepts only the keyed choice.

S4-M-Q2 · question page 71

LOGIC.CONSTRAINT.01 · Find the only candidate left after every clue

Which day of the week is it? Its name has no letter E. It comes after Wednesday in the week that starts on Monday. Its name has exactly 6 letters. Its name contains the letter U.

Answer: C. Sunday

Reasoning: After Wednesday: Thursday, Friday, Saturday, Sunday (none has an E). Exactly 6 letters: Friday, Sunday. Contains a U: Sunday.

- ✗ Friday: fits the first three clues but has no letter U
- ✗ Saturday: has 8 letters
- ✗ Thursday: has 8 letters

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the four clues applied to all 7 days leave only Sunday. Automatic: the check function accepts only the keyed choice.

S4-M-Q3 · question page 72

LOGIC.CONSTRAINT.01 · Find the only candidate left after every clue

A whole number from 1 to 50 leaves remainder 3 when divided by 7. It is greater than 20. It is even. It does not leave remainder 4 when divided by 5. What is the number?

Answer: 38

Reasoning: Remainder 3 mod 7: 3, 10, 17, 24, 31, 38, 45. Greater than 20: 24, 31, 38, 45. Even: 24, 38. 24 leaves remainder 4 when divided by 5, so the number is 38.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S4-M-Q4 · question page 73

LOGIC.CONSTRAINT.01 · Find the only candidate left after every clue

Three friends, Ana, Ben and Cai, each have exactly one pet: a cat, a dog or a fish, all different. Ana does not have the dog. Ben has neither the cat nor the dog. Which pet does

each friend have?

Answer: Ana: cat; Ben: fish; Cai: dog

Reasoning: Ben has neither cat nor dog, so Ben has the fish. Ana does not have the dog, so Ana has the cat, and Cai has the dog.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 6 assignments enumerated; 1 satisfy every clue.

S4-M-Q5 · question page 74

LOGIC.CONSTRAINT.01 · Find the only candidate left after every clue

A locker number is one of 12, 15, 18, 21, 24, 27. It is even. It is not a multiple of 4. Which locker is it?

Answer: D. 18

Reasoning: Even: 12, 18, 24. Not a multiple of 4: 18.

- ✗ 27: 27 is odd, so it is not even
- ✗ 12: a multiple of 4
- ✗ 24: a multiple of 4

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S4-B1-Q1 · question page 75

LOGIC.CONSTRAINT.01 · Strategy: cross out, one clue at a time

A number from 1 to 10 is even, greater than 7 and less than 10. What is it?

Answer: 8

Reasoning: Even: 2, 4, 6, 8, 10. Greater than 7: 8, 10. Less than 10: 8.

Grading: The game's answer checker compares the typed value with the key (equivalent

number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S4-B1-Q2 · question page 76

LOGIC.CONSTRAINT.01 · Strategy: cross out, one clue at a time

A day of the week (Monday to Sunday) starts with T and has more than 7 letters. Which day?

Answer: B. Thursday

Reasoning: T-days: Tuesday (7), Thursday (8).

- ✗ Tuesday: has exactly 7 letters
- ✗ Saturday: does not start with T
- ✗ Wednesday: does not start with T

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S4-B1-Q3 · question page 77

LOGIC.CONSTRAINT.01 · Strategy: cross out, one clue at a time

How many months have a name with exactly 4 letters?

Answer: 2

Reasoning: June and July.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S4-B1-Q4 · question page 78

LOGIC.CONSTRAINT.01 · Strategy: cross out, one clue at a time

If two candidates are left after every clue, the clues determine the answer.

Answer: False

Reasoning: Two candidates fit every clue, so the answer is not determined.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S4-B1-Q5 · question page 79

LOGIC.CONSTRAINT.01 · Strategy: cross out, one clue at a time

Candidates: 2, 3, 4, 5, 6. Clue 1: the number is prime. Clue 2: the number is odd. How many candidates are left?

Answer: C. 2

Reasoning: Primes: 2, 3, 5. Odd: 3, 5.

- ✗ 1: 3 and 5 are both odd primes
- ✗ 3: 2 is prime but even
- ✗ 0: 3 and 5 fit

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S4-F1-Q1 · question page 80

LOGIC.CONSTRAINT.01 · Eliminate with clues about neighbors and positions

Which month is it? The month before it has a name longer than 5 letters. The month after it does not end in "ber". The month after it contains the letter U. Its own name contains the letter A.

Answer: A. January

Reasoning: Months whose previous month has more than 5 letters:

January, February, March, September, October, November, December.
Next month not ending in "ber": January, February, March, December.
Next month containing U: January (February) and December (January).
Own name with an A: January.

- ✗ December: fits the three neighbor clues but has no letter A
- ✗ February: the month after it, March, has no letter U
- ✗ May: the month before it, April, has only 5 letters

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: all 12 months tested against the four neighbor clues. Automatic: the check function accepts only the keyed choice.

S4-F1-Q2 · question page 81

LOGIC.CONSTRAINT.01 · Eliminate with clues about neighbors and positions

In the week Monday to Sunday, exactly one day has a name whose day before and day after both contain the letter T.

Answer: False

Reasoning: Two days qualify: Wednesday (Tuesday and Thursday) and Friday (Thursday and Saturday).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S4-F1-Q3 · question page 82

LOGIC.CONSTRAINT.01 · Eliminate with clues about neighbors and positions

A whole number from 1 to 100 is a perfect square, is odd, has two digits, and its digits add to 13. What is it?

Answer: 49

Reasoning: Odd two-digit squares: 25, 49, 81. Digit sums 7, 13, 9. So 49.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S4-F1-Q4 · question page 83

LOGIC.CONSTRAINT.01 · Eliminate with clues about neighbors and positions

Four runners, Dev, Eva, Finn and Gia, finished 1st to 4th, no ties. Eva finished right after Dev. Gia finished before Finn. Finn finished before Dev. Assign each runner a place.

Answer: Dev: 3rd; Eva: 4th; Finn: 2nd; Gia: 1st

Reasoning: Gia is before Finn and Finn before Dev, so Dev is at least 3rd; Eva is right after Dev, so Dev is 3rd and Eva 4th. Then Gia is 1st and Finn 2nd.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 24 assignments enumerated; 1 satisfy every clue.

S4-F1-Q5 · question page 84

LOGIC.CONSTRAINT.01 · Eliminate with clues about neighbors and positions

Candidates are the whole numbers 1 to 30. Clues: A: the number is odd. B: it is a multiple of 3. C: it is greater than 10. D: it is a multiple of 5. Which number fits all four clues?

Answer: B. 15

Reasoning: Odd multiples of 15 up to 30: 15 only.

- ✗ 30: even, so it fails clue A
- ✗ 5: not a multiple of 3 and not greater than 10
- ✗ 25: not a multiple of 3

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S4-F2-Q1 · question page 85

LOGIC.CONSTRAINT.01 · Which clues are really needed?

Candidates are the whole numbers 1 to 30, with four clues: A: odd; B: a multiple of 3; C: greater than 10; D: a multiple of 5. Only 15 fits all four. Which ONE clue can be removed so that exactly one number is still left?

Answer: A. C (greater than 10)

Reasoning: Remove each clue in turn and recount. Without C the clues say: odd, multiple of 3 and multiple of 5, that is an odd multiple of 15, and the only one up to 30 is 15. Removing any other clue leaves two or more numbers.

- ✗ B (multiple of 3): without B, 15 and 25 both fit
- ✗ A (odd): without A, 15 and 30 both fit
- ✗ D (multiple of 5): without D, 15, 21 and 27 fit

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: each clue removed in turn and the survivors recounted. Automatic: the check function accepts only the keyed choice.

S4-F2-Q2 · question page 86

LOGIC.CONSTRAINT.01 · Which clues are really needed?

A number from 1 to 30 is a multiple of 4. What is the least number of extra yes/no questions of the form "Is it greater than k?" that always identify it?

Answer: 3

Reasoning: Multiples of 4 up to 30: 4, 8, 12, 16, 20, 24, 28 (7 candidates). Each question at best halves the list, so 2 questions separate at most 4; 3 questions separate up to 8, and halving 7 -> 4 -> 2 -> 1 works.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S4-F2-Q3 · question page 87

LOGIC.CONSTRAINT.01 · Which clues are really needed?

Adding a new clue to a puzzle can never increase the number of candidates left.

Answer: True

Reasoning: Each clue only removes candidates; the survivors of more clues are a subset of the survivors of fewer.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S4-F2-Q4 · question page 88

LOGIC.CONSTRAINT.01 · Which clues are really needed?

Candidates are the whole numbers 1 to 20. Write a number n that survives BOTH clues: n is prime, and n leaves remainder 1 when divided by 4.

Answer: any values that satisfy: isPrime(n) and n % 4 == 1 and n >= 1 and n <= 20 (example: n = 5)

Reasoning: Primes up to 20 leaving remainder 1 mod 4: 5, 13, 17. Any one is correct.

Tempting answers the rule rejects: n = 9; n = 7; n = 21

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 3 valid answers.

S4-F2-Q5 · question page 89

LOGIC.CONSTRAINT.01 · Which clues are really needed?

Six boxes are numbered 1 to 6. One holds a prize. Which single clue, added alone, would leave exactly one box?

Answer: D. "The box number is prime and even."

Reasoning: The only even prime is 2.

- ✗ "The box number is odd.": leaves 1, 3, 5
- ✗ "The box number is less than 4.": leaves 1, 2, 3
- ✗ "The box number is a multiple of 3.": leaves 3 and 6

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S5-M-Q1 · question page 91

LOGIC.CONNECTIVE.01 · Form negations and find truth values

What is the negation of "Sam is taller than Ana"?

Answer: C. Sam is not taller than Ana.

Reasoning: A negation is true exactly when the original is false. "Sam is taller" is false when Sam is shorter OR the same height; only "Sam is not taller than Ana" covers both.

- ✗ Ana is taller than Sam.: this leaves out the case where they are the same height
- ✗ Sam is shorter than Ana.: this also leaves out equal heights
- ✗ Sam and Ana are the same height.: this leaves out the case where Ana is taller

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S5-M-Q2 · question page 92

LOGIC.CONNECTIVE.01 · Form negations and find truth values

x stands for a real number. What is the negation of "x > 5"?

Answer: A. x ≤ 5

Reasoning: "x > 5" is false exactly when x is 5 or less: x ≤ 5.

- ✗ x < 5: misses x = 5, where "x > 5" is false
- ✗ x ≥ 5: includes values like 6 that make "x > 5" true
- ✗ x < -5: misses every x from -5 to 5

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S5-M-Q3 · question page 93

LOGIC.CONNECTIVE.01 · Form negations and find truth values

P is the statement "48 is a multiple of 9." The negation of P is true.

Answer: True

Reasoning: 48 = 9 x 5 + 3, so P is false and its negation, "48 is not a multiple of 9", is true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S5-M-Q4 · question page 94

LOGIC.CONNECTIVE.01 · Form negations and find truth values

Give a number x that makes the NEGATION of the statement "x ≤ 4" true.

Answer: any values that satisfy: x > 4 (example: x = 7)

Reasoning: The negation of "x ≤ 4" is "x > 4", so any number greater than 4 is correct.

Tempting answers the rule rejects: x = 4; x = 3; x = -5

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S5-M-Q5 · question page 95

LOGIC.CONNECTIVE.01 · Form negations and find truth values

Which pair of statements are negations of each other (in every situation, exactly one of the two is true)?

Answer: C. "The coin landed heads up." / "The coin did not land heads up."

Reasoning: Negations can never be both false or both true. Each other pair has a situation where both are false: they are opposites (contraries), not negations.

- ✗ "The cup is full." / "The cup is empty.": a half-full cup makes both false
- ✗ "Every student passed." / "Every student failed.": if some passed and some failed, both are false
- ✗ "x > 3" / "x < 3": x = 3 makes both false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S5-B1-Q1 · question page 96

LOGIC.ARGUMENT.01 · Truth values of simple statements

"17 is a prime number" is true.

Answer: True

Reasoning: 17 has no factors other than 1 and itself.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S5-B1-Q2 · question page 97

LOGIC.ARGUMENT.01 · Truth values of simple statements

"3 x 7 = 24" is true.

Answer: False

Reasoning: 3 x 7 = 21.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S5-B1-Q3 · question page 98

LOGIC.ARGUMENT.01 · Truth values of simple statements

"Every even number is a multiple of 4" is true.

Answer: False

Reasoning: 6 is even but not a multiple of 4, so the statement is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S5-B1-Q4 · question page 99

LOGIC.ARGUMENT.01 · Truth values of simple statements

Which statement is false?

Answer: D. 0.5 is greater than 0.75.

Reasoning: 0.5 < 0.75.

✗ 0.5 equals one half.: this is true: 0.5 = 1/2

✗ 0.75 is greater than 0.5.: this is true: 75 hundredths > 50 hundredths
✗ 0.5 + 0.25 = 0.75.: this is true: 50 + 25 = 75 hundredths

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S5-B1-Q5 · question page 100

LOGIC.ARGUMENT.01 · Truth values of simple statements

"10 is not an even number" is true.

Answer: False

Reasoning: 10 = 2 x 5 is even, so "10 is not even" is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S5-B2-Q1 · question page 101

LOGIC.CONNECTIVE.01 · Negation is not the same as the opposite

What is the negation of "The door is open"?

Answer: B. The door is not open.

Reasoning: Put "not" on the whole claim.

✗ The door is locked.: a door can be closed without being locked
✗ The window is open.: that is about the window
✗ The door is wide open.: that is not a negation

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S5-B2-Q2 · question page 102

LOGIC.CONNECTIVE.01 · Negation is not the same as the opposite

"The water is cold" and "The water is hot" are negations of each other.

Answer: False

Reasoning: Warm water makes both false; they are opposites, not negations.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S5-B2-Q3 · question page 103

LOGIC.CONNECTIVE.01 · Negation is not the same as the opposite

What is the negation of "Tom won the race"?

Answer: A. Tom did not win the race.

Reasoning: Negate the whole statement.

- ✗ Tom lost the race.: he may not have raced at all
- ✗ Tom came last.: too specific
- ✗ Someone else won.: there may have been no winner

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S5-B2-Q4 · question page 104

LOGIC.CONNECTIVE.01 · Negation is not the same as the opposite

If a statement is true, its negation is also true.

Answer: False

Reasoning: A statement and its negation always have opposite truth values.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S5-B2-Q5 · question page 105

LOGIC.CONNECTIVE.01 · Negation is not the same as the opposite

n is a whole number. What is the negation of "n is even"?

Answer: C. n is odd.

Reasoning: For whole numbers, "not even" means exactly "odd".

- ✗ n is 0.: too specific
- ✗ n is negative.: unrelated
- ✗ n is prime.: unrelated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S5-F1-Q1 · question page 106

LOGIC.CONNECTIVE.01 · Negate ranges, counts and double negatives

x stands for a real number. What is the negation of "$3 < x < 7$" (x is between 3 and 7)?

Answer: D. $x \leq 3$ or $x \geq 7$

Reasoning: x is not strictly between 3 and 7 exactly when it is at most 3 or at least 7.

- ✗ $x < 3$ or $x > 7$: misses $x = 3$ and $x = 7$
- ✗ $3 \geq x \geq 7$: no number satisfies this
- ✗ $x \leq 3$ and $x \geq 7$: no number satisfies this

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S5-F1-Q2 · question page 107

LOGIC.CONNECTIVE.01 · Negate ranges, counts and double negatives

Exactly 9 lamps are in a room. What is the negation of "At least 5 lamps are on"?

Answer: B. At most 4 lamps are on.

Reasoning: With whole numbers 0 to 9, "at least 5 on" is false exactly when 0 to 4 are on: at most 4.

- ✗ At most 5 lamps are on.: exactly 5 lamps on makes both statements true
- ✗ No lamps are on.: too strong: 1 to 4 lamps on also makes the original false
- ✗ At least 4 lamps are off.: with 5 on and 4 off, the original and this statement are both true

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S5-F1-Q3 · question page 108

LOGIC.CONNECTIVE.01 · Negate ranges, counts and double negatives

Complete the truth table: the columns are "not P" and "not (not P)".

Answer: F T / T F

Reasoning: "not P" flips each value; flipping again gives back P.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (2 rows).

S5-F1-Q4 · question page 109

LOGIC.CONNECTIVE.01 · Negate ranges, counts and double negatives

How many of these are true? (1) "It is not true that 8 is odd." (2) "It is false that 0 is not a whole number." (3) "It is not the case that 2 + 2 is not 4." (4) "It is not true that 5 is not less

than 3."

Answer: 3

Reasoning: (1) 8 is not odd: true. (2) says 0 IS a whole number: true. (3) says 2 + 2 is 4: true. (4) says 5 IS less than 3: false. Three are true.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S5-F1-Q5 · question page 110

LOGIC.CONNECTIVE.01 · Negate ranges, counts and double negatives

The negation of "Mia always walks to school" is "Mia never walks to school".

Answer: False

Reasoning: If Mia walks on some days and not others, both are false. The negation is "Mia does not always walk to school": on at least one day she does not walk.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S5-F2-Q1 · question page 111

LOGIC.CONNECTIVE.01 · Negation inside conditions and counts

Give a number x that makes the negation of "x < 2 or x > 5" true.

Answer: any values that satisfy: x >= 2 and x <= 5 (example: x = 3)

Reasoning: "x < 2 or x > 5" is false exactly when x is not less than 2 AND not greater than 5: 2 ≤ x ≤ 5.

Tempting answers the rule rejects: x = 1; x = 6; x = 5.5

Grading: The learner's values are put into the rule; ANY values that make the rule true are

correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S5-F2-Q2 · question page 112

LOGIC.CONNECTIVE.01 · Negation inside conditions and counts

There are 6 lights. What is the negation of "At most 2 lights are on"?

Answer: C. At least 3 lights are on.

Reasoning: "At most 2" is false exactly when 3 or more are on.

- ✗ At least 2 lights are on.: exactly 2 on makes both true
- ✗ Exactly 3 lights are on.: too narrow: 4, 5 or 6 on also make the original false
- ✗ At most 2 lights are off.: 3 lights on and 3 off makes both false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S5-F2-Q3 · question page 113

LOGIC.CONNECTIVE.01 · Negation inside conditions and counts

Ana says: "It is false that it is not true that the test is not on Friday." What is Ana saying?

Answer: D. The test is not on Friday.

Reasoning: Count: "false that" (1), "not true that" (2), "not on Friday" (3). Two negations cancel, leaving one: the test is not on Friday.

- ✗ The test is on Friday.: count the negations: there are three
- ✗ Nothing about the test.: it is a definite claim
- ✗ The test might be on Friday.: the statement is definite

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S5-F2-Q4 · question page 114

LOGIC.CONNECTIVE.01 · Negation inside conditions and counts

For every statement P, exactly one of P and "not P" is true.

Answer: True

Reasoning: That is what a negation means: it is true exactly when P is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S5-F2-Q5 · question page 115

LOGIC.CONNECTIVE.01 · Negation inside conditions and counts

Statements about a whole number n from 1 to 10: A: "n is even." B: "n is not greater than 6." C: "It is not true that n is a multiple of 3." For how many values of n are all three statements true?

Answer: 2

Reasoning: Even and at most 6: 2, 4, 6. Not a multiple of 3: 2, 4. Two values.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-M-Q1 · question page 117

LOGIC.CODE.01 · Decode and encode with a stated key, and find a key within a stated family

In a code, each letter has the value of its position in the alphabet (A = 1, B = 2, ..., Z = 26), and a word's value is the sum of its letters' values. For example, CAB = 3 + 1 + 2 = 6. What

is the value of BEAD?

Answer: 12

Reasoning: B = 2, E = 5, A = 1, D = 4: 2 + 5 + 1 + 4 = 12.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-M-Q2 · question page 118

LOGIC.CODE.01 · Decode and encode with a stated key, and find a key within a stated family

In a code, every letter is replaced by the letter 3 places later in the alphabet (A becomes D, B becomes E, and X, Y, Z wrap around to A, B, C). A coded word reads FDW. What was the original word?

Answer: B. CAT

Reasoning: Decoding moves each letter 3 places back: F -> C, D -> A, W -> T.

- ✗ IGZ: that shifts 3 more places forward instead of back
- ✗ CAW: only the first two letters were decoded
- ✗ DBX: each letter was moved back only 2 places

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S6-M-Q3 · question page 119

LOGIC.CODE.01 · Decode and encode with a stated key, and find a key within a stated family

A code adds the same whole number k to every digit and keeps only the last digit of each sum (so 8 + 3 gives 1). The code turns 2025 into 4247. What does it turn 1990 into?

Answer: 3112

Reasoning: From 2 -> 4, k = 2 (every digit agrees: 0 -> 2, 2 -> 4, 5 -> 7). Then 1 -> 3, 9 -> 1, 9 -> 1, 0 -> 2: 3112.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-M-Q4 · question page 120

LOGIC.CODE.01 · Decode and encode with a stated key, and find a key within a stated family

Each symbol stands for a whole number. Star + Star = 10. Star + Circle = 8. Circle + Square = 9. What number is Square?

Answer: 6

Reasoning: Star = 5, so Circle = 3, so Square = 6.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-M-Q5 · question page 121

LOGIC.CODE.01 · Decode and encode with a stated key, and find a key within a stated family

Two 3-letter words were interleaved: first letter of word 1, first letter of word 2, second letter of word 1, and so on. The result is CDAORT. Which two words were interleaved?

Answer: A. CAR and DOT

Reasoning: Letters 1, 3, 5 (C, A, R) form word 1 and letters 2, 4, 6 (D, O, T) form word 2.

✗ COD and ART: interleaving COD and ART gives CAORDT

✗ CAT and DOR: interleaving gives CDAOTR
✗ CAD and ORT: interleaving gives COARDT

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S6-B1-Q1 · question page 122

LOGIC.CODE.01 · Strategy: write the key as a table

A code replaces each letter by its position in the alphabet (A = 1, ..., Z = 26) and adds them. What is the value of DOG?

Answer: 26

Reasoning: D = 4, O = 15, G = 7: 26.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-B1-Q2 · question page 123

LOGIC.CODE.01 · Strategy: write the key as a table

A code moves each letter 1 place later (A -> B, ..., Z -> A). What is BOX in code?

Answer: A. CPY

Reasoning: B -> C, O -> P, X -> Y.

✗ ANW: moved back instead of forward
✗ CPX: the last letter was not coded
✗ DQZ: moved 2 places

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S6-B1-Q3 · question page 124

LOGIC.CODE.01 · Strategy: write the key as a table

A code replaces each digit d by 9 - d. What is the code for 306?

Answer: 693

Reasoning: 3 -> 6, 0 -> 9, 6 -> 3.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-B1-Q4 · question page 125

LOGIC.CODE.01 · Strategy: write the key as a table

If a code moves every letter 2 places later, then decoding moves every letter 2 places earlier.

Answer: True

Reasoning: Decoding undoes encoding.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S6-B1-Q5 · question page 126

LOGIC.CODE.01 · Strategy: write the key as a table

Triangle + Triangle + Triangle = 12. What number is Triangle?

Answer: 4

Reasoning: 3 x 4 = 12.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-F1-Q1 · question page 127

LOGIC.CODE.01 · Codes with structure: shapes of letters and combined rules

In this code, a letter's value is the number of closed spaces inside it in plain block capitals: A, D, O, P, Q and R have 1, B has 2, and every other letter has 0. A word's value is the sum of its letters' values. What is the value of BREAD?

Answer: 5

Reasoning: B 2 + R 1 + E 0 + A 1 + D 1 = 5.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-F1-Q2 · question page 128

LOGIC.CODE.01 · Codes with structure: shapes of letters and combined rules

In this code, a letter's value is the number of closed spaces inside it in plain block capitals: A, D, O, P, Q and R have 1, B has 2, and every other letter has 0. A word's value is the sum of its letters' values. Which word has value 0?

Answer: B. FLIES

Reasoning: F, L, I, E and S all have 0.

- ✗ TOWEL: O has 1
- ✗ MAZE: A has 1
- ✗ SHIRT: R has 1

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S6-F1-Q3 · question page 129

LOGIC.CODE.01 · Codes with structure: shapes of letters and combined rules

A code first writes a word backward and then moves each letter 1 place later in the alphabet. What is the code for MAP?

Answer: QBN

Reasoning: Backward: PAM. Shift by 1: QBN.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-F1-Q4 · question page 130

LOGIC.CODE.01 · Codes with structure: shapes of letters and combined rules

Each symbol stands for a whole number. Moon + Moon + Sun = 13. Sun + Sun = 6. Moon + Star = 9. What number is Star?

Answer: 4

Reasoning: Sun = 3, so 2 x Moon = 10 and Moon = 5, so Star = 4.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-F1-Q5 · question page 131

LOGIC.CODE.01 · Codes with structure: shapes of letters and combined rules

A code moves every letter 13 places later in the alphabet (wrapping around). Applying this code twice gives back the original word.

Answer: True

Reasoning: 13 + 13 = 26 places, a full trip around the alphabet.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S6-F2-Q1 · question page 132

LOGIC.CODE.01 · Find a key and prove when none exists

A code adds 3 to each digit of a 4-digit number and keeps only the last digit of each sum (so 8 becomes 1). Is there a 4-digit number whose code is the same number written backward?

Answer: C. No: the middle digits would need b + 6 to end in the same digit as b, which never happens.

Reasoning: If abcd codes to dcba, then b + 3 ends in c and c + 3 ends in b, so b + 6 ends in b, which is impossible (6 is not a multiple of 10).

- ✗ Yes: 1234 works.: 1234 codes to 4567, not 4321
- ✗ Yes: any number whose digits are all equal works.: 5555 codes to 8888, not 5555
- ✗ No: but only because the first digit cannot be 0.: the reason is the middle digits, not a leading zero

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: all 9000 four-digit numbers tested; none works. Automatic: the check function accepts only the keyed choice.

S6-F2-Q2 · question page 133

LOGIC.CODE.01 · Find a key and prove when none exists

A code moves every letter the same number of places later in the alphabet (wrapping around). It turns GO into MU. What does it turn LOGIC into?

Answer: RUMOI

Reasoning: G -> M is 6 places, and O -> U is also 6. Shift LOGIC by 6: RUMOI.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-F2-Q3 · question page 134

LOGIC.CODE.01 · Find a key and prove when none exists

Each symbol stands for a whole number from 1 to 9. Square + Square = Circle, and Circle + Square = 12. Find Square s and Circle c.

Answer: any values that satisfy: s + s == c and c + s == 12 and isInteger(s) and isInteger(c) (example: s = 4, c = 8)

Reasoning: 3 x Square = 12, so Square = 4 and Circle = 8. This is the only solution.

Tempting answers the rule rejects: s = 6, c = 6; s = 3, c = 9

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it. Search of the stated domain: 1 valid answers.

S6-F2-Q4 · question page 135

LOGIC.CODE.01 · Find a key and prove when none exists

In this code, a letter's value is the number of closed spaces inside it in plain block capitals: A, D, O, P, Q and R have 1, B has 2, and every other letter has 0. A word's value is the sum of its letters' values. Using only the letters B, O and T (each as often as you like), what is the greatest value of a 4-letter string?

Answer: 8

Reasoning: B has the most closed spaces (2), so BBBB = 8.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S6-F2-Q5 · question page 136

LOGIC.CODE.01 · Find a key and prove when none exists

A code sends each letter to the letter at the same distance from the other end of the alphabet (A <-> Z, B <-> Y, C <-> X, ...). Which word is coded as itself?

Answer: D. No word: no letter is coded as itself.

Reasoning: A letter at position p codes to position 27 - p. A letter would code to itself only if p = 27 - p, which needs p = 13.5: impossible. So no word codes to itself.

- ✗ MOON: M codes to N and O to L
- ✗ WIZARD: W -> D, I -> R, ...: it codes to DRAZIW
- ✗ NOON: N codes to M

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S7-M-Q1 · question page 138

LOGIC.ORDER.01 · Build an ordering that satisfies relational clues

Four books stand on a shelf. The atlas is left of the cookbook. The diary is right of the cookbook. The biography is at the far left. Order the books from left to right.

Answer: biography → atlas → cookbook → diary

Reasoning: The biography is first. The atlas, cookbook and diary must come in that order after it.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S7-M-Q2 · question page 139

LOGIC.ORDER.01 · Build an ordering that satisfies relational clues

Five friends arrived at a party one at a time. Gil arrived before Hana but after Ivo. Jo arrived last. Kai arrived right after Ivo. Order the arrivals, first to last.

Answer: Ivo → Kai → Gil → Hana → Jo

Reasoning: Jo is last. Ivo comes before Gil, who comes before Hana, and Kai is right after Ivo: Ivo, Kai, Gil, Hana, Jo.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 120 orderings of 5 items enumerated; 1 valid.

S7-M-Q3 · question page 140

LOGIC.ORDER.01 · Build an ordering that satisfies relational clues

Five runners finished a race with no ties. Uma beat Vic. Wes finished right after Uma. Xia beat Uma. Yan finished last. Who finished in the middle (3rd)?

Answer: C. Wes

Reasoning: Yan is 5th. Xia beat Uma and Uma beat Vic, with Wes right after Uma: Xia, Uma, Wes, Vic, Yan.

- ✗ Uma: Uma is 2nd
- ✗ Vic: Vic is 4th
- ✗ Xia: Xia is 1st

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: all 120 orders checked; only Xia, Uma, Wes, Vic, Yan fits. Automatic: the check function accepts only the keyed choice.

S7-M-Q4 · question page 141

LOGIC.ORDER.01 · Build an ordering that satisfies relational clues

Four flags hang on a pole, from top to bottom. The green flag is somewhere between the red and the blue flags. The red flag is not at the top. The yellow flag is directly below the blue flag. Order the flags from top to bottom.

Answer: blue → yellow → green → red

Reasoning: Blue must be above yellow with nothing between them, and

green lies between red and blue. If blue were below green, yellow would follow blue at the bottom and red would have to be on top, which is not allowed. So blue is on top, then yellow, then green, then red.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S7-M-Q5 · question page 142

LOGIC.ORDER.01 · Build an ordering that satisfies relational clues

Seven students stand in a line. Lee is 2nd from the left. Max is 3rd from the right. How many students stand between Lee and Max?

Answer: 2

Reasoning: Max is 3rd from the right of 7, so 5th from the left. Between positions 2 and 5 are positions 3 and 4: 2 students.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S7-B1-Q1 · question page 143

LOGIC.ORDER.01 · Strategy: place the most constrained item first

Three cups stand in a row. The red cup is between the other two. The blue cup is on the right. Order them from left to right.

Answer: white → red → blue

Reasoning: Blue on the right, red in the middle, white on the left.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 6 orderings of 3 items enumerated; 1 valid.

S7-B1-Q2 · question page 144

LOGIC.ORDER.01 · Strategy: place the most constrained item first

In a line of 9 people, Ola is 4th from the front. What is her position counted from the back?

Answer: 6

Reasoning: 9 - 4 + 1 = 6.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S7-B1-Q3 · question page 145

LOGIC.ORDER.01 · Strategy: place the most constrained item first

"Bea is right after Cy" means no one is between Cy and Bea.

Answer: True

Reasoning: "Right after" means directly next.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S7-B1-Q4 · question page 146

LOGIC.ORDER.01 · Strategy: place the most constrained item first

Ava is taller than Ben, and Ben is taller than Cal. Who is shortest?

Answer: B. Cal

Reasoning: Ava > Ben > Cal.

- ✗ Ava: Ava is tallest
- ✗ Ben: Ben is in the middle
- ✗ It cannot be known.: the clues give a full order

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S7-B1-Q5 · question page 147

LOGIC.ORDER.01 · Strategy: place the most constrained item first

Four racers: Dot finished first. Eli beat Fay. Gus finished last. Order them, first to last.

Answer: Dot → Eli → Fay → Gus

Reasoning: Dot first, Gus last, Eli before Fay.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S7-F1-Q1 · question page 148

LOGIC.ORDER.01 · Orderings with "next to" and "not next to"

Five houses stand in a row, numbered 1 to 5 from left to right. The bakery is next to the bank. The café is at one end. The dentist is directly right of the café. The florist is not next to the dentist. The bank is in house 4. Order the businesses from left to right.

Answer: café → dentist → bakery → bank → florist

Reasoning: The dentist is right of the café, so the café is house 1 and the dentist house 2. The bank is house 4. The florist is not next to the dentist, so not house 3: the florist is house 5 and the bakery house 3, next to the bank.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 120 orderings of 5 items enumerated; 1 valid.

S7-F1-Q2 · question page 149

LOGIC.ORDER.01 · Orderings with "next to" and "not next to"

Six people, A to F, sit in seats 1 to 6 in a row. C is in seat 1 and D is in seat 6. B sits right after A, and E sits right after B. F does not sit next to C. Which seat is F in?

Answer: 5

Reasoning: A, B, E form a block of three in seats 2-4 or 3-5. If the block were in 3-5, F would be in seat 2, next to C. So the block is in seats 2-4 and F is in seat 5.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S7-F1-Q3 · question page 150

LOGIC.ORDER.01 · Orderings with "next to" and "not next to"

Five students line up. Pat is in front of Quinn. Rae is right behind Sol. Tia is first. Quinn is last. Which statement must be true?

Answer: D. Tia is directly in front of Pat or of Sol.

Reasoning: Tia is 1st and Quinn 5th. The middle three are Pat and the block Sol-Rae in some order: Pat, Sol, Rae or Sol, Rae, Pat. In both, position 2 is Pat or Sol.

- ✗ Pat is 2nd.: Sol and Rae can be 2nd and 3rd
- ✗ Rae is 4th.: Rae can be 3rd
- ✗ Sol is right behind Pat.: Sol can be in front of Pat

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: all 120 orders checked. Automatic: the check function accepts only the keyed choice.

S7-F1-Q4 · question page 151

LOGIC.ORDER.01 · Orderings with "next to" and "not next to"

Four friends sit in seats 1 to 4 in a row. Ari is not next to Bo. Cy sits in seat 2. Bo is in a higher seat number than Cy. Di is not in seat 4. Give each friend's seat.

Answer: Ari: 1; Bo: 4; Cy: 2; Di: 3

Reasoning: Cy is in seat 2. Bo is in seat 3 or 4. If Bo were in 3, Ari (not next to Bo) would need seat 1, leaving seat 4 for Di, which is not allowed. So Bo is in 4; Ari is not in 3, so Ari is in 1 and Di in 3.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 24 assignments enumerated; 1 satisfy every clue.

S7-F1-Q5 · question page 152

LOGIC.ORDER.01 · Orderings with "next to" and "not next to"

If five people stand in a line and three of them must stand next to each other in a fixed order, there are exactly 6 possible lines.

Answer: True

Reasoning: Treat the three as one block: 3 units (the block and two people) can be ordered in 3 x 2 x 1 = 6 ways.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S7-F2-Q1 · question page 153

LOGIC.ORDER.01 · Count and compare orderings

Five students, P, Q, R, S and T, stand in a line. P is somewhere in front of Q, and R is somewhere in front of S. How many different lines are possible?

Answer: 30

Reasoning: There are 120 lines without clues. Half have P before Q, and of those half have R before S: 120 / 4 = 30.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S7-F2-Q2 · question page 154

LOGIC.ORDER.01 · Count and compare orderings

Six relay swimmers, A to F, swim in order. B swims right after E. D swims before A. C swims last. F swims first. A swims right after B. Give the swimming order.

Answer: F → D → E → B → A → C

Reasoning: F is first and C last. E, B, A form a block (B right after E, A right after B) in positions 2-4 or 3-5. If the block were in 2-4, D would be 5th, after A, which is not allowed. So the block is in 3-5 and D is 2nd: F, D, E, B, A, C.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 720 orderings of 6 items enumerated; 1 valid.

S7-F2-Q3 · question page 155

LOGIC.ORDER.01 · Count and compare orderings

Four people, W, X, Y and Z, stand in a line. Clues: (1) W is not first. (2) X is right after W. (3) Y is last. Which clue could be removed and still leave exactly one possible line?

Answer: A. No clue can be removed.

Reasoning: With all three clues the only line is Z, W, X, Y. Removing any one clue leaves at least two lines, so every clue is needed.

- ✗ Clue 1: without it, both W, X, Z, Y and Z, W, X, Y fit
- ✗ Clue 2: without it, four lines fit (for example X, W, Z, Y)
- ✗ Clue 3: without it, four lines fit (for example Z, W, X, Y and Y, W, X, Z)

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: each clue removed in turn and all 24 lines recounted. Automatic: the check function accepts only the keyed choice.

S7-F2-Q4 · question page 156

LOGIC.ORDER.01 · Count and compare orderings

Seven people stand in a line, numbered 1 to 7 from the front. Ana is in position a and Ben is in position b. Ana is in front of Ben, exactly two people stand between them, and Ben is not last. Give one possible pair of positions.

Answer: any values that satisfy: b - a == 3 and b <= 6 and a >= 1 and isInteger(a) and isInteger(b) (example: a = 2, b = 5)

Reasoning: Two people between means b = a + 3. Ben not last means b ≤ 6, so (1, 4), (2, 5) or (3, 6).

Tempting answers the rule rejects: a = 4, b = 7; a = 2, b = 4; a = 5, b = 2

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 3 valid answers.

S7-F2-Q5 · question page 157

LOGIC.ORDER.01 · Count and compare orderings

With the clues "J is before K" and "K is before L" for three people in a line, the order is fully determined.

Answer: True

Reasoning: J, K, L is the only order.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S8-M-Q1 · question page 159

LOGIC.CONNECTIVE.02 · Truth values of and, inclusive or, exclusive or

"7 is odd and 7 is greater than 10" is true.

Answer: False

Reasoning: An "and" statement needs both parts true; 7 > 10 is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S8-M-Q2 · question page 160

LOGIC.CONNECTIVE.02 · Truth values of and, inclusive or, exclusive or

"7 is odd or 7 is greater than 10" is true (with "or" meaning at least one).

Answer: True

Reasoning: An inclusive "or" needs at least one part true; 7 is odd.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S8-M-Q3 · question page 161

LOGIC.CONNECTIVE.02 · Truth values of and, inclusive or, exclusive or

A sign says: "Each meal comes with soup or salad, but not both." Which order follows the sign?

Answer: A. a meal with salad and no soup

Reasoning: This is an exclusive "or": exactly one of soup and salad.

- ✗ a meal with soup and salad: "not both" rules this out
- ✗ a meal with neither soup nor salad: the meal must come with one of them
- ✗ a meal with two salads and a soup: includes both soup and salad

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S8-M-Q4 · question page 162

LOGIC.CONNECTIVE.02 · Truth values of and, inclusive or, exclusive or

Give a whole number n that makes "n is even or n is greater than 20" true but "n is even and n is greater than 20" false.

Answer: any values that satisfy: (isEven(n) or n > 20) and not (isEven(n) and n > 20) and isInteger(n) (example: n = 4)

Reasoning: Exactly one part must be true: an even number up to 20 (like 4) or an odd number greater than 20 (like 25).

Tempting answers the rule rejects: n = 22; n = 7; n = 30

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 20 valid answers.

S8-M-Q5 · question page 163

LOGIC.CONNECTIVE.02 · Truth values of and, inclusive or, exclusive or

P is true and Q is false. Which statement is FALSE?

Answer: D. P and Q

Reasoning: "P and Q" needs both true, but Q is false.

- ✗ P or Q: one part is true, so the "or" is true
- ✗ P exclusive-or Q: exactly one part is true, so it is true
- ✗ P and not Q: P is true and not Q is true

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S8-B1-Q1 · question page 164

LOGIC.CONNECTIVE.01 · Truth values of the parts

"12 is a multiple of 5" is true.

Answer: False

Reasoning: 12 = 5 x 2 + 2.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S8-B1-Q2 · question page 165

LOGIC.CONNECTIVE.01 · Truth values of the parts

"-3 < 1" is true.

Answer: True

Reasoning: -3 is to the left of 1.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S8-B1-Q3 · question page 166

LOGIC.CONNECTIVE.01 · Truth values of the parts

"A triangle has 4 sides" is true.

Answer: False

Reasoning: A triangle has 3 sides.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S8-B1-Q4 · question page 167

LOGIC.CONNECTIVE.01 · Truth values of the parts

"0 is even" is true.

Answer: True

Reasoning: 0 = 2 x 0.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S8-B1-Q5 · question page 168

LOGIC.CONNECTIVE.01 · Truth values of the parts

Which part of "9 is odd and 9 is prime" is false?

Answer: B. 9 is prime

Reasoning: 9 = 3 x 3.

- ✗ 9 is odd: that part is true
- ✗ both parts: 9 is odd is true
- ✗ neither part: 9 = 3 x 3, so it is not prime

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S8-B2-Q1 · question page 169

LOGIC.CONNECTIVE.02 · Inclusive or exclusive?

With inclusive "or", "P or Q" is false when both P and Q are true.

Answer: False

Reasoning: Inclusive "or" needs at least one part true; with both true it is true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S8-B2-Q2 · question page 170

LOGIC.CONNECTIVE.02 · Inclusive or exclusive?

With exclusive "or", "P or Q" is true when both P and Q are true.

Answer: False

Reasoning: Exclusive means exactly one.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S8-B2-Q3 · question page 171

LOGIC.CONNECTIVE.02 · Inclusive or exclusive?

A rule says "You may bring a pencil or a pen (or both)." Which kind of "or" is this?

Answer: C. inclusive

Reasoning: "Or both" signals inclusive "or".

- ✗ exclusive: "or both" allows both
- ✗ neither kind: it is an "or"
- ✗ it cannot be told: "or both" tells us

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S8-B2-Q4 · question page 172

LOGIC.CONNECTIVE.02 · Inclusive or exclusive?

"The light is either on or off" describes two cases that can never both happen. Which "or" fits best?

Answer: B. exclusive

Reasoning: Exactly one of the two is true.

- ✗ inclusive: both cannot be true at once, so exactly one holds
- ✗ neither: it is an "or" statement

✗ "and": it is not a conjunction

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S8-B2-Q5 · question page 173

LOGIC.CONNECTIVE.02 · Inclusive or exclusive?

When P is false and Q is false, exclusive "P or Q" is true.

Answer: False

Reasoning: Exclusive "or" needs exactly one part true; with neither true it is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S8-F1-Q1 · question page 174

LOGIC.CONNECTIVE.02 · Compound statements with three parts

"(3 > 2 and 5 > 9) or 4 is even" is true.

Answer: True

Reasoning: The bracket is false (5 > 9 is false), but 4 is even, so the "or" is true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S8-F1-Q2 · question page 175

LOGIC.CONNECTIVE.02 · Compound statements with three parts

"3 > 2 and (5 > 9 or 4 is odd)" is true.

Answer: False

Reasoning: The bracket is false (both parts false), so the "and" is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S8-F1-Q3 · question page 176

LOGIC.CONNECTIVE.02 · Compound statements with three parts

For how many whole numbers n from 1 to 20 is "n is a multiple of 3 exclusive-or n is a multiple of 4" true?

Answer: 9

Reasoning: Multiples of 3: 6 numbers; of 4: 5 numbers; of both (12): 1 number. Exactly one: 6 + 5 - 2 x 1 = 9.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S8-F1-Q4 · question page 177

LOGIC.CONNECTIVE.02 · Compound statements with three parts

Complete the truth table for "P exclusive-or Q" and for "P or Q" (inclusive).

Answer: F T / T T / T T / F F

Reasoning: They differ only in the first row, where both parts are true.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S8-F1-Q5 · question page 178

LOGIC.CONNECTIVE.02 · Compound statements with three parts

Statement S: "P or Q" (inclusive). S is false. What can be concluded?

Answer: A. P is false and Q is false.

Reasoning: An inclusive "or" is false only when both parts are false.

- ✗ P is false or Q is false.: true, but it does not say enough: both must be false
- ✗ Exactly one of P and Q is false.: both are false
- ✗ P is true and Q is false.: then S would be true

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S8-F2-Q1 · question page 179

LOGIC.CONNECTIVE.02 · Design statements with given truth values

P, Q and R are statements. In how many of the 8 combinations of truth values is "(P or Q) and not R" true?

Answer: 3

Reasoning: R must be false (4 cases), and at least one of P, Q true: 3 of those 4.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S8-F2-Q2 · question page 180

LOGIC.CONNECTIVE.02 · Design statements with given truth values

Give whole numbers a and b with "a > b exclusive-or a + b is even" true and both numbers between 1 and 9.

Answer: any values that satisfy: ((a > b) != isEven(a + b)) and a >= 1 and a <= 9 and b >= 1 and b <= 9 and isInteger(a) and isInteger(b) (example: a = 5, b = 2)

Reasoning: Exactly one of "a > b" and "a + b is even" must hold, for example a = 5, b = 2 (a > b, sum 7 odd) or a = 2, b = 4 (a < b, sum even).

Tempting answers the rule rejects: a = 5, b = 3; a = 2, b = 5

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it. Search of the stated domain: 45 valid answers.

S8-F2-Q3 · question page 181

LOGIC.CONNECTIVE.02 · Design statements with given truth values

"P exclusive-or Q" has the same truth value as "(P or Q) and not (P and Q)" in every row of a truth table.

Answer: True

Reasoning: Both are true exactly when one part is true and the other false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S8-F2-Q4 · question page 182

LOGIC.CONNECTIVE.02 · Design statements with given truth values

Three lamps A, B and C are each on or off. Exactly one of these statements is true: (1) "A is on or B is on." (2) "C is on." (3) "A is on exclusive-or C is on." Which lamps are on?

Answer: D. B only

Reasoning: Check the 8 patterns. If C is on, (2) is true, so (1) and (3) must be false: (3) false needs A on, but then (1) is true. So C is off and (2) is false; exactly one of (1), (3) is true. (3) is true exactly when A is on, and then (1) is also true. So A is off, (3) is false, and (1) must be true: B is on.

✗ A only: then (1) and (3) are both true

✗ A and B: then (1) and (3) are both true

✗ none: then all three statements are false

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: all 8 on/off patterns checked; only B on fits. Automatic: the check function accepts only the keyed choice.

S8-F2-Q5 · question page 183

LOGIC.CONNECTIVE.02 · Design statements with given truth values

For how many whole numbers n from 1 to 30 is "n is odd and (n is a multiple of 3 or n is a multiple of 5)" true?

Answer: 7

Reasoning: Odd multiples of 3: 3, 9, 15, 21, 27. Odd multiples of 5: 5, 15, 25. Together, counting 15 once: 3, 5, 9, 15, 21, 25, 27, which is 7 numbers.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S9-M-Q1 · question page 185

LOGIC.SPATIAL.02 · Distinguish rotations from reflections

Which figure shows the shape TURNED (rotated in the page), not flipped over? (Only the shape matters, not where it sits in its box.)

Answer: C. Figure C

Reasoning: Turning never changes which way the shape "curls". Figure C is the shape after a quarter-turn clockwise; A, B and D are mirror images, which no turn can produce for this shape.

✗ Figure A: a left-right flip (reflection)

✗ Figure B: a top-bottom flip (reflection)
✗ Figure D: a flip across a diagonal (reflection)

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the panel check accepts only the keyed panel.

S9-M-Q2 · question page 186

LOGIC.SPATIAL.02 · Distinguish rotations from reflections

Three of the figures are the shape turned (rotated). One is the shape flipped over (a mirror image). Which figure is the flipped one?

Answer: D. Figure D

Reasoning: Turn each candidate back to the starting position: A, B and C match the shape; D can only be matched by flipping it over.

✗ Figure A: a quarter-turn clockwise
✗ Figure B: a half-turn
✗ Figure C: a quarter-turn counterclockwise

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the panel check accepts only the keyed panel.

S9-M-Q3 · question page 187

LOGIC.SPATIAL.02 · Distinguish rotations from reflections

The shape is turned a quarter-turn CLOCKWISE (90 degrees, like a clock hand moving from 12 to 3). Which figure shows the result?

Answer: B. Figure B

Reasoning: After a clockwise quarter-turn, the left column of the shape becomes its top row, read from left to right going upward; figure B is that result.

✗ Figure A: a quarter-turn counterclockwise

✗ Figure C: a half-turn

✗ Figure D: a mirror image, not a turn

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the panel check accepts only the keyed panel.

S9-M-Q4 · question page 188

LOGIC.SPATIAL.02 · Distinguish rotations from reflections

Capital letters are turned upside down by a half-turn (180 degrees in the page). In the block capitals used here, exactly these letters look the same after a half-turn: H I N O S X Z. Which letter looks the same after the half-turn?

Answer: C. N

Reasoning: N is on the stated list; M, E and A are not.

✗ M: a half-turn of M looks like W

✗ E: a half-turn of E faces the other way and is upside down

✗ A: a half-turn of A is upside down

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S9-M-Q5 · question page 189

LOGIC.SPATIAL.02 · Distinguish rotations from reflections

A 4 by 4 grid has shaded cells at (1,1), (1,2), (1,3) and (2,1), written as row,column with row 1 at the top. The whole grid is turned a quarter-turn clockwise about its center. Mark the cells where the shaded cells end up.

Answer: 1,3, 1,4, 2,4, 3,4

Reasoning: A clockwise quarter-turn of a 4 by 4 grid sends row r, column c to row c, column 5 - r. So (1,1) -> (1,4), (1,2) -> (2,4), (1,3) -> (3,4), (2,1) -> (1,3).

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each cell mapped by (r, c) -> (c, 5 - r).

S9-B1-Q1 · question page 190

LOGIC.SPATIAL.02 · Strategy: follow one marked corner

A quarter-turn clockwise followed by another quarter-turn clockwise is the same as a half-turn.

Answer: True

Reasoning: 90 + 90 = 180 degrees.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S9-B1-Q2 · question page 191

LOGIC.SPATIAL.02 · Strategy: follow one marked corner

An arrow points up. It is turned a quarter-turn clockwise. Which way does it point now?

Answer: A. right

Reasoning: Up turns to the right, like a clock hand from 12 to 3.

- ✗ left: that is a quarter-turn counterclockwise
- ✗ down: that is a half-turn
- ✗ up: it has turned

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S9-B1-Q3 · question page 192

LOGIC.SPATIAL.02 · Strategy: follow one marked corner

An arrow points right. It is reflected in a vertical mirror line. Which way does it point now?

Answer: B. left

Reasoning: Left and right swap in a vertical mirror.

- ✗ right: a vertical mirror swaps left and right
- ✗ up: reflections do not turn it sideways
- ✗ down: reflections in a vertical line do not change up and down

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S9-B1-Q4 · question page 193

LOGIC.SPATIAL.02 · Strategy: follow one marked corner

Flipping a shape over twice in the same mirror line brings it back to where it started.

Answer: True

Reasoning: The second flip undoes the first.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S9-B1-Q5 · question page 194

LOGIC.SPATIAL.02 · Strategy: follow one marked corner

How many quarter-turns clockwise bring a shape back to its starting position for the first time, if the shape has no symmetry at all?

Answer: 4

Reasoning: Four quarter-turns make a full turn.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S9-F1-Q1 · question page 195

LOGIC.SPATIAL.02 · Combine turns and flips

Reflecting a shape in a vertical line and then in a horizontal line gives the same result as a half-turn.

Answer: True

Reasoning: (x, y) -> (-x, y) -> (-x, -y), which is a half-turn.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S9-F1-Q2 · question page 196

LOGIC.SPATIAL.02 · Combine turns and flips

A shape is turned a quarter-turn clockwise three times. Which single move gives the same result?

Answer: C. a quarter-turn counterclockwise

Reasoning: 270 degrees clockwise is the same as 90 degrees counterclockwise.

- ✗ a half-turn: three quarter-turns are 270 degrees, not 180
- ✗ a quarter-turn clockwise: that is only one of the three turns
- ✗ a reflection: turns never produce a mirror image of this shape

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S9-F1-Q3 · question page 197

LOGIC.SPATIAL.02 · Combine turns and flips

A word written in capital letters is turned a half-turn (180 degrees). In the block capitals used here, exactly these letters look the same after a half-turn: H I N O S X Z. A half-turn also

reverses the order of the letters. Which word looks exactly the same after the turn?

Answer: D. SOS

Reasoning: The word must read the same backward and use only listed letters: S, O, S.

- ✗ NOW: W is not on the list
- ✗ ZOO: turned, it reads OOZ
- ✗ MOM: M turns into W

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S9-F1-Q4 · question page 198

LOGIC.SPATIAL.02 · Combine turns and flips

A 4 by 4 grid has shaded cells at (1,2), (2,2), (3,2) and (3,3), written as row,column with row 1 at the top. The grid is given a half-turn about its center. Mark where the shaded cells end up.

Answer: 2,2, 2,3, 3,3, 4,3

Reasoning: A half-turn sends (r, c) to (5 - r, 5 - c): (4,3), (3,3), (2,3), (2,2).

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each cell mapped by (r, c) -> (5 - r, 5 - c).

S9-F1-Q5 · question page 199

LOGIC.SPATIAL.02 · Combine turns and flips

A shape has no symmetry. How many different-looking positions can it be put in using turns AND flips (on a grid, counting each distinct picture once)?

Answer: 8

Reasoning: 4 turns of the shape and 4 turns of its mirror image, all

different when the shape has no symmetry.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S9-F2-Q1 · question page 200

LOGIC.SPATIAL.02 · Symmetry under turns and flips

Of the 8 moves of a square grid figure (4 turns: 0, 90, 180, 270 degrees; and 4 flips: vertical, horizontal and both diagonals), how many leave a plus-shaped figure of 5 cells (a center cell with one cell on each side) unchanged?

Answer: 8

Reasoning: The plus shape has every symmetry of the square.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S9-F2-Q2 · question page 201

LOGIC.SPATIAL.02 · Symmetry under turns and flips

For which figure can the mirror image be produced by turning the figure (without flipping it over)? Exactly two of the four figures have this property; choose the answer that names both.

Answer: A. B and D

Reasoning: A figure's mirror image is a turn of it exactly when the figure has a mirror line itself. B (a T) and D (a staircase) each have a mirror line; A (an L) and C (a Z) do not.

- ✗ A and C: their mirror images cannot be produced by any turn
- ✗ A and B: A's mirror image cannot be produced by a turn
- ✗ C and D: C's mirror image cannot be produced by a turn

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S9-F2-Q3 · question page 202

LOGIC.SPATIAL.02 · Symmetry under turns and flips

An S-shaped figure of 4 cells (two cells in a row, and two more cells one row down, shifted one place left) is turned and flipped. Which statement is true?

Answer: C. A half-turn leaves it unchanged, but no flip does.

Reasoning: The S shape has half-turn symmetry only.

- ✗ A quarter-turn leaves it unchanged.: a quarter-turn stands it upright
- ✗ A flip in a vertical line leaves it unchanged.: that turns the S into a Z
- ✗ No move other than doing nothing leaves it unchanged.: the half-turn also does

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S9-F2-Q4 · question page 203

LOGIC.SPATIAL.02 · Symmetry under turns and flips

If a figure looks the same after a quarter-turn, then it also looks the same after a half-turn.

Answer: True

Reasoning: A half-turn is two quarter-turns; each leaves the figure unchanged.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S9-F2-Q5 · question page 204

LOGIC.SPATIAL.02 · Symmetry under turns and flips

How many of the 7 shapes made of 4 grid cells joined edge to edge (counting a shape and its turns as the same, but its mirror image as different) look the same after a half-turn?

Answer: 4

Reasoning: The 7 shapes: straight line, square, T, L, J, S, Z. Those unchanged by a half-turn: straight line, square, S and Z.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S10-M-Q1 · question page 206

LOGIC.ARGUMENT.02 · Tell deductive conclusions from inductive ones

Which argument is DEDUCTIVE (the conclusion must be true if the premises are true)?

Answer: B. Every prime number greater than 2 is odd. 31 is a prime number greater than 2. So 31 is odd.

Reasoning: Only the first conclusion is forced by its premises; the others generalize from examples or from "most".

- ✗ The last 20 buses were on time. So the next bus will be on time.: past cases make the conclusion likely, not certain
- ✗ Every swan I have seen is white. So all swans are white.: a pattern in examples (induction)
- ✗ Most students like recess. So Ana likes recess.: "most" gives probability, not certainty

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S10-M-Q2 · question page 207

LOGIC.ARGUMENT.02 · Tell deductive conclusions from inductive ones

"1 + 3 = 4, 1 + 3 + 5 = 9, 1 + 3 + 5 + 7 = 16. So the sum of the first n odd numbers is always n x n." What kind of reasoning is this?

Answer: D. inductive: it generalizes from a few examples

Reasoning: The pattern is in fact true, but the argument only checks three cases: that is inductive reasoning.

- ✗ deductive: the examples prove it for every n: checking a few cases does not prove every case
- ✗ neither: it is not an argument: it gives reasons for a conclusion
- ✗ deductive: the conclusion is true: a true conclusion does not make the reasoning deductive

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S10-M-Q3 · question page 208

LOGIC.ARGUMENT.02 · Tell deductive conclusions from inductive ones

In an inductive argument, the premises can all be true while the conclusion is false.

Answer: True

Reasoning: Induction makes the conclusion likely, not certain; a new case can break the pattern.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S10-M-Q4 · question page 209

LOGIC.ARGUMENT.02 · Tell deductive conclusions from inductive ones

n x n + n + 11 is prime for n = 1, 2, 3, ..., 9. A student concludes it is prime for every whole number n. Which value of n shows the conclusion is false?

Answer: A. 10

Reasoning: 10 x 10 + 10 + 11 = 121 = 11 x 11, which is not prime, so the conclusion is false.

- ✗ 8: 64 + 8 + 11 = 83 is prime
- ✗ 4: 16 + 4 + 11 = 31 is prime
- ✗ 9: 81 + 9 + 11 = 101 is prime

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S10-M-Q5 · question page 210

LOGIC.ARGUMENT.02 · Tell deductive conclusions from inductive ones

Which conclusion follows DEDUCTIVELY from "All members of the chess club are in grade 10 or grade 11, and Jun is a member of the chess club"?

Answer: B. Jun is in grade 10 or grade 11.

Reasoning: Only this conclusion is forced by the premises.

- ✗ Jun is in grade 11.: he could be in grade 10
- ✗ Jun is good at chess.: not stated
- ✗ Most grade 10 students are in the chess club.: not stated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S10-B1-Q1 · question page 211

LOGIC.ARGUMENT.01 · Premises and conclusions

"All the cookies are gone, so someone ate them." Which part is the conclusion?

Answer: D. someone ate them

Reasoning: "So" introduces the conclusion.

- ✗ all the cookies are gone: the reason given
- ✗ cookies are tasty: not stated
- ✗ nobody ate them: the opposite of the conclusion

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S10-B1-Q2 · question page 212

LOGIC.ARGUMENT.01 · Premises and conclusions

A premise is the statement that an argument is trying to prove.

Answer: False

Reasoning: The statement an argument tries to prove is its conclusion; premises are the reasons offered for it.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S10-B1-Q3 · question page 213

LOGIC.ARGUMENT.01 · Premises and conclusions

"It rained every day this week. So it will rain tomorrow." What is the premise?

Answer: A. It rained every day this week.

Reasoning: The reason given first.

- ✗ It will rain tomorrow.: the conclusion
- ✗ Rain is good for plants.: not stated
- ✗ It never rains.: not stated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S10-B1-Q4 · question page 214

LOGIC.ARGUMENT.01 · Premises and conclusions

Every argument has at least one premise and one conclusion.

Answer: True

Reasoning: That is what makes it an argument.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S10-B1-Q5 · question page 215

LOGIC.ARGUMENT.01 · Premises and conclusions

"Since 12 is even, 12 is divisible by 2." What is the conclusion?

Answer: C. 12 is divisible by 2

Reasoning: The claim supported by the reason.

- ✗ 12 is even: the reason after "since"
- ✗ 12 is odd: false and not stated
- ✗ 2 is even: not stated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S10-B2-Q1 · question page 216

LOGIC.ARGUMENT.02 · Certain or only likely?

"Most tickets in the box are red, and I drew a ticket from the box" makes "my ticket is red" certain.

Answer: False

Reasoning: If only most tickets are red, the drawn ticket could be one of the others, so "my ticket is red" is likely but not certain.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S10-B2-Q2 · question page 217

LOGIC.ARGUMENT.02 · Certain or only likely?

"Most tickets in the box are red, and I drew a ticket" makes "my ticket is red" certain.

Answer: False

Reasoning: It is likely, not certain.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S10-B2-Q3 · question page 218

LOGIC.ARGUMENT.02 · Certain or only likely?

Which word signals that a conclusion is only likely?

Answer: B. probably

Reasoning: "Probably" marks a likely conclusion.

- ✗ must: signals certainty
- ✗ therefore: signals a conclusion, not its strength
- ✗ every: a premise word

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S10-B2-Q4 · question page 219

LOGIC.ARGUMENT.02 · Certain or only likely?

If a pattern holds for 100 examples, it must hold for the 101st.

Answer: False

Reasoning: Examples alone never guarantee the next case.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S10-B2-Q5 · question page 220

LOGIC.ARGUMENT.02 · Certain or only likely?

A die has been rolled 5 times and showed an even number each time. Which conclusion is certain?

Answer: C. Each of those 5 rolls was even.

Reasoning: Only the statement about what already happened is certain.

- ✗ The next roll will be even.: not certain
- ✗ The die only has even numbers.: not certain
- ✗ The die is unfair.: not certain

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S10-F1-Q1 · question page 221

LOGIC.ARGUMENT.02 · Patterns that break

The claim "n x n - n + 17 is prime for every whole number n" holds for n = 1 to 16. Find a whole number n that makes n x n - n + 17 NOT prime.

Answer: any values that satisfy: not isPrime(n*n - n + 17) and n >= 1 and isInteger(n) (example: n = 17)

Reasoning: For n = 17: 17 x 17 - 17 + 17 = 289 = 17 x 17. Other values also work (for example 18: 323 = 17 x 19).

Tempting answers the rule rejects: n = 1; n = 5; n = 16

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 8 valid answers.

S10-F1-Q2 · question page 222

LOGIC.ARGUMENT.02 · Patterns that break

Points are placed on a circle and every pair is joined by a straight line. With 1, 2, 3, 4, 5 points (in general position), the circle is cut into 1, 2, 4, 8, 16 regions. A student predicts 32 regions for 6 points. What is the actual number?

Answer: A. 31

Reasoning: The region count is 1 + C(n, 2) + C(n, 4): for n = 6, 1 + 15 + 15 = 31. The early pattern of doubling was only inductive evidence.

- ✗ 32: the doubling pattern breaks at 6 points
- ✗ 30: one too few
- ✗ 36: not the count

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S10-F1-Q3 · question page 223

LOGIC.ARGUMENT.02 · Patterns that break

A deductive argument with true premises can have a false conclusion.

Answer: False

Reasoning: In a valid deductive argument, true premises guarantee a

true conclusion.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S10-F1-Q4 · question page 224

LOGIC.ARGUMENT.02 · Patterns that break

Which statement is the strongest inductive argument?

Answer: D. All 500 randomly chosen bolts from the batch passed, so nearly all bolts in the batch pass.

Reasoning: A large random sample gives the strongest support for a general claim about the batch.

- ✗ My two friends like the song, so everyone does.: two hand-picked examples
- ✗ The first bolt passed, so all bolts pass.: one example
- ✗ All bolts in the batch passed, so this bolt passed.: that is deductive, not inductive

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S10-F1-Q5 · question page 225

LOGIC.ARGUMENT.02 · Patterns that break

Which reasoning is deductive?

Answer: B. All squares are rectangles; this figure is a square; so it is a rectangle.

Reasoning: Only the first is forced by its premises.

- ✗ This figure looks like a rectangle, so it probably is one.: based on

appearance

✗ Every rectangle I measured had equal diagonals, so all rectangles do.: from measured examples

✗ Most rectangles in the book are squares, so this one is.: from "most"

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S10-F2-Q1 · question page 226

LOGIC.ARGUMENT.02 · From pattern to proof

The expression n x n + n + 41 is prime for n = 0, 1, 2, ..., 39. What is the smallest whole number n for which it is NOT prime?

Answer: 40

Reasoning: n = 40 gives 1681 = 41 x 41.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S10-F2-Q2 · question page 227

LOGIC.ARGUMENT.02 · From pattern to proof

"For every whole number n, n x n - n is even, because 2 x 2 - 2 = 2, 3 x 3 - 3 = 6 and 4 x 4 - 4 = 12 are even." Which statement about this argument is correct?

Answer: A. The claim is true, but three examples do not prove it.

Reasoning: n x n - n = n(n - 1) is a product of two consecutive whole numbers, one of them even, so the claim is true; but the argument only checks examples, which does not prove it for every n.

✗ The argument proves the claim, because three cases are checked.: cases do not prove every n

✗ The claim is false, because 1 x 1 - 1 = 0.: 0 is even

✗ The claim is false for large n.: n(n - 1) is always even

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S10-F2-Q3 · question page 228

LOGIC.ARGUMENT.02 · From pattern to proof

If every case of a finite list has been checked, the conclusion about that list is deductive.

Answer: True

Reasoning: Checking ALL cases of a finite list is a complete proof (proof by cases).

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S10-F2-Q4 · question page 229

LOGIC.ARGUMENT.02 · From pattern to proof

Give a whole number n from 1 to 20 for which "2 to the power n, minus 1, is prime" is FALSE, even though n is prime. (2 to the power n is 2 multiplied by itself n times.)

Answer: any values that satisfy: isPrime(n) and not isPrime(2^n - 1) and n <= 20 (example: n = 11)

Reasoning: 2 to the 11th minus 1 = 2047 = 23 x 89. The pattern "n prime gives 2^n - 1 prime" holds for 2, 3, 5, 7 but fails at 11, the only such n up to 20.

Tempting answers the rule rejects: n = 7; n = 5; n = 4

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 1 valid answers.

S10-F2-Q5 · question page 230

LOGIC.ARGUMENT.02 · From pattern to proof

A student checks that every whole number from 4 to 100 that is even can be written as a sum of two primes. Which conclusion is justified?

Answer: C. Every even number from 4 to 100 is a sum of two primes.

Reasoning: Only the statement about the checked range is established.

- ✗ Every even number greater than 2 is a sum of two primes.: checking up to 100 does not cover all even numbers
- ✗ No odd number is a sum of two primes.: 2 + 3 = 5
- ✗ Every even number is a sum of two primes.: 2 is even and is not a sum of two primes

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S11-M-Q1 · question page 232

LOGIC.WORD.01 · Use letter counts, positions and order to test word patterns

Which word has its letters in alphabetical order (each letter the same as or later in the alphabet than the one before)?

Answer: D. ALMOST

Reasoning: A, L, M, O, S, T are in alphabetical order.

- ✗ BEGAN: A comes after G
- ✗ FLOWER: E comes after W in the word but before W in the alphabet
- ✗ CHART: A comes after H

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S11-M-Q2 · question page 233

LOGIC.WORD.01 · Use letter counts, positions and order to test word patterns

How many times does the letter S appear in the word ASSESSMENTS?

Answer: 5

Reasoning: A S S E S S M E N T S: positions 2, 3, 5, 6 and 11.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S11-M-Q3 · question page 234

LOGIC.WORD.01 · Use letter counts, positions and order to test word patterns

Number words are written in capital letters (ONE, TWO, ..., TEN). Which number word has exactly as many letters as the number it names?

Answer: A. FOUR

Reasoning: F-O-U-R has 4 letters.

- ✗ FIVE: 4 letters, value 5
- ✗ THREE: 5 letters, value 3
- ✗ TWO: 3 letters, value 2

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S11-M-Q4 · question page 235

LOGIC.WORD.01 · Use letter counts, positions and order to test word patterns

Take the first letter of each word: "Every Apple Tastes Sweet." What word do the first letters spell?

Answer: C. EATS

Reasoning: E, A, T, S in order: EATS.

- ✗ EAST: the order is E, A, T, S
- ✗ SEAT: the letters must stay in order
- ✗ TEAS: the letters must stay in order

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S11-M-Q5 · question page 236

LOGIC.WORD.01 · Use letter counts, positions and order to test word patterns

Write the letters of STEAM in alphabetical order. Which letter is in the middle position?

Answer: B. M

Reasoning: A, E, M, S, T: the 3rd letter is M.

- ✗ E: E is 2nd
- ✗ A: A is 1st
- ✗ S: S is 4th

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S11-B1-Q1 · question page 237

LOGIC.WORD.01 · Strategy: list the letters with their positions

The letters of BELLOW are in alphabetical order.

Answer: True

Reasoning: B, E, L, L, O, W: each the same as or later than the one

before.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S11-B1-Q2 · question page 238

LOGIC.WORD.01 · Strategy: list the letters with their positions

How many times does the letter E appear in REFERENCE?

Answer: 4

Reasoning: R E F E R E N C E.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S11-B1-Q3 · question page 239

LOGIC.WORD.01 · Strategy: list the letters with their positions

What is the 4th letter of the word PLANET?

Answer: D. N

Reasoning: P L A N.

- ✗ A: the 3rd letter
- ✗ E: the 5th letter
- ✗ L: the 2nd letter

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S11-B1-Q4 · question page 240

LOGIC.WORD.01 · Strategy: list the letters with their positions

Take the first letter of each word: "Owls Watch Lakes." What do the first letters spell?

Answer: B. OWL

Reasoning: O, W, L.

- ✗ LOW: the order changed
- ✗ OLW: the letters are out of order
- ✗ WOL: the order changed

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S11-B1-Q5 · question page 241

LOGIC.WORD.01 · Strategy: list the letters with their positions

How many letters does the word EIGHT have?

Answer: 5

Reasoning: E I G H T.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S11-F1-Q1 · question page 242

LOGIC.WORD.01 · Patterns across several words

In the list GO, CAT, LAMP, PLANT, each word has one more letter than the word before. Which word could come next?

Answer: C. GARDEN

Reasoning: GO has 2 letters, CAT 3, LAMP 4, PLANT 5, so the next word needs 6 letters.

✗ HOUSE: 5 letters
✗ BALLOON: 7 letters
✗ TREE: 4 letters

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S11-F1-Q2 · question page 243

LOGIC.WORD.01 · Patterns across several words

How many of the number words ONE, TWO, THREE, ..., TEN have their letters in alphabetical order?

Answer: 0

Reasoning: Check each: none of them is in alphabetical order (for example ONE: N comes before O).

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S11-F1-Q3 · question page 244

LOGIC.WORD.01 · Patterns across several words

Which word has its letters in strictly reverse alphabetical order (each letter earlier in the alphabet than the one before it)?

Answer: D. SPOKE

Reasoning: S, P, O, K, E: each letter comes earlier in the alphabet than the one before it.

✗ STONE: T comes after S
✗ TRACK: C comes after A
✗ SPELL: L comes after E

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S11-F1-Q4 · question page 245

LOGIC.WORD.01 · Patterns across several words

Write the letters of each number word ONE to TEN in alphabetical order; for example, ONE becomes ENO. Which number word becomes EENSV? Give its value as a number.

Answer: 7

Reasoning: SEVEN sorted: E, E, N, S, V. No other number word has these letters.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S11-F1-Q5 · question page 246

LOGIC.WORD.01 · Patterns across several words

Every word whose letters are in alphabetical order reads differently backward unless all its letters are the same.

Answer: True

Reasoning: Backward it is in reverse order; the two orders agree only if no two letters differ.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S11-F2-Q1 · question page 247

LOGIC.WORD.01 · Count and construct letter strings under rules

How many 3-letter strings can be made from the letters A, B, C, D (letters may repeat)

whose letters are in alphabetical order (each letter the same as or later than the one before)?

Answer: 20

Reasoning: Choosing a multiset of 3 letters from 4 kinds: C(6, 3) = 20.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S11-F2-Q2 · question page 248

LOGIC.WORD.01 · Count and construct letter strings under rules

How many 4-letter strings using only the letters X and Y have no two equal letters next to each other?

Answer: 2

Reasoning: They must alternate: XYXY or YXYX.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S11-F2-Q3 · question page 249

LOGIC.WORD.01 · Count and construct letter strings under rules

How many 4-letter strings using the letters A, B, C (letters may repeat) have no letter next to itself?

Answer: 24

Reasoning: 3 choices for the first letter, then 2 for each of the next three: 3 x 2 x 2 x 2 = 24.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S11-F2-Q4 · question page 250

LOGIC.WORD.01 · Count and construct letter strings under rules

A 6-letter string uses each of the letters A, B, C exactly twice, and no letter is next to itself. Which string follows the rule?

Answer: A. ABCABC

Reasoning: Check neighbors: A-B, B-C, C-A, A-B, B-C are all different.

- ✗ AABBCC: equal letters are next to each other
- ✗ ABBCAC: B is next to B
- ✗ ACBBCA: B is next to B

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S11-F2-Q5 · question page 251

LOGIC.WORD.01 · Count and construct letter strings under rules

The five letters of LEVEL (L, E, V, E, L) are rearranged into 5-letter strings. How many different strings have no two equal letters next to each other?

Answer: 12

Reasoning: Place V first. V at either end: the other four positions are in a row and must alternate (LELE or ELEL), 2 ways each, so 4. V in the middle: positions 1-2 and 4-5 each need one L and one E, 2 x 2 = 4 ways. V 2nd or 4th: the single end letter is L or E and the three positions on the other side must alternate starting with the other letter, 2 ways each, so 4. Total 4 + 4 + 4 = 12.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S12-M-Q1 · question page 253

LOGIC.CONDITIONAL.01 · Find when a conditional is true and the one case that makes it false

A rule: "If a package weighs more than 5 kg, then it has a yellow label." Which package BREAKS the rule?

Answer: B. a 7 kg package with a blue label

Reasoning: A conditional is false only when the "if" part is true and the "then" part is false: heavy and not yellow.

- ✗ a 3 kg package with a yellow label: the rule says nothing about light packages
- ✗ a 3 kg package with a blue label: light packages may have any label
- ✗ a 7 kg package with a yellow label: this follows the rule

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S12-M-Q2 · question page 254

LOGIC.CONDITIONAL.01 · Find when a conditional is true and the one case that makes it false

Complete the truth table for "if P then Q".

Answer: T / F / T / T

Reasoning: It is false only in the row P true, Q false.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S12-M-Q3 · question page 255

LOGIC.CONDITIONAL.01 · Find when a conditional is true and the one case that makes it false

Find a whole number n that is a counterexample to "If n is odd, then n is prime." (A counterexample makes the "if" part true and the "then" part false.)

Answer: any values that satisfy: isOdd(n) and not isPrime(n) and n >= 1 (example: n = 9)

Reasoning: Any odd number that is not prime works: 1, 9, 15, 21, 25, ...

Tempting answers the rule rejects: n = 7; n = 4; n = 2

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S12-M-Q4 · question page 256

LOGIC.CONDITIONAL.01 · Find when a conditional is true and the one case that makes it false

"If 5 > 8, then 2 + 2 = 5" is true.

Answer: True

Reasoning: The "if" part is false, so the conditional is true (it makes no promise when the "if" part is false).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S12-M-Q5 · question page 257

LOGIC.CONDITIONAL.01 · Find when a conditional is true and the one case that makes it false

"If it is a weekend day, the library opens at noon." Today the library opened at 9 a.m. What follows?

Answer: C. Today is not a weekend day.

Reasoning: If it were a weekend day, the library would have opened at noon. It did not, so it is not a weekend day.

- ✗ Today is a weekend day.: then it would have opened at noon
- ✗ The rule was broken.: the rule allows 9 a.m. on weekdays
- ✗ Nothing follows.: the contrapositive gives a conclusion

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S12-B1-Q1 · question page 258

LOGIC.CONDITIONAL.01 · Find the "if" part and the "then" part

In "If the sun sets, the street lights turn on," what is the "if" part (the hypothesis)?

Answer: D. the sun sets

Reasoning: The part after "if".

- ✗ the street lights turn on: that is the "then" part
- ✗ the lights are bright: not stated
- ✗ it is night: not stated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S12-B1-Q2 · question page 259

LOGIC.CONDITIONAL.01 · Find the "if" part and the "then" part

In "Students may enter the lab only if they wear goggles," which is the "then" part when it is written as "If ..., then ..."?

Answer: A. they wear goggles

Reasoning: "A only if B" means "if A, then B": if a student enters, the student wears goggles.

✗ students enter the lab: that becomes the "if" part

✗ the lab is open: not stated

✗ they leave the lab: not stated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S12-B1-Q3 · question page 260

LOGIC.CONDITIONAL.01 · Find the "if" part and the "then" part

Rewrite "Every multiple of 10 ends in 0" as a conditional.

Answer: C. If a number is a multiple of 10, then it ends in 0.

Reasoning: "Every A is B" means "if something is A, then it is B".

✗ If a number ends in 0, then it is a multiple of 10.: the parts are swapped

✗ If a number is not a multiple of 10, then it does not end in 0.: that changes the meaning

✗ A number ends in 0 and is a multiple of 10.: not a conditional

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S12-B1-Q4 · question page 261

LOGIC.CONDITIONAL.01 · Find the "if" part and the "then" part

In "If x = 3, then x + 1 = 4," the hypothesis is "x + 1 = 4".

Answer: False

Reasoning: The hypothesis is "x = 3".

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S12-B1-Q5 · question page 262

LOGIC.CONDITIONAL.01 · Find the "if" part and the "then" part

In "The alarm rings whenever the door opens," which part is the hypothesis?

Answer: A. the door opens

Reasoning: "Whenever the door opens" = "if the door opens".

- ✗ the alarm rings: the conclusion
- ✗ the alarm is loud: not stated
- ✗ the door closes: not stated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S12-B2-Q1 · question page 263

LOGIC.CONNECTIVE.01 · Truth values of the parts

"6 is a factor of 30" is true.

Answer: True

Reasoning: 30 = 6 x 5.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S12-B2-Q2 · question page 264

LOGIC.CONNECTIVE.01 · Truth values of the parts

"4 squared is 8" is true.

Answer: False

Reasoning: 4 x 4 = 16.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S12-B2-Q3 · question page 265

LOGIC.CONNECTIVE.01 · Truth values of the parts

"Every square is a triangle" is true.

Answer: False

Reasoning: A square has 4 sides.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S12-B2-Q4 · question page 266

LOGIC.CONNECTIVE.01 · Truth values of the parts

Which statement is true?

Answer: B. -5 is less than 2.

Reasoning: On a number line -5 is left of 2.

- ✗ 2 is less than -5.: 2 is to the right of -5 on the number line, so it is greater
- ✗ -5 is greater than 2.: -5 is to the left of 2 on the number line, so it is less
- ✗ -5 equals 5.: -5 and 5 are on opposite sides of 0

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S12-B2-Q5 · question page 267

LOGIC.CONNECTIVE.01 · Truth values of the parts

"1 is a prime number" is true.

Answer: False

Reasoning: A prime has exactly two different factors; 1 has one.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S12-F1-Q1 · question page 268

LOGIC.CONDITIONAL.01 · Conditionals in mathematics

Find a real number x that makes "If x squared is greater than 9, then x is greater than 3" FALSE.

Answer: any values that satisfy: x^2 > 9 and not (x > 3) (example: x = -4)

Reasoning: The conditional is false when x squared > 9 but x ≤ 3: any x < -3 works, for example -4.

Tempting answers the rule rejects: x = 4; x = 2; x = -3

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S12-F1-Q2 · question page 269

LOGIC.CONDITIONAL.01 · Conditionals in mathematics

"If a whole number is divisible by 6, then it is divisible by 3" is true for every whole number.

Answer: True

Reasoning: 6 = 2 x 3, so a multiple of 6 is a multiple of 3. No case has the "if" true and the "then" false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S12-F1-Q3 · question page 270

LOGIC.CONDITIONAL.01 · Conditionals in mathematics

For how many whole numbers n from 1 to 12 is "If n is even, then n is a multiple of 4" FALSE?

Answer: 3

Reasoning: False when n is even but not a multiple of 4: 2, 6, 10.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S12-F1-Q4 · question page 271

LOGIC.CONDITIONAL.01 · Conditionals in mathematics

P is false. Which statement MUST be true?

Answer: D. If P then Q

Reasoning: A conditional with a false "if" part is always true.

- ✗ If Q then P: false when Q is true
- ✗ P and Q: false because P is false
- ✗ Q: not determined

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S12-F1-Q5 · question page 272

LOGIC.CONDITIONAL.01 · Conditionals in mathematics

"If you score above 90, you earn a badge." Lin earned a badge. What follows?

Answer: C. Nothing about Lin's score follows.

Reasoning: The rule only says what happens above 90; badges may be earned in other ways.

- ✗ Lin scored above 90.: the rule does not say badges come only from scores above 90
- ✗ Lin scored 90 or below.: not supported
- ✗ The rule was broken.: earning a badge breaks nothing

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S12-F2-Q1 · question page 273

LOGIC.CONDITIONAL.01 · Nested and chained conditionals

Complete the truth table for "if P then (if Q then R)".

Answer: T T / F F / T T / T T / T T / F T / T T / T T

Reasoning: It is false only when P and Q are true and R is false.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (8 rows).

S12-F2-Q2 · question page 274

LOGIC.CONDITIONAL.01 · Nested and chained conditionals

In how many of the 8 rows of a truth table for P, Q, R is "(if P then Q) and (if Q then R)" true?

Answer: 4

Reasoning: Check all 8 rows; it is true in TTT, FTT, FFT, FFF.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S12-F2-Q3 · question page 275

LOGIC.CONDITIONAL.01 · Nested and chained conditionals

"If P then (if Q then R)" has the same truth value as "if (P and Q) then R" in every row.

Answer: True

Reasoning: Both are false exactly when P and Q are true and R is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S12-F2-Q4 · question page 276

LOGIC.CONDITIONAL.01 · Nested and chained conditionals

"If P then (if Q then R)" has the same truth value as "if (if P then Q) then R" in every row.

Answer: False

Reasoning: Row P false, Q false, R false: the first is true, but "if P then Q" is true and R false, so the second is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S12-F2-Q5 · question page 277

LOGIC.CONDITIONAL.01 · Nested and chained conditionals

Find whole numbers a and b that make the statement "If a + b is even, then a and b are both even" FALSE.

Answer: any values that satisfy: isEven(a + b) and not (isEven(a) and isEven(b)) and isInteger(a) and isInteger(b) (example: a = 3, b = 5)

Reasoning: Two odd numbers have an even sum: for example 3 + 5 = 8.

Tempting answers the rule rejects: a = 2, b = 4; a = 2, b = 3

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it. Search of the stated domain: 25 valid answers.

S13-M-Q1 · question page 279

LOGIC.CONDITIONAL.02 · Translate "only if", "unless", "necessary" and "sufficient"

"Being at least 16 years old is NECESSARY for getting a driving permit in this town." Which conditional says the same thing?

Answer: A. If a person has a driving permit, then the person is at least 16.

Reasoning: A necessary condition is required: whoever has the permit must meet it. "B is necessary for A" means "if A, then B".

- ✗ If a person is at least 16, then the person has a driving permit.: this makes age sufficient, which was not claimed
- ✗ If a person is under 16, then the person may still get a permit.: this contradicts the statement
- ✗ A person is at least 16 and has a permit.: not a conditional

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S13-M-Q2 · question page 280

LOGIC.CONDITIONAL.02 · Translate "only if", "unless", "necessary" and "sufficient"

"You may enter the museum only if you have a ticket." Which statement means the same?

Answer: D. If you enter the museum, then you have a ticket.

Reasoning: "A only if B" means "if A, then B".

- ✗ If you have a ticket, then you may enter the museum.: "only if" does not say a ticket is enough

✗ If you have no ticket, then you may still enter.: that contradicts the rule

✗ You have a ticket and you enter.: not a conditional

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S13-M-Q3 · question page 281

LOGIC.CONDITIONAL.02 · Translate "only if", "unless", "necessary" and "sufficient"

In logic, "P unless Q" means "if not Q, then P". Which statement means the same as "The picnic is held unless it rains"?

Answer: B. If it does not rain, the picnic is held.

Reasoning: With the stated meaning: if it does not rain, then the picnic is held.

✗ If it rains, the picnic is held.: swaps the condition

✗ If the picnic is held, it does not rain.: that is the converse of the stated meaning

✗ If it does not rain, the picnic is not held.: adds a negation

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S13-M-Q4 · question page 282

LOGIC.CONDITIONAL.02 · Translate "only if", "unless", "necessary" and "sufficient"

Being a square is a NECESSARY condition for being a rectangle.

Answer: False

Reasoning: A 2 by 5 rectangle is not a square, so a shape can be a rectangle without being a square. Being a square is sufficient, not necessary.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S13-M-Q5 · question page 283

LOGIC.CONDITIONAL.02 · Translate "only if", "unless", "necessary" and "sufficient"

n is a whole number. Which condition is NECESSARY for "n is divisible by 6"?

Answer: C. n is even

Reasoning: Every multiple of 6 is even, so being even is required (necessary). It is not sufficient: 4 is even.

- ✗ n is divisible by 12: sufficient but not necessary: 6 is divisible by 6 but not by 12
- ✗ n is divisible by 9: 12 is divisible by 6 but not by 9
- ✗ n ends in the digit 6: 12 is divisible by 6 but ends in 2

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: each condition tested against every n from 1 to 200. Automatic: the check function accepts only the keyed choice.

S13-B1-Q1 · question page 284

LOGIC.CONDITIONAL.01 · Write a statement as "if ..., then ..."

Rewrite "All birds have feathers" as a conditional.

Answer: A. If an animal is a bird, then it has feathers.

Reasoning: "All A are B" means "if A, then B".

- ✗ If an animal has feathers, then it is a bird.: parts swapped
- ✗ If an animal is not a bird, then it has no feathers.: changes the meaning
- ✗ Birds and feathers.: not a statement

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S13-B1-Q2 · question page 285

LOGIC.CONDITIONAL.01 · Write a statement as "if ..., then ..."

Rewrite "When the bell rings, class ends" as a conditional.

Answer: D. If the bell rings, then class ends.

Reasoning: "When" introduces the condition.

- ✗ If class ends, then the bell rings.: parts swapped
- ✗ The bell rings and class ends.: not a conditional
- ✗ If the bell does not ring, class does not end.: changes the meaning

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S13-B1-Q3 · question page 286

LOGIC.CONDITIONAL.01 · Write a statement as "if ..., then ..."

"If x = 2, then x + x = 4" has hypothesis "x + x = 4".

Answer: False

Reasoning: The hypothesis is the part after "if", which is "x = 2". "x + x = 4" is the conclusion.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S13-B1-Q4 · question page 287

LOGIC.CONDITIONAL.01 · Write a statement as "if ..., then ..."

What is the conclusion of "If a number ends in 5, it is divisible by 5"?

Answer: B. it is divisible by 5

Reasoning: The "then" part.

- ✗ a number ends in 5: the hypothesis
- ✗ it is odd: not stated
- ✗ it is 5: not stated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S13-B1-Q5 · question page 288

LOGIC.CONDITIONAL.01 · Write a statement as "if ..., then ..."

"Every even number is divisible by 2" can be written "If a number is even, then it is divisible by 2".

Answer: True

Reasoning: "Every A is B" = "if A, then B".

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S13-B2-Q1 · question page 289

LOGIC.CONDITIONAL.02 · Necessary means required; sufficient means enough

Having fuel is the only thing a gas car needs in order to run.

Answer: False

Reasoning: A car with fuel can still fail to run, for example with a dead battery. Fuel is necessary, not sufficient.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S13-B2-Q2 · question page 290

LOGIC.CONDITIONAL.02 · Necessary means required; sufficient means enough

Having fuel is sufficient for a gas car to run.

Answer: False

Reasoning: A car with fuel may still not run (for example, a dead battery).

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S13-B2-Q3 · question page 291

LOGIC.CONDITIONAL.02 · Necessary means required; sufficient means enough

Which is SUFFICIENT for a whole number to be even?

Answer: B. It is a multiple of 10.

Reasoning: Every multiple of 10 is even.

- ✗ It is greater than 2.: 3 is greater than 2 and odd
- ✗ It is a multiple of 3.: 3 is odd
- ✗ It ends in 1.: such numbers are odd

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S13-B2-Q4 · question page 292

LOGIC.CONDITIONAL.02 · Necessary means required; sufficient means enough

If A is sufficient for B, then whenever B happens, A happens.

Answer: False

Reasoning: Sufficient means A guarantees B. B can still happen without A.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S13-B2-Q5 · question page 293

LOGIC.CONDITIONAL.02 · Necessary means required; sufficient means enough

Being a mammal is ___ for being a dog.

Answer: A. necessary

Reasoning: Every dog is a mammal (required), but not every mammal is a dog.

- ✗ sufficient: a cat is a mammal but not a dog
- ✗ neither necessary nor sufficient: every dog is a mammal
- ✗ both necessary and sufficient: not sufficient

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S13-F1-Q1 · question page 294

LOGIC.CONDITIONAL.02 · Combine conditions

"n is even" is necessary for "n is divisible by 6". Give a whole number n that shows it is NOT sufficient: n is even but not divisible by 6.

Answer: any values that satisfy: isEven(n) and n % 6 != 0 and isInteger(n) (example: n = 4)

Reasoning: Any even number that is not a multiple of 6: 2, 4, 8, 10, ...

Tempting answers the rule rejects: n = 6; n = 12; n = 3

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S13-F1-Q2 · question page 295

LOGIC.CONDITIONAL.02 · Combine conditions

n is a whole number. Which condition is SUFFICIENT for "n is divisible by 4"?

Answer: D. n is divisible by 8

Reasoning: Every multiple of 8 is a multiple of 4.

- ✗ n is even: 2 is even but not divisible by 4
- ✗ n is divisible by 2 and by 3: 6 is not divisible by 4
- ✗ n is greater than 4: 5 is not divisible by 4

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: tested for every n from 1 to 200. Automatic: the check function accepts only the keyed choice.

S13-F1-Q3 · question page 296

LOGIC.CONDITIONAL.02 · Combine conditions

"A triangle is equilateral only if it is isosceles (has at least two equal sides)." Which statement is correct?

Answer: C. Being isosceles is necessary for being equilateral.

Reasoning: "A only if B": B is necessary for A.

- ✗ Being isosceles is sufficient for being equilateral.: a 5-5-8 triangle is isosceles, not equilateral
- ✗ Being equilateral is necessary for being isosceles.: reverses the relation
- ✗ The two conditions mean the same thing.: they do not

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S13-F1-Q4 · question page 297

LOGIC.CONDITIONAL.02 · Combine conditions

"You get the discount unless you pay late." Using "P unless Q" = "if not Q, then P", which person is guaranteed the discount?

Answer: D. someone who did not pay late

Reasoning: If you did not pay late, you get the discount.

- ✗ someone who paid late: the statement promises nothing to late payers
- ✗ everyone: late payers are not covered
- ✗ no one: on-time payers are covered

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S13-F1-Q5 · question page 298

LOGIC.CONDITIONAL.02 · Combine conditions

A condition can be both necessary and sufficient for another.

Answer: True

Reasoning: For example, for a whole number, "ends in 0 or 5" is necessary and sufficient for "divisible by 5".

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S13-F2-Q1 · question page 299

LOGIC.CONDITIONAL.02 · Chains of conditions

Facts: A is sufficient for B. B is sufficient for C. Which statement must be true?

Answer: A. C is necessary for A.

Reasoning: A -> B -> C, so A -> C: C is necessary for A (and A is sufficient for C).

- ✗ A is necessary for C.: C can hold without A
- ✗ B is sufficient for A.: reverses the first fact
- ✗ C is sufficient for A.: reverses the chain

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: all 8 truth assignments of A, B, C checked. Automatic: the check function accepts only the keyed choice.

S13-F2-Q2 · question page 300

LOGIC.CONDITIONAL.02 · Chains of conditions

For whole numbers n from 1 to 60, condition X: "n is divisible by 4"; condition Y: "n is divisible by 6". How many n satisfy Y but not X (showing X is not necessary for Y)?

Answer: 5

Reasoning: Multiples of 6 up to 60: 10 numbers. Those also divisible by 4 are multiples of 12: 5. So 10 - 5 = 5.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S13-F2-Q3 · question page 301

LOGIC.CONDITIONAL.02 · Chains of conditions

Which condition on a whole number n is BOTH necessary and sufficient for "n is divisible by 10"?

Answer: C. n is divisible by 2 and by 5

Reasoning: A number is divisible by 10 exactly when it is divisible by

both 2 and 5.

- ✗ n is divisible by 5: not sufficient: 15
- ✗ n is even: not sufficient: 4
- ✗ n is divisible by 20: not necessary: 10

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S13-F2-Q4 · question page 302

LOGIC.CONDITIONAL.02 · Chains of conditions

If A is necessary for B and B is necessary for A, then A and B are always both true or both false.

Answer: True

Reasoning: B -> A and A -> B together mean A and B have the same truth value.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S13-F2-Q5 · question page 303

LOGIC.CONDITIONAL.02 · Chains of conditions

Give a whole number n for which "n is a multiple of 3" holds but "n is a multiple of 9" fails, showing that being a multiple of 3 is not sufficient for being a multiple of 9.

Answer: any values that satisfy: n % 3 == 0 and n % 9 != 0 and isInteger(n) (example: n = 6)

Reasoning: Any multiple of 3 that is not a multiple of 9: 3, 6, 12, 15, ...

Tempting answers the rule rejects: n = 9; n = 18; n = 4

Grading: The learner's values are put into the rule; ANY values that make the rule true are

correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S14-M-Q1 · question page 305

LOGIC.GRID.01 · Solve a matching puzzle by combining clues

Three students, Ama, Bex and Cho, each play one instrument: drums, flute or violin (all different). Ama does not play the flute. Bex plays neither the drums nor the flute. Which instrument does each student play?

Answer: Ama: drums; Bex: violin; Cho: flute

Reasoning: Bex plays the violin (not drums, not flute). Ama does not play the flute, so Ama plays the drums, and Cho plays the flute.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 6 assignments enumerated; 1 satisfy every clue.

S14-M-Q2 · question page 306

LOGIC.GRID.01 · Solve a matching puzzle by combining clues

Four friends, Dara, Eli, Faye and Gus, each ordered a different drink: cocoa, juice, milk or tea. Dara did not order a hot drink (cocoa and tea are hot). Eli ordered tea. Faye ordered neither cocoa nor juice. Who ordered what?

Answer: Dara: juice; Eli: tea; Faye: milk; Gus: cocoa

Reasoning: Eli: tea. Faye: not cocoa, not juice, not tea, so milk. Dara: not hot, so juice. Gus: cocoa.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 24 assignments enumerated; 1 satisfy every clue.

S14-M-Q3 · question page 307

LOGIC.GRID.01 · Solve a matching puzzle by combining clues

Four runners wore bibs 1, 2, 3 and 4 (all different): Hal, Ivy, Jun and Kit. Ivy's number is

twice Hal's. Hal's number is even. Jun's number is odd. Kit's number is greater than Jun's. What number did each runner wear?

Answer: Hal: 2; Ivy: 4; Jun: 1; Kit: 3

Reasoning: Ivy = 2 x Hal with Hal even means Hal 2 and Ivy 4. Jun and Kit share 1 and 3; Jun is odd either way, and Kit is greater, so Jun 1 and Kit 3.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 24 assignments enumerated; 1 satisfy every clue.

S14-M-Q4 · question page 308

LOGIC.GRID.01 · Solve a matching puzzle by combining clues

Three pets, a cat, a dog and a rabbit, belong to Lea, Max and Noa (one each). The cat's owner sits next to Max. Lea owns neither the dog nor the cat. Noa does not own the dog. Who owns which pet?

Answer: Lea: rabbit; Max: dog; Noa: cat

Reasoning: Lea owns the rabbit. Noa does not own the dog, so Noa owns the cat and Max owns the dog. (The cat's owner sits next to Max, so the cat is not Max's.)

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 6 assignments enumerated; 1 satisfy every clue.

S14-M-Q5 · question page 309

LOGIC.GRID.01 · Solve a matching puzzle by combining clues

Four lockers, numbered 1 to 4 in a row, belong to Oz, Pia, Quin and Rae (one each). Pia's locker is directly to the right of Oz's (one number higher). Rae's locker is at an end of the row. Quin's locker number is even. Rae's number is lower than Quin's. Oz's locker is not number 2. Who has which locker?

Answer: Oz: 3; Pia: 4; Quin: 2; Rae: 1

Reasoning: Rae is at an end and lower than Quin, so Rae is 1. Oz is not 2, and Pia is directly after Oz, so Oz 3 and Pia 4. Quin gets 2, which is even.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 24 assignments enumerated; 1 satisfy every clue.

S14-B1-Q1 · question page 310

LOGIC.GRID.01 · Strategy: mark X for "no" and O for "yes" in a grid

Two friends, Ann and Bo, each pick a different color: red or blue. Ann does not pick red. Who picks which color?

Answer: Ann: blue; Bo: red

Reasoning: Ann: blue, so Bo: red.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 2 assignments enumerated; 1 satisfy every clue.

S14-B1-Q2 · question page 311

LOGIC.GRID.01 · Strategy: mark X for "no" and O for "yes" in a grid

In a 3 by 3 grid (3 people, 3 jobs, one each), a person's row has X in two cells. What follows?

Answer: B. The person has the job in the third cell.

Reasoning: Only one option is left in that row.

- ✗ The person has no job.: everyone has exactly one job
- ✗ Nothing follows yet.: only one option remains
- ✗ Another person has that job.: it must be this person's job

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S14-B1-Q3 · question page 312

LOGIC.GRID.01 · Strategy: mark X for "no" and O for "yes" in a grid

In a matching puzzle where each option is used once, if Kim has the bike, no one else has the bike.

Answer: True

Reasoning: Each option is used exactly once.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S14-B1-Q4 · question page 313

LOGIC.GRID.01 · Strategy: mark X for "no" and O for "yes" in a grid

Three cousins, Cy, Di and Ed, are ages 8, 9 and 10 (all different). Cy is the youngest. Ed is not 10. How old is each?

Answer: Cy: 8; Di: 10; Ed: 9

Reasoning: Cy 8; Ed not 10, so 9; Di 10.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 6 assignments enumerated; 1 satisfy every clue.

S14-B1-Q5 · question page 314

LOGIC.GRID.01 · Strategy: mark X for "no" and O for "yes" in a grid

Three people and three snacks, one each. Gil does not have the apple or the pear. Which snack does Gil have, if the snacks are apple, pear and plum?

Answer: D. plum

Reasoning: Two snacks are ruled out, so Gil has the plum.

✗ apple: ruled out

✗ pear: ruled out

✗ it cannot be known: only plum is left

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S14-F1-Q1 · question page 315

LOGIC.GRID.01 · Grids with relative clues

Four friends, Ari, Bea, Cal and Dev, finished a quiz with scores 6, 7, 8 and 9 (all different). Bea scored 1 more than Dev. Ari scored the most. Cal did not score 6. What did each score?

Answer: Ari: 9; Bea: 7; Cal: 8; Dev: 6

Reasoning: Ari 9. Bea and Dev are consecutive with Bea higher, among 6, 7, 8. Cal is not 6, so Dev is 6 and Bea 7, leaving Cal 8.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 24 assignments enumerated; 1 satisfy every clue.

S14-F1-Q2 · question page 316

LOGIC.GRID.01 · Grids with relative clues

Five boxes in a row, positions 1 to 5, hold a hat, a kite, a lamp, a mug and a net (one each). The kite is in an even position. The lamp is directly left of the mug. The hat is at position 5. The net is not next to the kite. Which item is in each position?

Answer: 1: net; 2: lamp; 3: mug; 4: kite; 5: hat

Reasoning: The hat is 5th. The kite is 2nd or 4th. Lamp-mug is a pair. If the kite were 2nd, the pair would be 3-4 and the net 1st, next to the kite: not allowed. So the kite is 4th, the pair is 1-2 or 2-3; the net is not next to the kite (not 3rd), so the pair is 2-3 and the net is 1st.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 120 assignments enumerated; 1 satisfy every clue.

S14-F1-Q3 · question page 317

LOGIC.GRID.01 · Grids with relative clues

Rae, Sol and Tom own a cat, a dog and a fish, one each. Clue 1: Rae does not own the cat. Clue 2: Sol does not own the dog. These two clues allow three matchings. Which additional clue leaves exactly one?

Answer: B. Tom owns the cat.

Reasoning: The three matchings allowed by clues 1 and 2 are (Rae, Sol, Tom) = (dog, cat, fish), (dog, fish, cat), (fish, cat, dog). Only "Tom owns the cat" keeps exactly one: (dog, fish, cat).

- ✗ Sol does not own the fish.: two matchings remain: Rae dog, Sol cat, Tom fish; and Rae fish, Sol cat, Tom dog
- ✗ Tom does not own the dog.: two remain: Rae dog with Sol cat or Sol fish
- ✗ Rae does not own the fish.: Rae owns the dog, and Sol and Tom can still split cat and fish two ways

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: each added clue tested against all 6 matchings. Automatic: the check function accepts only the keyed choice.

S14-F1-Q4 · question page 318

LOGIC.GRID.01 · Grids with relative clues

Three students, Uma, Vik and Wyn, each chose a different sport (golf, judo or swimming) and a different day (Monday, Tuesday, Wednesday). The judo student trains on Monday. Uma trains on Wednesday. Uma does not play golf. Wyn trains on Tuesday. Which sport did each choose?

Answer: Uma: swimming; Vik: judo; Wyn: golf

Reasoning: Uma trains Wednesday and Wyn Tuesday, so Vik trains Monday; the judo student trains Monday, so Vik does judo. Uma does not play golf, so Uma swims and Wyn plays golf.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 6 assignments enumerated; 1 satisfy every clue.

S14-F1-Q5 · question page 319

LOGIC.GRID.01 · Grids with relative clues

In a matching puzzle with 4 people and 4 options (one each), if three people's options are known, the fourth person's option is also known.

Answer: True

Reasoning: Only one option is left.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S14-F2-Q1 · question page 320

LOGIC.GRID.01 · Two-attribute grids and decisive clues

Four students, Al, Bo, Cy and Di, each have a different favorite number from 1 to 4 and a different color (red, blue, green, gold). The red student's number is 1 more than the blue student's number. Al is gold. Bo's number is 4. Cy's number is twice Di's number. What is each student's number?

Answer: Al: 3; Bo: 4; Cy: 2; Di: 1

Reasoning: Bo is 4. Cy = 2 x Di with 1, 2, 3 left gives Di 1 and Cy 2, so Al 3. (The colors are not asked, but they must also work: Al is gold; red is one more than blue, which among Bo 4, Cy 2, Di 1 means Cy red and Di blue, since 4 = 3 + 1 would need Al blue. So Bo is green.)

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 24 assignments enumerated; 1 satisfy every clue.

S14-F2-Q2 · question page 321

LOGIC.GRID.01 · Two-attribute grids and decisive clues

Three people wear a red, a blue and a green hat, one each, so there are 6 possible matchings. Clues have the form "person X does not wear color Y". What is the smallest number of such clues that can leave exactly one matching?

Answer: A. 3

Reasoning: Example with 3 clues: A not red, A not blue (so A green), B not red (so B blue): C red. With 2 clues, at least 2 matchings always remain (checked for every pair of clues).

- ✗ 1: one clue removes 2 matchings and leaves 4
- ✗ 2: two clues always leave at least 2 matchings
- ✗ 4: three well-chosen clues already suffice

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: every set of 1, 2 and 3 clues tested against the 6 matchings. Automatic: the check function accepts only the keyed choice.

S14-F2-Q3 · question page 322

LOGIC.GRID.01 · Two-attribute grids and decisive clues

Four people each get a different one of 4 prizes. How many matchings are possible if one person, Ash, cannot get prize 1 or prize 2?

Answer: 12

Reasoning: Ash has 2 choices (prize 3 or 4); the other three people take the remaining 3 prizes in 3 x 2 x 1 = 6 ways: 2 x 6 = 12.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S14-F2-Q4 · question page 323

LOGIC.GRID.01 · Two-attribute grids and decisive clues

In a 3-person, 3-option matching puzzle, the clues "A not x" and "B not x" together force C to

have x.

Answer: True

Reasoning: Option x must go to someone, and A and B are ruled out.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S14-F2-Q5 · question page 324

LOGIC.GRID.01 · Two-attribute grids and decisive clues

Five players wear the numbers 1 to 5 (all different). Al's number plus Bo's number is 9. Give one possible pair: Al's number a and Bo's number b.

Answer: any values that satisfy: a + b == 9 and a != b and a >= 1 and a <= 5 and b >= 1 and b <= 5 and isInteger(a) and isInteger(b) (example: a = 4, b = 5)

Reasoning: Only 4 + 5 = 9 works with different numbers up to 5, in either order.

Tempting answers the rule rejects: a = 3, b = 6; a = 5, b = 5

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it. Search of the stated domain: 2 valid answers.

S15-M-Q1 · question page 326

LOGIC.CONDITIONAL.03 · Form the converse, inverse and contrapositive and judge equivalence

What is the CONTRAPOSITIVE of "If a shape is a triangle, then it has three sides"?

Answer: C. If a shape does not have three sides, then it is not a triangle.

Reasoning: The contrapositive of "if P then Q" is "if not Q, then not P".

✗ If a shape has three sides, then it is a triangle.: that is the converse

✗ If a shape is not a triangle, then it does not have three sides.: that is the inverse

✗ A shape is a triangle and does not have three sides.: that is the negation of the conditional

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S15-M-Q2 · question page 327

LOGIC.CONDITIONAL.03 · Form the converse, inverse and contrapositive and judge equivalence

Which statement is logically EQUIVALENT to "If n is a multiple of 4, then n is even"?

Answer: D. If n is not even, then n is not a multiple of 4.

Reasoning: A conditional and its contrapositive always have the same truth value.

✗ If n is even, then n is a multiple of 4.: the converse is not equivalent (6 is even)

✗ If n is not a multiple of 4, then n is not even.: the inverse is not equivalent (6 is even)

✗ n is a multiple of 4 and n is even.: not a conditional

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S15-M-Q3 · question page 328

LOGIC.CONDITIONAL.03 · Form the converse, inverse and contrapositive and judge equivalence

The converse of "If a whole number is a multiple of 6, then it is even" is true.

Answer: False

Reasoning: The converse, "If a whole number is even, then it is a multiple of 6", is false: 4 is even but not a multiple of 6.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S15-M-Q4 · question page 329

LOGIC.CONDITIONAL.03 · Form the converse, inverse and contrapositive and judge equivalence

The statement "If x > 2, then x squared > 4" is true. Find a real number x that shows its CONVERSE is false.

Answer: any values that satisfy: x^2 > 4 and not (x > 2) (example: x = -3)

Reasoning: The converse is "If x squared > 4, then x > 2". It fails for any x < -2, for example -3.

Tempting answers the rule rejects: x = 3; x = 1; x = -2

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S15-M-Q5 · question page 330

LOGIC.CONDITIONAL.03 · Form the converse, inverse and contrapositive and judge equivalence

"If P, then Q" is true. Which statement MUST also be true?

Answer: A. If not Q, then not P.

Reasoning: Only the contrapositive is equivalent to the original.

- ✗ If Q, then P.: the converse can be false
- ✗ If not P, then not Q.: the inverse can be false

✗ P and Q.: P may be false

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: checked in every row where "if P then Q" is true. Automatic: the check function accepts only the keyed choice.

S15-B1-Q1 · question page 331

LOGIC.CONNECTIVE.01 · Negate each part

What is the negation of "the shape has three sides"?

Answer: C. the shape does not have three sides

Reasoning: Put "not" on the whole claim.

✗ the shape has four sides: too specific
✗ the shape has no sides: too specific
✗ the shape has at least three sides: not a negation

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S15-B1-Q2 · question page 332

LOGIC.CONNECTIVE.01 · Negate each part

What is the negation of "n is even" for a whole number n?

Answer: B. n is odd

Reasoning: Whole numbers that are not even are odd.

✗ n is zero: too specific
✗ n is prime: unrelated
✗ n is negative: unrelated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S15-B1-Q3 · question page 333

LOGIC.CONNECTIVE.01 · Negate each part

The negation of "x > 3" is "x < 3".

Answer: False

Reasoning: It is "x ≤ 3".

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S15-B1-Q4 · question page 334

LOGIC.CONNECTIVE.01 · Negate each part

What is the negation of "it is raining"?

Answer: A. it is not raining

Reasoning: Negate the whole claim.

- ✗ it is sunny: it could be cloudy and dry
- ✗ it rained yesterday: unrelated
- ✗ it will rain: unrelated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S15-B1-Q5 · question page 335

LOGIC.CONNECTIVE.01 · Negate each part

"Not (not P)" says the same as "not P".

Answer: False

Reasoning: Two negations cancel, so "not (not P)" says the same as P.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S15-B2-Q1 · question page 336

LOGIC.CONDITIONAL.01 · Hypothesis and conclusion

In "If it snows, school is closed," what is the conclusion?

Answer: B. school is closed

Reasoning: The "then" part.

- ✗ it snows: the hypothesis
- ✗ it is cold: not stated
- ✗ school is open: not stated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S15-B2-Q2 · question page 337

LOGIC.CONDITIONAL.01 · Hypothesis and conclusion

Swap the parts of "If P, then Q". What do you get?

Answer: D. If Q, then P.

Reasoning: Swapping gives the converse.

- ✗ If not P, then not Q.: that negates instead of swapping
- ✗ If not Q, then not P.: that swaps AND negates
- ✗ P and Q.: not a conditional

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S15-B2-Q3 · question page 338

LOGIC.CONDITIONAL.01 · Hypothesis and conclusion

In "If x = 5, then x + 1 = 6," the conclusion is "x = 5".

Answer: False

Reasoning: The conclusion is the part after "then": "x + 1 = 6". "x = 5" is the hypothesis.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S15-B2-Q4 · question page 339

LOGIC.CONDITIONAL.01 · Hypothesis and conclusion

Which statement has hypothesis "a figure is a square"?

Answer: C. If a figure is a square, then it is a rectangle.

Reasoning: The hypothesis is the "if" part: "a figure is a square".

- ✗ If a figure is a rectangle, then it is a square.: hypothesis is "rectangle"
- ✗ A figure is a rectangle only if it has four sides.: hypothesis is "a figure is a rectangle"
- ✗ Every rectangle is a square.: hypothesis is "rectangle"

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S15-B2-Q5 · question page 340

LOGIC.CONDITIONAL.01 · Hypothesis and conclusion

Swapping and negating both parts of "If P, then Q" gives "If not Q, then not P".

Answer: True

Reasoning: That is the contrapositive.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S15-F1-Q1 · question page 341

LOGIC.CONDITIONAL.03 · Which forms are equivalent?

Complete the truth table for "if P then Q" and its contrapositive "if not Q then not P".

Answer: T T / F F / T T / T T

Reasoning: The two columns are identical: a conditional and its contrapositive are equivalent.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S15-F1-Q2 · question page 342

LOGIC.CONDITIONAL.03 · Which forms are equivalent?

Which two statements are ALWAYS equivalent to each other?

Answer: A. the converse and the inverse

Reasoning: The inverse "if not P then not Q" is the contrapositive of the converse "if Q then P".

- ✗ the original and the converse: they differ when P is false and Q true
- ✗ the original and the inverse: they differ when P is false and Q true
- ✗ the converse and the contrapositive: they differ when P is false and Q true

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S15-F1-Q3 · question page 343

LOGIC.CONDITIONAL.03 · Which forms are equivalent?

For how many whole numbers n from 1 to 20 is the INVERSE of "If n is a multiple of 4, then n is even" false?

Answer: 5

Reasoning: The inverse is "If n is not a multiple of 4, then n is not even". It is false when n is even but not a multiple of 4: 2, 6, 10, 14, 18.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S15-F1-Q4 · question page 344

LOGIC.CONDITIONAL.03 · Which forms are equivalent?

If a conditional is false, its contrapositive is true.

Answer: False

Reasoning: A conditional and its contrapositive are equivalent, so they are always true or false together.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S15-F1-Q5 · question page 345

LOGIC.CONDITIONAL.03 · Which forms are equivalent?

"If a quadrilateral is a square, then its diagonals are equal." Which related statement is FALSE?

Answer: B. the converse: if the diagonals are equal, the quadrilateral is a square

Reasoning: A rectangle that is not a square has equal diagonals, so the converse is false.

✗ the contrapositive: if the diagonals are not equal, it is not a square: equivalent to the true original

✗ the original statement: true for every square

✗ "a square has equal diagonals": the original, restated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S15-F2-Q1 · question page 346

LOGIC.CONDITIONAL.03 · Converses that hold and fail

For whole numbers, both "If n is odd, then n squared is odd" and its converse are true.

Answer: True

Reasoning: An odd number squared is odd; and if n were even, n squared would be even, so n squared odd forces n odd.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S15-F2-Q2 · question page 347

LOGIC.CONDITIONAL.03 · Converses that hold and fail

The statement "If n is prime and greater than 2, then n is odd" is true. Its converse is "If n is odd, then n is prime and greater than 2". Give a whole number n that shows the converse is false.

Answer: any values that satisfy: isOdd(n) and not (isPrime(n) and n > 2) (example: n = 9)

Reasoning: Any odd number that is not a prime greater than 2: 1, 9, 15, 21, ...

Tempting answers the rule rejects: n = 7; n = 4; n = 3

Grading: The learner's values are put into the rule; ANY values that make the rule true are

correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S15-F2-Q3 · question page 348

LOGIC.CONDITIONAL.03 · Converses that hold and fail

A statement and its converse are both true. Which statement must be true?

Answer: C. P if and only if Q

Reasoning: If P -> Q and Q -> P, then P and Q always match: that is "P if and only if Q".

- ✗ P and Q: both could be false
- ✗ not P: P could be true
- ✗ P or Q: both could be false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S15-F2-Q4 · question page 349

LOGIC.CONDITIONAL.03 · Converses that hold and fail

In how many of the 4 rows of a truth table do "if P then Q" and its converse "if Q then P" have DIFFERENT truth values?

Answer: 2

Reasoning: They differ when exactly one of P, Q is true: rows TF and FT.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S15-F2-Q5 · question page 350

LOGIC.CONDITIONAL.03 · Converses that hold and fail

C is the conditional "if P then Q". Which statement is correct?

Answer: D. If C is true, its inverse may be true or false.

Reasoning: In the rows where C is true, the inverse is true for (T, T) and (F, F) and false for (F, T), so it may be either.

- ✗ If C is true, its converse is true.: when P is false and Q true, C is true but the converse is false
- ✗ If C is false, its converse is false.: C is false only when P is true and Q false; then the converse "if Q then P" is true
- ✗ If C is true, its contrapositive may be false.: never: they are equivalent

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: each claim tested in all four rows. Automatic: the check function accepts only the keyed choice.

S16-M-Q1 · question page 352

LOGIC.CONDITIONAL.04 · Find truth values of "if and only if" statements and use them as definitions

Complete the truth table for "P if and only if Q".

Answer: T / F / F / T

Reasoning: It is true exactly when P and Q have the same truth value.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S16-M-Q2 · question page 353

LOGIC.CONDITIONAL.04 · Find truth values of "if and only if" statements and use them as definitions

For real numbers x, "x squared = 9 if and only if x = 3" is true.

Answer: False

Reasoning: x = -3 makes the left part true and the right part false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S16-M-Q3 · question page 354

LOGIC.CONDITIONAL.04 · Find truth values of "if and only if" statements and use them as definitions

Find a real number x that makes "x > 5 if and only if x > 3" FALSE.

Answer: any values that satisfy: (x > 5) != (x > 3) (example: x = 4)

Reasoning: The two parts must differ: x > 3 but not x > 5, so any x with 3 < x ≤ 5.

Tempting answers the rule rejects: x = 6; x = 2; x = 3

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S16-M-Q4 · question page 355

LOGIC.CONDITIONAL.04 · Find truth values of "if and only if" statements and use them as definitions

Which statement is a correct DEFINITION of an even whole number?

Answer: B. A whole number is even if and only if it is divisible by 2.

Reasoning: A definition must work both ways: every even number is divisible by 2, and every number divisible by 2 is even.

✗ A whole number is even if it is divisible by 4.: true, but it leaves out 2, 6, 10, ...

✗ A whole number is even only if it is greater than 1.: 0 is even

✗ A whole number is even if and only if it ends in 0.: 4 is even and does not end in 0

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S16-M-Q5 · question page 356

LOGIC.CONDITIONAL.04 · Find truth values of "if and only if" statements and use them as definitions

"P if and only if Q" is true and P is false. What follows?

Answer: C. Q is false.

Reasoning: In a true biconditional the two parts have the same value.

✗ Q is true.: then P and Q would differ

✗ Nothing about Q.: Q must match P

✗ P and Q are both true.: P is false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S16-B1-Q1 · question page 357

LOGIC.CONDITIONAL.01 · Conditionals first

"If 2 > 3, then 4 > 5" is true.

Answer: True

Reasoning: The "if" part is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S16-B1-Q2 · question page 358

LOGIC.CONDITIONAL.01 · Conditionals first

"If 3 > 2, then 4 > 5" is true.

Answer: False

Reasoning: True "if", false "then".

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S16-B1-Q3 · question page 359

LOGIC.CONDITIONAL.01 · Conditionals first

When is "if P then Q" false?

Answer: A. when P is true and Q is false

Reasoning: Only one row is false.

- ✗ when P is false: it is true then
- ✗ when Q is true: it is true then
- ✗ when both are false: it is true then

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S16-B1-Q4 · question page 360

LOGIC.CONDITIONAL.01 · Conditionals first

"If 4 > 1, then 2 > 6" is true.

Answer: False

Reasoning: The "if" part is true and the "then" part is false, which is the

one case that makes a conditional false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S16-B1-Q5 · question page 361

LOGIC.CONDITIONAL.01 · Conditionals first

In how many of the 4 rows is "if P then Q" true?

Answer: 3

Reasoning: All rows except P true, Q false.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S16-B2-Q1 · question page 362

LOGIC.CONDITIONAL.03 · The converse

What is the converse of "If it is a dog, then it barks"?

Answer: D. If it barks, then it is a dog.

Reasoning: Swap the parts.

- ✗ If it is not a dog, then it does not bark.: the inverse
- ✗ If it does not bark, then it is not a dog.: the contrapositive
- ✗ It is a barking dog.: not a conditional

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S16-B2-Q2 · question page 363

LOGIC.CONDITIONAL.03 · The converse

The converse of "If n = 2, then n is even" is "If n is not 2, then n is not even".

Answer: False

Reasoning: That statement negates both parts; it is the inverse. The converse swaps the parts: "If n is even, then n = 2".

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S16-B2-Q3 · question page 364

LOGIC.CONDITIONAL.03 · The converse

The converse of a true statement is always true.

Answer: False

Reasoning: Example: "If n = 2, then n is even" is true, its converse is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S16-B2-Q4 · question page 365

LOGIC.CONDITIONAL.03 · The converse

"P if and only if Q" means "if P then Q" AND what?

Answer: A. if Q then P

Reasoning: The other direction is the converse.

- ✗ if not P then Q: not the converse
- ✗ P or Q: not part of it
- ✗ not P and not Q: not part of it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S16-B2-Q5 · question page 366

LOGIC.CONDITIONAL.03 · The converse

If a conditional is true but its converse is false, the biconditional is true.

Answer: False

Reasoning: A biconditional needs both directions true; a false converse makes it false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S16-F1-Q1 · question page 367

LOGIC.CONDITIONAL.04 · Biconditionals as definitions and tests

For how many whole numbers n from 1 to 20 is "n is even if and only if n is a multiple of 4" TRUE?

Answer: 15

Reasoning: It fails exactly when n is even but not a multiple of 4: 2, 6, 10, 14, 18 (5 numbers). The other 15 make it true (odd numbers make both parts false).

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S16-F1-Q2 · question page 368

LOGIC.CONDITIONAL.04 · Biconditionals as definitions and tests

Complete the table to compare "P if and only if Q" with "(if P then Q) and (if Q then P)".

Answer: T T / F F / F F / T T

Reasoning: The columns match in every row.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S16-F1-Q3 · question page 369

LOGIC.CONDITIONAL.04 · Biconditionals as definitions and tests

Which definition of a SQUARE is correct (works in both directions)?

Answer: B. A quadrilateral is a square if and only if it has four equal sides and four right angles.

Reasoning: Only the first condition is true of every square and of nothing else.

- ✗ A quadrilateral is a square if and only if it has four equal sides.: a rhombus with no right angles has four equal sides
- ✗ A quadrilateral is a square if and only if it has four right angles.: a non-square rectangle has four right angles
- ✗ A quadrilateral is a square if and only if its diagonals are equal.: a non-square rectangle has equal diagonals

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S16-F1-Q4 · question page 370

LOGIC.CONDITIONAL.04 · Biconditionals as definitions and tests

"A whole number is divisible by 10 if and only if it ends in 0" is true.

Answer: True

Reasoning: Both directions hold for whole numbers.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S16-F1-Q5 · question page 371

LOGIC.CONDITIONAL.04 · Biconditionals as definitions and tests

Find a whole number n that makes "n is divisible by 3 if and only if n is odd" FALSE.

Answer: any values that satisfy: (n % 3 == 0) != isOdd(n) and isInteger(n) (example: n = 6)

Reasoning: The parts must differ: divisible by 3 but even (6, 12, ...) or odd but not divisible by 3 (1, 5, ...).

Tempting answers the rule rejects: n = 9; n = 4; n = 15

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S16-F2-Q1 · question page 372

LOGIC.CONDITIONAL.04 · Chains and opposites of biconditionals

"P exclusive-or Q" has the opposite truth value of "P if and only if Q" in every row.

Answer: True

Reasoning: Exclusive-or is true exactly when the parts differ; the biconditional exactly when they match.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S16-F2-Q2 · question page 373

LOGIC.CONDITIONAL.04 · Chains and opposites of biconditionals

"P iff Q" and "Q iff R" are both true. Which must be true?

Answer: C. P iff R

Reasoning: P matches Q and Q matches R, so P matches R.

- ✗ P and R: all three could be false
- ✗ P exclusive-or R: P and R match
- ✗ not Q: Q could be true

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S16-F2-Q3 · question page 374

LOGIC.CONDITIONAL.04 · Chains and opposites of biconditionals

In how many of the 8 rows of a truth table for P, Q, R is "(P iff Q) iff R" true?

Answer: 4

Reasoning: Exactly half of the rows: the statement is true when an even number of P, Q, R are false.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S16-F2-Q4 · question page 375

LOGIC.CONDITIONAL.04 · Chains and opposites of biconditionals

Find whole numbers a and b, each from 1 to 9, that make "a + b is even if and only if a is even" TRUE and a different from b.

Answer: any values that satisfy: (isEven(a + b) == isEven(a)) and a != b and a >= 1 and a <= 9 and b >= 1 and b <= 9 (example: a = 2, b = 4)

Reasoning: The biconditional is true exactly when b is even (then a + b and a have the same parity). Any a, b with b even and a different from b

works, such as 2 and 4 or 3 and 8.

Tempting answers the rule rejects: a = 2, b = 3; a = 4, b = 4

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it. Search of the stated domain: 32 valid answers.

S16-F2-Q5 · question page 376

LOGIC.CONDITIONAL.04 · Chains and opposites of biconditionals

Which statement about whole numbers is TRUE?

Answer: D. n is odd if and only if n + 1 is even.

Reasoning: Adding 1 always changes parity, so n odd and n + 1 even always match.

- ✗ n is odd if and only if n is prime.: 9 is odd, not prime; 2 is prime, not odd
- ✗ n is even if and only if n is a multiple of 4.: 6 is even but not a multiple of 4
- ✗ n is a multiple of 3 if and only if n is a multiple of 9.: 3 is a multiple of 3 but not of 9

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S17-M-Q1 · question page 378

LOGIC.KNIGHTS.01 · Decide who tells the truth when each speaker always tells the truth or always lies

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Ana says: "Ben is a liar." Ben says: "Ana and I are both truth-tellers." Mark each person as a truth-teller or a liar.

Answer: Ana: truth-teller; Ben: liar

Reasoning: If Ben were truthful, both would be truthful, but then Ana's "Ben is a liar" would be false. So Ben lies, Ana's statement is true, and Ana is a truth-teller.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 4 assignments enumerated; 1 satisfy every clue.

S17-M-Q2 · question page 379

LOGIC.KNIGHTS.01 · Decide who tells the truth when each speaker always tells the truth or always lies

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Cal says: "Dee and I are different kinds." Dee says: "Cal is a liar." Mark each person as a truth-teller or a liar.

Answer: Cal: truth-teller; Dee: liar

Reasoning: Suppose Cal tells the truth: they are different, so Dee is a liar, and Dee's claim "Cal is a liar" is indeed false. Consistent. Suppose Cal lies: they are the same kind, so Dee lies too, but then "Cal is a liar" would be true, a contradiction. So Cal is truthful and Dee lies.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 4 assignments enumerated; 1 satisfy every clue.

S17-M-Q3 · question page 380

LOGIC.KNIGHTS.01 · Decide who tells the truth when each speaker always tells the truth or always lies

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Eve says: "Fin and Gia are both liars." Fin says: "Eve is a truth-teller." Gia says: "Fin is a liar." Mark each person as a truth-teller or a liar.

Answer: Eve: liar; Fin: liar; Gia: truth-teller

Reasoning: Fin and Eve stand or fall together (Fin says Eve is truthful). If both were truthful, Eve's claim would make Fin a liar: contradiction. So Eve and Fin are liars. Then Gia's "Fin is a liar" is true, so Gia is truthful (and Eve's claim is indeed false).

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 8 assignments enumerated; 1 satisfy every clue.

S17-M-Q4 · question page 381

LOGIC.KNIGHTS.01 · Decide who tells the truth when each speaker always tells the truth or always lies

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Jo says: "Kai is a liar." Kai says: "Lu is a liar." Lu says: "Jo and Kai are both liars." Mark each person as a truth-teller or a liar.

Answer: Jo: liar; Kai: truth-teller; Lu: liar

Reasoning: If Lu were truthful, Jo and Kai would both lie, but Jo lying means Kai is truthful: contradiction. So Lu lies; then Kai's "Lu is a liar" is true, Kai is truthful, and Jo (who calls Kai a liar) lies.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 8 assignments enumerated; 1 satisfy every clue.

S17-M-Q5 · question page 382

LOGIC.KNIGHTS.01 · Decide who tells the truth when each speaker always tells the truth or always lies

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Mo says: "Exactly one of Ned and Ola is a liar." Ned says: "Mo is a liar." Ola says: "Ned and I are the same kind." Who is a liar?

Answer: B. Mo only

Reasoning: Check the 8 possibilities: only Mo lying, with Ned and Ola truthful, is consistent. Ned truthfully says Mo lies; Ola truthfully says Ned

and Ola match; Mo's claim "exactly one of Ned and Ola lies" is false.

✗ Ned only: then Mo and Ola are truthful, but Ola would be claiming Ned and Ola are the same kind

✗ Ola only: then Ned is truthful and calls Mo a liar, but Mo would be truthful

✗ Mo and Ned: then Ola is truthful, but Ola says Ned and Ola are the same kind

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: all 8 assignments checked. Automatic: the check function accepts only the keyed choice.

S17-B1-Q1 · question page 383

LOGIC.KNIGHTS.01 · Strategy: suppose, then test

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. A truth-teller can say "I am a liar."

Answer: False

Reasoning: That would be a false statement.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S17-B1-Q2 · question page 384

LOGIC.KNIGHTS.01 · Strategy: suppose, then test

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. A liar can say "I am a truth-teller."

Answer: True

Reasoning: That is a false statement, which a liar can make.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S17-B1-Q3 · question page 385

LOGIC.KNIGHTS.01 · Strategy: suppose, then test

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Vic says: "2 + 2 = 5." What is Vic?

Answer: C. a liar

Reasoning: A false statement comes from a liar.

- ✗ a truth-teller: the statement is false
- ✗ either: only liars make false statements
- ✗ neither: everyone is one or the other

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S17-B1-Q4 · question page 386

LOGIC.KNIGHTS.01 · Strategy: suppose, then test

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Wes is a truth-teller and says: "Xia is a liar." What is Xia?

Answer: D. a liar

Reasoning: Wes tells the truth.

- ✗ a truth-teller: Wes speaks truly
- ✗ either: Wes's true statement decides it
- ✗ neither: everyone is one or the other

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S17-B1-Q5 · question page 387

LOGIC.KNIGHTS.01 · Strategy: suppose, then test

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Yan is a liar and says: "Zed is a liar." What is Zed?

Answer: A. a truth-teller

Reasoning: The statement is false, so Zed is not a liar.

- ✗ a liar: Yan's statement is false
- ✗ either: it is decided
- ✗ neither: everyone is one or the other

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S17-F1-Q1 · question page 388

LOGIC.KNIGHTS.01 · Statements about counts

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Ash says: "Exactly one of us three is a truth-teller." Bly says: "Exactly two of us three are truth-tellers." Cru says: "None of us three is a truth-teller." Mark each person as a truth-teller or a liar.

Answer: Ash: truth-teller; Bly: liar; Cru: liar

Reasoning: The three claims contradict each other, so at most one is true. Cru cannot be truthful (that would make none truthful). If none were truthful, Cru's claim would be true: contradiction. So exactly one is truthful, which makes Ash's claim true: Ash is the truth-teller.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 8 assignments enumerated; 1 satisfy every clue.

S17-F1-Q2 · question page 389

LOGIC.KNIGHTS.01 · Statements about counts

On this island every person is either a truth-teller, who always tells the truth, or a liar, who

always lies. Dov says: "At least one of Ela and Fyn is a liar." Ela says: "Dov and Fyn are both truth-tellers." Fyn says: "Dov is a liar." Mark each person as a truth-teller or a liar.

Answer: Dov: truth-teller; Ela: liar; Fyn: liar

Reasoning: Suppose Dov truthful: then Fyn ("Dov is a liar") lies, so Ela's claim is false and Ela lies; Dov's claim (one of Ela, Fyn lies) is true. Consistent. Suppose Dov lies: then Ela and Fyn are both truthful, but Ela says Dov is truthful: contradiction. So Dov is truthful, Ela and Fyn lie.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 8 assignments enumerated; 1 satisfy every clue.

S17-F1-Q3 · question page 390

LOGIC.KNIGHTS.01 · Statements about counts

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Pat says: "Exactly two of us three are truth-tellers." Quy says: "Pat is a liar." Ren says: "Quy and Pat are the same kind." Mark each person as a truth-teller or a liar.

Answer: Pat: liar; Quy: truth-teller; Ren: liar

Reasoning: Quy and Pat are always different kinds (Quy calls Pat a liar), so Ren's claim is false: Ren lies. Then exactly one of Pat, Quy is truthful, so the number of truth-tellers is 1, Pat's claim is false, Pat lies and Quy is truthful.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 8 assignments enumerated; 1 satisfy every clue.

S17-F1-Q4 · question page 391

LOGIC.KNIGHTS.01 · Statements about counts

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. If two people each say "The other one is a liar," then exactly one of them is a liar.

Answer: True

Reasoning: If A is truthful, B lies; if A lies, B is truthful. Either way they are different.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S17-F1-Q5 · question page 392

LOGIC.KNIGHTS.01 · Statements about counts

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Four people each say exactly the same sentence: "Exactly one of us four is a truth-teller." How many of them are truth-tellers?

Answer: 0

Reasoning: They say the same sentence, so they are all the same kind. If all were truthful, there would be four truth-tellers, not one: impossible. So all four lie, and indeed the sentence is false (there are none).

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S17-F2-Q1 · question page 393

LOGIC.KNIGHTS.01 · Larger systems and impossible statements

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Uri says: "Exactly three of us four are liars." Vee says: "Uri is a truth-teller." Wil says: "Vee is a liar." Xan says: "Wil is a truth-teller and Uri is a liar." Mark each person as a truth-teller or a liar.

Answer: Uri: liar; Vee: liar; Wil: truth-teller; Xan: truth-teller

Reasoning: Vee and Uri are the same kind; Wil is the opposite of Vee. If Uri were truthful, Vee would be too, and at most two could lie:

contradiction with "exactly three". So Uri and Vee lie and Wil is truthful. Xan's claim (Wil truthful, Uri lying) is true, so Xan is truthful. Two liars: Uri's claim is false, as it must be.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 16 assignments enumerated; 1 satisfy every clue.

S17-F2-Q2 · question page 394

LOGIC.KNIGHTS.01 · Larger systems and impossible statements

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Which statement can NO person on the island say?

Answer: A. "I am a liar."

Reasoning: A truth-teller saying it would be lying; a liar saying it would be telling the truth.

- ✗ "I am a truth-teller.": anyone can say it: true for a truth-teller, false for a liar
- ✗ "2 + 2 = 4.": a truth-teller can say it
- ✗ "2 + 2 = 5.": a liar can say it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S17-F2-Q3 · question page 395

LOGIC.KNIGHTS.01 · Larger systems and impossible statements

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Five people stand in a circle. Each says: "The person on my left is a liar." In how many ways can the five be truth-tellers and liars so that every statement fits?

Answer: 0

Reasoning: Each statement makes neighbors different kinds, so the types must alternate around the circle. An odd circle cannot alternate

(going once around returns to the start after 5 changes), so no assignment works.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S17-F2-Q4 · question page 396

LOGIC.KNIGHTS.01 · Larger systems and impossible statements

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. A person says: "I am a liar or 2 + 2 = 4." This person must be a truth-teller.

Answer: True

Reasoning: The statement is true (2 + 2 = 4), so the speaker tells the truth. A liar could not say a true statement.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S17-F2-Q5 · question page 397

LOGIC.KNIGHTS.01 · Larger systems and impossible statements

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. You meet Gil and Hal. Gil says: "At least one of us two is a truth-teller." Hal says nothing. Which statement MUST be true?

Answer: D. If Hal is a truth-teller, then Gil is a truth-teller.

Reasoning: Three cases fit: both truthful; Gil truthful and Hal lying; both lying. In every case where Hal is truthful, Gil's claim is true, so Gil is truthful.

- ✗ Gil is a truth-teller.: Gil and Hal could both be liars: then Gil's claim is false, as a liar's must be
- ✗ Hal is a liar.: Gil and Hal could both be truth-tellers

✗ Both are liars.: Gil could be a truth-teller

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: all 4 assignments checked; 3 fit. Automatic: the check function accepts only the keyed choice.

S18-M-Q1 · question page 399

LOGIC.TRUTH.01 · Complete truth tables for compound statements

Complete the truth table for "P and (not Q)".

Answer: F F / T T / F F / T F

Reasoning: First fill "not Q", then combine with P: true only when P is true and Q is false.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S18-M-Q2 · question page 400

LOGIC.TRUTH.01 · Complete truth tables for compound statements

Complete the truth table for "(not P) or Q".

Answer: F T / F F / T T / T T

Reasoning: False only when P is true and Q is false (it matches "if P then Q").

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S18-M-Q3 · question page 401

LOGIC.TRUTH.01 · Complete truth tables for compound statements

Complete the truth table for "(P or Q) and not (P and Q)".

Answer: T T F / T F T / T F T / F F F

Reasoning: True exactly when one of P, Q is true: exclusive or.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S18-M-Q4 · question page 402

LOGIC.TRUTH.01 · Complete truth tables for compound statements

A truth table for three statements P, Q and R lists every combination of truth values. How many rows does it have?

Answer: 8

Reasoning: Each statement has 2 values: 2 x 2 x 2 = 8.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S18-M-Q5 · question page 403

LOGIC.TRUTH.01 · Complete truth tables for compound statements

Complete the truth table for "(P and Q) or R".

Answer: T T / T T / F T / F F / F T / F F / F T / F F

Reasoning: True when R is true or both P and Q are true.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (8 rows).

S18-B1-Q1 · question page 404

LOGIC.CONNECTIVE.02 · The basic connectives

Complete the truth table for "P and Q".

Answer: T / F / F / F

Reasoning: True only when both are true.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S18-B1-Q2 · question page 405

LOGIC.CONNECTIVE.02 · The basic connectives

Complete the truth table for "P or Q" (inclusive).

Answer: T / T / T / F

Reasoning: False only when both are false.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S18-B1-Q3 · question page 406

LOGIC.CONNECTIVE.02 · The basic connectives

Complete the truth table for "not P".

Answer: F / T

Reasoning: Flip the value.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (2 rows).

S18-B1-Q4 · question page 407

LOGIC.CONNECTIVE.02 · The basic connectives

"P and Q" is true when P is true and Q is false.

Answer: False

Reasoning: Both must be true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S18-B1-Q5 · question page 408

LOGIC.CONNECTIVE.02 · The basic connectives

"P or Q" is true when P is false and Q is true.

Answer: True

Reasoning: At least one is true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S18-B2-Q1 · question page 409

LOGIC.TRUTH.01 · Work from the inside out

In "not (P and Q)", which part is computed first?

Answer: C. P and Q

Reasoning: Inside the brackets first.

- ✗ not P: "not" applies to the whole bracket
- ✗ not Q: same reason
- ✗ P or Q: not in the statement

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S18-B2-Q2 · question page 410

LOGIC.TRUTH.01 · Work from the inside out

"not (P and Q)" and "(not P) and Q" always have the same truth value.

Answer: False

Reasoning: P true, Q true: the first is false and the second false; P false, Q false: the first is true and the second false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S18-B2-Q3 · question page 411

LOGIC.TRUTH.01 · Work from the inside out

P is true and Q is false. Is "not (P and Q)" true or false? Enter 1 for true or 0 for false.

Answer: 1

Reasoning: P and Q is false, so its negation is true.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S18-B2-Q4 · question page 412

LOGIC.TRUTH.01 · Work from the inside out

P is false, Q is true. What is the value of "(P or Q) and P"?

Answer: B. false

Reasoning: P or Q is true; true and false is false.

- ✗ true: the final "and P" is false
- ✗ it cannot be found: all parts are known
- ✗ both: a statement has one value

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S18-B2-Q5 · question page 413

LOGIC.TRUTH.01 · Work from the inside out

Complete the truth table for "not (P or Q)".

Answer: T F / T F / T F / F T

Reasoning: Compute the bracket, then negate.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S18-F1-Q1 · question page 414

LOGIC.TRUTH.01 · Three-variable tables

Complete the truth table for "P and (Q or R)".

Answer: T T / T T / T T / F F / T F / T F / T F / F F

Reasoning: True when P is true and at least one of Q, R is true.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (8 rows).

S18-F1-Q2 · question page 415

LOGIC.TRUTH.01 · Three-variable tables

Complete the truth table for "(P or Q) and (not R)".

Answer: T F / T T / T F / T T / T F / T T / F F / F F

Reasoning: True when R is false and at least one of P, Q is true.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (8 rows).

S18-F1-Q3 · question page 416

LOGIC.TRUTH.01 · Three-variable tables

In how many of the 8 rows is "P or (Q and R)" true?

Answer: 5

Reasoning: True whenever P is true (4 rows) plus the row P false, Q true, R true: 5.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S18-F1-Q4 · question page 417

LOGIC.TRUTH.01 · Three-variable tables

A statement in P and Q is true only in the row P true, Q false. Which statement is it?

Answer: A. P and (not Q)

Reasoning: Check each candidate in the four rows.

- ✗ P or (not Q): also true when both are true and when both are false
- ✗ (not P) and Q: true only when P is false and Q true
- ✗ not (P and Q): true in three rows

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S18-F1-Q5 · question page 418

LOGIC.TRUTH.01 · Three-variable tables

A truth table for 4 statements has 16 rows.

Answer: True

Reasoning: 2 x 2 x 2 x 2 = 16.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S18-F2-Q1 · question page 419

LOGIC.TRUTH.01 · Read and build statements from tables

A statement in P, Q, R is true in exactly the rows where an odd number of P, Q, R are true. Which statement is it?

Answer: C. P exclusive-or (Q exclusive-or R)

Reasoning: Exclusive-or flips the value each time a true part is added.

- ✗ P or Q or R: true also when two are true
- ✗ P and Q and R: true only when all three are true
- ✗ (P and Q) or R: true in rows with two true

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S18-F2-Q2 · question page 420

LOGIC.TRUTH.01 · Read and build statements from tables

How many different possible output columns can a truth table with two statement letters P and Q have (4 rows, each output T or F)?

Answer: 16

Reasoning: Each of the 4 rows can be T or F: 2 to the 4th = 16.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S18-F2-Q3 · question page 421

LOGIC.TRUTH.01 · Read and build statements from tables

Complete the truth table for "(P and not Q) or (not P and Q)".

Answer: F F F / T F T / F T T / F F F

Reasoning: Each row where exactly one is true makes one bracket true.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S18-F2-Q4 · question page 422

LOGIC.TRUTH.01 · Read and build statements from tables

In how many of the 16 rows of a table for P, Q, R, S is "(P and Q) or (R and S)" true?

Answer: 7

Reasoning: P and Q true: 4 rows; R and S true: 4 rows; both: 1 row. 4 + 4 - 1 = 7.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S18-F2-Q5 · question page 423

LOGIC.TRUTH.01 · Read and build statements from tables

Every truth-table column for P and Q can be produced by some statement using only "not", "and", "or".

Answer: True

Reasoning: For each true row write a bracket like "P and not Q", then join the brackets with "or" (for the all-F column use "P and not P").

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S19-M-Q1 · question page 425

LOGIC.TRUTH.02 · Classify statements as tautologies, contradictions or contingencies, and test equivalence

Which statement is a TAUTOLOGY (true in every row of its truth table)?

Answer: B. (P and Q) or not P or not Q

Reasoning: If P and Q are both true, the first bracket is true; otherwise "not P or not Q" is true. So it is always true.

- ✗ P or Q: false when both are false
- ✗ if P then Q: false when P is true and Q false
- ✗ P and Q and not Q: false in every row

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S19-M-Q2 · question page 426

LOGIC.TRUTH.02 · Classify statements as tautologies, contradictions or contingencies, and test equivalence

Which statement is a CONTRADICTION (false in every row)?

Answer: D. P and Q and not Q

Reasoning: Q and not Q can never both be true.

- ✗ P or not P: a tautology
- ✗ if P then Q: a contingency
- ✗ P or Q: a contingency

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S19-M-Q3 · question page 427

LOGIC.TRUTH.02 · Classify statements as tautologies, contradictions or contingencies, and test equivalence

Complete the truth table for "(if P then Q) or (if Q then P)".

Answer: T T T / F T T / T F T / T T T

Reasoning: In every row at least one of the two conditionals is true, so the last column is all T: a tautology.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S19-M-Q4 · question page 428

LOGIC.TRUTH.02 · Classify statements as tautologies, contradictions or contingencies, and test equivalence

"not (P and Q)" is logically equivalent to "(not P) or (not Q)".

Answer: True

Reasoning: Both are false only when P and Q are both true (De Morgan's law).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S19-M-Q5 · question page 429

LOGIC.TRUTH.02 · Classify statements as tautologies, contradictions or contingencies, and test equivalence

How is "if P then Q" classified?

Answer: A. a contingency: true in some rows and false in one

Reasoning: It is false only in the row P true, Q false, so it is a contingency.

- ✗ a tautology: it is false when P is true and Q false
- ✗ a contradiction: it is true in three rows
- ✗ neither true nor false: every row has a value

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S19-B1-Q1 · question page 430

LOGIC.TRUTH.01 · Fill in truth tables

Complete the truth table for "P or not P".

Answer: F T / T T

Reasoning: Always true.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (2 rows).

S19-B1-Q2 · question page 431

LOGIC.TRUTH.01 · Fill in truth tables

Complete the truth table for "P and not P".

Answer: F F / T F

Reasoning: Always false.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (2 rows).

S19-B1-Q3 · question page 432

LOGIC.TRUTH.01 · Fill in truth tables

Complete the truth table for "not P or Q".

Answer: T / F / T / T

Reasoning: False only when P true and Q false.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S19-B1-Q4 · question page 433

LOGIC.TRUTH.01 · Fill in truth tables

How many rows does a truth table for P and Q have?

Answer: 4

Reasoning: 2 x 2.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S19-B1-Q5 · question page 434

LOGIC.TRUTH.01 · Fill in truth tables

A truth table for three statements P, Q and R has 6 rows.

Answer: False

Reasoning: Each of the 3 statements can be true or false: 2 x 2 x 2 = 8 rows.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S19-B2-Q1 · question page 435

LOGIC.TRUTH.02 · Always, never, sometimes

A statement is true in every row of its table. What is it called?

Answer: B. a tautology

Reasoning: Always true = tautology.

- ✗ a contradiction: that is never true
- ✗ a contingency: that is sometimes true
- ✗ a premise: a role in an argument, not a class

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S19-B2-Q2 · question page 436

LOGIC.TRUTH.02 · Always, never, sometimes

A statement is false in every row. What is it called?

Answer: C. a contradiction

Reasoning: Never true = contradiction.

- ✗ a tautology: always true
- ✗ a contingency: sometimes true
- ✗ a conclusion: not a class

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S19-B2-Q3 · question page 437

LOGIC.TRUTH.02 · Always, never, sometimes

A contingency is true in every row of its truth table.

Answer: False

Reasoning: A statement true in every row is a tautology. A contingency is true in some rows and false in others.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S19-B2-Q4 · question page 438

LOGIC.TRUTH.02 · Always, never, sometimes

A column reads T, T, F, T. How is the statement classified?

Answer: D. a contingency

Reasoning: Mixed values: contingency.

- ✗ a tautology: it has an F
- ✗ a contradiction: it has a T
- ✗ it cannot be told: the column decides it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S19-B2-Q5 · question page 439

LOGIC.TRUTH.02 · Always, never, sometimes

"7 is odd and 7 is not odd" is true.

Answer: False

Reasoning: It has the form "P and not P", a contradiction: it is false in every case.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S19-F1-Q1 · question page 440

LOGIC.TRUTH.02 · Equivalences by truth table

Complete the table to compare "not (P or Q)" with "(not P) and (not Q)".

Answer: F F / F F / F F / T T

Reasoning: The columns match: De Morgan's law.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S19-F1-Q2 · question page 441

LOGIC.TRUTH.02 · Equivalences by truth table

"if P then Q" is equivalent to "if Q then P".

Answer: False

Reasoning: They differ when exactly one of P, Q is true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S19-F1-Q3 · question page 442

LOGIC.TRUTH.02 · Equivalences by truth table

Which statement is equivalent to "not (if P then Q)"?

Answer: B. P and not Q

Reasoning: "if P then Q" is false exactly when P is true and Q false.

✗ not P and not Q: true when both are false, but the original is false there

✗ if not P then not Q: that is the inverse

✗ not P or Q: that is the conditional itself

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S19-F1-Q4 · question page 443

LOGIC.TRUTH.02 · Equivalences by truth table

How many of these are tautologies? "P or not P"; "P and not P"; "if P then Q"; "(if P then Q) or (if Q then P)"; "P iff not P".

Answer: 2

Reasoning: "P or not P" and "(if P then Q) or (if Q then P)".

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S19-F1-Q5 · question page 444

LOGIC.TRUTH.02 · Equivalences by truth table

The negation of a tautology is also a tautology.

Answer: False

Reasoning: If a statement is true in every row, its negation is false in every row.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S19-F2-Q1 · question page 445

LOGIC.TRUTH.02 · Laws of logic

Which statement is equivalent to "P or (P and Q)"?

Answer: A. P

Reasoning: If P is true, the statement is true; if P is false, both parts are false. So it equals P (absorption).

- ✗ Q: differs when P true, Q false
- ✗ P and Q: differs when P true, Q false
- ✗ P or Q: differs when P false, Q true

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S19-F2-Q2 · question page 446

LOGIC.TRUTH.02 · Laws of logic

"((if P then Q) and P) implies Q", written "if ((if P then Q) and P) then Q", is a tautology.

Answer: True

Reasoning: This is modus ponens written as one statement; it is true in all four rows.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S19-F2-Q3 · question page 447

LOGIC.TRUTH.02 · Laws of logic

"if ((if P then Q) and Q) then P" is a tautology.

Answer: False

Reasoning: Row P false, Q true: the "if" part is true and P is false, so the statement is false (affirming the consequent).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S19-F2-Q4 · question page 448

LOGIC.TRUTH.02 · Laws of logic

Of the 16 possible truth-table columns for statements in P and Q, how many are contingencies?

Answer: 14

Reasoning: All 16 columns except all-T (tautology) and all-F (contradiction).

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S19-F2-Q5 · question page 449

LOGIC.TRUTH.02 · Laws of logic

Which pair are equivalent?

Answer: D. "if P then (Q and R)" and "(if P then Q) and (if P then R)"

Reasoning: Both sides of the first pair are false exactly when P is true and at least one of Q, R is false.

- ✗ "if (P or Q) then R" and "(if P then R) or (if Q then R)": P true, Q false, R false: the first is false, the second true
- ✗ "P and (Q or R)" and "(P and Q) or R": P false, R true: the first is false, the second true
- ✗ "not (P and Q)" and "not P and not Q": P true, Q false: the first is true, the second false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S20-M-Q1 · question page 451

LOGIC.SPATIAL.03 · Infer stacking order from what is visible

Rectangular paper sheets are lying flat, one on top of another, seen from directly above. Each sheet is white with a dark outline and a letter in its corner. A sheet hides the parts of the sheets under it. This stack has 3 sheets: A, B, C. Put the sheets in order from the BOTTOM sheet to the TOP sheet.

Answer: sheet B → sheet C → sheet A

Reasoning: Overlaps: B is under C, C is under A, B is under A. A's outline is complete, so A is on top; then C; B is at the bottom: B, C, A.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: every stacking order was drawn on a fine grid and compared with the picture. Automatic: all 6 orderings of 3 items enumerated; 1 valid.

S20-M-Q2 · question page 452

LOGIC.SPATIAL.03 · Infer stacking order from what is visible

Rectangular paper sheets are lying flat, one on top of another, seen from directly above. Each sheet is white with a dark outline and a letter in its corner. A sheet hides the parts of the sheets under it. This stack has 4 sheets: A, B, C, D. List the sheets starting with the one at the BOTTOM and ending with the one on TOP.

Answer: sheet B → sheet C → sheet A → sheet D

Reasoning: At each overlap the cut-off outline is underneath: B is under C, A and D; C is under A; A is under D. So the chain is B, C, A, D from bottom to top.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: every stacking order was drawn on a fine grid and compared with the picture. Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S20-M-Q3 · question page 453

LOGIC.SPATIAL.03 · Infer stacking order from what is visible

Four round discs are stacked with their centers lined up, from bottom to top: W (radius 5 cm), X (radius 3 cm), Y (radius 4 cm), Z (radius 1 cm). Looking straight down from above, which discs can be seen?

Answer: C. W, Y and Z

Reasoning: A disc can be seen if every disc above it is smaller. W: all above are smaller. X: Y (4 cm) is above and larger, so X is hidden. Y: only Z (1 cm) is above. Z: on top.

- ✗ W, X, Y and Z: X is entirely covered by the larger disc Y above it
- ✗ Z only: larger discs below Z show around it
- ✗ W and Z: Y shows as a ring between radius 1 and 4 cm

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S20-M-Q4 · question page 454

LOGIC.SPATIAL.03 · Infer stacking order from what is visible

Rectangular paper sheets are lying flat, one on top of another, seen from directly above. Each sheet is white with a dark outline and a letter in its corner. A sheet hides the parts of the sheets under it. This stack has 4 sheets: A, B, C, D. Put the sheets in order from the BOTTOM sheet to the TOP sheet.

Answer: sheet C → sheet D → sheet A → sheet B

Reasoning: Overlaps: C is under D and A; D is under A; A is under B. From bottom to top: C, D, A, B.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: every stacking order was drawn on a fine grid and compared with the picture. Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S20-M-Q5 · question page 455

LOGIC.SPATIAL.03 · Infer stacking order from what is visible

Square cards are placed one at a time on an 8 by 8 grid, each covering a block of cells (rows and columns numbered 1 to 8; "rows 1-4, columns 1-4" means a 4 by 4 block). In order: A covers rows 1-4, columns 1-4; B covers rows 3-6, columns 3-6; C covers rows 2-5, columns 2-5; D covers rows 1-3, columns 4-7; E covers rows 5-7, columns 1-3. Seen from above, how many of the five cards can be seen at l

Answer: 5

Reasoning: Check each card for a cell not covered by a later card. A: cell (1,1). B: cell (6,6). C: cell (4,4). D and E are near the top of the pile and show fully or partly. All 5 can be seen.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S20-B1-Q1 · question page 456

LOGIC.SPATIAL.03 · Strategy: the cut-off sheet is underneath

Where sheets X and Y overlap, the outline of X runs straight through and the outline of Y stops at X's edge. What follows?

Answer: C. Y is under X.

Reasoning: The sheet whose outline is cut off is underneath.

- ✗ X is under Y.: the cut-off outline is the lower one
- ✗ They are side by side.: they overlap
- ✗ It cannot be told.: the outlines decide it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S20-B1-Q2 · question page 457

LOGIC.SPATIAL.03 · Strategy: the cut-off sheet is underneath

Pairs found: P under Q, Q under R, P under R. Order the sheets from bottom to top.

Answer: sheet P → sheet Q → sheet R

Reasoning: P, Q, R.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 6 orderings of 3 items enumerated; 1 valid.

S20-B1-Q3 · question page 458

LOGIC.SPATIAL.03 · Strategy: the cut-off sheet is underneath

A sheet whose whole outline can be seen must be on top of every sheet it overlaps.

Answer: True

Reasoning: If any sheet above it overlapped it, part of its outline would be hidden.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S20-B1-Q4 · question page 459

LOGIC.SPATIAL.03 · Strategy: the cut-off sheet is underneath

Discs are stacked with centers lined up, from bottom to top: radius 2 cm, then 6 cm. Which can be seen from above?

Answer: D. only the 6 cm disc

Reasoning: The 6 cm disc covers the 2 cm disc completely.

- ✗ both discs: the larger disc above covers the smaller one
- ✗ only the 2 cm disc: it is underneath
- ✗ neither disc: the top disc is visible

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S20-B1-Q5 · question page 460

LOGIC.SPATIAL.03 · Strategy: the cut-off sheet is underneath

Five sheets are stacked. Exactly one of them is on top of all four others. How many sheets are under it?

Answer: 4

Reasoning: All four others are under the top sheet.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S20-F1-Q1 · question page 461

LOGIC.SPATIAL.03 · Larger stacks

Rectangular paper sheets are lying flat, one on top of another, seen from directly above. Each sheet is white with a dark outline and a letter in its corner. A sheet hides the parts of the sheets under it. This stack has 5 sheets: A, B, C, D, E. Put the five sheets in order from the BOTTOM sheet to the TOP sheet.

Answer: sheet E → sheet D → sheet A → sheet C → sheet B

Reasoning: Every pair overlaps here. B is cut off by no sheet (top); C is cut off only by B; A by C and B; D by A, C and B; E by all four. From bottom to top: E, D, A, C, B.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: every stacking order was drawn on a fine grid and compared with the picture. Automatic: all 120 orderings of 5 items enumerated; 1 valid.

S20-F1-Q2 · question page 462

LOGIC.SPATIAL.03 · Larger stacks

Rectangular paper sheets are lying flat, one on top of another, seen from directly above. Each sheet is white with a dark outline and a letter in its corner. A sheet hides the parts of the sheets under it. This stack has 5 sheets: A, B, C, D, E. List the five sheets starting with the BOTTOM sheet and ending with the TOP sheet.

Answer: sheet B → sheet E → sheet A → sheet C → sheet D

Reasoning: Every pair overlaps. D is cut off by no sheet (top); C only by D; A by C and D; E by A, C and D; B by all four. From bottom to top: B, E, A, C, D.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: every stacking order was drawn on a fine grid and compared with the picture. Automatic: all 120 orderings of 5 items enumerated; 1 valid.

S20-F1-Q3 · question page 463

LOGIC.SPATIAL.03 · Larger stacks

Cards are placed one at a time on an 8 by 8 grid: A covers rows 2-3, columns 2-3; B covers rows 1-4, columns 1-4; C covers rows 4-6, columns 4-6; D covers rows 1-2, columns 5-6; E covers rows 3-6, columns 1-4. Seen from above, how many of the five cards can be seen at least partly?

Answer: 4

Reasoning: A lies inside B and B was placed later, so A is hidden. B still shows in rows 1-2, columns 1-4. C shows in columns 5-6 (E only covers column 4 of C). D and E show. So B, C, D and E: 4 cards.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S20-F1-Q4 · question page 464

LOGIC.SPATIAL.03 · Larger stacks

Four discs with centers lined up are stacked, bottom to top: radius 6, 2, 5, 3 (cm). How many can be seen from above?

Answer: A. 3

Reasoning: Visible: 6 (nothing larger above), 5 (only 3 above), 3 (top). The 2 cm disc is under the 5 cm disc.

- ✗ 4: the 2 cm disc is covered by the 5 cm disc above it
- ✗ 2: the 5 cm disc shows as a ring outside the 3 cm disc
- ✗ 1: larger discs below show around smaller ones

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S20-F1-Q5 · question page 465

LOGIC.SPATIAL.03 · Larger stacks

If two sheets do not overlap at all, the picture from above cannot show which one is higher.

Answer: True

Reasoning: Nothing in the picture depends on their order.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S20-F2-Q1 · question page 466

LOGIC.SPATIAL.03 · How much does a picture tell?

Three sheets: A and B do not overlap each other, and sheet C overlaps both and is on top of both. How many different bottom-to-top orders are consistent with the picture from above?

Answer: 2

Reasoning: C must be on top. A and B can be in either order below it: A,

B, C or B, A, C.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S20-F2-Q2 · question page 467

LOGIC.SPATIAL.03 · How much does a picture tell?

Discs of radius 1, 2, 3 and 4 cm are stacked with centers lined up. In which stacking order (bottom to top) can ALL four discs be seen from above?

Answer: B. 4, 3, 2, 1

Reasoning: Each disc must be larger than every disc above it: largest at the bottom.

- ✗ 1, 2, 3, 4: only the top 4 cm disc shows
- ✗ 4, 2, 3, 1: the 2 cm disc is under the 3 cm disc
- ✗ 3, 4, 2, 1: the 3 cm disc is under the 4 cm disc

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S20-F2-Q3 · question page 468

LOGIC.SPATIAL.03 · How much does a picture tell?

Discs of radius 1, 2, 3 and 4 cm are stacked in some order with centers lined up. Over all 24 stacking orders, in how many is EXACTLY one disc visible from above?

Answer: 6

Reasoning: Exactly one disc shows when the 4 cm disc is on top. The other three can be in any order below: 3 x 2 x 1 = 6.

Grading: The game's answer checker compares the typed value with the key (equivalent

number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S20-F2-Q4 · question page 469

LOGIC.SPATIAL.03 · How much does a picture tell?

From a picture of a stack of sheets, the TOP sheet can always be identified if every sheet overlaps at least one other sheet.

Answer: False

Reasoning: Two separate piles (A under B, and C under D, with no overlap between the piles) have two sheets with complete outlines, B and D, and the picture cannot tell which of them is the highest.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S20-F2-Q5 · question page 470

LOGIC.SPATIAL.03 · How much does a picture tell?

Cards on a grid: P covers rows 1-3, columns 1-3; Q covers rows 2-4, columns 2-4. P was placed first. Which cell shows card P?

Answer: A. row 1, column 2

Reasoning: Q covers rows 2-4 and columns 2-4, so P shows in row 1 and in column 1.

- ✗ row 3, column 3: covered by Q
- ✗ row 2, column 2: covered by Q
- ✗ row 4, column 4: that cell is only Q

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S21-M-Q1 · question page 472

LOGIC.INFERENCE.01 · Draw valid conclusions with modus ponens, modus tollens, chains and disjunctive syllogism

Premises: "If the reservoir is full, the dam gates open." "The reservoir is full." What conclusion follows?

Answer: D. The dam gates open.

Reasoning: Modus ponens: if P then Q; P; therefore Q.

- ✗ The reservoir is not full.: contradicts a premise
- ✗ The dam gates do not open.: contradicts the conditional
- ✗ Nothing follows.: modus ponens gives a conclusion

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the form was checked by truth table.

S21-M-Q2 · question page 473

LOGIC.INFERENCE.01 · Draw valid conclusions with modus ponens, modus tollens, chains and disjunctive syllogism

Premises: "If a number is divisible by 9, it is divisible by 3." "The number 52 is not divisible by 3." What follows?

Answer: C. 52 is not divisible by 9.

Reasoning: Modus tollens: if P then Q; not Q; therefore not P.

- ✗ 52 is divisible by 9.: contradicts the premises
- ✗ 52 is divisible by 3.: contradicts a premise
- ✗ Nothing follows.: modus tollens gives a conclusion

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S21-M-Q3 · question page 474

LOGIC.INFERENCE.01 · Draw valid conclusions with modus ponens, modus tollens, chains and disjunctive syllogism

Arrange the steps of the argument so that each line follows from earlier ones and the conclusion comes last.

Answer: If the trail is icy, the park closes. → The trail is icy. → So the park closes. → If the park closes, the race is cancelled. → So the race is cancelled.

Also accepted: If the trail is icy, the park closes. → If the park closes, the race is cancelled. → The trail is icy. → So the park closes. → So the race is cancelled. | The trail is icy. → If the trail is icy, the park closes. → So the park closes. → If the park closes, the race is cancelled. → So the race is cancelled.

Reasoning: Use the first rule with the fact to get "the park closes", then the second rule to get the conclusion. The two rules may also be stated first.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 120 orderings of 5 items enumerated; 3 valid.

S21-M-Q4 · question page 475

LOGIC.INFERENCE.01 · Draw valid conclusions with modus ponens, modus tollens, chains and disjunctive syllogism

Premises: "The key is in the drawer or in the coat." "The key is not in the drawer." What follows?

Answer: B. The key is in the coat.

Reasoning: Disjunctive syllogism: P or Q; not P; therefore Q.

- ✗ The key is in the drawer.: contradicts a premise
- ✗ The key is in neither place.: contradicts the first premise
- ✗ Nothing follows.: disjunctive syllogism gives a conclusion

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S21-M-Q5 · question page 476

LOGIC.INFERENCE.01 · Draw valid conclusions with modus ponens, modus tollens, chains and disjunctive syllogism

This argument is valid: "If P then Q. If Q then R. So if P then R."

Answer: True

Reasoning: If P is true, the first premise gives Q and the second gives R (a chain, or hypothetical syllogism).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S21-B1-Q1 · question page 477

LOGIC.CONDITIONAL.01 · When is a conditional true?

"If P then Q" is false when P is false and Q is true.

Answer: False

Reasoning: With P false the conditional makes no promise, so it is true. It is false only when P is true and Q is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S21-B1-Q2 · question page 478

LOGIC.CONDITIONAL.01 · When is a conditional true?

"If P then Q" is false when P is false.

Answer: False

Reasoning: It is true whenever P is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S21-B1-Q3 · question page 479

LOGIC.CONDITIONAL.01 · When is a conditional true?

"If P then Q" is true and P is true. What is Q?

Answer: A. true

Reasoning: The only row with P true and the conditional true has Q true.

- ✗ false: then the conditional would be false
- ✗ either: Q is forced
- ✗ neither: Q has a value

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S21-B1-Q4 · question page 480

LOGIC.CONDITIONAL.01 · When is a conditional true?

"If P then Q" is true and Q is false. What is P?

Answer: C. false

Reasoning: The only row with Q false and the conditional true has P false.

- ✗ true: then the conditional would be false
- ✗ either: P is forced
- ✗ neither: P has a value

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S21-B1-Q5 · question page 481

LOGIC.CONDITIONAL.01 · When is a conditional true?

"If P then Q" is true and Q is true. P must be true.

Answer: False

Reasoning: P can be false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S21-B2-Q1 · question page 482

LOGIC.CONNECTIVE.02 · Using "or" and "not"

"P or Q" is true and P is true. Then Q must be false.

Answer: False

Reasoning: Inclusive "or" allows both parts to be true, so Q could be true or false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S21-B2-Q2 · question page 483

LOGIC.CONNECTIVE.02 · Using "or" and "not"

"P or Q" is true and P is true. Then Q is false.

Answer: False

Reasoning: Inclusive or allows both.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S21-B2-Q3 · question page 484

LOGIC.CONNECTIVE.02 · Using "or" and "not"

"The meeting is on Monday or Tuesday." It is not on Monday. When is it?

Answer: B. Tuesday

Reasoning: Eliminate Monday.

- ✗ Monday: ruled out
- ✗ neither day: the premise says one of them
- ✗ it cannot be told: only Tuesday is left

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S21-B2-Q4 · question page 485

LOGIC.CONNECTIVE.02 · Using "or" and "not"

Which is the negation of "the light is on"?

Answer: D. the light is not on

Reasoning: Put "not" on the claim.

- ✗ the light is dim: not a negation
- ✗ the lamp is broken: unrelated
- ✗ the light is bright: not a negation

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S21-B2-Q5 · question page 486

LOGIC.CONNECTIVE.02 · Using "or" and "not"

"P or Q" is true and both P and Q could be true.

Answer: True

Reasoning: Inclusive or.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S21-F1-Q1 · question page 487

LOGIC.INFERENCE.01 · Longer chains of valid steps

Premises: "If it is a holiday, the bank is closed." "If the bank is closed, Mara cannot deposit." "Mara deposited." What follows?

Answer: B. It is not a holiday.

Reasoning: Mara deposited, so the bank was not closed (modus tollens); so it is not a holiday (modus tollens again).

- ✗ It is a holiday.: contradicts the premises
- ✗ The bank is closed.: contradicts the premises
- ✗ Nothing follows.: two steps of modus tollens give a conclusion

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S21-F1-Q2 · question page 488

LOGIC.INFERENCE.01 · Longer chains of valid steps

Valid: "P or Q. If P then R. If Q then R. So R."

Answer: True

Reasoning: Either way R follows (proof by cases).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S21-F1-Q3 · question page 489

LOGIC.INFERENCE.01 · Longer chains of valid steps

Premises: "Ana or Ben will host." "If Ben hosts, the party is on Friday." "The party is not on Friday." Who hosts?

Answer: C. Ana

Reasoning: Not Friday, so Ben does not host (modus tollens); Ana or Ben hosts, so Ana does (disjunctive syllogism).

- ✗ Ben: then the party would be on Friday
- ✗ both: Ben does not host
- ✗ it cannot be told: the premises decide it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S21-F1-Q4 · question page 490

LOGIC.INFERENCE.01 · Longer chains of valid steps

Arrange into a valid argument ending with the conclusion "The fountain is off."

Answer: If the fountain is on, the pump runs. → The pump does not run. → So the fountain is off.

Also accepted: The pump does not run. → If the fountain is on, the pump runs. → So the fountain is off.

Reasoning: Modus tollens: the rule and the fact (in either order), then the conclusion.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 6 orderings of 3 items enumerated; 2 valid.

S21-F1-Q5 · question page 491

LOGIC.INFERENCE.01 · Longer chains of valid steps

Valid: "If P then Q. Not Q. So not P."

Answer: True

Reasoning: Modus tollens.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S21-F2-Q1 · question page 492

LOGIC.INFERENCE.01 · Valid forms with several variables

Valid: "If P then (Q and R). Not R. So not P."

Answer: True

Reasoning: If P were true, R would be true; R is false, so P is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S21-F2-Q2 · question page 493

LOGIC.INFERENCE.01 · Valid forms with several variables

Valid: "If P then (Q or R). Not Q. So R."

Answer: False

Reasoning: P may be false: P false, Q false, R false makes both premises true and the conclusion false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S21-F2-Q3 · question page 494

LOGIC.INFERENCE.01 · Valid forms with several variables

Premises: "If A then B." "If C then D." "A or C." Which conclusion follows?

Answer: A. B or D

Reasoning: If A, then B; if C, then D; one of A, C holds, so one of B, D holds (constructive dilemma).

- ✗ B and D: only one of A and C need be true
- ✗ A and C: only one is given
- ✗ not B: contradicts a possible case

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S21-F2-Q4 · question page 495

LOGIC.INFERENCE.01 · Valid forms with several variables

An argument has premises "P or Q" and "if P then R" and conclusion "Q or R". In how many of the 8 rows of its truth table are both premises true?

Answer: 4

Reasoning: Count the rows where P or Q is true and P -> R is true: TTT, FTT, FTF, TFT.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S21-F2-Q5 · question page 496

LOGIC.INFERENCE.01 · Valid forms with several variables

The argument "P or Q. If P then R. So Q or R." is valid.

Answer: True

Reasoning: In each row where both premises are true, Q or R is true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S22-M-Q1 · question page 498

LOGIC.INFERENCE.02 · Recognize affirming the consequent and denying the antecedent

"If a phone is waterproof, it survives a fall into a pool. Leo's phone survived a fall into a pool. So Leo's phone is waterproof." What is wrong with this argument?

Answer: D. It affirms the consequent: the phone may have survived for another reason.

Reasoning: From "if P then Q" and Q, P does not follow.

- ✗ Nothing: it is modus ponens.: modus ponens starts from the "if" part, not the "then" part
- ✗ It denies the antecedent.: no premise says "not waterproof"
- ✗ A premise is false.: the problem is the form, whatever the premises

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S22-M-Q2 · question page 499

LOGIC.INFERENCE.02 · Recognize affirming the consequent and denying the antecedent

"If n is a multiple of 4, then n is even. 6 is not a multiple of 4. So 6 is not even." What is wrong?

Answer: A. It denies the antecedent: not P does not give not Q.

Reasoning: The conclusion is false (6 is even) even though both premises are true, so the form is invalid.

- ✗ Nothing: it is modus tollens.: modus tollens starts from not Q
- ✗ It affirms the consequent.: no premise says "6 is even"
- ✗ The conditional is false.: it is true

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S22-M-Q3 · question page 500

LOGIC.INFERENCE.02 · Recognize affirming the consequent and denying the antecedent

This form is valid: "If P then Q. Q. So P."

Answer: False

Reasoning: P false, Q true makes both premises true and the conclusion false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S22-M-Q4 · question page 501

LOGIC.INFERENCE.02 · Recognize affirming the consequent and denying the antecedent

Use whole numbers to show that "If n is a multiple of 6, then n is even. n is even. So n is a multiple of 6." is invalid: give a value of n that makes both premises true and the conclusion false.

Answer: any values that satisfy: isEven(n) and n % 6 != 0 (example: n = 4)

Reasoning: Any even number that is not a multiple of 6 (2, 4, 8, 10, ...): the conditional is always true, n is even, and n is not a multiple of 6.

Tempting answers the rule rejects: n = 6; n = 3; n = 12

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S22-M-Q5 · question page 502

LOGIC.INFERENCE.02 · Recognize affirming the consequent and denying the antecedent

Which argument is VALID?

Answer: D. "If it rains, the game is off. The game is not off. So it is not raining."

Reasoning: Only the modus tollens argument is valid.

- ✗ "If it rains, the game is off. The game is off. So it is raining.": affirms the consequent
- ✗ "If it rains, the game is off. It is not raining. So the game is not off.": denies the antecedent
- ✗ "If it rains, the game is off. So if the game is off, it rains.": concludes the converse

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S22-B1-Q1 · question page 503

LOGIC.INFERENCE.01 · Valid forms first

"If P then Q. P. So Q." What is this form called?

Answer: C. modus ponens

Reasoning: Affirm the "if" part.

- ✗ modus tollens: that uses not Q
- ✗ affirming the consequent: that uses Q as a premise

✗ denying the antecedent: that uses not P

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S22-B1-Q2 · question page 504

LOGIC.INFERENCE.01 · Valid forms first

"If P then Q. Not Q. So not P." What is this form called?

Answer: B. modus tollens

Reasoning: Deny the "then" part.

✗ modus ponens: that uses P
✗ denying the antecedent: that uses not P
✗ affirming the consequent: that uses Q

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S22-B1-Q3 · question page 505

LOGIC.INFERENCE.01 · Valid forms first

Modus ponens is valid.

Answer: True

Reasoning: Truth table check.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S22-B1-Q4 · question page 506

LOGIC.INFERENCE.01 · Valid forms first

Modus tollens is valid.

Answer: True

Reasoning: Truth table check.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S22-B1-Q5 · question page 507

LOGIC.INFERENCE.01 · Valid forms first

In modus ponens, which part of the conditional is given as a fact?

Answer: A. the "if" part

Reasoning: P is given.

- ✗ the "then" part: that would affirm the consequent
- ✗ neither part: one part is given
- ✗ the negation of the "if" part: that would deny the antecedent

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S22-B2-Q1 · question page 508

LOGIC.QUANT.03 · A counterexample shows invalidity

An argument form is invalid if there is one case with all premises true and the conclusion false.

Answer: True

Reasoning: One such case is enough.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S22-B2-Q2 · question page 509

LOGIC.QUANT.03 · A counterexample shows invalidity

"If x > 5, then x > 2. x > 2. So x > 5." Which value of x shows it is invalid?

Answer: D. 3

Reasoning: x = 3: both premises true, conclusion false.

- ✗ 6: the conclusion is true
- ✗ 1: the premise x > 2 is false
- ✗ 10: the conclusion is true

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S22-B2-Q3 · question page 510

LOGIC.QUANT.03 · A counterexample shows invalidity

A valid argument can have true premises and a false conclusion.

Answer: False

Reasoning: That is what validity rules out.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S22-B2-Q4 · question page 511

LOGIC.QUANT.03 · A counterexample shows invalidity

"If an animal is a robin, it has wings. A bat has wings. So a bat is a robin." Which fact shows the form is invalid?

Answer: C. A bat has wings but is not a robin.

Reasoning: True premises, false conclusion.

- ✗ Robins have wings.: that is a premise
- ✗ Bats fly at night.: irrelevant
- ✗ Some birds cannot fly.: irrelevant

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S22-B2-Q5 · question page 512

LOGIC.QUANT.03 · A counterexample shows invalidity

If one case with true premises has a true conclusion, the argument is valid.

Answer: False

Reasoning: Validity needs EVERY case with true premises to have a true conclusion; one good case is not enough.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S22-F1-Q1 · question page 513

LOGIC.INFERENCE.02 · Spot invalid forms in disguise

"The alarm sounds only if there is smoke. There is smoke. So the alarm sounds." Is this valid?

Answer: B. No: "A only if S" means "if A then S"; from S, A does not follow.

Reasoning: This affirms the consequent of "if the alarm sounds, there is smoke".

- ✗ Yes: it is modus ponens.: "only if" puts the alarm in the "if" part
- ✗ Yes: smoke always sets off alarms.: the premises do not say that

✗ No: the conclusion is false.: validity is about form, not the truth of the conclusion

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S22-F1-Q2 · question page 514

LOGIC.INFERENCE.02 · Spot invalid forms in disguise

"Every square is a rectangle. This shape is not a square. So it is not a rectangle." This argument is valid.

Answer: False

Reasoning: Denying the antecedent: a non-square rectangle is a counterexample.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S22-F1-Q3 · question page 515

LOGIC.INFERENCE.02 · Spot invalid forms in disguise

Show that "If $x > 10$, then $x > 3$. x is not greater than 10. So x is not greater than 3." is invalid: give a number x that makes the premises true and the conclusion false.

Answer: any values that satisfy: not $(x > 10)$ and $x > 3$ (example: $x = 5$)

Reasoning: Any x with $3 < x \leq 10$.

Tempting answers the rule rejects: $x = 2$; $x = 11$; $x = 3$

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S22-F1-Q4 · question page 516

LOGIC.INFERENCE.02 · Spot invalid forms in disguise

Which argument has the SAME form as "If it is a cat, it is a mammal. It is a mammal. So it is a cat."?

Answer: A. "If n is 2, n is even. n is even. So n is 2."

Reasoning: Both affirm the consequent.

- ✗ "If n is 2, n is even. n is 2. So n is even.": modus ponens
- ✗ "If n is 2, n is even. n is odd. So n is not 2.": modus tollens
- ✗ "n is 2 or n is 3. n is not 3. So n is 2.": disjunctive syllogism

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S22-F1-Q5 · question page 517

LOGIC.INFERENCE.02 · Spot invalid forms in disguise

Denying the antecedent and affirming the consequent are both invalid.

Answer: True

Reasoning: Both have counterexample rows.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S22-F2-Q1 · question page 518

LOGIC.INFERENCE.02 · Invalid or just unconvincing?

"If a number is divisible by 4, it is even. 12 is even. So 12 is divisible by 4." The conclusion is true, so the argument is valid.

Answer: False

Reasoning: The conclusion happens to be true, but the form (affirming the consequent) is invalid; 6 shows the form can fail.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S22-F2-Q2 · question page 519

LOGIC.INFERENCE.02 · Invalid or just unconvincing?

For whole numbers n from 1 to 30, the argument "If n is a multiple of 6, then n is even. n is even. So n is a multiple of 6." has true premises for every even n. For how many of these n is the conclusion false?

Answer: 10

Reasoning: Even numbers up to 30: 15. Multiples of 6: 5. So 10 even numbers are not multiples of 6.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S22-F2-Q3 · question page 520

LOGIC.INFERENCE.02 · Invalid or just unconvincing?

Which argument is INVALID, even though every premise and the conclusion are true?

Answer: C. "If 9 is odd, then 9 is a whole number. 9 is a whole number. So 9 is odd."

Reasoning: The keyed argument affirms the consequent, so it is invalid even though every sentence in it is true.

- ✗ "If 9 is odd, then 9 is a whole number. 9 is odd. So 9 is a whole number.": valid (modus ponens)
- ✗ "9 is odd or 9 is even. 9 is not even. So 9 is odd.": valid (disjunctive syllogism)
- ✗ "If 9 is even, then 9 is a whole number. 9 is even. So 9 is a whole number.": valid (modus ponens); its second premise is false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S22-F2-Q4 · question page 521

LOGIC.INFERENCE.02 · Invalid or just unconvincing?

Valid: "If P then Q. If not P then Q. So Q."

Answer: True

Reasoning: Q follows whether P is true or false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S22-F2-Q5 · question page 522

LOGIC.INFERENCE.02 · Invalid or just unconvincing?

Valid: "If P then Q. If Q then P. Not P. So not Q."

Answer: True

Reasoning: The premises make P and Q equivalent; not P then gives not Q.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S23-M-Q1 · question page 524

LOGIC.SPATIAL.04 · Match a cube to its net: opposite and adjacent faces

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. Which letter is on the face OPPOSITE the face A?

Answer: B. E

Reasoning: In the cross-shaped net, the column A, C, E, F folds around the cube: A and E are two squares apart in the column, so they are opposite. (C is opposite F, and B is opposite D.)

- ✗ C: C shares an edge with A in the net, so they touch on the cube
- ✗ B: B folds up next to A, so they touch
- ✗ F: F folds over the top and touches A

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S23-M-Q2 · question page 525

LOGIC.SPATIAL.04 · Match a cube to its net: opposite and adjacent faces

Each figure is made of six squares. Which figure can be folded along the lines between squares into a closed cube (with no overlapping squares)?

Answer: C. Figure C

Reasoning: Fold C: its row of four squares wraps around to make four sides, and the two single squares (one above and one below the row) close the top and the bottom. The others fold with two squares on the same face.

- ✗ Figure A: five squares in a row: the 1st and 5th squares would land on the same face
- ✗ Figure B: a 2 by 3 block: the squares fold into a ring of four with two squares overlapping
- ✗ Figure D: four squares meet around one point, so two faces of the cube would overlap when folded

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the panel check accepts only the keyed panel.

S23-M-Q3 · question page 526

LOGIC.SPATIAL.04 · Match a cube to its net: opposite and adjacent faces

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. For each face P, Q and R of this net, which letter is on the opposite face?

Answer: P: U; Q: S; R: T

Reasoning: In the row Q, R, S, T, squares two apart are opposite: Q-S and R-T. P and U, attached at opposite ends, are opposite each other.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 6 assignments enumerated; 1 satisfy every clue.

S23-M-Q4 · question page 527

LOGIC.SPATIAL.04 · Match a cube to its net: opposite and adjacent faces

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. Which two letters can NEVER be seen at the same time when you look at the cube?

Answer: D. G and J

Reasoning: Two faces can never be seen together only if they are opposite. In this staircase net, G is opposite J, H is opposite K, and I is opposite L.

- ✗ G and H: they share an edge in the net
- ✗ H and I: they share an edge
- ✗ H and J: H is opposite K, so H and J share an edge on the cube

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S23-M-Q5 · question page 528

LOGIC.SPATIAL.04 · Match a cube to its net: opposite and adjacent faces

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. On the cube made from this net, faces M and W share an edge.

Answer: True

Reasoning: Folding this net makes M opposite O, N opposite W, and V opposite X. M and W are not an opposite pair, so they share an edge.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S23-B1-Q1 · question page 529

LOGIC.SPATIAL.04 · Strategy: skip one square to find the opposite face

A straight row of four squares, numbered 1 to 4, is part of a cube net. Which square is opposite square 1?

Answer: B. 3

Reasoning: Exactly one square between them.

- ✗ 2: shares an edge with 1
- ✗ 4: 4 is opposite 2
- ✗ 1: a face is not opposite itself

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S23-B1-Q2 · question page 530

LOGIC.SPATIAL.04 · Strategy: skip one square to find the opposite face

Two squares that share an edge in a net always share an edge on the cube.

Answer: True

Reasoning: Folding keeps shared edges together.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S23-B1-Q3 · question page 531

LOGIC.SPATIAL.04 · Strategy: skip one square to find the opposite face

How many pairs of opposite faces does a cube have?

Answer: 3

Reasoning: 6 faces in pairs: 3.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S23-B1-Q4 · question page 532

LOGIC.SPATIAL.04 · Strategy: skip one square to find the opposite face

On a cube, how many faces share an edge with any one face?

Answer: A. 4

Reasoning: All faces except itself and the opposite face.

- ✗ 5: the opposite face does not touch it
- ✗ 3: four sides surround it
- ✗ 6: a face does not touch itself

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S23-B1-Q5 · question page 533

LOGIC.SPATIAL.04 · Strategy: skip one square to find the opposite face

Two faces that are opposite on the cube can be seen at the same time.

Answer: False

Reasoning: You can see at most three faces at once, and they all meet at one corner; opposite faces never meet.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S23-F1-Q1 · question page 534

LOGIC.SPATIAL.04 · Harder nets

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. Which letter is opposite H?

Answer: D. K

Reasoning: In the staircase, H is opposite K (H and K are two steps apart along the stair).

- ✗ G: shares an edge with H
- ✗ I: shares an edge with H
- ✗ J: J is opposite G

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S23-F1-Q2 · question page 535

LOGIC.SPATIAL.04 · Harder nets

Each of the four figures is made of six squares. How many of them can be folded into a closed cube?

Answer: 2

Reasoning: Folding each figure: in A and C two squares land on the

same face of the cube, so they do not make a closed cube. B (a staircase) and D (a row of four with one square above and one below) fold into a cube. So 2.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S23-F1-Q3 · question page 536

LOGIC.SPATIAL.04 · Harder nets

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. For each face M, N and V of this net, which letter is on the opposite face?

Answer: M: O; N: W; V: X

Reasoning: M, N, O are a row of three, so M and O (one square apart) are opposite. V, W, X are a row of three, so V and X are opposite. That leaves N and W as the third pair.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 6 assignments enumerated; 1 satisfy every clue.

S23-F1-Q4 · question page 537

LOGIC.SPATIAL.04 · Harder nets

Every net of a cube has exactly 6 squares.

Answer: True

Reasoning: A cube has 6 faces.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S23-F1-Q5 · question page 538

LOGIC.SPATIAL.04 · Harder nets

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. In the cross-shaped net (A on top; B, C, D across; E and F below C), which two faces each share an edge with all of B, C and D?

Answer: C. A and E

Reasoning: The opposite pairs are A-E, B-D and C-F. A face touches B, C and D unless it is one of them or opposite one of them; that leaves A and E.

- ✗ A and F: F is opposite C, so F does not touch C
- ✗ E and F: F does not touch C
- ✗ B and D: B and D are opposite each other

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S23-F2-Q1 · question page 539

LOGIC.SPATIAL.04 · Counting and designing nets

Count the different nets of a cube: shapes of six squares joined edge to edge that fold into a cube, where shapes that are turns or flips of each other count as the same. How many are there?

Answer: 11

Reasoning: There are 35 shapes of six squares; folding each one shows that exactly 11 make a cube.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S23-F2-Q2 · question page 540

LOGIC.SPATIAL.04 · Counting and designing nets

The faces of a cube are colored so that faces sharing an edge always have different colors. What is the least number of colors needed?

Answer: 3

Reasoning: Opposite faces never touch, so each opposite pair can share a color: 3 colors. Two colors are not enough: the three faces at a corner all touch each other.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S23-F2-Q3 · question page 541

LOGIC.SPATIAL.04 · Counting and designing nets

On a standard die, opposite faces add to 7. In a cross-shaped net with 1, 2, 3, 4 in the long column and 5, 6 on the two sides, the numbers can follow this rule.

Answer: False

Reasoning: In the long column, the 1st and 3rd squares are opposite (1 and 3) and so are the 2nd and 4th (2 and 4); neither pair adds to 7.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S23-F2-Q4 · question page 542

LOGIC.SPATIAL.04 · Counting and designing nets

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. The numbers 1 to 6 are written on the cross-shaped net with A = 1, B = 2, C = 3, D = 5, E = 6 and F = 4. Which statement is true?

Answer: A. Every pair of opposite faces adds to 7.

Reasoning: Opposite pairs of this net are A-E (1, 6), B-D (2, 5) and C-F (3, 4): each adds to 7.

- ✗ Faces 1 and 2 are opposite.: they share an edge
- ✗ Faces 3 and 4 touch.: they are opposite
- ✗ Faces 2 and 6 are opposite.: 2 is opposite 5

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S23-F2-Q5 · question page 543

LOGIC.SPATIAL.04 · Counting and designing nets

A cube net can contain four squares in a 2 by 2 block.

Answer: False

Reasoning: Four squares around one point would fold onto only three faces at that corner, so two would overlap.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S24-M-Q1 · question page 545

LOGIC.INFERENCE.03 · Tell whether an argument is valid, and whether it is also sound

"All mammals breathe air. Whales are mammals. So whales breathe air." How is this argument best described?

Answer: D. valid and sound

Reasoning: The form is valid and both premises are true, so the argument is sound.

- ✗ valid but not sound: both premises are true
- ✗ sound but not valid: a sound argument must be valid

✗ invalid: the conclusion is forced by the premises

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S24-M-Q2 · question page 546

LOGIC.INFERENCE.03 · Tell whether an argument is valid, and whether it is also sound

"All birds can swim. A sparrow is a bird. So a sparrow can swim." How is this argument best described?

Answer: B. valid but not sound

Reasoning: The form is valid; the first premise is false, so the argument is not sound.

✗ valid and sound: "All birds can swim" is false
✗ invalid: IF the premises were true, the conclusion would have to be true
✗ sound but not valid: sound arguments are always valid

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S24-M-Q3 · question page 547

LOGIC.INFERENCE.03 · Tell whether an argument is valid, and whether it is also sound

An argument can be sound even if its conclusion is false.

Answer: False

Reasoning: Sound means valid with true premises, and then the conclusion must be true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S24-M-Q4 · question page 548

LOGIC.INFERENCE.03 · Tell whether an argument is valid, and whether it is also sound

"If a number is divisible by 10, it is even. 14 is even. So 14 is divisible by 10." How is this described?

Answer: D. invalid, so it cannot be sound

Reasoning: Affirming the consequent is invalid; an invalid argument is never sound.

- ✗ valid and sound: the form affirms the consequent
- ✗ valid but not sound: the form is not valid
- ✗ sound but not valid: impossible

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S24-M-Q5 · question page 549

LOGIC.INFERENCE.03 · Tell whether an argument is valid, and whether it is also sound

Which argument is SOUND?

Answer: B. "Every prime greater than 2 is odd. 13 is a prime greater than 2. So 13 is odd."

Reasoning: Only the first is valid with true premises.

- ✗ "Every prime is odd. 2 is prime. So 2 is odd.": the first premise is false (2 is even)
- ✗ "If 13 is odd, 13 is prime. 13 is prime. So 13 is odd.": invalid form (affirming the consequent)
- ✗ "Every odd number is prime. 9 is odd. So 9 is prime.": the first premise is false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S24-B1-Q1 · question page 550

LOGIC.INFERENCE.01 · Is the form valid?

"If P then Q. P. So Q." is a valid form.

Answer: True

Reasoning: Modus ponens.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S24-B1-Q2 · question page 551

LOGIC.INFERENCE.01 · Is the form valid?

"If P then Q. Q. So P." is a valid form.

Answer: False

Reasoning: Affirming the consequent.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S24-B1-Q3 · question page 552

LOGIC.INFERENCE.01 · Is the form valid?

A valid argument guarantees what?

Answer: C. If the premises are true, the conclusion is true.

Reasoning: Definition.

✗ The premises are true.: validity says nothing about whether premises are true

✗ The conclusion is true.: only if the premises are true

✗ The argument is convincing.: persuasiveness is a different question

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S24-B1-Q4 · question page 553

LOGIC.INFERENCE.01 · Is the form valid?

"P or Q. P. So not Q." is a valid form.

Answer: False

Reasoning: With inclusive "or", P and Q can both be true, so the premises can be true while "not Q" is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S24-B1-Q5 · question page 554

LOGIC.INFERENCE.01 · Is the form valid?

An argument with a true conclusion is always valid.

Answer: False

Reasoning: An invalid form can reach a true conclusion by luck.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S24-B2-Q1 · question page 555

LOGIC.ARGUMENT.01 · Are the premises true?

"Every rectangle has four equal sides" is true.

Answer: False

Reasoning: A 2 by 5 rectangle does not have four equal sides.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S24-B2-Q2 · question page 556

LOGIC.ARGUMENT.01 · Are the premises true?

"Every rectangle is a square" is true.

Answer: False

Reasoning: A 2 by 5 rectangle is not a square.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S24-B2-Q3 · question page 557

LOGIC.ARGUMENT.01 · Are the premises true?

"Every multiple of 2 is a multiple of 4" is true.

Answer: False

Reasoning: 6 is a multiple of 2 but not of 4.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S24-B2-Q4 · question page 558

LOGIC.ARGUMENT.01 · Are the premises true?

"Every prime number is odd" is true.

Answer: False

Reasoning: 2 is prime and even.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S24-B2-Q5 · question page 559

LOGIC.ARGUMENT.01 · Are the premises true?

"Water freezes at 10 degrees Celsius at normal air pressure" is true.

Answer: False

Reasoning: At normal air pressure water freezes at 0 degrees Celsius; at 10 degrees it is liquid.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S24-F1-Q1 · question page 560

LOGIC.INFERENCE.03 · Classify harder arguments

"Every whole number that is divisible by 4 is divisible by 8. 12 is divisible by 4. So 12 is divisible by 8." How is it described?

Answer: A. valid but not sound

Reasoning: Valid form; false premise; and here the conclusion is also false.

- ✗ valid and sound: the first premise is false: 4 is not divisible by 8
- ✗ invalid: the form is valid
- ✗ sound but not valid: impossible

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S24-F1-Q2 · question page 561

LOGIC.INFERENCE.03 · Classify harder arguments

A valid argument with a false conclusion can have all of its premises true.

Answer: False

Reasoning: In a valid argument, true premises force a true conclusion. So a false conclusion means some premise is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S24-F1-Q3 · question page 562

LOGIC.INFERENCE.03 · Classify harder arguments

"No reptile has fur. A snake is a reptile. So a snake has no fur." How is it described?

Answer: D. valid and sound

Reasoning: Valid form, true premises.

- ✗ valid but not sound: both premises are true
- ✗ invalid: the conclusion follows
- ✗ sound but not valid: impossible

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S24-F1-Q4 · question page 563

LOGIC.INFERENCE.03 · Classify harder arguments

If an argument is invalid, its conclusion must be false.

Answer: False

Reasoning: Its conclusion may be true by accident; invalid only means it is not guaranteed.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S24-F1-Q5 · question page 564

LOGIC.INFERENCE.03 · Classify harder arguments

Which statement is TRUE?

Answer: A. Every sound argument is valid.

Reasoning: Sound = valid + true premises.

- ✗ Every valid argument is sound.: valid arguments can have false premises
- ✗ Every argument with true premises is sound.: it must also be valid
- ✗ Every argument with a true conclusion is sound.: it could be invalid

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S24-F2-Q1 · question page 565

LOGIC.INFERENCE.03 · Validity, soundness and truth together

Which combination is IMPOSSIBLE for an argument?

Answer: C. valid, all premises true, conclusion false

Reasoning: Validity rules out exactly this combination.

- ✗ valid, a false premise, conclusion true: possible: "All fish fly; robins are fish; so robins fly"
- ✗ invalid, all premises true, conclusion true: possible: affirming the consequent with true parts

✗ invalid, all premises true, conclusion false: possible: that is what invalid allows

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S24-F2-Q2 · question page 566

LOGIC.INFERENCE.03 · Validity, soundness and truth together

An argument whose premises can never all be true at the same time (they contradict each other) is valid.

Answer: True

Reasoning: There is no case with all premises true and the conclusion false, so it is valid (but never sound).

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S24-F2-Q3 · question page 567

LOGIC.INFERENCE.03 · Validity, soundness and truth together

Of the 8 combinations (valid or invalid; premises all true or not; conclusion true or false), how many are impossible for an argument?

Answer: 1

Reasoning: Only "valid, all premises true, conclusion false" is impossible; every other combination has an example (for instance "valid, false premise, false conclusion": "All cats fly; Tom is a cat; so Tom flies").

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author

S24-F2-Q4 · question page 568

LOGIC.INFERENCE.03 · Validity, soundness and truth together

"If x = 3, then x squared = 9. x squared = 9. So x = 3." For real x, this argument is:

Answer: B. invalid: x = -3 makes the premises true and the conclusion false

Reasoning: Affirming the consequent.

- ✗ valid: squaring is unique: squaring is not one-to-one
- ✗ sound: both premises can be true: soundness needs validity
- ✗ valid but not sound: the form is not valid

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S24-F2-Q5 · question page 569

LOGIC.INFERENCE.03 · Validity, soundness and truth together

Adding a false premise to a sound argument keeps it valid.

Answer: True

Reasoning: Adding premises never turns a valid argument invalid; it only stops it from being sound.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S25-M-Q1 · question page 571

LOGIC.SEQUENCE.01 · Carry out stated moves on a row of tokens and find moves that reach a target

Tiles in a row each show a different letter. The row reads T, E, A, M. Move the leftmost tile to the far right end. Then swap the first two tiles. What is the row now, from left to right?

Answer: A → E → M → T

Reasoning: Moving T to the end gives E, A, M, T. Swapping the first two gives A, E, M, T.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S25-M-Q2 · question page 572

LOGIC.SEQUENCE.01 · Carry out stated moves on a row of tokens and find moves that reach a target

Tiles in a row each show a different letter. One move swaps two tiles that are next to each other. What is the least number of moves that turn D, C, B, A into A, B, C, D?

Answer: 6

Reasoning: Each move fixes at most one pair that is in the wrong order. D, C, B, A has 6 such pairs (every pair), so at least 6 moves are needed, and 6 moves suffice (bubble the letters into place).

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S25-M-Q3 · question page 573

LOGIC.SEQUENCE.01 · Carry out stated moves on a row of tokens and find moves that reach a target

Tiles in a row each show a different letter. Which single move turns R, O, P, E into O, P, E, R?

Answer: D. Move R from the left end to the right end.

Reasoning: Taking R off the left and putting it on the right leaves O, P, E, R.

- ✗ Swap R and O.: that gives O, R, P, E
- ✗ Swap the two middle tiles.: that gives R, P, O, E

✗ Move E from the right end to the left end.: that gives E, R, O, P

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S25-M-Q4 · question page 574

LOGIC.SEQUENCE.01 · Carry out stated moves on a row of tokens and find moves that reach a target

Tiles in a row each show a different letter. One move swaps two tiles that are next to each other. What is the least number of moves that turn C, A, B, E, D into A, B, C, D, E?

Answer: 3

Reasoning: Pairs out of order: (C, A), (C, B), (E, D): 3. Each swap fixes one pair, so 3 moves are needed and enough: swap C and A, swap C and B, swap E and D.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S25-M-Q5 · question page 575

LOGIC.SEQUENCE.01 · Carry out stated moves on a row of tokens and find moves that reach a target

Tiles in a row each show a different letter. Reversing the whole row twice puts every tile back where it started.

Answer: True

Reasoning: The second reversal undoes the first.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S25-B1-Q1 · question page 576

LOGIC.SEQUENCE.01 · Strategy: number the positions

Tiles in a row each show a different letter. Row: K, I, T. Swap the first two tiles. What is the row?

Answer: I → K → T

Reasoning: K and I trade places.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 6 orderings of 3 items enumerated; 1 valid.

S25-B1-Q2 · question page 577

LOGIC.SEQUENCE.01 · Strategy: number the positions

Tiles in a row each show a different letter. Row: P, A, N. Move the leftmost tile to the far right. What is the row?

Answer: A → N → P

Reasoning: P goes to the end: A, N, P.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 6 orderings of 3 items enumerated; 1 valid.

S25-B1-Q3 · question page 578

LOGIC.SEQUENCE.01 · Strategy: number the positions

Tiles in a row each show a different letter. How many pairs of tiles are in the wrong alphabetical order in C, B, A?

Answer: 3

Reasoning: (C, B), (C, A), (B, A).

Grading: The game's answer checker compares the typed value with the key (equivalent

number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S25-B1-Q4 · question page 579

LOGIC.SEQUENCE.01 · Strategy: number the positions

Tiles in a row each show a different letter. Swapping two neighboring tiles twice returns the row to how it was.

Answer: True

Reasoning: The second swap undoes the first.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S25-B1-Q5 · question page 580

LOGIC.SEQUENCE.01 · Strategy: number the positions

Tiles in a row each show a different letter. One move swaps neighboring tiles. Least number of moves from B, A, C to A, B, C?

Answer: 1

Reasoning: Swap B and A.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S25-F1-Q1 · question page 581

LOGIC.SEQUENCE.01 · Fewest moves under different rules

Tiles in a row each show a different letter. One move takes ANY tile and puts it at the far right end. What is the least number of moves that turn B, C, D, A into A, B, C, D?

Answer: 3

Reasoning: A must end up first, so B, C and D must each be moved behind it: move B, then C, then D. Fewer is impossible because each of B, C, D is in front of A and only a move can put it after A.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S25-F1-Q2 · question page 582

LOGIC.SEQUENCE.01 · Fewest moves under different rules

Tiles in a row each show a different letter. One move takes ANY tile and puts it at the far right end. What is the least number of moves that turn A, C, D, B into A, B, C, D?

Answer: 2

Reasoning: Move C, then D: A, D, B, C, then A, B, C, D.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S25-F1-Q3 · question page 583

LOGIC.SEQUENCE.01 · Fewest moves under different rules

Tiles in a row each show a different letter. Row: S, T, O, P. Move the tile in position 3 to the far left end, then swap the last two tiles. What is the row?

Answer: O → S → P → T

Reasoning: Moving O to the front gives O, S, T, P; swapping the last two gives O, S, P, T.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S25-F1-Q4 · question page 584

LOGIC.SEQUENCE.01 · Fewest moves under different rules

Tiles in a row each show a different letter. One move swaps neighboring tiles. What is the least number of moves that turn E, D, C, B, A into A, B, C, D, E?

Answer: 10

Reasoning: Every one of the 10 pairs is out of order.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S25-F1-Q5 · question page 585

LOGIC.SEQUENCE.01 · Fewest moves under different rules

Tiles in a row each show a different letter. If one move swaps neighboring tiles, any arrangement of 4 tiles can be reached from any other.

Answer: True

Reasoning: Neighbor swaps can sort any row (as in bubble sort), and reversing the steps reaches any arrangement.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S25-F2-Q1 · question page 586

LOGIC.SEQUENCE.01 · Moves that can never work

Tiles in a row each show a different letter. One move reverses any three tiles that are next to each other (for example A, B, C becomes C, B, A). Starting from A, B, C, D, E, can you reach B, A, C, D, E?

Answer: A. No: each move keeps every tile in an odd or an even position, and A would need to move from position 1 to position 2.

Reasoning: Reversing three tiles swaps the 1st and 3rd of them and

keeps the middle one: every tile moves by 0 or 2 places, so it stays on odd or on even positions. A starts in position 1 (odd) and would need position 2.

- ✗ Yes: reverse A, B, C, then reverse B, A, ... in a few moves.: no sequence works
- ✗ Yes: any arrangement can be reached with enough moves.: only 12 of the 120 arrangements can be reached
- ✗ No: because each move changes three tiles at once.: that alone does not rule it out

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: all positions reachable by the move were searched. Automatic: the check function accepts only the keyed choice.

S25-F2-Q2 · question page 587

LOGIC.SEQUENCE.01 · Moves that can never work

Tiles in a row each show a different letter. One move reverses any three neighboring tiles. Starting from A, B, C, D, E, how many different arrangements can be reached (counting the start)?

Answer: 12

Reasoning: Tiles in odd positions (A, C, E) can be put in any order among positions 1, 3, 5 (6 ways), and B, D in positions 2, 4 (2 ways): 6 x 2 = 12.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S25-F2-Q3 · question page 588

LOGIC.SEQUENCE.01 · Moves that can never work

Tiles in a row each show a different letter. One move swaps neighboring tiles. The row C, A, D, B, E must become A, B, C, D, E. Give the least number of moves m.

Answer: any values that satisfy: m == 3 (example: m = 3)

Reasoning: Pairs out of order: (C, A), (C, B), (D, B): 3, so 3 moves are needed and enough.

Tempting answers the rule rejects: m = 2; m = 4

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it.

S25-F2-Q4 · question page 589

LOGIC.SEQUENCE.01 · Moves that can never work

Tiles in a row each show a different letter. One move swaps ANY two tiles (not only neighbors). D, C, B, A can be turned into A, B, C, D in 2 moves.

Answer: True

Reasoning: Swap D and A, then swap C and B.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S25-F2-Q5 · question page 590

LOGIC.SEQUENCE.01 · Moves that can never work

Tiles in a row each show a different letter. One move swaps ANY two tiles. What is the least number of moves that turn B, C, D, E, A into A, B, C, D, E?

Answer: 4

Reasoning: Each swap can put at most one tile into its final place unless it fixes two at once, which needs two tiles in each other's places; here the five tiles form one cycle, so 5 - 1 = 4 swaps are needed.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S26-M-Q1 · question page 592

LOGIC.QUANT.01 · Decide the truth of "all", "some" and "no" statements over a stated domain

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). "All chess members have a bus pass."

Answer: True

Reasoning: The chess members are Ava and Dee, and both have a bus pass.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S26-M-Q2 · question page 593

LOGIC.QUANT.01 · Decide the truth of "all", "some" and "no" statements over a stated domain

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). "Some member aged 16 has a bus pass."

Answer: False

Reasoning: The members aged 16 are Ben and Eli; neither has a bus pass.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S26-M-Q3 · question page 594

LOGIC.QUANT.01 · Decide the truth of "all", "some" and "no" statements over a stated domain

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). Which statement is TRUE?

Answer: C. No art member has a bus pass.

Reasoning: The only art member, Eli, has no bus pass.

- ✗ All robotics members have a bus pass.: Ben does not
- ✗ Some chess member is 16.: the chess members are 15 and 17
- ✗ Every member is at least 16.: Ava and Cy are 15

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S26-M-Q4 · question page 595

LOGIC.QUANT.01 · Decide the truth of "all", "some" and "no" statements over a stated domain

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). How many members make "the member is 15 or has no bus pass" true?

Answer: 4

Reasoning: Aged 15: Ava, Cy. No bus pass: Ben, Eli. Together: 4 members (Dee is the only exception).

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S26-M-Q5 · question page 596

LOGIC.QUANT.01 · Decide the truth of "all", "some" and "no" statements over a stated domain

The domain is the whole numbers 1 to 10. Which statement is FALSE?

Answer: B. Every number in the domain is less than 10.

Reasoning: 10 is in the domain and is not less than 10.

- ✗ Some number in the domain is prime.: 2, 3, 5 and 7 are prime
- ✗ No number in the domain is negative.: this is true: 1 to 10 are all positive
- ✗ Some number in the domain is a multiple of 7.: this is true: 7 is in the domain

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S26-B1-Q1 · question page 597

LOGIC.QUANT.01 · Read "all", "some" and "no"

"Some students walk to school" is true when ...

Answer: B. at least one student walks to school

Reasoning: "Some" means at least one.

- ✗ every student walks to school: that is "all"
- ✗ exactly one student walks: "some" allows more
- ✗ most students walk: "some" does not mean most

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S26-B1-Q2 · question page 598

LOGIC.QUANT.01 · Read "all", "some" and "no"

"No student walks to school" is true when ...

Answer: C. not even one student walks

Reasoning: "No" means zero.

- ✗ some students do not walk: that is "not all"
- ✗ most students ride: still allows walkers
- ✗ exactly one student walks: one walker makes it false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S26-B1-Q3 · question page 599

LOGIC.QUANT.01 · Read "all", "some" and "no"

"All the lights are on" is true if exactly one light is off.

Answer: False

Reasoning: One light that is off is an exception, and one exception makes an "all" statement false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S26-B1-Q4 · question page 600

LOGIC.QUANT.01 · Read "all", "some" and "no"

"Some numbers in {2, 4, 6} are odd" is true.

Answer: False

Reasoning: None of them is odd.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S26-B1-Q5 · question page 601

LOGIC.QUANT.01 · Read "all", "some" and "no"

"Every number in {3, 5, 7} is odd" is true.

Answer: True

Reasoning: All three are odd.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S26-B2-Q1 · question page 602

LOGIC.QUANT.01 · Stay inside the stated domain

The domain is {1, 2, 3}. "Every number in the domain is less than 4" is true.

Answer: True

Reasoning: Only 1, 2, 3 count.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S26-B2-Q2 · question page 603

LOGIC.QUANT.01 · Stay inside the stated domain

The domain is {10, 20, 30}. "Some number in the domain is odd" is true.

Answer: False

Reasoning: All three are even.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S26-B2-Q3 · question page 604

LOGIC.QUANT.01 · Stay inside the stated domain

The domain is the days Monday to Friday. Which is true?

Answer: A. No day in the domain is a weekend day.

Reasoning: Saturday and Sunday are outside the domain.

- ✗ Some day in the domain is Sunday.: Sunday is not in the domain
- ✗ Every day in the domain starts with M.: only Monday
- ✗ Some day in the domain is Saturday.: not in the domain

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S26-B2-Q4 · question page 605

LOGIC.QUANT.01 · Stay inside the stated domain

"All prime numbers are odd" is true when the domain is {3, 5, 7, 11}.

Answer: True

Reasoning: Within this domain, every prime is odd.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S26-B2-Q5 · question page 606

LOGIC.QUANT.01 · Stay inside the stated domain

"All prime numbers are odd" is true when the domain is all whole numbers.

Answer: False

Reasoning: 2 is prime and even.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S26-F1-Q1 · question page 607

LOGIC.QUANT.01 · Quantifiers with two conditions

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). "Every member without a bus pass is 16."

Answer: True

Reasoning: Ben and Eli have no bus pass; both are 16.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S26-F1-Q2 · question page 608

LOGIC.QUANT.01 · Quantifiers with two conditions

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). "Some member with a bus pass is older than every robotics member."

Answer: True

Reasoning: Dee (17) has a bus pass and is older than Ben (16) and Cy (15).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S26-F1-Q3 · question page 609

LOGIC.QUANT.01 · Quantifiers with two conditions

The domain is the whole numbers 1 to 20. How many numbers make "the number is even and not a multiple of 3" true?

Answer: 7

Reasoning: Even: 10 numbers. Even multiples of 3 (6, 12, 18): 3. So 7.

Grading: The game's answer checker compares the typed value with the key (equivalent

number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S26-F1-Q4 · question page 610

LOGIC.QUANT.01 · Quantifiers with two conditions

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). Which statement is FALSE?

Answer: D. No member aged 15 is in robotics.

Reasoning: Cy is 15 and in robotics.

- ✗ Every chess member is 15 or 17.: true: Ava 15, Dee 17
- ✗ Some member has no bus pass.: Ben, Eli
- ✗ All members aged 17 play chess.: Dee is the only 17-year-old

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S26-F1-Q5 · question page 611

LOGIC.QUANT.01 · Quantifiers with two conditions

The domain is empty (it has no members). "Every member of the domain is green" is true.

Answer: True

Reasoning: There is no member to break the claim, so an "every" statement about an empty domain is true ("vacuously" true).

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S26-F2-Q1 · question page 612

LOGIC.QUANT.01 · Nested quantifiers

The domain is the whole numbers 1 to 5. "For every number x, there is a number y with y > x" is true.

Answer: False

Reasoning: For x = 5 there is no larger number in the domain.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S26-F2-Q2 · question page 613

LOGIC.QUANT.01 · Nested quantifiers

The domain is the whole numbers 1 to 5. "There is a number y such that for every number x, y ≥ x" is true.

Answer: True

Reasoning: y = 5 works for every x.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S26-F2-Q3 · question page 614

LOGIC.QUANT.01 · Nested quantifiers

The domain is all whole numbers. Which statement is TRUE?

Answer: C. For every x, there is a y with y > x.

Reasoning: Given any x, y = x + 1 works. The order of "for every" and "there is" matters.

- ✗ There is a y such that for every x, y > x.: no whole number is larger than every whole number (including itself)
- ✗ For every x and every y, y > x.: false for x = 3, y = 2

✗ There is an x such that for every y, y > x.: x would have to be less than itself

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S26-F2-Q4 · question page 615

LOGIC.QUANT.01 · Nested quantifiers

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). For how many members x is the statement "some other member is the same age as x" true?

Answer: 4

Reasoning: Ava and Cy share 15; Ben and Eli share 16; Dee (17) has no partner.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S26-F2-Q5 · question page 616

LOGIC.QUANT.01 · Nested quantifiers

The domain is the whole numbers 1 to 9. Give a number x that makes "for every y in the domain, x + y > 5" true.

Answer: any values that satisfy: x >= 5 and x <= 9 and isInteger(x) (example: x = 6)

Reasoning: The smallest y is 1, so x + 1 > 5, x > 4: any x from 5 to 9.

Tempting answers the rule rejects: x = 4; x = 10; x = 2

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 5 valid answers.

S27-M-Q1 · question page 618

LOGIC.QUANT.02 · Negate statements with "all", "some", "no", and nested quantifiers

What is the negation of "All the tickets were sold"?

Answer: A. At least one ticket was not sold.

Reasoning: The negation of "all are" is "some (at least one) is not".

- ✗ No ticket was sold.: too strong: one unsold ticket is enough to make the original false
- ✗ All the tickets were unsold.: same as "no ticket was sold"
- ✗ Some tickets were sold.: compatible with the original

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S27-M-Q2 · question page 619

LOGIC.QUANT.02 · Negate statements with "all", "some", "no", and nested quantifiers

What is the negation of "Some student brought a calculator"?

Answer: D. No student brought a calculator.

Reasoning: The negation of "at least one is" is "none is".

- ✗ Some student did not bring a calculator.: can be true together with the original
- ✗ Every student brought a calculator.: that makes the original true
- ✗ Exactly one student brought a calculator.: that makes the original true

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S27-M-Q3 · question page 620

LOGIC.QUANT.02 · Negate statements with "all", "some", "no", and nested quantifiers

What is the negation of "No bridge in the town is closed"?

Answer: B. Some bridge in the town is closed.

Reasoning: The negation of "none is" is "at least one is".

- ✗ Every bridge in the town is closed.: too strong
- ✗ Some bridge in the town is open.: compatible with the original
- ✗ No bridge in the town is open.: not the negation

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S27-M-Q4 · question page 621

LOGIC.QUANT.02 · Negate statements with "all", "some", "no", and nested quantifiers

The domain is the whole numbers. The statement "Every whole number greater than 1 has an even square" is false. Give a number n that makes its negation true: a whole number greater than 1 whose square is odd.

Answer: any values that satisfy: n > 1 and isOdd(n*n) and isInteger(n) (example: n = 3)

Reasoning: The negation is "Some whole number greater than 1 has an odd square": any odd n > 1.

Tempting answers the rule rejects: n = 2; n = 1; n = 4

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S27-M-Q5 · question page 622

LOGIC.QUANT.02 · Negate statements with "all", "some", "no", and nested quantifiers

The negation of "Every square is a rectangle" is "No square is a rectangle".

Answer: False

Reasoning: The negation is "Some square is not a rectangle".

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S27-B1-Q1 · question page 623

LOGIC.QUANT.01 · The meaning of each quantifier

"Some cards are red" is false if exactly one card is red.

Answer: False

Reasoning: "Some" means at least one, so one red card makes the statement true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S27-B1-Q2 · question page 624

LOGIC.QUANT.01 · The meaning of each quantifier

"All cards are red" is false if one card is blue.

Answer: True

Reasoning: One exception is enough.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S27-B1-Q3 · question page 625

LOGIC.QUANT.01 · The meaning of each quantifier

"No card is red" is true if one card is red.

Answer: False

Reasoning: "No" means zero.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S27-B1-Q4 · question page 626

LOGIC.QUANT.01 · The meaning of each quantifier

"Not all cards are red" means ...

Answer: D. at least one card is not red

Reasoning: One non-red card is enough.

- ✗ no card is red: too strong
- ✗ exactly one card is red: unrelated
- ✗ every card is blue: too strong

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S27-B1-Q5 · question page 627

LOGIC.QUANT.01 · The meaning of each quantifier

"Some cards are red" and "some cards are not red" can both be true.

Answer: True

Reasoning: A mix of red and blue.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S27-B2-Q1 · question page 628

LOGIC.CONNECTIVE.01 · Negating simple statements

What is the negation of "the bus is late"?

Answer: A. the bus is not late

Reasoning: Negate the whole claim.

- ✗ the bus is early: it could be on time
- ✗ the train is late: unrelated
- ✗ the bus is very late: not a negation

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S27-B2-Q2 · question page 629

LOGIC.CONNECTIVE.01 · Negating simple statements

The negation of "$x \geq 4$" is "$x \leq 4$".

Answer: False

Reasoning: $x = 4$ makes both "$x \geq 4$" and "$x \leq 4$" true, so they are not negations. The negation is "$x < 4$".

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S27-B2-Q3 · question page 630

LOGIC.CONNECTIVE.01 · Negating simple statements

What is the negation of "the number is even and positive"?

Answer: C. the number is odd or not positive

Reasoning: Not (A and B) = not A or not B.

- ✗ the number is odd and negative: too strong
- ✗ the number is even or positive: not a negation
- ✗ the number is zero: too narrow

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S27-B2-Q4 · question page 631

LOGIC.CONNECTIVE.01 · Negating simple statements

"Not (the door is locked)" means the door is not locked.

Answer: True

Reasoning: Plain negation.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S27-B2-Q5 · question page 632

LOGIC.CONNECTIVE.01 · Negating simple statements

What is the negation of "it rained on Monday or Tuesday"?

Answer: B. it rained on neither Monday nor Tuesday

Reasoning: Not (A or B) = not A and not B.

- ✗ it did not rain on Monday: it could have rained on Tuesday
- ✗ it rained on both days: not a negation
- ✗ it rained on Wednesday: unrelated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S27-F1-Q1 · question page 633

LOGIC.QUANT.02 · Negate quantified statements with conditions

What is the negation of "Every student who studied passed"?

Answer: B. Some student who studied did not pass.

Reasoning: Negation of "every A is B": some A is not B.

- ✗ Every student who did not study failed.: unrelated to the original claim
- ✗ No student who studied passed.: too strong
- ✗ Some student who passed did not study.: swaps the condition

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S27-F1-Q2 · question page 634

LOGIC.QUANT.02 · Negate quantified statements with conditions

The domain is the real numbers. What is the negation of "For every x, x squared ≥ 0"?

Answer: C. There is an x with x squared < 0.

Reasoning: Negation: "there is an x" with the opposite inequality.

- ✗ For every x, x squared < 0.: too strong
- ✗ There is an x with x squared ≥ 0.: compatible with the original
- ✗ For every x, x squared ≤ 0.: not the negation

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S27-F1-Q3 · question page 635

LOGIC.QUANT.02 · Negate quantified statements with conditions

The negation of "Some prime number is even" is "Every prime number is odd" (for whole

numbers greater than 1).

Answer: True

Reasoning: "No prime is even" means every prime is not even; for whole numbers greater than 1, not even means odd.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S27-F1-Q4 · question page 636

LOGIC.QUANT.02 · Negate quantified statements with conditions

The statement "Every whole number from 1 to 30 that ends in 9 is prime" is false. Give a number that shows it (a witness for its negation).

Answer: any values that satisfy: n % 10 == 9 and n >= 1 and n <= 30 and not isPrime(n) (example: n = 9)

Reasoning: The numbers ending in 9 are 9, 19 and 29; 19 and 29 are prime, but 9 = 3 x 3 is not. So n = 9.

Tempting answers the rule rejects: n = 19; n = 29; n = 39

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 1 valid answers.

S27-F1-Q5 · question page 637

LOGIC.QUANT.02 · Negate quantified statements with conditions

The negation of "No dog is a cat" is "Every dog is a cat".

Answer: False

Reasoning: The negation is "Some dog is a cat".

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S27-F2-Q1 · question page 638

LOGIC.QUANT.02 · Nested quantifiers and their negations

What is the negation of "For every student, there is a club the student likes"?

Answer: A. There is a student who likes no club.

Reasoning: Negate step by step: "there is a student such that NOT (there is a club the student likes)" = "there is a student who likes no club".

- ✗ For every student, there is a club the student does not like.: only flips the inner part
- ✗ There is a club that no student likes.: swaps the roles
- ✗ No student likes any club.: too strong

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S27-F2-Q2 · question page 639

LOGIC.QUANT.02 · Nested quantifiers and their negations

The domain is the whole numbers. What is the negation of "There is a number y such that for every x, $y > x$"?

Answer: D. For every y, there is an x with $y \leq x$.

Reasoning: Flip each quantifier and negate the inside: for every y, there is an x such that not $(y > x)$.

- ✗ For every y and every x, $y \leq x$.: too strong
- ✗ There is a y with $y \leq x$ for some x.: not the negation
- ✗ There is no number y.: not the negation

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S27-F2-Q3 · question page 640

LOGIC.QUANT.02 · Nested quantifiers and their negations

The negation of "For every x there is a y with x + y = 0" is "There is an x such that for every y, x + y ≠ 0".

Answer: True

Reasoning: Flip both quantifiers and negate the equation.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S27-F2-Q4 · question page 641

LOGIC.QUANT.02 · Nested quantifiers and their negations

The domain is the whole numbers 1 to 10. The statement "For every x there is a y with y = 2x" is false in this domain. Give an x that shows it.

Answer: any values that satisfy: x >= 6 and x <= 10 and isInteger(x) (example: x = 6)

Reasoning: For x from 6 to 10, 2x is at least 12, outside the domain. Any of 6 to 10 works.

Tempting answers the rule rejects: x = 5; x = 1; x = 11

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 5 valid answers.

S27-F2-Q5 · question page 642

LOGIC.QUANT.02 · Nested quantifiers and their negations

For any statement about a domain, either the statement or its negation is true.

Answer: True

Reasoning: A statement and its negation always have opposite truth values.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S28-M-Q1 · question page 644

LOGIC.GRAPH.01 · Trace routes in a network and decide which routes exist

Roads are the straight lines; towns are the circles. Roads meet only at towns. What is the least number of roads on a route from A to F?

Answer: 3

Reasoning: A to B (or C), then to D, then to F: 3 roads. No route uses fewer, because A is not joined to D or F, and no town next to A is joined to F.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S28-M-Q2 · question page 645

LOGIC.GRAPH.01 · Trace routes in a network and decide which routes exist

Roads are the straight lines; towns are the circles. Roads meet only at towns. How many different routes go from A to F without visiting any town twice?

Answer: 8

Reasoning: Every route goes through D (the only way to E and F). From A to D: A-B-D, A-C-D, A-B-C-D, A-C-B-D (4 ways). From D to F: D-F or D-E-F (2 ways). 4 x 2 = 8.

Grading: The game's answer checker compares the typed value with the key (equivalent

number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S28-M-Q3 · question page 646

LOGIC.GRAPH.01 · Trace routes in a network and decide which routes exist

Roads are the straight lines; towns are the circles. Roads meet only at towns. Which town, if it were closed (with all its roads), would cut the network into two parts that cannot reach each other?

Answer: D. D

Reasoning: Every route from A, B or C to E or F passes through D.

- ✗ B: A can still reach D through C
- ✗ C: A can still reach D through B
- ✗ E: F is still joined to D

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S28-M-Q4 · question page 647

LOGIC.GRAPH.01 · Trace routes in a network and decide which routes exist

Walkers 1, 2 and 3 each follow their own line from the left to a gate on the right. Where lines cross, there is no turning: each walker keeps going straight along their own line. Which gate does walker 2 reach?

Answer: A. Z

Reasoning: Follow line 2: it goes right, climbs to the top, crosses down to the bottom right and ends at gate Z.

- ✗ X: that is where another walker's line ends
- ✗ Y: that is where another walker's line ends
- ✗ none of the gates: every line ends at a gate

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S28-M-Q5 · question page 648

LOGIC.GRAPH.01 · Trace routes in a network and decide which routes exist

Roads are the straight lines; towns are the circles. Roads meet only at towns. A walk can use every road exactly once (towns may be visited more than once).

Answer: True

Reasoning: Count the roads at each town: A 2, B 3, C 3, D 4, E 2, F 2. Exactly two towns (B and C) have an odd number, so a walk using every road once exists: it must start at B and end at C (or the reverse).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S28-B1-Q1 · question page 649

LOGIC.GRAPH.01 · Strategy: mark the route as you trace it

Roads are the straight lines; towns are the circles. Roads meet only at towns. How many roads meet at town D?

Answer: 4

Reasoning: D-B, D-C, D-E, D-F.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S28-B1-Q2 · question page 650

LOGIC.GRAPH.01 · Strategy: mark the route as you trace it

Roads are the straight lines; towns are the circles. Roads meet only at towns. Which towns

can be reached from A using exactly one road?

Answer: C. B and C

Reasoning: A is joined only to B and C.

- ✗ B, C and D: D needs two roads
- ✗ only B: C is also joined to A
- ✗ D: not joined to A

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S28-B1-Q3 · question page 651

LOGIC.GRAPH.01 · Strategy: mark the route as you trace it

Roads are the straight lines; towns are the circles. Roads meet only at towns. There is a road directly from A to D.

Answer: False

Reasoning: A is joined only to B and C.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S28-B1-Q4 · question page 652

LOGIC.GRAPH.01 · Strategy: mark the route as you trace it

Roads are the straight lines; towns are the circles. Roads meet only at towns. How many routes go from E to F without visiting a town twice?

Answer: 2

Reasoning: E-F directly, or E-D-F.

Grading: The game's answer checker compares the typed value with the key (equivalent

number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S28-B1-Q5 · question page 653

LOGIC.GRAPH.01 · Strategy: mark the route as you trace it

In a network, a town with 3 roads must be the start or end of a walk that uses every road once.

Answer: True

Reasoning: At a town in the middle of the walk, roads are used in pairs (in and out); an odd number leaves one unpaired road, so the walk must start or end there.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S28-F1-Q1 · question page 654

LOGIC.GRAPH.01 · Routes with conditions

Roads are the straight lines; towns are the circles. Roads meet only at towns. How many routes go from P to U without visiting a town twice?

Answer: 6

Reasoning: Listing them in order: P-Q-R-U, P-Q-U, P-Q-T-U, P-S-T-U, P-S-T-Q-R-U, P-S-T-Q-U.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S28-F1-Q2 · question page 655

LOGIC.GRAPH.01 · Routes with conditions

Roads are the straight lines; towns are the circles. Roads meet only at towns. How many routes from P to U (no town twice) pass through T?

Answer: 4

Reasoning: P-Q-T-U, P-S-T-U, P-S-T-Q-R-U, P-S-T-Q-U.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S28-F1-Q3 · question page 656

LOGIC.GRAPH.01 · Routes with conditions

Roads are the straight lines; towns are the circles. Roads meet only at towns. A walk can use every road exactly once.

Answer: True

Reasoning: Roads at each town: P 2, Q 4, R 2, S 2, T 3, U 3. Exactly two odd towns (T and U), so yes: start at T and end at U.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S28-F1-Q4 · question page 657

LOGIC.GRAPH.01 · Routes with conditions

Roads are the straight lines; towns are the circles. Roads meet only at towns. Which single road, if closed, makes T unreachable from P by any route?

Answer: B. none of these roads

Reasoning: T has three roads (to Q, S and U), and P reaches T through more than one of them, so no single road cuts T off.

✗ P-S: T can still be reached through Q

✗ Q-T: T can still be reached through S

✗ S-T: T can still be reached through Q

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S28-F1-Q5 · question page 658

LOGIC.GRAPH.01 · Routes with conditions

Roads are the straight lines; towns are the circles. Roads meet only at towns. What is the least number of roads on a route from S to R?

Answer: 3

Reasoning: S-T-U-R or S-P-Q-R or S-T-Q-R: 3 roads.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S28-F2-Q1 · question page 659

LOGIC.GRAPH.01 · Every road, every town

In a network of 6 towns, the numbers of roads at the towns are 1, 2, 2, 3, 3 and 3. Every road has two ends, each at a town. How many roads are there?

Answer: 7

Reasoning: Adding the numbers counts every road twice (once at each end): 1 + 2 + 2 + 3 + 3 + 3 = 14, so there are 14 / 2 = 7 roads.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S28-F2-Q2 · question page 660

LOGIC.GRAPH.01 · Every road, every town

Some network has exactly three towns with an odd number of roads.

Answer: False

Reasoning: Every road adds 2 to the total count of road-ends, so the total is even; that needs an even number of odd towns.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S28-F2-Q3 · question page 661

LOGIC.GRAPH.01 · Every road, every town

Roads are the straight lines; towns are the circles. Roads meet only at towns. What is the least number of roads that must be ADDED so that a walk can use every road exactly once and return to its starting town?

Answer: B. 1

Reasoning: A closed walk needs every town to have an even number of roads. Only B and C are odd; adding one road B-C fixes both.

- ✗ 0: B and C have an odd number of roads, so a closed walk is impossible now
- ✗ 2: one new road between B and C makes every town even
- ✗ 3: far fewer are needed

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S28-F2-Q4 · question page 662

LOGIC.GRAPH.01 · Every road, every town

Roads are the straight lines; towns are the circles. Roads meet only at towns. How many routes from A to F visit EVERY town exactly once?

Answer: 2

Reasoning: A route through all six towns must go A, then B and C (in some order), then D, then E, then F: A-B-C-D-E-F and A-C-B-D-E-F.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S28-F2-Q5 · question page 663

LOGIC.GRAPH.01 · Every road, every town

A network has towns with these numbers of roads: 3, 3, 2, 2 and k. For a walk that uses every road exactly once to be possible in a connected network, at most two towns may have an odd number of roads. Give a whole number k from 1 to 6 that keeps this possible.

Answer: any values that satisfy: isEven(k) and k >= 1 and k <= 6 (example: k = 2)

Reasoning: There are already two odd towns (the two 3s), so k must be even: 2, 4 or 6.

Tempting answers the rule rejects: k = 3; k = 1; k = 5

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 3 valid answers.

S29-M-Q1 · question page 665

LOGIC.QUANT.03 · Give a witness for a "some" statement and a counterexample to an "all" statement

Give a witness for the true statement "Some whole number between 20 and 30 is prime."

Answer: any values that satisfy: isPrime(n) and n > 20 and n < 30 (example: n = 23)

Reasoning: The primes between 20 and 30 are 23 and 29.

Tempting answers the rule rejects: n = 21; n = 25; n = 31

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 2 valid answers.

S29-M-Q2 · question page 666

LOGIC.QUANT.03 · Give a witness for a "some" statement and a counterexample to an "all" statement

Give a counterexample to "Every whole number that is divisible by 3 is odd."

Answer: any values that satisfy: n % 3 == 0 and isEven(n) and isInteger(n) (example: n = 6)

Reasoning: A counterexample is a multiple of 3 that is even: 6, 12, 18, ...

Tempting answers the rule rejects: n = 9; n = 4; n = 15

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S29-M-Q3 · question page 667

LOGIC.QUANT.03 · Give a witness for a "some" statement and a counterexample to an "all" statement

Which number is a counterexample to "Every number that ends in 1 is prime"?

Answer: A. 21

Reasoning: 21 ends in 1 and 21 = 3 x 7.

- ✗ 11: 11 is prime, so it agrees with the claim
- ✗ 31: 31 is prime

✗ 12: does not end in 1

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S29-M-Q4 · question page 668

LOGIC.QUANT.03 · Give a witness for a "some" statement and a counterexample to an "all" statement

Give real numbers a and b that are a counterexample to "For all real numbers a and b, (a + b) squared = a squared + b squared."

Answer: any values that satisfy: (a + b)^2 != a^2 + b^2 (example: a = 1, b = 1)

Reasoning: (1 + 1) squared = 4 but 1 + 1 = 2. Any a and b with a x b not 0 works.

Tempting answers the rule rejects: a = 0, b = 5; a = 3, b = 0

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it.

S29-M-Q5 · question page 669

LOGIC.QUANT.03 · Give a witness for a "some" statement and a counterexample to an "all" statement

One example where a statement is true proves "Every number has this property."

Answer: False

Reasoning: An example (a witness) proves only a "some" statement. An "every" statement needs an argument for all cases; one counterexample is enough to disprove it.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S29-B1-Q1 · question page 670

LOGIC.QUANT.01 · Read "every" and "some" statements

What kind of example disproves "Every bird can fly"?

Answer: C. a bird that cannot fly

Reasoning: A counterexample has the "every" property but not the claimed one.

- ✗ a bird that can fly: agrees with the claim
- ✗ an animal that is not a bird: irrelevant
- ✗ a flying insect: irrelevant

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S29-B1-Q2 · question page 671

LOGIC.QUANT.01 · Read "every" and "some" statements

What kind of example proves "Some square numbers are even"?

Answer: D. an even square number, such as 4

Reasoning: One witness proves "some".

- ✗ an odd square number: does not help
- ✗ an even number that is not a square: not a square
- ✗ every square number: one example is enough

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S29-B1-Q3 · question page 672

LOGIC.QUANT.01 · Read "every" and "some" statements

"All cats are black" is shown true by one black cat.

Answer: False

Reasoning: One black cat shows that some cats are black. Showing that all cats are black needs every cat to be black.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S29-B1-Q4 · question page 673

LOGIC.QUANT.01 · Read "every" and "some" statements

"All cats are black" is shown false by one white cat.

Answer: True

Reasoning: One counterexample.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S29-B1-Q5 · question page 674

LOGIC.QUANT.01 · Read "every" and "some" statements

"No number is negative" is shown false by the number 1.

Answer: False

Reasoning: 1 is not negative, so it does not contradict the statement. A negative number such as -1 would.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S29-B2-Q1 · question page 675

LOGIC.QUANT.03 · Test a number against a property

15 is prime.

Answer: False

Reasoning: 15 = 3 x 5.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S29-B2-Q2 · question page 676

LOGIC.QUANT.03 · Test a number against a property

27 is divisible by 3 and odd.

Answer: True

Reasoning: 27 = 3 x 9, odd.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S29-B2-Q3 · question page 677

LOGIC.QUANT.03 · Test a number against a property

What is 7 squared minus 7?

Answer: 42

Reasoning: 49 - 7.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S29-B2-Q4 · question page 678

LOGIC.QUANT.03 · Test a number against a property

2 is both prime and even.

Answer: True

Reasoning: 2 has exactly two factors and is even.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S29-B2-Q5 · question page 679

LOGIC.QUANT.03 · Test a number against a property

Which number is a multiple of 4 but not of 8?

Answer: A. 12

Reasoning: 12 = 4 x 3.

- ✗ 16: a multiple of 8
- ✗ 8: a multiple of 8
- ✗ 10: not a multiple of 4

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S29-F1-Q1 · question page 680

LOGIC.QUANT.03 · Counterexamples in algebra

Give a real number x that is a counterexample to "For every real number x, x squared ≥ x."

Answer: any values that satisfy: x^2 < x (example: x = 0.5)

Reasoning: Any x strictly between 0 and 1, such as 0.5 (0.25 < 0.5).

Tempting answers the rule rejects: x = 2; x = -1; x = 1

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S29-F1-Q2 · question page 681

LOGIC.QUANT.03 · Counterexamples in algebra

Give positive whole numbers a and b that are a counterexample to "If a divides b x c, then a divides b or a divides c", using c = 2.

Answer: any values that satisfy: divides(a, b * 2) and not divides(a, b) and not divides(a, 2) and a > 0 and b > 0 and isInteger(a) and isInteger(b) (example: a = 4, b = 2)

Reasoning: a = 4, b = 2: 4 divides 2 x 2 = 4, but 4 divides neither 2 nor 2. (a = 6, b = 3 also works.)

Tempting answers the rule rejects: a = 3, b = 3; a = 2, b = 5

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it.

S29-F1-Q3 · question page 682

LOGIC.QUANT.03 · Counterexamples in algebra

Which value of n is a counterexample to "n squared + n + 1 is prime for every whole number n ≥ 1"?

Answer: B. 4

Reasoning: 4: 16 + 4 + 1 = 21 = 3 x 7.

- ✗ 1: 3 is prime
- ✗ 2: 7 is prime
- ✗ 3: 13 is prime

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S29-F1-Q4 · question page 683

LOGIC.QUANT.03 · Counterexamples in algebra

The statement "For every real x, if x > 1 then x squared > x" has no counterexample.

Answer: True

Reasoning: If x > 1, multiplying x > 1 by the positive x gives x squared > x.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S29-F1-Q5 · question page 684

LOGIC.QUANT.03 · Counterexamples in algebra

Give whole numbers a and b, each from 1 to 9, that show "a - b = b - a" can be false.

Answer: any values that satisfy: a - b != b - a and a >= 1 and a <= 9 and b >= 1 and b <= 9 (example: a = 5, b = 2)

Reasoning: Any a different from b.

Tempting answers the rule rejects: a = 4, b = 4

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 1 listed mistakes fail it.

S29-F2-Q1 · question page 685

LOGIC.QUANT.03 · Witnesses that are hard to find

Give a witness for "Some whole number from 1 to 100 leaves remainder 2 when divided by

3, remainder 3 when divided by 4, and remainder 4 when divided by 5."

Answer: any values that satisfy: n % 3 == 2 and n % 4 == 3 and n % 5 == 4 and n >= 1 and n <= 100 (example: n = 59)

Reasoning: Each remainder is one less than the divisor, so n + 1 is divisible by 3, 4 and 5, that is by 60. n + 1 = 60 gives n = 59, the only such number up to 100.

Tempting answers the rule rejects: n = 29; n = 119; n = 14

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 1 valid answers.

S29-F2-Q2 · question page 686

LOGIC.QUANT.03 · Witnesses that are hard to find

Is there a whole number n from 1 to 100 with n squared ending in the digit 2?

Answer: D. No: squares end only in 0, 1, 4, 5, 6 or 9.

Reasoning: The last digit of n squared depends only on the last digit of n; checking 0-9 gives 0, 1, 4, 9, 6, 5, 6, 9, 4, 1.

- ✗ Yes: n = 12.: 144 ends in 4
- ✗ Yes: n = 22.: 484 ends in 4
- ✗ No: but only because 2 is even.: that is not the reason; 4 and 6 are even and do appear

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S29-F2-Q3 · question page 687

LOGIC.QUANT.03 · Witnesses that are hard to find

Give whole numbers a, b and c (each from 1 to 20) with a squared + b squared = c squared and a < b.

Answer: any values that satisfy: a^2 + b^2 == c^2 and a < b and a >= 1 and c <= 20 and isInteger(a) and isInteger(b) and isInteger(c) (example: a = 3, b = 4, c = 5)

Reasoning: For example 3, 4, 5; 5, 12, 13; 6, 8, 10; 8, 15, 17; 9, 12, 15; 12, 16, 20.

Tempting answers the rule rejects: a = 2, b = 3, c = 4; a = 4, b = 3, c = 5

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it.

S29-F2-Q4 · question page 688

LOGIC.QUANT.03 · Witnesses that are hard to find

How many counterexamples to "Every odd number from 1 to 30 is prime" are there in that range?

Answer: 6

Reasoning: Odd numbers 1 to 29 that are not prime: 1, 9, 15, 21, 25, 27.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S29-F2-Q5 · question page 689

LOGIC.QUANT.03 · Witnesses that are hard to find

To disprove "There is a whole number n with n + 5 = n", one example is enough.

Answer: False

Reasoning: A "there is" statement is disproved by showing NO number

works: here n + 5 = n would give 5 = 0, which is impossible for every n. A single example cannot do that.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S30-M-Q1 · question page 691

LOGIC.SETS.01 · Mark subset, intersection and complement relations on Venn diagrams

Circle A is the set of students in soccer and circle B the set of students in band. Mark the region for students in soccer but NOT in band.

Answer: A

Reasoning: Inside A, outside B: the part of A that does not overlap B.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S30-M-Q2 · question page 692

LOGIC.SETS.01 · Mark subset, intersection and complement relations on Venn diagrams

Circle A is the set of even numbers and circle B the set of multiples of 3 (whole numbers 1 to 30 make up the outside box). Mark the region that contains 12.

Answer: AB

Reasoning: 12 is even and a multiple of 3: the overlap.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S30-M-Q3 · question page 693

LOGIC.SETS.01 · Mark subset, intersection and complement relations on Venn diagrams

Three circles: A (likes math), B (likes art), C (likes music). Mark every region for people who like EXACTLY two of the three subjects.

Answer: AB, AC, BC

Reasoning: The three regions where exactly two circles overlap, not the center.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 8 regions tested against the condition.

S30-M-Q4 · question page 694

LOGIC.SETS.01 · Mark subset, intersection and complement relations on Venn diagrams

In a class of 30 students, 18 play soccer, 12 play in the band, and 5 do both. How many students do neither?

Answer: 5

Reasoning: Soccer or band: 18 + 12 - 5 = 25. Neither: 30 - 25 = 5.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S30-M-Q5 · question page 695

LOGIC.SETS.01 · Mark subset, intersection and complement relations on Venn diagrams

Every member of set A is also in set B (A is a subset of B). On a two-circle diagram, mark the region that must be EMPTY.

Answer: A

Reasoning: If every A is a B, nothing can be in A outside B.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S30-B1-Q1 · question page 696

LOGIC.SETS.01 · Regions of a two-circle diagram

Mark the region for things in BOTH A and B.

Answer: AB

Reasoning: The overlap.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S30-B1-Q2 · question page 697

LOGIC.SETS.01 · Regions of a two-circle diagram

Mark the region for things in NEITHER A nor B.

Answer: O

Reasoning: Outside both circles.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S30-B1-Q3 · question page 698

LOGIC.SETS.01 · Regions of a two-circle diagram

Mark every region for things in A (whether or not they are in B).

Answer: A, AB

Reasoning: A only and the overlap.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S30-B1-Q4 · question page 699

LOGIC.SETS.01 · Regions of a two-circle diagram

How many regions does a two-circle Venn diagram have, counting the outside?

Answer: 4

Reasoning: A only, B only, both, neither.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S30-B1-Q5 · question page 700

LOGIC.SETS.01 · Regions of a two-circle diagram

Mark every region for things in B but not in A.

Answer: B

Reasoning: B only.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S30-B2-Q1 · question page 701

LOGIC.CONNECTIVE.02 · Set words: and, or, not

The intersection of A and B contains things that are ...

Answer: C. in A and in B

Reasoning: Intersection = "and".

- ✗ in A or in B: that is the union
- ✗ in neither: outside both
- ✗ in A but not B: part of A

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S30-B2-Q2 · question page 702

LOGIC.CONNECTIVE.02 · Set words: and, or, not

The union of A and B contains things that are ...

Answer: A. in A or in B (or both)

Reasoning: Union = inclusive "or".

- ✗ in A and in B: that is the intersection
- ✗ in exactly one: that excludes the overlap
- ✗ in neither: outside both

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S30-B2-Q3 · question page 703

LOGIC.CONNECTIVE.02 · Set words: and, or, not

The complement of A contains everything in the box that is not in A.

Answer: True

Reasoning: Complement = "not".

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S30-B2-Q4 · question page 704

LOGIC.CONNECTIVE.02 · Set words: and, or, not

A = {1, 2, 3, 4}, B = {3, 4, 5}. How many numbers are in the union of A and B?

Answer: 5

Reasoning: {1, 2, 3, 4, 5}.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S30-B2-Q5 · question page 705

LOGIC.CONNECTIVE.02 · Set words: and, or, not

A = {1, 2, 3, 4}, B = {3, 4, 5}. How many numbers are in the intersection?

Answer: 2

Reasoning: {3, 4}.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S30-F1-Q1 · question page 706

LOGIC.SETS.01 · Three-circle diagrams and counts

Three circles A, B, C. Mark every region for things in A or B but NOT in C.

Answer: A, B, AB

Reasoning: A only, B only and the A-B overlap outside C.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 8 regions tested against the condition.

S30-F1-Q2 · question page 707

LOGIC.SETS.01 · Three-circle diagrams and counts

Three circles A, B, C. Mark every region for things in at least two of the sets.

Answer: AB, AC, BC, ABC

Reasoning: The three two-set overlaps and the center.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 8 regions tested against the condition.

S30-F1-Q3 · question page 708

LOGIC.SETS.01 · Three-circle diagrams and counts

Of 40 students, 22 take French, 17 take Spanish and 9 take both. How many take exactly one of the two?

Answer: 21

Reasoning: French only 22 - 9 = 13; Spanish only 17 - 9 = 8; 13 + 8 = 21.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S30-F1-Q4 · question page 709

LOGIC.SETS.01 · Three-circle diagrams and counts

Every member of A is in B, and no member of B is in C. On a three-circle diagram (A, B, C), mark every region that must be EMPTY.

Answer: A, AC, BC, ABC

Reasoning: "Every A is B" empties A outside B (A only and A-C only). "No B is C" empties every region in both B and C (B-C only and the center).

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 8 regions tested against the condition.

S30-F1-Q5 · question page 710

LOGIC.SETS.01 · Three-circle diagrams and counts

If A is a subset of B, then the intersection of A and B is A.

Answer: True

Reasoning: Everything in A is also in B.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S30-F2-Q1 · question page 711

LOGIC.SETS.01 · Inclusion and exclusion

In a group of 50 people, 25 like tea, 30 like coffee and 15 like both. How many like neither?

Answer: 10

Reasoning: Tea or coffee: 25 + 30 - 15 = 40. Neither: 10.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S30-F2-Q2 · question page 712

LOGIC.SETS.01 · Inclusion and exclusion

Sets A, B, C have 10, 12 and 8 members. A and B share 4, A and C share 3, B and C share 2, and all three share 1. How many members are in the union?

Answer: 22

Reasoning: 10 + 12 + 8 - 4 - 3 - 2 + 1 = 22.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S30-F2-Q3 · question page 713

LOGIC.SETS.01 · Inclusion and exclusion

In a club of 20, s members swim and r members run; every member does at least one. Give possible numbers s and r if exactly 5 do both.

Answer: any values that satisfy: s + r - 5 == 20 and s >= 5 and r >= 5 and s <= 20 and r <= 20 and isInteger(s) and isInteger(r) (example: s = 12, r = 13)

Reasoning: Everyone does at least one, so s + r - 5 = 20, s + r = 25, with each at least 5.

Tempting answers the rule rejects: s = 10, r = 10; s = 20, r = 10

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it. Search of the stated domain: 16 valid answers.

S30-F2-Q4 · question page 714

LOGIC.SETS.01 · Inclusion and exclusion

Two sets can have 8 and 6 members and a union of 15 members.

Answer: False

Reasoning: The union has at most 8 + 6 = 14 members.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S30-F2-Q5 · question page 715

LOGIC.SETS.01 · Inclusion and exclusion

Sets A and B are inside a box of 30 things. A has 12 things, B has 9, and A and B share 4. How many things are in the box but in neither A nor B?

Answer: 13

Reasoning: 30 - (12 + 9 - 4) = 13.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S31-M-Q1 · question page 717

LOGIC.GRAPH.02 · Turn a map into a neighbor graph and reason about neighbors and colorings

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many neighbors does region F have?

Answer: 6

Reasoning: F shares borders with B, D, E, G, H and I: 6 neighbors.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S31-M-Q2 · question page 718

LOGIC.GRAPH.02 · Turn a map into a neighbor graph and reason about neighbors and colorings

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Which region has the most neighbors?

Answer: B. F

Reasoning: Counting each region's neighbors: F has 6, more than any other.

- ✗ B: B has 5 neighbors
- ✗ D: D has 4 neighbors
- ✗ C: C has 3 neighbors

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the

check function accepts only the keyed choice.

S31-M-Q3 · question page 719

LOGIC.GRAPH.02 · Turn a map into a neighbor graph and reason about neighbors and colorings

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Regions A and I are neighbors.

Answer: False

Reasoning: A borders only B, D and G.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S31-M-Q4 · question page 720

LOGIC.GRAPH.02 · Turn a map into a neighbor graph and reason about neighbors and colorings

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Each region gets one color and neighbors must get different colors. What is the least number of colors needed for this map?

Answer: 3

Reasoning: R, S and T all border each other, so at least 3 colors are needed. Three are enough: P red, Q blue, R green, S blue, T red, U blue.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S31-M-Q5 · question page 721

LOGIC.GRAPH.02 · Turn a map into a neighbor graph and reason about neighbors and colorings

Regions are neighbors when they share a border of some length; touching only at a corner

point does not count. Color each region red, blue or green so that neighbors differ. P is red and Q is blue. Give every region's color.

Answer: P: red; Q: blue; R: green; S: blue; T: red; U: blue

Also accepted: P: red; Q: blue; R: green; S: blue; T: red; U: green

Reasoning: R borders P and Q, so R is green. S borders P and R, so S is blue. T borders Q, R and S, so T is red. U borders only T, so U is blue or green; both are accepted.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 729 assignments enumerated; 2 satisfy every clue.

S31-B1-Q1 · question page 722

LOGIC.GRAPH.02 · Strategy: draw the neighbor graph

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many neighbors does region C have?

Answer: 3

Reasoning: C borders B, E and H.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S31-B1-Q2 · question page 723

LOGIC.GRAPH.02 · Strategy: draw the neighbor graph

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Regions B and D are neighbors.

Answer: True

Reasoning: They share a border.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S31-B1-Q3 · question page 724

LOGIC.GRAPH.02 · Strategy: draw the neighbor graph

Two regions touch only at a single corner point. Are they neighbors under this rule?

Answer: C. No

Reasoning: A border must have length.

- ✗ Yes: a corner point is not a shared border
- ✗ Only if they are the same color: irrelevant
- ✗ It depends on their size: irrelevant

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S31-B1-Q4 · question page 725

LOGIC.GRAPH.02 · Strategy: draw the neighbor graph

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many neighbors does region U have?

Answer: 1

Reasoning: U borders only T.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S31-B1-Q5 · question page 726

LOGIC.GRAPH.02 · Strategy: draw the neighbor graph

If three regions all border each other, a coloring with different colors for neighbors needs at least 3 colors.

Answer: True

Reasoning: Each of the three must differ from the other two.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S31-F1-Q1 · question page 727

LOGIC.GRAPH.02 · Neighbor counts as clues

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. What is the total of the neighbor counts of all nine regions?

Answer: 36

Reasoning: Each shared border is counted twice (once from each side), so the total is twice the number of shared borders.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S31-F1-Q2 · question page 728

LOGIC.GRAPH.02 · Neighbor counts as clues

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many pairs of regions share a border?

Answer: 18

Reasoning: Half of the total of the neighbor counts.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S31-F1-Q3 · question page 729

LOGIC.GRAPH.02 · Neighbor counts as clues

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. What is the least number of colors needed so that neighbors differ?

Answer: 3

Reasoning: B, D and F border each other, so at least 3 colors; a 3-coloring exists (checked over all colorings).

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S31-F1-Q4 · question page 730

LOGIC.GRAPH.02 · Neighbor counts as clues

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. In any coloring of this map with 3 colors where neighbors differ, regions A and F have the same color.

Answer: True

Reasoning: A borders B, D and G, and so does F. D borders both B and G. If B and G had different colors, B, D and G would use all three colors and A could not be colored. So B and G match, B, D, G use two colors, and A and F must both take the third color.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S31-F1-Q5 · question page 731

LOGIC.GRAPH.02 · Neighbor counts as clues

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Which ONE region could be removed so that only 2 colors are needed

for the rest?

Answer: D. R

Reasoning: Every group of three regions that all border each other contains R (P-Q-R, P-R-S, Q-R-T, R-S-T). Without R, the rest can alternate: P and T one color, Q, S and U the other.

- ✗ T: P, Q and R still border each other
- ✗ U: R, S and T still border each other
- ✗ S: P, Q and R still border each other

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S31-F2-Q1 · question page 732

LOGIC.GRAPH.02 · Colorings: count and prove

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many ways can the map be colored with red, blue and green so that neighbors differ?

Answer: 12

Reasoning: Fix colors for P and Q (6 ways), then the rest follow as in the MAIN puzzle with 2 choices for U: 6 x 2 = 12.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S31-F2-Q2 · question page 733

LOGIC.GRAPH.02 · Colorings: count and prove

Some map drawn on a page needs at least 4 colors so that neighbors differ.

Answer: True

Reasoning: For example, one region surrounded by three regions that all border each other: the four all border each other, so they need 4 different colors.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S31-F2-Q3 · question page 734

LOGIC.GRAPH.02 · Colorings: count and prove

A map has regions in a ring: region 1 borders 2, 2 borders 3, and so on, and the last region borders region 1 (and there are no other borders). Give a number of regions n, from 3 to 10, for which 2 colors are NOT enough.

Answer: any values that satisfy: isOdd(n) and n >= 3 and n <= 10 (example: n = 5)

Reasoning: Going around the ring the colors must alternate, which closes up only when n is even. Any odd n: 3, 5, 7 or 9.

Tempting answers the rule rejects: n = 4; n = 6; n = 2

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 4 valid answers.

S31-F2-Q4 · question page 735

LOGIC.GRAPH.02 · Colorings: count and prove

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many ways can this map be colored with red, blue and green so that neighbors differ, if A is red?

Answer: 2

Reasoning: Counted by checking all colorings with A red.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S31-F2-Q5 · question page 736

LOGIC.GRAPH.02 · Colorings: count and prove

If a map can be colored with 4 colors so that neighbors differ, it can also be colored with 3 colors.

Answer: False

Reasoning: Four regions that all border each other need 4 different colors, so 3 colors are not always enough.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S32-M-Q1 · question page 738

LOGIC.SETS.02 · Test categorical arguments with Venn diagrams, including the existence assumption

Premise: "No reptile is warm-blooded." Circle A is reptiles and circle B is warm-blooded animals. Mark the region this premise says is EMPTY.

Answer: AB

Reasoning: "No A is B" means the overlap is empty.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S32-M-Q2 · question page 739

LOGIC.SETS.02 · Test categorical arguments with Venn diagrams, including the existence assumption

"All violinists are musicians. All musicians practice. So all violinists practice." Is this valid?

Answer: C. Yes: every violinist is inside the musicians, who are all inside those who practice.

Reasoning: Nested circles: violinists inside musicians inside practicers.

- ✗ No: some violinists may not practice.: the premises rule that out
- ✗ Yes, but only if every musician is a violinist.: not needed
- ✗ No: the conclusion reverses a premise.: it does not

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S32-M-Q3 · question page 740

LOGIC.SETS.02 · Test categorical arguments with Venn diagrams, including the existence assumption

"All squares are rectangles. Some rectangles are not rhombuses. So some squares are not rhombuses." Is this valid?

Answer: A. No: the rectangles that are not rhombuses may all lie outside the squares.

Reasoning: In fact every square is a rhombus, so the premises are true and the conclusion false: invalid.

- ✗ Yes: squares are rectangles, so they share every property.: subsets do not share every property
- ✗ Yes: the conclusion is true.: validity is about form; and in fact every square IS a rhombus
- ✗ No: no rectangle is a square.: that is false: every square is a rectangle

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S32-M-Q4 · question page 741

LOGIC.SETS.02 · Test categorical arguments with Venn diagrams, including the existence assumption

"All A are B. Some C are A. So some C are B." This form is valid.

Answer: True

Reasoning: The C that is an A is inside B, so some C is B.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S32-M-Q5 · question page 742

LOGIC.SETS.02 · Test categorical arguments with Venn diagrams, including the existence assumption

"All dragons breathe fire. So some dragons breathe fire." When does this follow?

Answer: D. Only if at least one dragon exists (the existence assumption).

Reasoning: From "all" to "some" needs at least one member.

- ✗ Always: "all" includes "some".: if there are no dragons, "all dragons breathe fire" is true but "some dragon breathes fire" is false
- ✗ Never: the premise is about all dragons.: it follows once a dragon exists
- ✗ Only if dragons are real animals.: it is about existence, not about being an animal

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S32-B1-Q1 · question page 743

LOGIC.SETS.01 · Mark statements on a Venn diagram

"All A are B." Mark the region that must be EMPTY.

Answer: A

Reasoning: Nothing in A outside B.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S32-B1-Q2 · question page 744

LOGIC.SETS.01 · Mark statements on a Venn diagram

"No A is B." Mark the region that must be EMPTY.

Answer: AB

Reasoning: The overlap.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S32-B1-Q3 · question page 745

LOGIC.SETS.01 · Mark statements on a Venn diagram

"Some A is B." Mark the region where something must be.

Answer: AB

Reasoning: At least one thing in the overlap.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S32-B1-Q4 · question page 746

LOGIC.SETS.01 · Mark statements on a Venn diagram

"Some A is not B." Mark the region where something must be.

Answer: A

Reasoning: At least one thing in A outside B.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S32-B1-Q5 · question page 747

LOGIC.SETS.01 · Mark statements on a Venn diagram

"All A are B" and "No A is B" can both be true if A has no members.

Answer: True

Reasoning: With A empty, both statements are true (vacuously).

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S32-B2-Q1 · question page 748

LOGIC.INFERENCE.03 · Valid means no counterexample

An argument is valid if a situation exists with true premises and a false conclusion.

Answer: False

Reasoning: Such a situation shows it is invalid.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S32-B2-Q2 · question page 749

LOGIC.INFERENCE.03 · Valid means no counterexample

"All cats are mammals. All dogs are mammals. So all cats are dogs." Which situation shows it is invalid?

Answer: B. the real world: cats and dogs are both mammals but cats are not dogs

Reasoning: True premises, false conclusion.

- ✗ a world with no cats: then the conclusion is true
- ✗ a world where all mammals are dogs: then cats would be dogs
- ✗ a world with no mammals: then there are no cats

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S32-B2-Q3 · question page 750

LOGIC.INFERENCE.03 · Valid means no counterexample

"All A are B. All B are C. So all A are C." is valid.

Answer: True

Reasoning: Nested sets.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S32-B2-Q4 · question page 751

LOGIC.INFERENCE.03 · Valid means no counterexample

"Some A are B. Some B are C. So some A are C." is valid.

Answer: False

Reasoning: The A-and-B things and the B-and-C things may be different.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S32-B2-Q5 · question page 752

LOGIC.INFERENCE.03 · Valid means no counterexample

Which premise pair makes "So no A is C" follow?

Answer: C. "All A are B." and "No B is C."

Reasoning: A inside B, and B outside C, so A is outside C.

- ✗ "Some A are B." and "No B is C.": only some A are known to be outside C
- ✗ "All A are B." and "Some B are not C.": A may still overlap C
- ✗ "No A is B." and "No B is C.": A and C may overlap outside B

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S32-F1-Q1 · question page 753

LOGIC.SETS.02 · Harder categorical forms

"No A is B. All C are B. So no C is A." This form is valid.

Answer: True

Reasoning: C lies inside B and B is outside A.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S32-F1-Q2 · question page 754

LOGIC.SETS.02 · Harder categorical forms

"All A are B. No C is A. So no C is B." This form is valid.

Answer: False

Reasoning: C can overlap the part of B outside A.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S32-F1-Q3 · question page 755

LOGIC.SETS.02 · Harder categorical forms

"Some athletes are vegetarians. All vegetarians eat beans. So ..." Which conclusion follows?

Answer: D. some athletes eat beans

Reasoning: The athletes who are vegetarians eat beans.

- ✗ all athletes eat beans: only some are known to be vegetarians
- ✗ all bean eaters are athletes: reverses the relations
- ✗ no athlete eats meat: not supported

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S32-F1-Q4 · question page 756

LOGIC.SETS.02 · Harder categorical forms

Premises: "All A are B" and "No B is C". On a three-circle diagram (A, B, C), mark every region that the premises make EMPTY.

Answer: A, AC, BC, ABC

Reasoning: "All A are B" empties A outside B; "No B is C" empties B inside C.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 8 regions tested against the condition.

S32-F1-Q5 · question page 757

LOGIC.SETS.02 · Harder categorical forms

"All A are B. Some A are C. So some B are C." This form is valid.

Answer: True

Reasoning: The A that is C is also B.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S32-F2-Q1 · question page 758

LOGIC.SETS.02 · Existence and hidden assumptions

"All A are B. All A are C. So some B are C." This form is valid without any existence assumption.

Answer: False

Reasoning: If A is empty, both premises are true and B and C may not overlap.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S32-F2-Q2 · question page 759

LOGIC.SETS.02 · Existence and hidden assumptions

Which added premise makes "All A are B. All A are C. So some B are C." valid?

Answer: A. At least one A exists.

Reasoning: An A would be both B and C.

- ✗ At least one B exists.: B may exist outside C
- ✗ At least one C exists.: C may exist outside B
- ✗ All B are A.: still fails if A is empty

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S32-F2-Q3 · question page 760

LOGIC.SETS.02 · Existence and hidden assumptions

How many of these forms are valid (with no existence assumption)? (1) All A are B; all B are C; so all A are C. (2) No A is B; all C are A; so no C is B. (3) All A are B; some C are not B; so some C are not A. (4) Some A are B; all B are C; so some A are C.

Answer: 4

Reasoning: All four are valid: (1) nesting; (2) C inside A, A outside B; (3) a C outside B cannot be in A; (4) the A that is B is also C.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S32-F2-Q4 · question page 761

LOGIC.SETS.02 · Existence and hidden assumptions

"All unicorns have one horn" is true if no unicorns exist.

Answer: True

Reasoning: An "all" statement about an empty set is vacuously true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S32-F2-Q5 · question page 762

LOGIC.SETS.02 · Existence and hidden assumptions

"No square is a circle. Some circles are shaded. So some shaded things are not squares." Is it valid?

Answer: B. Yes: a shaded circle is not a square.

Reasoning: The shaded circle is a witness.

- ✗ No: shaded things may all be squares.: the shaded circles are not squares
- ✗ No: it needs squares to exist.: it does not
- ✗ Yes, but only if all circles are shaded.: not needed

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S33-M-Q1 · question page 764

LOGIC.WORD.02 · Find shortest one-letter-change paths through a supplied word list

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Put these four words in ladder order, starting with SAND, so that each word differs from the ne

Answer: SAND → SEND → SENT → RENT

Reasoning: SAND -> SEND (A to E) -> SENT (D to T) -> RENT (S to R).

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S33-M-Q2 · question page 765

LOGIC.WORD.02 · Find shortest one-letter-change paths through a supplied word list

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. What is the least number of steps in a ladder from BAND to BOAT?

Answer: 4

Reasoning: BAND -> BEND -> BENT -> BEAT -> BOAT (or through BEAD). Three steps are impossible: BAND has no neighbor on the list that is two steps from BOAT.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S33-M-Q3 · question page 766

LOGIC.WORD.02 · Find shortest one-letter-change paths through a supplied word list

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Which word fills the gap in the ladder SENT, ?, RANT?

Answer: B. RENT

Reasoning: SENT -> RENT -> RANT.

- ✗ BENT: BENT to RANT changes two letters
- ✗ SEND: SEND to RANT changes three letters
- ✗ PANT: SENT to PANT changes two letters

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S33-M-Q4 · question page 767

LOGIC.WORD.02 · Find shortest one-letter-change paths through a supplied word list

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Give a word from the list that is one step from both BEND and BEAT.

Answer: any values that satisfy: w == 'bent' or w == 'bead' (example: w = BENT)

Reasoning: BENT (BEND -> BENT, BEAT -> BENT) and BEAD (BEND -> BEAD, BEAT -> BEAD) both work.

Tempting answers the rule rejects: w = BELT; w = BOAT; w = BEST

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S33-M-Q5 · question page 768

LOGIC.WORD.02 · Find shortest one-letter-change paths through a supplied word list

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. A ladder from LAND to BELT can be done in 3 steps.

Answer: False

Reasoning: LAND and BELT differ in all 4 positions (L/B, A/E, N/L, D/T), and each step changes only one letter, so at least 4 steps are needed: LAND -> BAND -> BEND -> BENT -> BELT.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S33-B1-Q1 · question page 769

LOGIC.WORD.02 · Strategy: change the letters that must change, one at a time

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Which word is one step from BEST?

Answer: A. REST

Reasoning: BEST -> REST changes only B to R.

- ✗ RENT: two letters differ from BEST
- ✗ POST: two letters differ
- ✗ BOAT: two letters differ

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S33-B1-Q2 · question page 770

LOGIC.WORD.02 · Strategy: change the letters that must change, one at a time

In how many positions do the words MOST and PART differ?

Answer: 3

Reasoning: M/P, O/A, S/R differ; T matches.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S33-B1-Q3 · question page 771

LOGIC.WORD.02 · Strategy: change the letters that must change, one at a time

A ladder from MAST to LAST can be done in one step.

Answer: True

Reasoning: Only M changes to L.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S33-B1-Q4 · question page 772

LOGIC.WORD.02 · Strategy: change the letters that must change, one at a time

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. What is the least number of steps from PORT to MOST?

Answer: 2

Reasoning: PORT -> POST -> MOST.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S33-B1-Q5 · question page 773

LOGIC.WORD.02 · Strategy: change the letters that must change, one at a time

If two words differ in 3 positions, every ladder between them has at least 3 steps.

Answer: True

Reasoning: Each step fixes at most one position.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S33-F1-Q1 · question page 774

LOGIC.WORD.02 · Longer ladders and detours

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. What is the least number of steps from BEND to PORT?

Answer: 6

Reasoning: BEND -> BENT -> RENT -> RANT -> PANT -> PART -> PORT. BEND and PORT differ in 4 positions, but the list forces a longer route.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S33-F1-Q2 · question page 775

LOGIC.WORD.02 · Longer ladders and detours

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. How many different shortest ladders go from BAND to BOAT?

Answer: 2

Reasoning: BAND-BEND-BENT-BEAT-BOAT and BAND-BEND-BEAD-BEAT-BOAT.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S33-F1-Q3 · question page 776

LOGIC.WORD.02 · Longer ladders and detours

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Which word needs the MOST steps to reach from SAND (using shortest ladders)?

Answer: D. LAST

Reasoning: LAST is 11 steps from SAND: the chain RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST is the only way out to it, and LAST is at its far end.

✗ MOST: MOST takes 9 steps; LAST is two steps farther, through MAST

✗ BOLT: BOLT takes only 5 steps: SAND, BAND, BEND, BENT, BELT, BOLT

✗ PORT: PORT takes 7 steps; the chain continues past it to POST, MOST, MAST, LAST

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S33-F1-Q4 · question page 777

LOGIC.WORD.02 · Longer ladders and detours

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Give a word from the list that is exactly one step from LAND and also one step from SEND.

Answer: any values that satisfy: w == 'sand' or w == 'lend' (example: w = SAND)

Reasoning: SAND (LAND -> SAND, SEND -> SAND) and LEND (LAND -> LEND, SEND -> LEND) both work.

Tempting answers the rule rejects: w = BAND; w = BEND

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it.

S33-F1-Q5 · question page 778

LOGIC.WORD.02 · Longer ladders and detours

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. BOAT is one step from BOLT.

Answer: True

Reasoning: B, O and T stay in place; only the third letter changes (A to L).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S33-F2-Q1 · question page 779

LOGIC.WORD.02 · Prove a ladder is shortest or impossible

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. What is the least number of steps from SAND to MOST?

Answer: 9

Reasoning: Breadth-first search over the list gives 9 steps: SAND, SEND, SENT, RENT, RANT, PANT, PART, PORT, POST, MOST.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S33-F2-Q2 · question page 780

LOGIC.WORD.02 · Prove a ladder is shortest or impossible

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST,

MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Suppose PANT is removed from the list. What is the least number of steps from RANT to PART now?

Answer: 0

Reasoning: Without PANT, PART's only neighbor is PORT. The words PART, PORT, POST, MOST, MAST, LAST then link only to each other, and none of them is one step from RANT or any word connected to RANT, so no ladder exists.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S33-F2-Q3 · question page 781

LOGIC.WORD.02 · Prove a ladder is shortest or impossible

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Every ladder from BAND to BOAT passes through BEND.

Answer: False

Reasoning: BAND -> SAND -> SEND -> SENT -> BENT -> BEAT -> BOAT avoids BEND. It is longer (6 steps) than the shortest ladders, but the statement is about every ladder.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S33-F2-Q4 · question page 782

LOGIC.WORD.02 · Prove a ladder is shortest or impossible

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST,

MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. How many words on the list are exactly one step from BENT?

Answer: 6

Reasoning: BEND, SENT, BEAT, BELT, BEST, RENT.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S33-F2-Q5 · question page 783

LOGIC.WORD.02 · Prove a ladder is shortest or impossible

A ladder must go from a 4-letter word to another 4-letter word that differs in all 4 positions. What is the least possible number of steps?

Answer: C. 4

Reasoning: Each of the 4 positions must change at least once.

- ✗ 3: each step changes only one position
- ✗ 2: each step changes only one position
- ✗ 8: that is not the least possible

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S34-M-Q1 · question page 785

LOGIC.CONSTRAINT.02 · Translate puzzle clues into exact conditions and test candidate arrangements against them

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Which condition says exactly the same thing as the clue "Exactly one locker lies between Cal and Dee"?

Answer: D. |C - D| = 2

Reasoning: One locker between them means their numbers differ by 2, in either order, so |C - D| = 2.

✗ |C - D| = 1: that says Cal and Dee are next to each other, with no locker between
✗ C - D = 2: that only allows Cal to the right of Dee; the clue allows either order
✗ |C - D| >= 2: that also allows two or more lockers between them

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S34-M-Q2 · question page 786

LOGIC.CONSTRAINT.02 · Translate puzzle clues into exact conditions and test candidate arrangements against them

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Clues: (1) Ava is somewhere left of Ben. (2) Exactly one locker lies between Cal and Dee. (3) Eli is not at either end. (4) Ben and Cal are next to each other. (5) Dee is

Answer: Ava → Dee → Eli → Cal → Ben

Reasoning: Dee is in 2 or 4 (clue 5), and Cal is two away from Dee (clue 2). Testing each case with clues 1, 3 and 4 leaves only Ava 1, Dee 2, Eli 3, Cal 4, Ben 5.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 120 orderings of 5 items enumerated; 1 valid.

S34-M-Q3 · question page 787

LOGIC.CONSTRAINT.02 · Translate puzzle clues into exact conditions and test candidate arrangements against them

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Clues: (1) Ava is somewhere left of Ben. (2) Exactly one locker lies between Cal and Dee. (3) Eli is not at either end. (4) Ben and Cal are next to each other. (5) Dee is

Answer: True

Reasoning: Ava 1, Cal 2, Eli 3, Dee 4, Ben 5: clue 1 holds (1 < 5), clue 2 holds (|2 - 4| = 2), clue 3 holds, clue 4 fails (|5 - 2| = 3), clue 5 holds. Exactly one clue is broken.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S34-M-Q4 · question page 788

LOGIC.CONSTRAINT.02 · Translate puzzle clues into exact conditions and test candidate arrangements against them

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Clues: (1) Ava is somewhere left of Ben. (2) Exactly one locker lies between Cal and Dee. (3) Eli is not at either end. (4) Ben and Cal are next to each other. (5) Dee is

Answer: A. Clue 2

Reasoning: Cal is in 5 and Dee in 2: |5 - 2| = 3, so two lockers lie between them, not one.

- ✗ Clue 1: Ava 1 is left of Ben 4, so clue 1 holds
- ✗ Clue 4: Ben 4 and Cal 5 are next to each other, so clue 4 holds
- ✗ Clue 5: Dee is in locker 2, which is even, so clue 5 holds

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S34-M-Q5 · question page 789

LOGIC.CONSTRAINT.02 · Translate puzzle clues into exact conditions and test candidate arrangements against them

A row has nine lockers numbered 1 to 9. Eli's locker must meet every condition: it is odd-numbered; it is not at either end (not 1 and not 9); and it is at least 2 lockers away from locker 5 (|E - 5| >= 2). Give one locker number Eli could have.

Answer: any values that satisfy: isInteger(e) and isOdd(e) and e > 1 and e < 9 and abs(e - 5) >= 2 (example: e = 3)

Reasoning: Odd and not an end: 3, 5 or 7. At least 2 from 5 removes 5. Eli could be in 3 or 7.

Tempting answers the rule rejects: e = 5; e = 9; e = 4; e = 1

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 4 listed mistakes fail it. Search of the stated domain: 2 valid answers.

S34-B1-Q1 · question page 790

LOGIC.CONSTRAINT.02 · Strategy: write each clue as a condition on numbers

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Which condition matches the clue "Ava is somewhere left of Ben"?

Answer: C. A < B

Reasoning: "Somewhere left of" only says Ava's number is smaller.

- ✗ A = B - 1: that says Ava is directly left of Ben; the clue allows a gap
- ✗ A > B: that puts Ava to the right of Ben
- ✗ A + 1 < B: that requires at least one locker between them

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S34-B1-Q2 · question page 791

LOGIC.CONSTRAINT.02 · Strategy: write each clue as a condition on numbers

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Which condition matches the clue "Ben and Cal are next to each other"?

Answer: B. |B - C| = 1

Reasoning: Next to each other means the numbers differ by 1, in either order.

- ✗ B - C = 1: that only allows Ben directly right of Cal
- ✗ |B - C| = 2: that puts one locker between them
- ✗ |B - C| <= 2: that also allows one locker between them

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S34-B1-Q3 · question page 792

LOGIC.CONSTRAINT.02 · Strategy: write each clue as a condition on numbers

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. With five lockers, the clue "Eli is not at either end" means E is 2, 3 or 4.

Answer: True

Reasoning: The ends are lockers 1 and 5, so Eli is in 2, 3 or 4.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S34-B1-Q4 · question page 793

LOGIC.CONSTRAINT.02 · Strategy: write each clue as a condition on numbers

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. The clue "Dee is directly left of Ava" means the same as D < A.

Answer: False

Reasoning: "Directly left" means the next locker: D = A - 1. D < A also allows a gap.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S34-B1-Q5 · question page 794

LOGIC.CONSTRAINT.02 · Strategy: write each clue as a condition on numbers

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Dee is in locker 2, and exactly one locker lies between Cal and Dee. Which locker is Cal in?

Answer: 4

Reasoning: |C - 2| = 2 gives C = 4 or C = 0, and there is no locker 0.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S34-F1-Q1 · question page 795

LOGIC.CONSTRAINT.02 · Harder clue words

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and

E stand for the friends' locker numbers. Which condition matches the clue "At least two lockers lie between Ava and Eli"?

Answer: A. |A - E| >= 3

Reasoning: Two lockers between means the numbers differ by 3; "at least" allows more, in either order.

- ✗ |A - E| >= 2: that allows only one locker between them
- ✗ |A - E| = 3: that allows exactly two between them, not more
- ✗ A - E >= 3: that only allows Ava to the right of Eli

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S34-F1-Q2 · question page 796

LOGIC.CONSTRAINT.02 · Harder clue words

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. In these puzzles, "X unless Y" means "if Y is false, then X is true." Which condition matches the clue "Ben is in locker 3 unless Cal is"?

Answer: B. B = 3 or C = 3

Reasoning: If Cal is not in 3, Ben is. So one of them is in locker 3: B = 3 or C = 3.

- ✗ B = 3 and C is not 3: that rules out Cal in locker 3, but the clue allows it
- ✗ B is not 3 and C is not 3: that is the opposite: the clue needs one of them in locker 3
- ✗ C = 3: that rules out Ben in locker 3, but the clue allows it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S34-F1-Q3 · question page 797

LOGIC.CONSTRAINT.02 · Harder clue words

Six friends, Ava, Ben, Cal, Dee, Eli and Fay, each use one of six lockers numbered 1 to 6 from left to right. "Left of" means a lower number. Clues: (1) Ava is somewhere left of Ben. (2) Exactly two lockers lie between Cal and Dee. (3) Fay is somewhere right of Cal. (4) Dee is in an odd-numbered locker. (5) Cal is somewhere left of Ava. (6) Ben is not in locker 6. Put the friends in locker order,

Answer: Eli → Cal → Ava → Ben → Dee → Fay

Reasoning: Dee is odd and three away from Cal, so (Cal, Dee) is (2, 5), (4, 1) or (6, 3). Cal is left of Ava and Fay, so Cal is not 6. If Cal is 4, Ava and Fay need lockers 5 and 6, and Ben must be right of Ava, so Ava 5 and Ben 6, which clue 6 forbids. So Cal 2 and Dee 5. Ava, Ben and Fay take 3, 4 and 6 with Ava left of Ben and Ben not in 6: Ava 3, Ben 4, Fay 6, and Eli 1.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 720 orderings of 6 items enumerated; 1 valid.

S34-F1-Q4 · question page 798

LOGIC.CONSTRAINT.02 · Harder clue words

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. The two conditions $A < B$ and $B < A + 2$ together say that Ben is directly right of Ava.

Answer: True

Reasoning: Locker numbers are whole numbers, so $A < B < A + 2$ forces $B = A + 1$.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S34-F1-Q5 · question page 799

LOGIC.CONSTRAINT.02 · Harder clue words

Seven lockers are numbered 1 to 7 from left to right. Give lockers for Ava and Ben that meet every clue: Ava is somewhere left of Ben; at least two lockers lie between them; Ben's locker is even-numbered.

Answer: any values that satisfy: isInteger(a) and isInteger(b) and a >= 1 and b <= 7 and a < b and b - a >= 3 and isEven(b) (example: a = 1, b = 4)

Reasoning: For example A = 1, B = 4. Other answers: (1, 6), (2, 6), (3, 6).

Tempting answers the rule rejects: a = 2, b = 4; a = 6, b = 2; a = 1, b = 5

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 4 valid answers.

S34-F2-Q1 · question page 800

LOGIC.CONSTRAINT.02 · Test what each clue contributes

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Clues: (1) Ava is somewhere left of Ben. (2) Exactly one locker lies between Cal and Dee. (3) Eli is not at either end. (4) Ben and Cal are next to each other. (5) Dee is

Answer: 4

Reasoning: Enumerating with clues 1, 2, 4 and 5 gives 4 arrangements, so clue 3 is needed for a single answer.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S34-F2-Q2 · question page 801

LOGIC.CONSTRAINT.02 · Test what each clue contributes

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. How many arrangements fit both clues "Eli is not at either end" and "Ben and Cal are next to each other" (and no other clues)?

Answer: 24

Reasoning: Treat Ben-and-Cal as a block. Eli is in 2, 3 or 4; for each, count the places for the block in either order and the arrangements of Ava and Dee in the last two lockers: 24 in all.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S34-F2-Q3 · question page 802

LOGIC.CONSTRAINT.02 · Test what each clue contributes

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Clues: (1) Ava is somewhere left of Ben. (2) Exactly one locker lies between Cal and Dee. (3) Eli is not at either end. (4) Ben and Cal are next to each other. (5) Dee is

Answer: True

Reasoning: The answer to the original puzzle has Dee in locker 2, and enumerating clues 1-4 with D = 2 leaves only Ava, Dee, Eli, Cal, Ben.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S34-F2-Q4 · question page 803

LOGIC.CONSTRAINT.02 · Test what each clue contributes

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Which condition matches the clue "Ava and Ben are not both at the ends"?

Answer: D. At most one of Ava and Ben is in locker 1 or 5

Reasoning: "Not both" only rules out the case where both are at the ends; zero or one at an end is allowed.

- ✗ Neither Ava nor Ben is in locker 1 or 5: that forbids even one of them at an end
- ✗ Exactly one of Ava and Ben is in locker 1 or 5: that forbids both being away from the ends
- ✗ Ava and Ben are both in lockers 2, 3 or 4: that is the same as "neither is at an end"

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S34-F2-Q5 · question page 804

LOGIC.CONSTRAINT.02 · Test what each clue contributes

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. In these puzzles, "X unless Y" means "if Y is false, then X is true." Under the clue "Ben is in locker 3 unless Cal is", an arrangement with Ava in locker 3 is allowed.

Answer: False

Reasoning: If Ava is in 3, Cal is not in 3, so the clue says Ben must be in 3. Ben cannot share the locker, so the arrangement breaks the clue.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S35-M-Q1 · question page 806

LOGIC.CONSTRAINT.03 · Decide what must be true, what could be true and what cannot be true in a constrained arrangement

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Which statement MUST be true?

Answer: C. Gears is on Monday or Tuesday.

Reasoning: Gears must be before Jazz, and Jazz is not on Friday, so Gears is not on Thursday or Friday. If Gears were on Wednesday, Jazz would be Thursday and Honey Friday, leaving Monday and Tuesday (consecutive) for Fossils and Kites. So Gears is on Monday or Tuesday.

- ✗ Honey is on Friday.: Fossils, Gears, Honey, Jazz, Kites (Monday to Friday) works with Honey on Wednesday
- ✗ Fossils is on Monday.: Gears, Kites, Honey, Jazz, Fossils works with Fossils on Friday
- ✗ Kites is on Friday.: Fossils, Gears, Jazz, Kites, Honey works with Kites on Thursday

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S35-M-Q2 · question page 807

LOGIC.CONSTRAINT.03 · Decide what must be true, what could be true and what cannot be true in a constrained arrangement

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Which statement CANNOT be true?

Answer: A. Jazz is on Tuesday.

Reasoning: If Jazz is on Tuesday, Gears is on Monday. Honey takes Wednesday or Friday, and Fossils and Kites take the other two of Wednesday, Thursday, Friday. If Honey is on Friday they are on Wednesday and Thursday (consecutive); if Honey is on Wednesday, they are on Thursday and Friday (consecutive). Either way condition 3 fails.

- ✗ Fossils is on Friday.: Gears, Kites, Honey, Jazz, Fossils meets every condition
- ✗ Kites is on Wednesday.: Fossils, Gears, Kites, Jazz, Honey meets every condition
- ✗ Honey is on Wednesday.: Fossils, Gears, Honey, Jazz, Kites meets every condition

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S35-M-Q3 · question page 808

LOGIC.CONSTRAINT.03 · Decide what must be true, what could be true and what cannot be true in a constrained arrangement

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Fossils could be on Friday.

Answer: True

Reasoning: Gears, Kites, Honey, Jazz, Fossils (Monday to Friday) meets all four conditions.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S35-M-Q4 · question page 809

LOGIC.CONSTRAINT.03 · Decide what must be true, what could be true and what cannot

be true in a constrained arrangement

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. How many complete schedules meet all four conditions?

Answer: 10

Reasoning: Case by Honey's day. Honey Wednesday: 4 schedules. Honey Friday: 6 schedules. Total 10.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S35-M-Q5 · question page 810

LOGIC.CONSTRAINT.03 · Decide what must be true, what could be true and what cannot be true in a constrained arrangement

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Suppose Honey is on Wednesday. Using Monday = 1, Tuesday = 2, Wednesday = 3, Thursday = 4 and Friday = 5, give a day number Ki

Answer: any values that satisfy: k == 1 or k == 2 or k == 5 (example: k = 5)

Reasoning: With Honey on Wednesday the schedules are F G H J K, G F H J K, G K H J F and K G H J F, so Kites could be on day 1, 2 or 5.

Tempting answers the rule rejects: k = 3; k = 4

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it.

S35-B1-Q1 · question page 811

LOGIC.CONSTRAINT.03 · Strategy: list every case, then read must, could, cannot

Three talks, P, Q and R, are on days 1, 2 and 3, one per day. Conditions: P is not on day 1; Q is earlier than R. How many schedules meet both conditions?

Answer: 2

Reasoning: P on day 2: Q, P, R. P on day 3: Q, R, P.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S35-B1-Q2 · question page 812

LOGIC.CONSTRAINT.03 · Strategy: list every case, then read must, could, cannot

Three talks, P, Q and R, are on days 1, 2 and 3, one per day. Conditions: P is not on day 1; Q is earlier than R. Then Q must be on day 1.

Answer: True

Reasoning: The two schedules are Q, P, R and Q, R, P. Q is on day 1 in both.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S35-B1-Q3 · question page 813

LOGIC.CONSTRAINT.03 · Strategy: list every case, then read must, could, cannot

Three talks, P, Q and R, are on days 1, 2 and 3, one per day. Conditions: P is not on day 1; Q is earlier than R. Then R must be on day 3.

Answer: False

Reasoning: In Q, R, P, R is on day 2. R could be on day 3 but need not be.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S35-B1-Q4 · question page 814

LOGIC.CONSTRAINT.03 · Strategy: list every case, then read must, could, cannot

In a scheduling puzzle, what does it mean to say a statement CANNOT be true?

Answer: B. It is false in every schedule that meets the conditions.

Reasoning: Cannot be true means no valid schedule makes it true.

- ✗ It is false in at least one schedule that meets the conditions.: that means the statement need not be true; it could still be true in another schedule
- ✗ It is true in exactly one schedule that meets the conditions.: then the statement could be true
- ✗ It is not mentioned in any of the conditions.: a statement can be forced false by conditions that never mention it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S35-B1-Q5 · question page 815

LOGIC.CONSTRAINT.03 · Strategy: list every case, then read must, could, cannot

Three talks, P, Q and R, are on days 1, 2 and 3, one per day. Conditions: P is not on day 1; Q is earlier than R. Which statement COULD be true?

Answer: C. P is on day 2.

Reasoning: Q, P, R meets both conditions and has P on day 2.

- ✗ P is on day 1.: the first condition forbids it
- ✗ R is on day 1.: Q must be earlier than R, so R is never first
- ✗ Q is on day 3.: Q must be earlier than R, so Q is never last

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S35-F1-Q1 · question page 816

LOGIC.CONSTRAINT.03 · Must, could, cannot under an added condition

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. If Kites is on Monday, which statement MUST be true?

Answer: D. Gears is on Tuesday.

Reasoning: Fossils cannot be on Tuesday (next to Kites), Jazz cannot be on Tuesday (Gears must come first), and Honey is on Wednesday or Friday, so Gears is on Tuesday.

- ✗ Honey is on Friday.: Kites, Gears, Honey, Jazz, Fossils puts Honey on Wednesday
- ✗ Fossils is on Wednesday.: Kites, Gears, Jazz, Fossils, Honey puts Fossils on Thursday
- ✗ Jazz is on Thursday.: Kites, Gears, Jazz, Fossils, Honey puts Jazz on Wednesday

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S35-F1-Q2 · question page 817

LOGIC.CONSTRAINT.03 · Must, could, cannot under an added condition

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. If Kites is on Friday, how many schedules are possible?

Answer: 2

Reasoning: Honey must be on Wednesday, and Fossils is not on Thursday (next to Kites). Fossils, Gears and Jazz fill Monday, Tuesday and Thursday with Gears before Jazz, so Jazz is on Thursday: F G H J K and G F H J K.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S35-F1-Q3 · question page 818

LOGIC.CONSTRAINT.03 · Must, could, cannot under an added condition

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. If Jazz is on Wednesday, then Honey must be on Friday.

Answer: True

Reasoning: Honey is on Wednesday or Friday, and Wednesday is taken by Jazz.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S35-F1-Q4 · question page 819

LOGIC.CONSTRAINT.03 · Must, could, cannot under an added condition

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Which ONE added condition would leave exactly one possible schedule?

Answer: A. Fossils is on Wednesday.

Reasoning: With Fossils on Wednesday, only Kites, Gears, Fossils, Jazz, Honey works.

✗ Fossils is on Friday.: two schedules remain: G K H J F and K G H J F

✗ Kites is on Thursday.: two schedules remain: F G J K H and G F J K H

✗ Honey is on Wednesday.: four schedules remain

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S35-F1-Q5 · question page 820

LOGIC.CONSTRAINT.03 · Must, could, cannot under an added condition

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Suppose Honey is on Friday. Using Monday = 1 to Friday = 5, give the day numbers of Gears and Jazz in one schedule that meets

Answer: any values that satisfy: (g == 2 and (j == 3 or j == 4)) or (g == 1 and j == 3) (example: g = 2, j = 3)

Reasoning: Listing the schedules with Honey on Friday gives (Gears, Jazz) = (1, 3), (2, 3) or (2, 4). The tempting (1, 4) fails: Fossils and Kites would be on Tuesday and Wednesday, which are consecutive.

Tempting answers the rule rejects: g = 1, j = 4; g = 3, j = 4; g = 4, j = 2

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S35-F2-Q1 · question page 821

LOGIC.CONSTRAINT.03 · Change the conditions and re-test

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. How many of the five talks could be on Friday?

Answer: 3

Reasoning: Fossils, Honey and Kites each appear on Friday in some schedule. Jazz cannot (condition 4) and Gears cannot (it must be before Jazz).

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S35-F2-Q2 · question page 822

LOGIC.CONSTRAINT.03 · Change the conditions and re-test

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Suppose condition 3 is replaced by "Fossils is on Monday." Which statement must be true?

Answer: C. Jazz is on Wednesday or Thursday.

Reasoning: The four schedules are F G H J K, F G J K H, F G K J H and F K G J H. Jazz is on Wednesday or Thursday in each.

- ✗ Gears is on Tuesday.: Fossils, Kites, Gears, Jazz, Honey has Gears on Wednesday
- ✗ Kites is on Friday.: Fossils, Gears, Jazz, Kites, Honey has Kites on Thursday
- ✗ Honey is on Friday.: Fossils, Gears, Honey, Jazz, Kites has Honey on Wednesday

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S35-F2-Q3 · question page 823

LOGIC.CONSTRAINT.03 · Change the conditions and re-test

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Removing condition 4 adds exactly two new schedules.

Answer: False

Reasoning: Without condition 4 there are 14 schedules, which is 4 more than 10.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S35-F2-Q4 · question page 824

LOGIC.CONSTRAINT.03 · Change the conditions and re-test

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Add a fifth condition: no two of Gears, Honey and Jazz are on consecutive days. How many schedules remain?

Answer: 2

Reasoning: Gears, Jazz and Honey must be on days 1, 3 and 5, so Gears is Monday, Jazz Wednesday, Honey Friday. Fossils and Kites take Tuesday and Thursday in either order: 2 schedules.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S35-F2-Q5 · question page 825

LOGIC.CONSTRAINT.03 · Change the conditions and re-test

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from

Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. If Gears is on Monday, then Fossils and Kites could be on Tuesday and Thursday.

Answer: True

Reasoning: Gears, Fossils, Jazz, Kites, Honey meets every condition.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S36-M-Q1 · question page 827

LOGIC.SEARCH.01 · Organize a complete case analysis and count or find every solution

How many three-digit whole numbers (100 to 999) have digits that add up to 5?

Answer: 15

Reasoning: Case by the hundreds digit: 1 leaves 4 for two digits (5 ways), 2 leaves 3 (4 ways), 3 leaves 2 (3 ways), 4 leaves 1 (2 ways), 5 leaves 0 (1 way): 15.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S36-M-Q2 · question page 828

LOGIC.SEARCH.01 · Organize a complete case analysis and count or find every solution

A 12-hour digital clock shows times from 1:00 to 12:59: the hour (1 to 12, no leading zero) and then two minute digits (00 to 59). At how many times do the shown digits add up to exactly 22?

Answer: 3

Reasoning: The minute digits add to at most 5 + 9 = 14, so the hour digits must add to at least 8: hours 8 and 9. Hour 8 needs minutes adding to 14: 59. Hour 9 needs 13: 49 and 58. Three times: 8:59, 9:49, 9:58.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S36-M-Q3 · question page 829

LOGIC.SEARCH.01 · Organize a complete case analysis and count or find every solution

Find the two-digit whole number that equals 3 times the sum of its digits.

Answer: any values that satisfy: isInteger(n) and n >= 10 and n <= 99 and n == 3 * (floor(n / 10) + mod(n, 10)) (example: n = 27)

Reasoning: 27 = 3 x (2 + 7). Writing n = 10a + b, the condition 10a + b = 3a + 3b gives 7a = 2b, so a = 2, b = 7 is the only digit solution.

Tempting answers the rule rejects: n = 18; n = 72; n = 30

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 1 valid answers.

S36-M-Q4 · question page 830

LOGIC.SEARCH.01 · Organize a complete case analysis and count or find every solution

In how many ways can you make 30 cents using nickels (5 cents), dimes (10 cents) and quarters (25 cents)? You may use any number of each coin, including none; the order of the coins does not matter.

Answer: 5

Reasoning: With no quarter: 0, 1, 2 or 3 dimes, the rest nickels (4 ways). With one quarter: one nickel (1 way). Total 5.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S36-M-Q5 · question page 831

LOGIC.SEARCH.01 · Organize a complete case analysis and count or find every solution

Exactly 6 of the 24 orderings of the letters A, B, C and D have A before B and also C before D.

Answer: True

Reasoning: Each of the conditions "A before B" and "C before D" holds in half of the orderings, independently: 24 / 2 / 2 = 6. Listing them confirms 6.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S36-B1-Q1 · question page 832

LOGIC.SEARCH.01 · Strategy: make an organized list

How many two-digit numbers have digits that add up to 6?

Answer: 6

Reasoning: 15, 24, 33, 42, 51, 60: six.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S36-B1-Q2 · question page 833

LOGIC.SEARCH.01 · Strategy: make an organized list

In how many ways can you make 20 cents using only nickels and dimes? The order of the coins does not matter.

Answer: 3

Reasoning: 0 dimes and 4 nickels, 1 dime and 2 nickels, 2 dimes: 3

ways.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S36-B1-Q3 · question page 834

LOGIC.SEARCH.01 · Strategy: make an organized list

There are exactly 4 two-digit numbers whose digits multiply to 12.

Answer: True

Reasoning: Tens digit 2, 3, 4, 6 gives 26, 34, 43, 62. Other tens digits do not divide 12 into a single digit.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S36-B1-Q4 · question page 835

LOGIC.SEARCH.01 · Strategy: make an organized list

Ana lists the ways to make 15 cents from nickels and dimes: "3 nickels; 1 dime and 1 nickel." Is her list complete? The order of the coins does not matter.

Answer: D. Yes: she used 0 dimes and 1 dime, and 2 dimes is too much

Reasoning: Organize by the number of dimes: 0 dimes needs 3 nickels, 1 dime needs 1 nickel, 2 dimes is too much. Her two ways are all of them.

- ✗ No: 2 dimes and no nickels is missing: 2 dimes make 20 cents, not 15
- ✗ No: 1 dime alone is missing: 1 dime makes only 10 cents
- ✗ No: a nickel then a dime is a third way: the order of the coins does not matter

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S36-B1-Q5 · question page 836

LOGIC.SEARCH.01 · Strategy: make an organized list

There are 5 two-digit numbers whose digits add up to 3.

Answer: False

Reasoning: By tens digit: 12, 21, 30. There are 3.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S36-F1-Q1 · question page 837

LOGIC.SEARCH.01 · Counting with extra conditions

How many three-digit numbers have three different digits that add up to 6?

Answer: 14

Reasoning: The digit sets are {0, 1, 5}, {0, 2, 4} and {1, 2, 3}. Each set with a 0 gives 4 numbers (0 cannot lead), and {1, 2, 3} gives 6: 4 + 4 + 6 = 14.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S36-F1-Q2 · question page 838

LOGIC.SEARCH.01 · Counting with extra conditions

In how many ways can 10 be written as a sum of 1s and 2s if the order matters? (For example, 3 can be written 3 ways: 1 + 1 + 1, 1 + 2 and 2 + 1.)

Answer: 89

Reasoning: Organize by the first number: ways(n) = ways(n - 1) + ways(n - 2). Starting 1, 2, 3, 5, 8, ... gives 89 for 10.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S36-F1-Q3 · question page 839

LOGIC.SEARCH.01 · Counting with extra conditions

Find a three-digit number whose three digits are all odd, all different, and add up to 15.

Answer: any values that satisfy: isInteger(n) and n >= 100 and n <= 999 and isOdd(floor(n / 100)) and isOdd(mod(floor(n / 10), 10)) and isOdd(mod(n, 10)) and floor(n / 100) + mod(floor(n / 10), 10) + mod(n, 10) == 15 and floor(n / 100) != mod(floor(n / 10), 10) and floor(n / 100) != mod(n, 10) and mod(floor(n / 10), 10) != mod(n, 10) (example: n = 159)

Reasoning: The digit sets are {1, 5, 9} and {3, 5, 7}; any order works, for example 159 or 357.

Tempting answers the rule rejects: n = 555; n = 168; n = 339

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 12 valid answers.

S36-F1-Q4 · question page 840

LOGIC.SEARCH.01 · Counting with extra conditions

There are more ways to make 50 cents from nickels and dimes than from dimes and quarters. The order of the coins does not matter.

Answer: True

Reasoning: Nickels and dimes: 0 to 5 dimes, 6 ways. Dimes and

quarters: 5 dimes, or 2 quarters, 2 ways (1 quarter leaves 25 cents, which dimes cannot make).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S36-F1-Q5 · question page 841

LOGIC.SEARCH.01 · Counting with extra conditions

How many rectangles of any size (including squares) can be traced along the lines of a grid that is 2 squares tall and 3 squares wide?

Answer: B. 18

Reasoning: By size: 1 by 1: 6; 1 tall by 2 wide: 4; 1 by 3: 2; 2 tall by 1: 3; 2 by 2: 2; 2 by 3: 1. Total 18.

- ✗ 6: that counts only the single squares
- ✗ 15: that leaves out the three rectangles 2 squares tall and 1 wide
- ✗ 17: that leaves out the whole grid

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S36-F2-Q1 · question page 842

LOGIC.SEARCH.01 · Searches with many cases

A 12-hour digital clock shows times from 1:00 to 12:59: the hour (1 to 12, no leading zero) and then two minute digits (00 to 59). At how many times do the shown digits add up to exactly 6?

Answer: 36

Reasoning: Case by the hour: its digit sum s leaves 6 - s for the minutes; count minute pairs (tens 0 to 5) for each hour and add.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S36-F2-Q2 · question page 843

LOGIC.SEARCH.01 · Searches with many cases

In how many ways can you make 1 dollar (100 cents) using nickels, dimes and quarters? The order of the coins does not matter.

Answer: 29

Reasoning: Case by quarters. 0 quarters: 0 to 10 dimes (11 ways). 1 quarter leaves 75: 0 to 7 dimes (8). 2 quarters leave 50: 6. 3 quarters leave 25: 3. 4 quarters: 1. Total 29.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S36-F2-Q3 · question page 844

LOGIC.SEARCH.01 · Searches with many cases

How many positive whole numbers less than 1000 have digits that add up to 4?

Answer: 15

Reasoning: One digit: 4 (1). Two digits: 13, 22, 31, 40 (4). Three digits: leading digit 1 to 4 leaves 3, 2, 1, 0 for two digits: 4 + 3 + 2 + 1 = 10. Total 15.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S36-F2-Q4 · question page 845

LOGIC.SEARCH.01 · Searches with many cases

Every whole number of cents from 8 to 30 can be paid exactly using only 3-cent and 5-cent stamps.

Answer: True

Reasoning: 8 = 3 + 5, 9 = 3 + 3 + 3, 10 = 5 + 5. Once three amounts in a row work, adding a 3-cent stamp reaches every larger amount.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S36-F2-Q5 · question page 846

LOGIC.SEARCH.01 · Searches with many cases

Give a whole number of cents from 1 to 10 that CANNOT be paid exactly using only 3-cent and 5-cent stamps.

Answer: any values that satisfy: n == 1 or n == 2 or n == 4 or n == 7 (example: n = 7)

Reasoning: Checking 1 to 10: 3, 5, 6, 8, 9 and 10 can be made; 1, 2, 4 and 7 cannot.

Tempting answers the rule rejects: n = 8; n = 6; n = 10; n = 3

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 4 listed mistakes fail it.

S37-M-Q1 · question page 848

LOGIC.PROOF.01 · Build a direct argument about even, odd and multiples from definitions

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Put the steps of a direct proof that the sum of two odd integers is even in order.

Answer: Let m = 2a + 1 and n = 2b + 1, where a and b are integers. → Then m + n = 2a + 2b + 2. → So m + n = 2(a + b + 1). → Since a + b + 1 is an integer, m + n is even.

Reasoning: Start from the definitions, add, factor out 2, then name the integer that shows the sum is even.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S37-M-Q2 · question page 849

LOGIC.PROOF.01 · Build a direct argument about even, odd and multiples from definitions

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Which first line correctly begins a direct proof that the product of any two odd integers is odd?

Answer: C. Let m = 2a + 1 and n = 2b + 1, where a and b are integers.

Reasoning: Two general odd integers need two separate integers a and b.

- ✗ Let m = 2a + 1 and n = 2a + 1, where a is an integer.: using the same a makes m and n equal, so it proves the claim only for a number times itself
- ✗ Let m = 3 and n = 5.: one example cannot prove a claim about all odd integers
- ✗ Assume that m times n is odd.: that assumes the conclusion the proof must reach

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S37-M-Q3 · question page 850

LOGIC.PROOF.01 · Build a direct argument about even, odd and multiples from definitions

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. A proof that an even integer plus an odd integer is odd reaches the line m + n = 2a + 2b + 1. Which line correctly finishes it?

Answer: D. m + n = 2(a + b) + 1, and a + b is an integer, so m + n is odd.

Reasoning: The definition of odd needs the form 2(integer) + 1.

✗ m + n is odd because the expression 2a + 2b + 1 contains a 1.: containing a 1 is not the definition of odd

✗ Since 4 + 3 = 7 is odd, m + n is odd.: an example does not finish a general proof

✗ m + n = 2(a + b + 1), so m + n is even.: 2a + 2b + 1 is not equal to 2(a + b + 1)

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S37-M-Q4 · question page 851

LOGIC.PROOF.01 · Build a direct argument about even, odd and multiples from definitions

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Checking that 3 + 5, 7 + 9 and 11 + 13 are all even proves that the sum of any two odd integers is even.

Answer: False

Reasoning: Examples can support a guess but cannot prove a claim about infinitely many integers. A proof must use general odd integers 2a + 1 and 2b + 1.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S37-M-Q5 · question page 852

LOGIC.PROOF.01 · Build a direct argument about even, odd and multiples from definitions

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. To show that 6n + 4 is even for every integer n, write 6n + 4 = 2(pn + q). Give the numbers p and q.

Answer: any values that satisfy: p == 3 and q == 2 (example: p = 3, q = 2)

Reasoning: 6n + 4 = 2(3n + 2), and 3n + 2 is an integer, so 6n + 4 is even.

Tempting answers the rule rejects: p = 6, q = 4; p = 3, q = 4; p = 2, q = 3

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S37-B1-Q1 · question page 853

LOGIC.PROOF.01 · Use the definitions of even and odd

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Which expression is odd for every integer k?

Answer: A. 2k + 7

Reasoning: 2k + 7 = 2(k + 3) + 1, which has the odd form.

- ✗ 2k + 4: 2k + 4 = 2(k + 2), which is even
- ✗ 4k: 4k = 2(2k), which is even
- ✗ k + 1: k + 1 is odd only when k is even

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S37-B1-Q2 · question page 854

LOGIC.PROOF.01 · Use the definitions of even and odd

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some

integer k. An integer n is a multiple of m if n = mk for some integer k. 2k + 6 is even for every integer k.

Answer: True

Reasoning: 2k + 6 = 2(k + 3).

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S37-B1-Q3 · question page 855

LOGIC.PROOF.01 · Use the definitions of even and odd

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Write 19 in the form 2k + 1. What is k?

Answer: 9

Reasoning: 19 = 2(9) + 1.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S37-B1-Q4 · question page 856

LOGIC.PROOF.01 · Use the definitions of even and odd

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. If n = 2k + 1, then n + 1 is odd.

Answer: False

Reasoning: n + 1 = 2k + 2 = 2(k + 1), which is even.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S37-B1-Q5 · question page 857

LOGIC.PROOF.01 · Use the definitions of even and odd

Which statement is the definition of an even integer?

Answer: D. n = 2k for some integer k

Reasoning: Even means 2 times some integer.

- ✗ n ends in the digit 0: that describes multiples of 10; 4 is even but ends in 4
- ✗ n = 2k + 1 for some integer k: that is the definition of odd
- ✗ n leaves remainder 1 when divided by 2: that also describes odd numbers

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S37-F1-Q1 · question page 858

LOGIC.PROOF.01 · Direct proofs about multiples

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Put the steps of a direct proof that if n is a multiple of 3, then n squared is a multiple of 9 in a correct order. (Two orders are correct.)

Answer: Let n = 3k, where k is an integer. → Then n squared = 9k squared. → k squared is an integer, because k is an integer. → So n squared is 9 times an integer, a multiple of 9.

Also accepted: Let n = 3k, where k is an integer. → k squared is an integer, because k is an integer. → Then n squared = 9k squared. → So n squared is 9 times an integer, a multiple of 9.

Reasoning: Begin with the definition (A) and end with the conclusion (D). B and C can come in either order.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 2 valid.

S37-F1-Q2 · question page 859

LOGIC.PROOF.01 · Direct proofs about multiples

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Which is a correct direct proof that the sum of three consecutive integers is a multiple of 3?

Answer: A. n + (n + 1) + (n + 2) = 3n + 3 = 3(n + 1), and n + 1 is an integer.

Reasoning: Write three general consecutive integers, add, and factor out 3.

- ✗ 1 + 2 + 3 = 6, and 6 is a multiple of 3.: one example does not prove the general claim
- ✗ n + n + n = 3n, which is a multiple of 3.: n, n, n are not consecutive integers
- ✗ n + (n + 1) + (n + 2) = 3n + 3, which is odd.: 3n + 3 is not always odd, and odd is not the claim

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S37-F1-Q3 · question page 860

LOGIC.PROOF.01 · Direct proofs about multiples

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. If n is odd, then n squared minus 1 is a multiple of 8.

Answer: True

Reasoning: n = 2k + 1 gives n squared - 1 = 4k(k + 1). One of k and k + 1 is even, so k(k + 1) = 2j and the expression is 8j.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S37-F1-Q4 · question page 861

LOGIC.PROOF.01 · Direct proofs about multiples

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. (2a + 1)(2b + 1) = 2(2ab + a + b) + 1. For a = 3 and b = 4, what number is 2ab + a + b?

Answer: any values that satisfy: x == 31 (example: x = 31)

Reasoning: 2(3)(4) + 3 + 4 = 31. Check: 7 x 9 = 63 = 2(31) + 1.

Tempting answers the rule rejects: x = 63; x = 24; x = 32

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S37-F1-Q5 · question page 862

LOGIC.PROOF.01 · Direct proofs about multiples

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. A student proves that every multiple of 4 is even by writing n = 4k = 2(2k). What justifies the last sentence, "so n is even"?

Answer: B. 2k is an integer, so n is 2 times an integer.

Reasoning: The definition needs n = 2 times an integer, and 2k is an integer.

- ✗ 4 is an even number.: that alone does not put n in the form 2 times an integer
- ✗ n can be any number at all.: the claim is about multiples of 4 only
- ✗ k must be an even number.: k can be any integer; nothing makes it

even

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S37-F2-Q1 · question page 863

LOGIC.PROOF.01 · Judge and complete direct proofs

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Put the steps of a direct proof that the square of an odd integer leaves remainder 1 when divided by 4 in order.

Answer: Let n = 2k + 1, where k is an integer. → Then n squared = 4k squared + 4k + 1. → So n squared = 4(k squared + k) + 1. → Since k squared + k is an integer, n squared leaves remainder 1 when divided by 4.

Reasoning: Definition, expand, factor out 4, then name the integer.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S37-F2-Q2 · question page 864

LOGIC.PROOF.01 · Judge and complete direct proofs

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Claim: the sum of two even integers is a multiple of 4. "Proof: Let m = 2k and n = 2k. Then m + n = 4k." What is wrong?

Answer: C. It uses the same k for both, so it only covers the case m = n.

Reasoning: With separate integers, m + n = 2(a + b), which is a multiple of 4 only when a + b is even. 2 + 4 = 6 shows the claim is false.

✗ It should have started with m = 4k.: the given numbers are only known to be even

✗ Nothing is wrong; the proof is correct.: 2 + 4 = 6 is not a multiple of 4, so the claim is false

✗ It should check an example before starting.: checking examples is not the flaw

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S37-F2-Q3 · question page 865

LOGIC.PROOF.01 · Judge and complete direct proofs

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. The statement "n squared + n is even for every integer n" can be proved by two cases: n even and n odd.

Answer: True

Reasoning: If n = 2k, n squared + n = 2(2k squared + k). If n = 2k + 1, n squared + n = (2k + 1)(2k + 2) = 2(2k + 1)(k + 1). Both are even.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S37-F2-Q4 · question page 866

LOGIC.PROOF.01 · Judge and complete direct proofs

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. For odd n = 2k + 1, n squared + 4n + 3 = 2(2k squared + 6k + 4). For k = 5, what number is 2k squared + 6k + 4?

Answer: 84

Reasoning: 2(25) + 30 + 4 = 84. Check: n = 11 gives 121 + 44 + 3 = 168 = 2 x 84.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S37-F2-Q5 · question page 867

LOGIC.PROOF.01 · Judge and complete direct proofs

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Give an integer n from 1 to 20 for which n squared + n + 1 is even. If no such n exists, enter 0.

Answer: any values that satisfy: n == 0 (example: n = 0)

Reasoning: n squared + n = n(n + 1) is always even, so n squared + n + 1 is always odd. No such n exists.

Tempting answers the rule rejects: n = 1; n = 2; n = 7

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S38-M-Q1 · question page 869

LOGIC.SPATIAL.05 · Decide which pieces cover a region exactly, and prove when no covering exists

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. In how many different ways can a rectangle 2 squares tall and 4 squares wide be tiled with dominoes?

Answer: 5

Reasoning: Look at the left end: one upright domino leaves a 2 by 3 (3 ways), or two flat dominoes leave a 2 by 2 (2 ways): 3 + 2 = 5.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S38-M-Q2 · question page 870

LOGIC.SPATIAL.05 · Decide which pieces cover a region exactly, and prove when no covering exists

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. The 3 by 3 board with its center square removed (shown) can be tiled by 4 dominoes.

Answer: True

Reasoning: The 8 remaining squares form a ring; going around it, pair the squares two at a time.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S38-M-Q3 · question page 871

LOGIC.SPATIAL.05 · Decide which pieces cover a region exactly, and prove when no covering exists

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. Which region can be tiled by 4 dominoes?

Answer: B. Figure B

Reasoning: Region B: the top-left square must pair with the square below it, and the rest then pairs off in one way.

- ✗ Figure A: color it like a checkerboard: 5 squares of one color and 3 of the other, but every domino covers one of each
- ✗ Figure C: color it like a checkerboard: 5 squares of one color and 3 of the other
- ✗ Figure D: the right end of the long row and the square below the row each touch only one square, the same one, so both cannot be covered

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the panel check accepts only the keyed panel.

S38-M-Q4 · question page 872

LOGIC.SPATIAL.05 · Decide which pieces cover a region exactly, and prove when no covering exists

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. One square is removed from a 5 by 5 board, and the other 24 squares are to be tiled by 12 dominoes. Rows and columns are numbered 1 to 5. Give the row r and column c of a square whose removal makes a tiling possible.

Answer: any values that satisfy: isInteger(r) and isInteger(c) and r >= 1 and r <= 5 and c >= 1 and c <= 5 and isEven(r + c) (example: r = 1, c = 1)

Reasoning: Color the board like a checkerboard with the corners dark: 13 dark, 12 light. A tiling needs 12 of each, so the removed square must be dark (r + c even). Every dark square works; a computer search of all 25 removals confirms it.

Tempting answers the rule rejects: r = 1, c = 2; r = 2, c = 3; r = 6, c = 2

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Tilings counted for every one of the 25 removals: exactly the 13 squares with r + c even leave a tileable board. Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 13 valid answers.

S38-M-Q5 · question page 873

LOGIC.SPATIAL.05 · Decide which pieces cover a region exactly, and prove when no covering exists

An L-tromino is three squares in an L shape (a 2 by 2 square with one square missing). It may be turned any way. In how many ways can a rectangle 2 squares tall and 3 wide be tiled by two L-trominoes?

Answer: 2

Reasoning: One L must cover the top-left square. Of its placements, only two leave an L-shaped gap; each gives one tiling.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S38-B1-Q1 · question page 874

LOGIC.SPATIAL.05 · Strategy: color the board like a checkerboard

A 4 by 4 board is colored like a checkerboard. How many of its squares are dark?

Answer: 8

Reasoning: Each row has 2 dark and 2 light squares: 4 x 2 = 8.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S38-B1-Q2 · question page 875

LOGIC.SPATIAL.05 · Strategy: color the board like a checkerboard

Every domino placed on a checkerboard covers one dark square and one light square.

Answer: True

Reasoning: The two squares share a side, and squares sharing a side have different colors.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S38-B1-Q3 · question page 876

LOGIC.SPATIAL.05 · Strategy: color the board like a checkerboard

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. A region has 7 dark and 5 light squares on a checkerboard coloring. According to the color count, what is the greatest number of dominoes that could be placed on it without overlapping?

Answer: 5

Reasoning: Each domino uses one light square, and there are only 5.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S38-B1-Q4 · question page 877

LOGIC.SPATIAL.05 · Strategy: color the board like a checkerboard

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. A region has 6 dark and 6 light squares. What does the color count tell you?

Answer: C. It does not rule out a tiling; you still have to find one or show none exists.

Reasoning: Equal counts pass the test, but a region such as region D in this set can still fail.

- ✗ It can certainly be tiled by 6 dominoes.: equal colors are needed, but they are not enough
- ✗ It cannot be tiled by dominoes.: equal colors do not rule out a tiling
- ✗ It can be tiled in exactly 6 ways.: the color count says nothing about the number of ways

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S38-B1-Q5 · question page 878

LOGIC.SPATIAL.05 · Strategy: color the board like a checkerboard

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. A region with an even number of squares can always be tiled by dominoes.

Answer: False

Reasoning: The T shape in the worked example has 4 squares and cannot be tiled.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S38-F1-Q1 · question page 879

LOGIC.SPATIAL.05 · Count tilings and use other pieces

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. In how many ways can a strip 2 squares tall and 6 squares wide be tiled by dominoes?

Answer: 13

Reasoning: Let w(n) be the number for width n. The left end is one upright domino (leaving width n - 1) or two flat ones (leaving n - 2): w(n) = w(n - 1) + w(n - 2). 1, 2, 3, 5, 8, 13.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S38-F1-Q2 · question page 880

LOGIC.SPATIAL.05 · Count tilings and use other pieces

A straight tromino covers three squares in a row. A 3 by 3 square can be tiled by three straight trominoes in exactly 2 ways.

Answer: True

Reasoning: If the top-left tromino lies flat, the top row is filled and so are the other rows: all flat. If it stands upright, all must stand. 2 ways.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S38-F1-Q3 · question page 881

LOGIC.SPATIAL.05 · Count tilings and use other pieces

A straight tromino covers three squares in a row, flat or upright. One corner square is removed from a 4 by 4 board. Can the other 15 squares be tiled by 5 straight trominoes?

Answer: A. Yes, and there is more than one way.

Reasoning: For example, the row with the gap holds one flat tromino in its other three squares, and the remaining 3 by 4 block can be filled with upright trominoes. A full search finds 4 tilings.

- ✗ No: 15 squares cannot be split into trominoes.: 15 = 5 x 3, so the count is fine
- ✗ No: the corner gap blocks every tiling.: tilings exist; the search finds 4
- ✗ Yes, but in exactly one way.: a search finds 4 different tilings

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S38-F1-Q4 · question page 882

LOGIC.SPATIAL.05 · Count tilings and use other pieces

A straight tromino covers three squares in a row, flat or upright. In how many ways can a rectangle 3 squares tall and 4 wide be tiled by straight trominoes?

Answer: 3

Reasoning: All flat does not fit (rows of 4). Each column can hold an upright tromino; a flat block of three flat trominoes fills 3 by 3. Upright x 4; flat block then upright; upright then flat block: 3.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S38-F1-Q5 · question page 883

LOGIC.SPATIAL.05 · Count tilings and use other pieces

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. A strip 2 squares tall and n squares wide can be tiled by dominoes in exactly 8 ways. What is n?

Answer: any values that satisfy: n == 5 (example: n = 5)

Reasoning: The counts for widths 1, 2, 3, 4, 5 are 1, 2, 3, 5, 8.

Tempting answers the rule rejects: n = 4; n = 8; n = 6

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Counts by width: [1, 2, 3, 5, 8, 13]. Automatic: the example passes the rule and 3 listed mistakes fail it.

S38-F2-Q1 · question page 884

LOGIC.SPATIAL.05 · Prove impossibility and count harder tilings

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. In how many ways can a rectangle 3 squares tall and 4 wide be tiled by dominoes?

Answer: 11

Reasoning: An organized search by the placement covering the top-left square, then the next uncovered square, finds 11 tilings.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S38-F2-Q2 · question page 885

LOGIC.SPATIAL.05 · Prove impossibility and count harder tilings

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. A board 4 rows by 5 columns has two squares removed: row 1, column 1 and row 3, column 3. The other 18 squares can be tiled by 9 dominoes.

Answer: False

Reasoning: Both removed squares have the same checkerboard color (1 + 1 and 3 + 3 are even). That leaves 8 of that color and 10 of the other, so no tiling exists.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S38-F2-Q3 · question page 886

LOGIC.SPATIAL.05 · Prove impossibility and count harder tilings

An L-tromino is three squares in an L shape, turned any way. In how many ways can a strip 2 squares tall and 9 wide be tiled by L-trominoes?

Answer: B. 8

Reasoning: The leftmost L-tromino and its partner always fill a 2 by 3 block, which has 2 tilings. Three blocks: 2 x 2 x 2 = 8.

- ✗ 6: the strip splits into three 2 by 3 blocks with 2 choices each, so multiply, not add
- ✗ 3: that counts the blocks, not the ways
- ✗ 9: that is the width, not a count of tilings

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S38-F2-Q4 · question page 887

LOGIC.SPATIAL.05 · Prove impossibility and count harder tilings

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. One square is removed from a 3 by 3 board. Rows and columns are numbered 1 to 3. Give a square (row r, column c) whose removal leaves 8 squares that can be tiled by dominoes.

Answer: any values that satisfy: isInteger(r) and isInteger(c) and r >= 1 and r <= 3 and c >= 1 and c <= 3 and isEven(r + c) (example: r = 2, c = 2)

Reasoning: The corners and the center are the majority color (5 of 9). Removing one of them leaves 4 of each, and each such board can be tiled.

Tempting answers the rule rejects: r = 1, c = 2; r = 2, c = 3

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Tilings counted for all 9 removals; the 5 with r + c even work. Automatic: the example passes the rule and 2 listed mistakes fail it. Search of the stated domain: 5 valid answers.

S38-F2-Q5 · question page 888

LOGIC.SPATIAL.05 · Prove impossibility and count harder tilings

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. One square is removed from a board 3 rows by 5 columns. For how many of the 15 squares does removing it leave 14 squares that can be tiled by dominoes?

Answer: 8

Reasoning: The 8 squares of the majority color (row + column even) are the only candidates, and a search shows each of them works.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S39-M-Q1 · question page 890

LOGIC.PROOF.02 · Explain each step in solving an equation and justify the solution method

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Put the steps for solving 4x + 7 = 2x + 19 in order.

Answer: 2x + 7 = 19 (subtract 2x from both sides) → 2x = 12 (subtract 7 from both sides) → x = 6 (divide both sides by 2) → Check: 4(6) + 7 = 31 and 2(6) + 19 = 31

Reasoning: Each line uses the line before it; the check comes last, once x is known.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S39-M-Q2 · question page 891

LOGIC.PROOF.02 · Explain each step in solving an equation and justify the solution method

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Which reason justifies going from 5x - 4 = 21 to 5x = 25?

Answer: D. Add 4 to both sides; equal numbers plus 4 are still equal.

Reasoning: Adding the same number to two equal numbers gives equal results: 21 + 4 = 25.

- ✗ Use the distributive property on 5x - 4.: nothing is distributed in this step
- ✗ Multiply both sides by 4.: multiplying by 4 would give 20x - 16 = 84
- ✗ Rewrite 5x - 4 as 4 - 5x.: subtraction cannot be reordered that way, and the step does not do it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S39-M-Q3 · question page 892

LOGIC.PROOF.02 · Explain each step in solving an equation and justify the solution method

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. A student writes: Line 1: 2(x + 3) = 14. Line 2: 2x + 3 = 14. Line 3: 2x = 11. Line 4: x = 5.5. Which line is the first one that does NOT follow from the line before it?

Answer: A. Line 2

Reasoning: 2(x + 3) = 2x + 6, not 2x + 3. The correct steps give 2x = 8 and x = 4.

- ✗ Line 3: from Line 2, subtracting 3 does give 2x = 11
- ✗ Line 4: from Line 3, dividing by 2 does give x = 5.5
- ✗ None; every line follows: Line 2 is wrong, and x = 5.5 does not check: 2(8.5) = 17

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S39-M-Q4 · question page 893

LOGIC.PROOF.02 · Explain each step in solving an equation and justify the solution method

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. From 6 - x = 10, the next line "x = 10 - 6" follows correctly.

Answer: False

Reasoning: Subtracting 6 gives -x = 4, so x = -4. The line x = 4 does not follow; 6 - 4 = 2, not 10.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S39-M-Q5 · question page 894

LOGIC.PROOF.02 · Explain each step in solving an equation and justify the solution method

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Solve 7 - 2(x - 1) = x - 3. Give x.

Answer: any values that satisfy: x == 4 (example: x = 4)

Reasoning: 7 - 2x + 2 = x - 3 (distribute), 9 - 2x = x - 3 (combine), 12 = 3x (add 2x and 3), x = 4 (divide by 3). Check: 7 - 2(3) = 1 and 4 - 3 = 1.

Tempting answers the rule rejects: x = 2; x = -4; x = 6

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S39-B1-Q1 · question page 895

LOGIC.PROOF.02 · Properties of equality

If a = b, which statement must also be true?

Answer: C. a - 5 = b - 5

Reasoning: Doing the same thing to two equal numbers keeps them equal.

- ✗ a - 5 = b + 5: different amounts were changed on the two sides
- ✗ 5a = b: only one side was multiplied
- ✗ a / 5 = b / 3: the two sides were divided by different numbers

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S39-B1-Q2 · question page 896

LOGIC.PROOF.02 · Properties of equality

If 3x = 12, then x = 4 follows by dividing both sides by 3.

Answer: True

Reasoning: 12 / 3 = 4.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S39-B1-Q3 · question page 897

LOGIC.PROOF.02 · Properties of equality

Which step uses the distributive property?

Answer: B. 5(x - 2) = 13 becomes 5x - 10 = 13

Reasoning: The distributive property rewrites 5(x - 2) as 5x - 10.

- ✗ x + 3 = 8 becomes x = 5: that subtracts 3 from both sides
- ✗ 4x = 20 becomes x = 5: that divides both sides by 4
- ✗ x + 2 = 9 becomes 2 + x = 9: that uses the commutative property

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S39-B1-Q4 · question page 898

LOGIC.PROOF.02 · Properties of equality

Multiplying both sides of an equation by 0 always gives an equation with the same solutions.

Answer: False

Reasoning: It gives 0 = 0, which every number satisfies, so solutions can be gained.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S39-B1-Q5 · question page 899

LOGIC.PROOF.02 · Properties of equality

To solve x / 4 = 9 in one step, by what number do you multiply both sides?

Answer: 4

Reasoning: Multiplying both sides by 4 gives x = 36.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S39-F1-Q1 · question page 900

LOGIC.PROOF.02 · Steps that need care

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. A student solves x squared = 5x by dividing both sides by x and gets x = 5. What is wrong?

Answer: D. Dividing by x assumes x is not 0, and x = 0 is also a solution.

Reasoning: x squared - 5x = 0 gives x(x - 5) = 0, so x = 0 or x = 5.

- ✗ Nothing; x = 5 is the only solution.: x = 0 also works: 0 = 0
- ✗ The answer should be x = 0 only.: x = 5 also works: 25 = 25
- ✗ x squared = 5x has no solutions.: both 0 and 5 are solutions

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S39-F1-Q2 · question page 901

LOGIC.PROOF.02 · Steps that need care

The equation 3(x + 2) = 3x + 6 is true for every number x.

Answer: True

Reasoning: By the distributive property, 3(x + 2) and 3x + 6 are the same expression.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S39-F1-Q3 · question page 902

LOGIC.PROOF.02 · Steps that need care

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Solve 2(x - 4) + 3 = 5x + 7. What is x?

Answer: -4

Reasoning: 2x - 8 + 3 = 5x + 7, so 2x - 5 = 5x + 7, so -12 = 3x, so x = -4. Check: 2(-8) + 3 = -13 and 5(-4) + 7 = -13.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S39-F1-Q4 · question page 903

LOGIC.PROOF.02 · Steps that need care

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Solving 4x + 1 = 4x - 3 leads to the line 1 = -3. What does this show?

Answer: C. The equation has no solution.

Reasoning: If some x solved it, then 1 = -3 would follow, which is false. So no x solves it.

- ✗ The solution is x = 0.: x = 0 gives 1 = -3, which is false
- ✗ Every number is a solution.: that would be shown by a true line such as 1 = 1
- ✗ The solution is x = -3.: x = -3 gives -11 = -15, which is false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S39-F1-Q5 · question page 904

LOGIC.PROOF.02 · Steps that need care

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Put the lines of this solution of (x + 5) / 3 = 4 in order.

Answer: Assume x is a number with (x + 5) / 3 = 4. → x + 5 = 12 (multiply both sides by 3) → x = 7 (subtract 5 from both sides) → Check: (7 + 5) / 3 = 4

Reasoning: Start from the assumption that a solution exists, then undo the division, then the addition, then check.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S39-F2-Q1 · question page 905

LOGIC.PROOF.02 · Justify a solution method

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. To solve the equation (square root of (x + 2)) = x, a student squares both sides: x + 2 = x squared, so x = 2 or x = -1. Which answer is correct?

Answer: A. Only x = 2; x = -1 gives square root of 1 = 1, not -1.

Reasoning: Squaring both sides can add solutions, so each candidate must be checked in the original equation.

- ✗ Both x = 2 and x = -1.: x = -1 fails the original equation
- ✗ Only x = -1.: x = -1 fails and x = 2 works: square root of 4 = 2
- ✗ Neither one.: x = 2 works: square root of 4 = 2

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S39-F2-Q2 · question page 906

LOGIC.PROOF.02 · Justify a solution method

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. If each line of a solution follows from the line before it, then every number found at

the end must solve the original equation.

Answer: False

Reasoning: The steps show that any solution is among the numbers found, not that each number found works. Squaring, for example, can add a number that fails the original equation.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S39-F2-Q3 · question page 907

LOGIC.PROOF.02 · Justify a solution method

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Solve 3(2x - 1) = 4(x + 2) - 1. Give x.

Answer: any values that satisfy: x == 5 (example: x = 5)

Reasoning: 6x - 3 = 4x + 7, so 2x = 10, so x = 5. Check: 3(9) = 27 and 4(7) - 1 = 27.

Tempting answers the rule rejects: x = 3; x = 2; x = -5

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S39-F2-Q4 · question page 908

LOGIC.PROOF.02 · Justify a solution method

Which step can give an equation with MORE solutions than the one before it?

Answer: D. Squaring both sides

Reasoning: Squaring cannot be undone: a = b and a = -b both lead to a squared = b squared.

✗ Adding 7 to both sides: subtracting 7 undoes it, so the solutions stay the same

✗ Dividing both sides by 2: multiplying by 2 undoes it, so the solutions stay the same

✗ Multiplying both sides by -3: dividing by -3 undoes it, so the solutions stay the same

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S39-F2-Q5 · question page 909

LOGIC.PROOF.02 · Justify a solution method

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Put the lines of this solution of 5 - 3(x - 1) = 2x - 2 in order.

Answer: 5 - 3x + 3 = 2x - 2 (distributive property) → 8 - 3x = 2x - 2 (combine 5 and 3) → 10 = 5x (add 3x and 2 to both sides) → x = 2 (divide both sides by 5)

Reasoning: Distribute, combine like terms, collect x on one side, divide. Check: 5 - 3(1) = 2 and 2(2) - 2 = 2.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S40-M-Q1 · question page 911

LOGIC.PROOF.03 · Set up and complete an argument by contradiction

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. To prove "there is no largest even integer" by contradiction, how should the proof begin?

Answer: B. Assume there is a largest even integer; call it N.

Reasoning: Assume the opposite. Then N + 2 is even and larger than N,

which is impossible.

- ✗ Assume there is no largest even integer.: that assumes the statement itself, not its opposite
- ✗ Show that 100 is an even integer.: one example says nothing about a largest even integer
- ✗ Assume every even integer is less than 100.: that is a different, narrower assumption

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S40-M-Q2 · question page 912

LOGIC.PROOF.03 · Set up and complete an argument by contradiction

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Put the steps of a proof that if n squared is even, then n is even, in order.

Answer: Assume n squared is even, but n is odd. → Then n = 2k + 1 for some integer k. → So n squared = 2(2k squared + 2k) + 1, which is odd. → This contradicts n squared being even, so n must be even.

Reasoning: Assume the opposite of the conclusion, use the definition, compute, then point to the contradiction.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S40-M-Q3 · question page 913

LOGIC.PROOF.03 · Set up and complete an argument by contradiction

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Reaching a statement that contradicts a known fact shows that the assumption was false.

Answer: True

Reasoning: If the assumption were true, a false statement would follow from it. So the assumption is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S40-M-Q4 · question page 914

LOGIC.PROOF.03 · Set up and complete an argument by contradiction

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Claim: no integer is both even and odd. Suppose n = 2a and n = 2b + 1 for integers a and b. Which line gives the contra

Answer: C. 2a = 2b + 1, so 2(a - b) = 1, but 1 is not 2 times an integer.

Reasoning: a - b would be an integer equal to 1/2, which is impossible.

- ✗ 2a = 2b + 1, so a = b.: that algebra is wrong, and it is not a contradiction
- ✗ n = 4 is even and not odd.: one example does not treat a general n
- ✗ 2a + 2b + 1 is an odd number.: adding the two forms is true but contradicts nothing

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S40-M-Q5 · question page 915

LOGIC.PROOF.03 · Set up and complete an argument by contradiction

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. To show that 15 is not the sum of two even integers, assume 15 = 2a + 2b for

integers a and b. Then a + b must equal wh

Answer: any values that satisfy: x == 7.5 (example: x = 7.5)

Reasoning: 2(a + b) = 15 gives a + b = 7.5. But a + b is an integer, so the assumption is impossible.

Tempting answers the rule rejects: x = 7; x = 8; x = 15

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S40-B1-Q1 · question page 916

LOGIC.PROOF.03 · Strategy: assume the opposite and look for the impossible

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. To prove "n is not divisible by 4" by contradiction, what do you assume?

Answer: A. n is divisible by 4.

Reasoning: Assume the exact opposite of the statement.

- ✗ n is not divisible by 4.: that is the statement itself
- ✗ n is divisible by 2.: that is a different statement
- ✗ n is an odd number.: that is a different statement

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S40-B1-Q2 · question page 917

LOGIC.PROOF.03 · Strategy: assume the opposite and look for the impossible

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. To prove "there is no integer x with 2x = 7" by contradiction, what do you assume?

Answer: D. There is an integer x with 2x = 7.

Reasoning: The opposite of "there is no such x" is "there is such an x".

✗ Every integer x has 2x = 7.: the opposite of "no x" is "some x", not "every x"
✗ There is an integer x with 2x not equal to 7.: that is true and is not the opposite
✗ There is no integer x with 2x = 7.: that is the statement itself

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S40-B1-Q3 · question page 918

LOGIC.PROOF.03 · Strategy: assume the opposite and look for the impossible

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. A proof by contradiction begins by assuming that the statement to be proved is true.

Answer: False

Reasoning: It begins by assuming the statement is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S40-B1-Q4 · question page 919

LOGIC.PROOF.03 · Strategy: assume the opposite and look for the impossible

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. If assuming P leads to 1 = 0, then P is false.

Answer: True

Reasoning: A true assumption cannot lead to a false statement by valid steps.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S40-B1-Q5 · question page 920

LOGIC.PROOF.03 · Strategy: assume the opposite and look for the impossible

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Assume there is a largest odd integer M. Which number gives the contradiction?

Answer: B. M + 2

Reasoning: M + 2 is odd and larger than M.

- ✗ M + 1: M + 1 is even, so it does not contradict M being the largest odd integer
- ✗ M - 2: M - 2 is smaller than M
- ✗ 2M: 2M is even, so it does not contradict anything

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S40-F1-Q1 · question page 921

LOGIC.PROOF.03 · Contradiction with divisibility

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. To prove that no integers a and b satisfy 6a + 9b = 10, which fact gives the contradiction?

Answer: A. 6a + 9b = 3(2a + 3b) is a multiple of 3, but 10 is not.

Reasoning: If 6a + 9b = 10, then 10 would be 3 times an integer, which it is not.

- ✗ 6 and 9 are both odd numbers.: 6 is even
- ✗ 10 is even, and 6a + 9b is always odd.: a = 1, b = 0 gives 6, which is

even

✗ a and b would have to be fractions.: that is what is to be shown; it is not a contradiction

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S40-F1-Q2 · question page 922

LOGIC.PROOF.03 · Contradiction with divisibility

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Put the steps of the proof that the square root of 2 is not a fraction in order.

Answer: Assume the square root of 2 = p / q, with p and q integers in lowest terms. → Then p squared = 2q squared, so p squared is even, so p is even: p = 2m. → Then 4m squared = 2q squared, so q squared = 2m squared, and q is even. → Both p and q are even, which contradicts lowest terms.

Reasoning: Each step uses the one before it, and D clashes with the assumption in A.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S40-F1-Q3 · question page 923

LOGIC.PROOF.03 · Contradiction with divisibility

Among any 5 integers, two of them leave the same remainder when divided by 4.

Answer: True

Reasoning: Assume all 5 remainders are different. But only 4 remainders (0, 1, 2, 3) exist, a contradiction.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S40-F1-Q4 · question page 924

LOGIC.PROOF.03 · Contradiction with divisibility

A perfect square is divided by 4. What is the largest remainder (0, 1, 2 or 3) that can NEVER occur?

Answer: 3

Reasoning: Even squares are (2k) squared = 4k squared (remainder 0); odd squares are 4(k squared + k) + 1 (remainder 1). So 2 and 3 never occur; the larger is 3.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S40-F1-Q5 · question page 925

LOGIC.PROOF.03 · Contradiction with divisibility

Give an integer n from 1 to 50 for which n squared + 1 is a multiple of 3. If no such n exists, enter 0.

Answer: any values that satisfy: n == 0 (example: n = 0)

Reasoning: Assume some n works. n leaves remainder 0, 1 or 2 when divided by 3, so n squared leaves 0, 1 or 1; then n squared + 1 leaves 1 or 2, never 0. Contradiction, so none exists.

Tempting answers the rule rejects: n = 1; n = 2; n = 5

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked: no n from 1 to 50 works. Automatic: the example passes the rule and 3 listed mistakes fail it.

S40-F2-Q1 · question page 926

LOGIC.PROOF.03 · Contradiction in larger arguments

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Assume there are only finitely many primes, p1, p2, ..., pk, and let N = p1 x p2 x ... x pk + 1. Why is this a contradiction?

Answer: D. N leaves remainder 1 when divided by each listed prime, so a prime factor of N is not on the list.

Reasoning: Every integer greater than 1 has a prime factor, and none of the listed primes divides N.

- ✗ N is always a prime number.: not always: 2 x 3 x 5 x 7 x 11 x 13 + 1 = 30031 = 59 x 509
- ✗ N is always an even number.: N is one more than an even product, so it is odd
- ✗ N is larger than every prime, so N is not prime.: being large does not make a number composite

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S40-F2-Q2 · question page 927

LOGIC.PROOF.03 · Contradiction in larger arguments

2 x 3 x 5 x 7 x 11 x 13 + 1 is a prime number.

Answer: False

Reasoning: 30031 = 59 x 509.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S40-F2-Q3 · question page 928

LOGIC.PROOF.03 · Contradiction in larger arguments

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Put the steps of a proof that if a + b is at least 20, then a is at least 10 or b is at least 10, in order.

Answer: Assume a + b is at least 20, but a < 10 and b < 10. → Adding the two inequalities gives a + b < 20. → That contradicts a + b being at least 20. → So a is at least 10 or b is at least 10.

Reasoning: The opposite of "a at least 10 or b at least 10" is "a < 10 and b < 10".

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S40-F2-Q4 · question page 929

LOGIC.PROOF.03 · Contradiction in larger arguments

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Claim: every integer is even. "Proof: Assume some integer is odd, say 3. Then 3 = 2k for no integer k. Contradiction." What is wrong?

Answer: B. Nothing impossible was reached; 3 not being 2k agrees with 3 being odd.

Reasoning: A contradiction must clash with the assumption or a known fact. Here everything found is consistent.

- ✗ It should assume that every integer is odd.: the opposite of "every integer is even" is "some integer is not even"
- ✗ It should use 5 instead of 3.: any odd example has the same problem
- ✗ Nothing; the proof is correct.: the claim is false, since 3 is odd

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S40-F2-Q5 · question page 930

LOGIC.PROOF.03 · Contradiction in larger arguments

At least how many integers must you choose to be sure that two of them leave the same remainder when divided by 6?

Answer: 7

Reasoning: Six integers can have the six different remainders 0 to 5. With 7, assuming all remainders differ needs 7 remainders, but only 6 exist.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S41-M-Q1 · question page 932

LOGIC.INVARIANT.01 · Use odd/even and alternation to predict outcomes and rule out possibilities

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Seven gears form a closed ring, each touching the gears on either side (figure). Gear 1 is turned clockwise. What happens?

Answer: C. The gears jam: going around the ring, gear 1 would have to turn both ways.

Reasoning: Going around: 1 clockwise, 2 counter, 3 clockwise, ..., 7 clockwise. Gear 7 touches gear 1, so they must turn opposite ways, but both are clockwise. An odd ring cannot turn.

- ✗ Every gear turns, and the directions alternate around the ring.: alternating around 7 gears ends with gear 7 clockwise, touching gear 1, which is also clockwise
- ✗ Every gear turns clockwise.: touching gears must turn opposite ways
- ✗ Only gears 1, 3, 5 and 7 turn.: gears that touch a turning gear are forced to turn

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S41-M-Q2 · question page 933

LOGIC.INVARIANT.01 · Use odd/even and alternation to predict outcomes and rule out possibilities

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Twelve gears stand in a row (not a ring), each touching the next. Gear 1 turns clockwise. How many of the 12 gears turn counterclockwise?

Answer: 6

Reasoning: Odd-numbered gears turn clockwise and even-numbered gears counterclockwise: 2, 4, ..., 12 is 6 gears.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S41-M-Q3 · question page 934

LOGIC.INVARIANT.01 · Use odd/even and alternation to predict outcomes and rule out possibilities

A frog starts on 0 on a number line and makes exactly 9 jumps, each 1 unit left or right. It can end on 4.

Answer: False

Reasoning: Each jump changes the position by 1, so it switches between even and odd. After 9 jumps (odd) the frog is on an odd number, and 4 is even.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S41-M-Q4 · question page 935

LOGIC.INVARIANT.01 · Use odd/even and alternation to predict outcomes and rule out possibilities

A lamp starts OFF. Each press of its switch changes it (OFF to ON, or ON to OFF). Give a number of presses from 1 to 20 that leaves the lamp ON.

Answer: any values that satisfy: isInteger(n) and n >= 1 and n <= 20 and isOdd(n) (example: n = 7)

Reasoning: After an odd number of presses the lamp is ON; after an even number it is OFF.

Tempting answers the rule rejects: n = 8; n = 20; n = 0

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 10 valid answers.

S41-M-Q5 · question page 936

LOGIC.INVARIANT.01 · Use odd/even and alternation to predict outcomes and rule out possibilities

A token moves along a row of squares numbered 1 to 15. Each move jumps exactly 2 squares left or right (staying on the row). It starts on square 4. How many of the 15 squares can it ever reach, counting square 4?

Answer: 7

Reasoning: Jumps of 2 keep the token on even squares: 2, 4, 6, 8, 10, 12, 14. That is 7 squares, and all of them can be reached.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S41-B1-Q1 · question page 937

LOGIC.INVARIANT.01 · Strategy: keep track of odd and even only

The sum of 5 odd numbers is odd.

Answer: True

Reasoning: Pairs of odd numbers add to even numbers; 5 odd numbers make 2 pairs and 1 left over, so the sum is even + odd = odd.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S41-B1-Q2 · question page 938

LOGIC.INVARIANT.01 · Strategy: keep track of odd and even only

A coin starts heads up. It is turned over 13 times. Which side is up at the end?

Answer: B. Tails

Reasoning: 13 is odd, so the side switches an odd number of times: tails.

- ✗ Heads: heads is up only after an even number of turns
- ✗ Either side, depending on how it is turned: every turn switches the side, so the result is fixed
- ✗ It cannot be known without trying: the parity of 13 decides it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S41-B1-Q3 · question page 939

LOGIC.INVARIANT.01 · Strategy: keep track of odd and even only

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Nine gears stand in a row, each touching the next. Gear 1 turns clockwise. What is the number of the last gear in the row that turns clockwise?

Answer: 9

Reasoning: Odd-numbered gears turn clockwise, and 9 is odd.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S41-B1-Q4 · question page 940

LOGIC.INVARIANT.01 · Strategy: keep track of odd and even only

An even number plus an odd number is even.

Answer: False

Reasoning: 2 + 3 = 5. Even + odd is always odd.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S41-B1-Q5 · question page 941

LOGIC.INVARIANT.01 · Strategy: keep track of odd and even only

Which total CANNOT be the sum of three odd numbers?

Answer: C. 10

Reasoning: odd + odd + odd = even + odd = odd, so the total is always odd.

✗ 9: 3 + 3 + 3 = 9
✗ 15: 5 + 5 + 5 = 15
✗ 3: 1 + 1 + 1 = 3

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S41-F1-Q1 · question page 942

LOGIC.INVARIANT.01 · Parity on boards and rings

A token starts on a corner square of a 5 by 5 board. Each move goes to a square sharing a side (going back is allowed). After exactly 7 moves, how many different squares could the token be on?

Answer: 12

Reasoning: Color the board like a checkerboard. Each move changes color, so after 7 moves the token is on the other color from the corner: the 12 such squares. All 12 are within 7 moves.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S41-F1-Q2 · question page 943

LOGIC.INVARIANT.01 · Parity on boards and rings

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Ten gears form a closed ring, each touching the gears on either side. They can all turn.

Answer: True

Reasoning: Alternating around 10 gears ends with gear 10 counterclockwise, opposite to gear 1, which it touches. No conflict.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S41-F1-Q3 · question page 944

LOGIC.INVARIANT.01 · Parity on boards and rings

Seven cups stand upside down. A move turns over exactly 4 cups (any 4). Can all seven cups end right side up?

Answer: A. No: each move changes the number of upside-down cups by an even amount, so it stays odd.

Reasoning: Turning 4 cups changes the upside-down count by -4, -2, 0, 2 or 4. Starting from 7 (odd), it stays odd and can never be 0.

- ✗ Yes: turn over 4 cups, then 3 cups.: a move must turn exactly 4 cups
- ✗ Yes, after exactly 7 moves.: no number of moves works
- ✗ No: 7 is not a multiple of 4.: the true reason is parity; for example 6 cups could all be turned right side up

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S41-F1-Q4 · question page 945

LOGIC.INVARIANT.01 · Parity on boards and rings

Put a + or - sign in front of each of the numbers 1, 2, 3, ..., 10 and add. Give a total from 0 to 10 that can be made this way.

Answer: any values that satisfy: isInteger(t) and t >= 0 and t <= 10 and isOdd(t) (example: t = 1)

Reasoning: 1 + 2 + ... + 10 = 55 is odd. Changing a + to a - lowers the total by an even amount, so every total is odd. Each odd total from 1 to 9 can be made.

Tempting answers the rule rejects: t = 0; t = 4; t = 10

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked: the totals from 0 to 10 that can be made are [1, 3, 5, 7, 9]. Automatic: the example passes the rule and 3 listed mistakes fail it.

S41-F1-Q5 · question page 946

LOGIC.INVARIANT.01 · Parity on boards and rings

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Nine gears are arranged in a 3 by 3 square, and each gear touches the

gears directly beside, above and below it. A corner gear turns clockwise. How many gears turn clockwise, including the corner gear?

Answer: 5

Reasoning: Color like a checkerboard: the corners and center share the corner's color (5 gears) and turn clockwise; the other 4 turn counterclockwise.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S41-F2-Q1 · question page 947

LOGIC.INVARIANT.01 · Prove impossibility with parity

A token must visit every square of a 5 by 5 board exactly once, each move going to a square that shares a side. It starts on the square next to a corner (row 1, column 2). Is this possible?

Answer: D. No: the path must start and end on the corner color, and this square has the other color.

Reasoning: With checkerboard colors there are 13 corner-colored and 12 other squares. A path alternates colors, so a 25-square path must have 13 of one color at positions 1, 3, ..., 25: it starts on the corner color. A full search confirms no path starts at row 1, column 2.

- ✗ Yes: go along row 1, then snake back and forth.: that path misses the corner square or repeats one
- ✗ Yes, but only if it ends on a corner.: no such path exists from this square at all
- ✗ No: 25 is not a multiple of 5.: 25 = 5 x 5; the count is not the problem

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S41-F2-Q2 · question page 948

LOGIC.INVARIANT.01 · Prove impossibility with parity

Put a + or - sign in front of each of the numbers 1, 2, 3, ..., 10 and add. How many different totals can be made?

Answer: 56

Reasoning: Totals are odd, from -55 to 55. Every odd number in that range can be made: 56 totals.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S41-F2-Q3 · question page 949

LOGIC.INVARIANT.01 · Prove impossibility with parity

On a square grid, a path that moves one square up, down, left or right at each step, and ends where it started, always has an even number of steps.

Answer: True

Reasoning: Each step changes the checkerboard color. To return to the starting color, the number of changes must be even.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S41-F2-Q4 · question page 950

LOGIC.INVARIANT.01 · Prove impossibility with parity

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Give a number of gears n from 3 to 12 for which a closed ring of n gears, each touching its two neighbors, would jam.

Answer: any values that satisfy: isInteger(n) and n >= 3 and n <= 12 and

isOdd(n) (example: n = 5)

Reasoning: A ring jams exactly when the number of gears is odd: 3, 5, 7, 9 or 11.

Tempting answers the rule rejects: n = 4; n = 12; n = 13

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 5 valid answers.

S41-F2-Q5 · question page 951

LOGIC.INVARIANT.01 · Prove impossibility with parity

A bug walks along the edges of a cube from corner to corner. It starts at one corner and makes exactly 5 moves (going back is allowed). How many of the 8 corners could it end on?

Answer: 4

Reasoning: Color the corners in two colors so every edge joins different colors (4 and 4). After an odd number of moves the bug is on the other color: 4 corners, and each can be reached.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S42-M-Q1 · question page 953

LOGIC.PROOF.04 · Find a counterexample that disproves a general mathematical claim

A counterexample to a claim about every case is one case where the claim is false. Claim: for every whole number n, n squared + n + 11 is prime. Which n is a counterexample?

Answer: B. 10

Reasoning: 100 + 10 + 11 = 121 = 11 x 11, which is not prime.

✗ 1: 1 + 1 + 11 = 13, which is prime

✗ 2: 4 + 2 + 11 = 17, which is prime

✗ 3: 9 + 3 + 11 = 23, which is prime

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S42-M-Q2 · question page 954

LOGIC.PROOF.04 · Find a counterexample that disproves a general mathematical claim

A counterexample to a claim about every case is one case where the claim is false. Claim: for all numbers a and b, if a > b, then a squared > b squared. Give a counterexample.

Answer: any values that satisfy: a > b and a^2 <= b^2 (example: a = 1, b = -2)

Reasoning: For example a = 1, b = -2: 1 > -2, but 1 is not greater than 4. A negative b works.

Tempting answers the rule rejects: a = 3, b = 2; a = 2, b = 3; a = 5, b = 0

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S42-M-Q3 · question page 955

LOGIC.PROOF.04 · Find a counterexample that disproves a general mathematical claim

A counterexample to a claim about every case is one case where the claim is false. The claim "every odd number greater than 1 is prime" is disproved by the number 9.

Answer: True

Reasoning: 9 is odd and greater than 1, but 9 = 3 x 3.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S42-M-Q4 · question page 956

LOGIC.PROOF.04 · Find a counterexample that disproves a general mathematical claim

A counterexample to a claim about every case is one case where the claim is false. Claim: 2 to the power n, plus 1, is prime for every whole number n from 1 to 6. What is the smallest n that disproves it?

Answer: 3

Reasoning: n = 1: 3, prime. n = 2: 5, prime. n = 3: 9 = 3 x 3, not prime.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S42-M-Q5 · question page 957

LOGIC.PROOF.04 · Find a counterexample that disproves a general mathematical claim

A counterexample to a claim about every case is one case where the claim is false. Claim: every multiple of 4 is a multiple of 8. Which number is NOT a counterexample?

Answer: A. 16

Reasoning: 16 is a multiple of 8, so the claim holds for 16.

- ✗ 4: 4 is a multiple of 4 but not of 8, so it is a counterexample
- ✗ 12: 12 is a multiple of 4 but not of 8, so it is a counterexample
- ✗ 20: 20 is a multiple of 4 but not of 8, so it is a counterexample

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S42-B1-Q1 · question page 958

LOGIC.PROOF.04 · What a counterexample is

What is a counterexample to the claim "every number with property P has property Q"?

Answer: D. A number with property P that does not have property Q

Reasoning: The claim fails exactly when P holds and Q does not.

- ✗ A number with property Q that does not have property P: that does not test the claim
- ✗ A number with neither property: the claim says nothing about numbers without P
- ✗ A number with both properties: that agrees with the claim

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S42-B1-Q2 · question page 959

LOGIC.PROOF.04 · What a counterexample is

One counterexample is enough to show that a claim about every case is false.

Answer: True

Reasoning: The claim said every case works; one failing case shows it does not.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S42-B1-Q3 · question page 960

LOGIC.PROOF.04 · What a counterexample is

If a claim works for five examples, it is proved true for every case.

Answer: False

Reasoning: A later case could still fail.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S42-B1-Q4 · question page 961

LOGIC.PROOF.04 · What a counterexample is

A counterexample to a claim about every case is one case where the claim is false. Which number is a counterexample to "all prime numbers are odd"?

Answer: C. 2

Reasoning: 2 is prime and even.

- ✗ 9: 9 is odd and not prime, so it does not test the claim
- ✗ 3: 3 is prime and odd, so it agrees with the claim
- ✗ 4: 4 is not prime

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S42-B1-Q5 · question page 962

LOGIC.PROOF.04 · What a counterexample is

A counterexample to a claim about every case is one case where the claim is false. Claim: every whole number divisible by 3 is divisible by 6. Give a counterexample from 1 to 30.

Answer: any values that satisfy: isInteger(n) and n >= 1 and n <= 30 and mod(n, 3) == 0 and mod(n, 6) != 0 (example: n = 9)

Reasoning: Any odd multiple of 3 works: 3, 9, 15, 21, 27.

Tempting answers the rule rejects: n = 6; n = 10; n = 12

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the

stated domain: 5 valid answers.

S42-F1-Q1 · question page 963

LOGIC.PROOF.04 · Counterexamples to algebra rules

A counterexample to a claim about every case is one case where the claim is false. Claim: (a + b) squared = a squared + b squared for all numbers a and b. Give whole numbers a and b that disprove it.

Answer: any values that satisfy: (a + b)^2 != a^2 + b^2 (example: a = 1, b = 1)

Reasoning: For a = 1, b = 1: 4 is not 2. The claim fails whenever a and b are both nonzero.

Tempting answers the rule rejects: a = 0, b = 5; a = 3, b = 0

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it.

S42-F1-Q2 · question page 964

LOGIC.PROOF.04 · Counterexamples to algebra rules

A counterexample to a claim about every case is one case where the claim is false. Claim: the square root of (a + b) equals the square root of a plus the square root of b, for all numbers a and b that are 0 or more. Which pair is a counterexample?

Answer: B. a = 9, b = 16

Reasoning: Square root of 25 is 5, but 3 + 4 = 7.

- ✗ a = 0, b = 4: square root of 4 = 0 + 2, so the claim holds here
- ✗ a = 0, b = 0: 0 = 0 + 0, so the claim holds here
- ✗ a = 1, b = 0: 1 = 1 + 0, so the claim holds here

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the

check function accepts only the keyed choice.

S42-F1-Q3 · question page 965

LOGIC.PROOF.04 · Counterexamples to algebra rules

A counterexample to a claim about every case is one case where the claim is false. Claim: n factorial plus 1 is prime for every positive whole number n. (n factorial is 1 x 2 x ... x n.) What is the smallest n that disproves it?

Answer: 4

Reasoning: 1! + 1 = 2, 2! + 1 = 3, 3! + 1 = 7 are prime; 4! + 1 = 25 = 5 x 5 is not.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S42-F1-Q4 · question page 966

LOGIC.PROOF.04 · Counterexamples to algebra rules

A counterexample to a claim about every case is one case where the claim is false. The claim "if n squared is a multiple of 4, then n is a multiple of 4" is disproved by n = 2.

Answer: True

Reasoning: 2 squared = 4 is a multiple of 4, but 2 is not.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S42-F1-Q5 · question page 967

LOGIC.PROOF.04 · Counterexamples to algebra rules

A counterexample to a claim about every case is one case where the claim is false. Claim: every rectangle with perimeter 20 has area 25. Which rectangle is a counterexample?

Answer: D. 4 by 6

Reasoning: A 4 by 6 rectangle has perimeter 20 but area 24.

- ✗ 5 by 5: perimeter 20 and area 25, so it agrees with the claim
- ✗ 2 by 10: its perimeter is 24, so the claim says nothing about it
- ✗ 3 by 8: its perimeter is 22, so the claim says nothing about it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S42-F2-Q1 · question page 968

LOGIC.PROOF.04 · Counterexamples that take a search

A counterexample to a claim about every case is one case where the claim is false. Claim: for whole numbers a, b and c, if a divides b x c, then a divides b or a divides c. Give a counterexample with a, b and c each from 2 to 12.

Answer: any values that satisfy: isInteger(a) and isInteger(b) and isInteger(c) and a >= 2 and a <= 12 and b >= 2 and b <= 12 and c >= 2 and c <= 12 and mod(b * c, a) == 0 and mod(b, a) != 0 and mod(c, a) != 0 (example: a = 4, b = 2, c = 6)

Reasoning: For example a = 4, b = 2, c = 6: 4 divides 12, but 4 divides neither 2 nor 6. (The claim is true when a is prime.)

Tempting answers the rule rejects: a = 3, b = 3, c = 5; a = 5, b = 2, c = 3

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it. Search of the stated domain: 77 valid answers.

S42-F2-Q2 · question page 969

LOGIC.PROOF.04 · Counterexamples that take a search

A counterexample to a claim about every case is one case where the claim is false. Claim: 2 to the power n, minus 1, is prime whenever n is prime. The smallest counterexample is n = 11.

Answer: True

Reasoning: n = 2, 3, 5, 7 give 3, 7, 31, 127, all prime. n = 11 gives 2047 = 23 x 89.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S42-F2-Q3 · question page 970

LOGIC.PROOF.04 · Counterexamples that take a search

Which claim is FALSE?

Answer: C. If n is prime, then n + 2 is prime.

Reasoning: Counterexample: 7 is prime, but 9 is not.

- ✗ If n is odd, then n squared is odd.: true: (2k + 1) squared = 2(2k squared + 2k) + 1
- ✗ The sum of two even numbers is even.: true: 2a + 2b = 2(a + b)
- ✗ If n squared is odd, then n is odd.: true: if n were even, n squared would be even

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S42-F2-Q4 · question page 971

LOGIC.PROOF.04 · Counterexamples that take a search

A counterexample to a claim about every case is one case where the claim is false. Claim: every whole number from 1 to 30 is a sum of two square numbers (0, 1, 4, 9, 16, 25, ...; a square may be used twice). What is the smallest counterexample?

Answer: 3

Reasoning: 1 = 1 + 0, 2 = 1 + 1, but 3 cannot be made: the squares up to 3 are 0 and 1.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S42-F2-Q5 · question page 972

LOGIC.PROOF.04 · Counterexamples that take a search

A counterexample to a claim about every case is one case where the claim is false. Claim: for every positive whole number n, 2 to the power n is greater than n. Why is n = -1 NOT a valid counterexample?

Answer: A. The claim is only about positive whole numbers, and -1 is not one.

Reasoning: A counterexample must be a case the claim is about.

- ✗ Because 2 to the power -1 is greater than -1.: that is true, but it is not why -1 cannot count
- ✗ Because a counterexample must be a large number.: any case covered by the claim may serve
- ✗ Because counterexamples cannot be negative.: they can be, when the claim covers negative numbers

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S43-M-Q1 · question page 974

LOGIC.PROOF.05 · Order and justify the steps of proofs about angles formed by intersecting and parallel lines

Lines meet at a point, forming angle 1 and angle 2 as vertical angles, and angle 3 next to both (angle 3 forms a linear pair with angle 1 and with angle 2). Put the steps of the proof that angle 1 = angle 2 in a correct order. (Two orders are correct.)

Answer: angle 1 + angle 3 = 180 degrees (linear pair) → angle 2 + angle 3 = 180 degrees (linear pair) → angle 1 + angle 3 = angle 2 + angle 3 (both sums equal 180 degrees) → angle 1 = angle 2 (subtract angle 3

from both sides)

Also accepted: angle 2 + angle 3 = 180 degrees (linear pair) → angle 1 + angle 3 = 180 degrees (linear pair) → angle 1 + angle 3 = angle 2 + angle 3 (both sums equal 180 degrees) → angle 1 = angle 2 (subtract angle 3 from both sides)

Reasoning: Both linear pairs come first (in either order), then substitution, then subtraction.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 2 valid.

S43-M-Q2 · question page 975

LOGIC.PROOF.05 · Order and justify the steps of proofs about angles formed by intersecting and parallel lines

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Which pair of angles are alternate interior angles?

Answer: B. Angles 4 and 5

Reasoning: Angles 4 and 5 lie between p and q, on opposite sides of t.

- ✗ Angles 1 and 5: they are corresponding angles: same position at each line
- ✗ Angles 2 and 3: they are vertical angles at line p
- ✗ Angles 4 and 8: they are corresponding angles: same position at each line

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S43-M-Q3 · question page 976

LOGIC.PROOF.05 · Order and justify the steps of proofs about angles formed by

intersecting and parallel lines

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Lines p and q are parallel. If angle 2 = 70 degrees, then angle 7 = 70 degrees.

Answer: True

Reasoning: Angle 2 = angle 3 (vertical), angle 3 = angle 6 (alternate interior, p parallel to q), and angle 6 = angle 7 (vertical). So angle 7 = 70 degrees.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S43-M-Q4 · question page 977

LOGIC.PROOF.05 · Order and justify the steps of proofs about angles formed by intersecting and parallel lines

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Lines p and q are parallel, and angle 1 = 115 degrees. Give the measure of angle 6 in degrees.

Answer: any values that satisfy: x == 65 (example: x = 65)

Reasoning: Angle 1 = angle 5 (corresponding), and angle 5 + angle 6 = 180 degrees (linear pair), so angle 6 = 65 degrees.

Tempting answers the rule rejects: x = 115; x = 75; x = 245

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S43-M-Q5 · question page 978

LOGIC.PROOF.05 · Order and justify the steps of proofs about angles formed by

intersecting and parallel lines

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Put the steps of the proof that angle 3 = angle 6 when p is parallel to q in a correct order. (Two orders are correct.)

Answer: p is parallel to q, cut by transversal t (given) → angle 2 = angle 6 (corresponding angles, p parallel to q) → angle 2 = angle 3 (vertical angles) → angle 3 = angle 6 (both equal angle 2)

Also accepted: p is parallel to q, cut by transversal t (given) → angle 2 = angle 3 (vertical angles) → angle 2 = angle 6 (corresponding angles, p parallel to q) → angle 3 = angle 6 (both equal angle 2)

Reasoning: Start with the given. The corresponding-angles step and the vertical-angles step can come in either order; the conclusion comes last.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 2 valid.

S43-B1-Q1 · question page 979

LOGIC.PROOF.05 · Name the angle pairs

Two angles form a linear pair (together they make a straight line). One measures 48 degrees. What does the other measure, in degrees?

Answer: 132

Reasoning: 180 - 48 = 132.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S43-B1-Q2 · question page 980

LOGIC.PROOF.05 · Name the angle pairs

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Angles 1 and 4 are vertical angles.

Answer: True

Reasoning: They are opposite each other at the same crossing.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S43-B1-Q3 · question page 981

LOGIC.PROOF.05 · Name the angle pairs

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). What kind of angle pair are angles 3 and 7?

Answer: C. Corresponding angles

Reasoning: Both are in the lower-left position at their crossings.

- ✗ Vertical angles: vertical angles are at the same crossing
- ✗ Alternate interior angles: angle 7 is below q, outside the two lines
- ✗ A linear pair: a linear pair is at one crossing and shares a side

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S43-B1-Q4 · question page 982

LOGIC.PROOF.05 · Name the angle pairs

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Angles 3 and 5 are alternate interior angles.

Answer: False

Reasoning: Both are on the left of t; they are same-side interior angles. The alternate interior pairs are 3 and 6, and 4 and 5.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S43-B1-Q5 · question page 983

LOGIC.PROOF.05 · Name the angle pairs

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Angle 3 = 2x + 10 degrees and angle 2 = 3x - 15 degrees. They are vertical angles. Find x.

Answer: any values that satisfy: x == 25 (example: x = 25)

Reasoning: Vertical angles are equal: 2x + 10 = 3x - 15, so x = 25.

Tempting answers the rule rejects: x = 5; x = 37

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it.

S43-F1-Q1 · question page 984

LOGIC.PROOF.05 · Perpendicular bisectors and converses

M is the midpoint of segment AB, and line PM is perpendicular to AB, so P is on the perpendicular bisector of AB. Put the steps of the proof that PA = PB in a correct order. (Two orders are correct.)

Answer: AM = BM, and angles PMA and PMB are right angles (midpoint; perpendicular) → PM = PM (a segment equals itself) → Triangle PMA is congruent to triangle PMB (SAS) → PA = PB (corresponding parts of congruent triangles)

Also accepted: PM = PM (a segment equals itself) → AM = BM, and angles PMA and PMB are right angles (midpoint; perpendicular) →

Triangle PMA is congruent to triangle PMB (SAS) → PA = PB (corresponding parts of congruent triangles)

Reasoning: The two side facts and the angle fact come first, in either order; then SAS, then the conclusion.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 2 valid.

S43-F1-Q2 · question page 985

LOGIC.PROOF.05 · Perpendicular bisectors and converses

Theorem: a point is on the perpendicular bisector of AB exactly when it is the same distance from A and B. Point P has PA = PB = 7. What follows?

Answer: D. P is on the perpendicular bisector of AB.

Reasoning: The "exactly when" part means equal distances put P on the perpendicular bisector.

- ✗ P is the midpoint of AB.: P need not be on segment AB at all
- ✗ AB = 7.: the distances to A and B do not fix the length of AB
- ✗ Nothing follows without more information.: the theorem works in both directions

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S43-F1-Q3 · question page 986

LOGIC.PROOF.05 · Perpendicular bisectors and converses

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Lines p and q are parallel. Then angle 4 + angle 6 = 180 degrees.

Answer: True

Reasoning: Angle 4 = angle 5 (alternate interior), and angle 5 + angle 6 = 180 degrees (linear pair).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S43-F1-Q4 · question page 987

LOGIC.PROOF.05 · Perpendicular bisectors and converses

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Lines p and q are parallel. Angle 4 = 3x + 5 degrees and angle 5 = 5x - 35 degrees. Find x.

Answer: any values that satisfy: x == 20 (example: x = 20)

Reasoning: Alternate interior angles are equal: 3x + 5 = 5x - 35, so x = 20 (both angles are 65 degrees).

Tempting answers the rule rejects: x = 15; x = 65

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it.

S43-F1-Q5 · question page 988

LOGIC.PROOF.05 · Perpendicular bisectors and converses

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Here p and q are NOT known to be parallel. Which fact would prove that they are parallel?

Answer: A. angle 3 = angle 6

Reasoning: If alternate interior angles are equal, the lines are parallel (the converse theorem).

✗ angle 1 = angle 4: vertical angles are always equal, parallel or not

✗ angle 1 + angle 2 = 180 degrees: a linear pair always adds to 180 degrees

✗ angle 5 = angle 8: vertical angles are always equal, parallel or not

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S43-F2-Q1 · question page 989

LOGIC.PROOF.05 · Complete and check angle proofs

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Put the steps of the proof that angle 4 + angle 6 = 180 degrees when p is parallel to q in a correct order. (Two orders are correct.)

Answer: angle 4 = angle 5 (alternate interior angles, p parallel to q) → angle 5 + angle 6 = 180 degrees (linear pair) → angle 4 + angle 6 = 180 degrees (substitute angle 4 for angle 5)

Also accepted: angle 5 + angle 6 = 180 degrees (linear pair) → angle 4 = angle 5 (alternate interior angles, p parallel to q) → angle 4 + angle 6 = 180 degrees (substitute angle 4 for angle 5)

Reasoning: A and B in either order, then the substitution.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 6 orderings of 3 items enumerated; 2 valid.

S43-F2-Q2 · question page 990

LOGIC.PROOF.05 · Complete and check angle proofs

A student writes: "Angle 1 = angle 4 because they look the same size in the diagram." What is wrong with this step?

Answer: A. A diagram is not a reason; the step needs the vertical angles theorem.

Reasoning: Each step of a proof must cite a definition, given or theorem.

- ✗ Angles 1 and 4 are never equal.: vertical angles are always equal
- ✗ The step should say angle 1 + angle 4 = 180 degrees.: that is false for vertical angles in general
- ✗ Nothing; looking at the diagram is enough.: diagrams can be drawn inaccurately

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S43-F2-Q3 · question page 991

LOGIC.PROOF.05 · Complete and check angle proofs

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Lines p and q are parallel. Angle 1 = 2x + 30 degrees and angle 6 = x + 15 degrees. Find x.

Answer: any values that satisfy: x == 45 (example: x = 45)

Reasoning: Angle 1 = angle 5 (corresponding), and angle 5 + angle 6 = 180. So 2x + 30 + x + 15 = 180, 3x = 135, x = 45.

Tempting answers the rule rejects: x = 15; x = -15

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it.

S43-F2-Q4 · question page 992

LOGIC.PROOF.05 · Complete and check angle proofs

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Without knowing that p and q are parallel, one can conclude that angle 3 = angle 6.

Answer: False

Reasoning: Alternate interior angles are equal only when the lines are parallel. If p is tilted, angle 3 and angle 6 differ.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S43-F2-Q5 · question page 993

LOGIC.PROOF.05 · Complete and check angle proofs

P is on the perpendicular bisector of segment AB. PA = 3x - 4 and PB = x + 10. What is PA?

Answer: 17

Reasoning: Points on the perpendicular bisector are equidistant from A and B: 3x - 4 = x + 10, so x = 7 and PA = 17.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S44-M-Q1 · question page 995

LOGIC.INVARIANT.02 · Find a quantity that never changes under the allowed moves and use it to prove a target is impossible

The numbers 1, 2, 3, ..., 10 are written on a board. A move erases any two numbers and writes their sum. After 9 moves one number is left. What is it?

Answer: B. 55

Reasoning: Erasing a and b and writing a + b does not change the total. The total starts at 55, so the last number is 55.

- ✗ 10: the moves combine numbers; they do not count them
- ✗ 45: that is the sum of 1 to 9; the 10 is still there
- ✗ It depends on the moves chosen: every move keeps the total the

same

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S44-M-Q2 · question page 996

LOGIC.INVARIANT.02 · Find a quantity that never changes under the allowed moves and use it to prove a target is impossible

The numbers 1, 2, ..., 10 are written on a board. A move erases two numbers and writes their difference (larger minus smaller). After 9 moves one number is left. It can be 0.

Answer: False

Reasoning: Replacing a and b by a - b changes the total by 2b, an even amount, so the total stays odd (it starts at 55). The last number is odd, so it cannot be 0.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S44-M-Q3 · question page 997

LOGIC.INVARIANT.02 · Find a quantity that never changes under the allowed moves and use it to prove a target is impossible

Seven coins lie in a row: H H T H T T T (H = heads, T = tails). A move turns over any two coins that are next to each other. What is the smallest number of heads that can ever be showing?

Answer: 1

Reasoning: A move changes the number of heads by -2, 0 or +2, so it stays odd (it starts at 3). It can reach 1 but never 0.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S44-M-Q4 · question page 998

LOGIC.INVARIANT.02 · Find a quantity that never changes under the allowed moves and use it to prove a target is impossible

Seven coins lie in a row: H H T H T T T. A move turns over any two coins that are next to each other. Give a number of heads (0 to 7) that can be showing after some moves.

Answer: any values that satisfy: isInteger(h) and h >= 0 and h <= 7 and isOdd(h) (example: h = 5)

Reasoning: The number of heads stays odd: 1, 3, 5 or 7 are possible, and all four can be reached.

Tempting answers the rule rejects: h = 0; h = 4; h = 2

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Breadth-first search over all reachable rows: head counts [1, 3, 5, 7]. Automatic: the example passes the rule and 3 listed mistakes fail it.

S44-M-Q5 · question page 999

LOGIC.INVARIANT.02 · Find a quantity that never changes under the allowed moves and use it to prove a target is impossible

A robot starts at the point (0, 0). Each move takes it from (x, y) to (x + 1, y + 2) or to (x + 2, y + 1). Which point can it reach?

Answer: C. (4, 5)

Reasoning: x + y is always a multiple of 3. (4, 5): two moves of (1, 2) and one of (2, 1).

- ✗ (5, 5): each move adds 3 to x + y, but 5 + 5 = 10 is not a multiple of 3
- ✗ (2, 3): 2 + 3 = 5 is not a multiple of 3
- ✗ (7, 0): 7 + 0 = 7 is not a multiple of 3, and y can never stay 0

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S44-B1-Q1 · question page 1000

LOGIC.INVARIANT.02 · Strategy: look for something that never changes

A counter shows 12. A move adds 4 or subtracts 6. Can it ever show 15?

Answer: D. No: every move keeps the number even, and 15 is odd.

Reasoning: Adding 4 or subtracting 6 keeps an even number even.

- ✗ Yes: add 4, then subtract 1.: subtracting 1 is not an allowed move
- ✗ Yes: after enough moves.: no number of moves can change even to odd
- ✗ No: 15 is not a multiple of 4.: 12 + 4 + 4 - 6 = 14 is reachable and is not a multiple of 4 either; the true reason is parity

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S44-B1-Q2 · question page 1001

LOGIC.INVARIANT.02 · Strategy: look for something that never changes

Three numbers on a board are 2, 4 and 9. A move adds 2 to any one of them. The total of the three numbers always stays odd.

Answer: True

Reasoning: The total starts at 15 (odd), and each move adds 2, which keeps it odd.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S44-B1-Q3 · question page 1002

LOGIC.INVARIANT.02 · Strategy: look for something that never changes

A move erases two numbers a and b on a board and writes a + b. The board starts with 3, 5, 6 and 8. What number is left after 3 moves?

Answer: 22

Reasoning: The total never changes: 3 + 5 + 6 + 8 = 22.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S44-B1-Q4 · question page 1003

LOGIC.INVARIANT.02 · Strategy: look for something that never changes

If a quantity is the same after every allowed move, it can never change, however many moves are made.

Answer: True

Reasoning: Each move leaves it unchanged, so a chain of moves does too.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S44-B1-Q5 · question page 1004

LOGIC.INVARIANT.02 · Strategy: look for something that never changes

A robot starts at (0, 0). Each move goes from (x, y) to (x + 2, y) or to (x, y + 2). Which point can it NEVER reach?

Answer: B. (3, 4)

Reasoning: Both coordinates always stay even, and 3 is odd.

✗ (2, 4): reached by (2, 0), then (2, 2), then (2, 4)

✗ (0, 6): three moves of (0, 2)

✗ (4, 0): two moves of (2, 0)

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S44-F1-Q1 · question page 1005

LOGIC.INVARIANT.02 · Invariants with remainders

The numbers 1, 2, ..., 10 are on a board. A move erases two numbers a and b and writes a + b + 1. After 9 moves one number is left. What is it?

Answer: 64

Reasoning: Each move raises the total by exactly 1. The total starts at 55, and 9 moves add 9: 64.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S44-F1-Q2 · question page 1006

LOGIC.INVARIANT.02 · Invariants with remainders

A robot starts at (0, 0). Each move goes from (x, y) to (x + 1, y + 2) or to (x + 2, y + 1). It can reach (10, 10).

Answer: False

Reasoning: x + y must be a multiple of 3, and 20 is not.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S44-F1-Q3 · question page 1007

LOGIC.INVARIANT.02 · Invariants with remainders

Three numbers are on a board: 1, 2, 3. A move picks two of them, adds 1 to one and subtracts 1 from the other. Which quantity never changes?

Answer: C. The total of the three numbers

Reasoning: The move adds 1 and subtracts 1, so the total is always 6.

- ✗ The largest number: adding 1 to the largest number changes it
- ✗ The product of the three numbers: 1, 2, 3 can become 0, 3, 3, whose product is 0
- ✗ The number of odd numbers: 1, 2, 3 can become 2, 2, 2, which has no odd numbers

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S44-F1-Q4 · question page 1008

LOGIC.INVARIANT.02 · Invariants with remainders

Ten coins all show heads. A move turns over any four coins (they need not be next to each other). Give a number of heads from 0 to 10 that can NEVER be showing.

Answer: any values that satisfy: isInteger(h) and h >= 0 and h <= 10 and isOdd(h) (example: h = 7)

Reasoning: If k of the four turned coins were heads, the heads count changes by (4 - k) - k = 4 - 2k, an even number. It starts at 10, so it stays even: 1, 3, 5, 7 and 9 are impossible.

Tempting answers the rule rejects: h = 0; h = 6; h = 10

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Search of all reachable head counts: [0, 2, 4, 6, 8, 10]. Automatic: the example passes the rule and 3 listed mistakes fail it.

S44-F1-Q5 · question page 1009

LOGIC.INVARIANT.02 · Invariants with remainders

A counter shows 100. A move subtracts 7 or adds 3. What is the smallest number of moves that makes it show 0?

Answer: 20

Reasoning: If a moves subtract 7 and b moves add 3, then 7a - 3b = 100. So 7a is at least 100 and 7a - 100 is a multiple of 3: a = 16 gives 112 - 100 = 12, so b = 4. Total 16 + 4 = 20, and any other solution has larger a and b.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S44-F2-Q1 · question page 1010

LOGIC.INVARIANT.02 · Prove targets impossible

The numbers 1, 2, 3, 4 and 5 are on a board. A move erases two numbers a and b and writes a x b + a + b. After 4 moves one number is left. What is it?

Answer: 719

Reasoning: a x b + a + b = (a + 1)(b + 1) - 1. So the product of (each number + 1) never changes: 2 x 3 x 4 x 5 x 6 = 720. The last number n has n + 1 = 720, so n = 719.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S44-F2-Q2 · question page 1011

LOGIC.INVARIANT.02 · Prove targets impossible

A 4 by 4 grid of lights starts all off. A move switches every light in one whole row, or every light in one whole column. Some sequence of moves leaves exactly one light on.

Answer: False

Reasoning: Look at any 2 by 2 block of lights. A row or column move switches either none or exactly 2 of its 4 lights, so the number of lit lights in the block stays even. With exactly one light on, a block containing it would have 1 lit light, which is odd.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S44-F2-Q3 · question page 1012

LOGIC.INVARIANT.02 · Prove targets impossible

A robot starts at (0, 0). Each move goes from (x, y) to (x + 3, y), to (x, y + 3), or to (x + 1, y + 1). Which point can it reach?

Answer: D. (5, 2)

Reasoning: x - y is always a multiple of 3. (5, 2): two moves of (1, 1) and one of (3, 0).

- ✗ (4, 0): x - y changes only by 3, -3 or 0, so it stays a multiple of 3; here x - y = 4
- ✗ (2, 3): here x - y = -1, not a multiple of 3
- ✗ (1, 5): here x - y = -4, not a multiple of 3

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S44-F2-Q4 · question page 1013

LOGIC.INVARIANT.02 · Prove targets impossible

A 4 by 4 grid of lights starts all off. A move switches every light in one whole row, or every

light in one whole column. Give a number of lit lights from 1 to 15 that some sequence of moves can leave.

Answer: any values that satisfy: h == 4 or h == 6 or h == 8 or h == 10 or h == 12 (example: h = 6)

Reasoning: Only the rows and columns switched an odd number of times matter. With r such rows and c such columns, the lit count is 4r + 4c - 2rc. For r and c from 0 to 4 this gives 0, 4, 6, 8, 10, 12 or 16.

Tempting answers the rule rejects: h = 1; h = 2; h = 5; h = 14

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Breadth-first search of all reachable patterns: lit counts [0, 4, 6, 8, 10, 12, 16]. Automatic: the example passes the rule and 4 listed mistakes fail it.

S44-F2-Q5 · question page 1014

LOGIC.INVARIANT.02 · Prove targets impossible

The numbers 1, 2, 3, ..., 10 are on a board. A move erases two numbers a and b and writes a + b + 1. The last number left does not depend on which moves are made.

Answer: True

Reasoning: Each move raises the total by exactly 1, and there are always exactly 9 moves, so the last number is 55 + 9 = 64.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S45-M-Q1 · question page 1016

LOGIC.GRID.02 · Fill and analyze letter grids under stated rules and a supplied word list

Rule: each row and each column must contain each of the letters A, B, C and D exactly once. The grid in the figure has six letters filled in. Which letter belongs in the shaded cell (bottom row, first column)?

Answer: D

Reasoning: Column 1 already has A (row 1) and C (row 2), so rows 3 and 4 start with B and D in some order. Row 3 already has D (in column 4), so row 3 starts with B and row 4 starts with D. (The whole grid is ADBC, CADB, BCAD, DBCA.)

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S45-M-Q2 · question page 1017

LOGIC.GRID.02 · Fill and analyze letter grids under stated rules and a supplied word list

Word list (the ONLY words allowed): CAT, ARE, TEN, ART, EAR, NET, TAN, ATE, RAT, ERA, TEA, ACE, ONE, NAP, APE, PEN, OAR, ORE, ROE, TOE, TON, NOT, ANT. In a word square, the three rows, read left to right, are the same words as the three columns, read top to bottom, in the same order. The first row of a word square is TEN. Give the second and third rows.

Answer: any values that satisfy: w2 == 'era' and w3 == 'nap' (example: w2 = ERA, w3 = NAP)

Reasoning: Column 1 must be TEN, so row 2 starts with E and row 3 with N. Row 2 = column 2, which starts with E: ERA fits (E, R, A). Then column 3 is N, A, ?, and row 3 = NA?: NAP. Check: columns TEN, ERA, NAP.

Tempting answers the rule rejects: w2 = EAR, w3 = NAP; w2 = ERA, w3 = NET

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Search of every pair of listed words: 1 word square(s) start with TEN. Automatic: the example passes the rule and 2 listed mistakes fail it.

S45-M-Q3 · question page 1018

LOGIC.GRID.02 · Fill and analyze letter grids under stated rules and a supplied word list

Rule: a 3 by 3 grid is filled with the letters A and B so that no two cells that share a side hold the same letter. The center cell holds A. Then all four corner cells hold the same letter.

Answer: True

Reasoning: The center's four side neighbors must be B, and each corner touches two of those B cells, so each corner is A.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S45-M-Q4 · question page 1019

LOGIC.GRID.02 · Fill and analyze letter grids under stated rules and a supplied word list

Rule: each row and each column must contain each of the letters A, B and C exactly once. The top-left cell holds A and the center cell holds A. Which letter must go in the bottom-right cell?

Answer: A. A

Reasoning: Row 3 needs an A, but columns 1 and 2 already have one. So A goes in column 3 of row 3.

- ✗ B: the grid can be completed only with A there
- ✗ C: the grid can be completed only with A there
- ✗ It could be B or C: A is already in rows 1 and 2 and in columns 1 and 2, so A must go in row 3, column 3

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S45-M-Q5 · question page 1020

LOGIC.GRID.02 · Fill and analyze letter grids under stated rules and a supplied word list

Rule: each row and each column must contain each of the letters A, B and C exactly once. The first row is A B C. How many ways are there to complete the grid?

Answer: 2

Reasoning: The second row must avoid the letter above each cell: B C A or C A B. Each choice forces the third row. So 2 ways.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S45-B1-Q1 · question page 1021

LOGIC.GRID.02 · Strategy: fill the most constrained cell first

Rule: each row and each column must contain each of the letters A, B and C exactly once. A row reads A _ C. Which letter fills the gap?

Answer: B

Reasoning: Each row holds A, B and C once, and only B is missing.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S45-B1-Q2 · question page 1022

LOGIC.GRID.02 · Strategy: fill the most constrained cell first

Rule: each row and each column must contain each of the letters A, B and C exactly once. If column 1 holds A and B, its third cell must hold C.

Answer: True

Reasoning: C is the only letter not yet used in that column.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S45-B1-Q3 · question page 1023

LOGIC.GRID.02 · Strategy: fill the most constrained cell first

Rule: each row and each column must contain each of the letters A, B, C and D exactly once. A cell's row already holds A and D, and its column holds B. Which letter must go in the cell?

Answer: D. C

Reasoning: Only C is used by neither the row nor the column.

- ✗ A: A is already in the row
- ✗ B: B is already in the column
- ✗ D: D is already in the row

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S45-B1-Q4 · question page 1024

LOGIC.GRID.02 · Strategy: fill the most constrained cell first

Rule: each row and each column must contain each of the letters A, B and C exactly once. A grid can have the row A B C directly above the row A C B.

Answer: False

Reasoning: Both rows put A in column 1, so column 1 would have A twice.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S45-B1-Q5 · question page 1025

LOGIC.GRID.02 · Strategy: fill the most constrained cell first

Rule: each row and each column must contain each of the letters A, B and C exactly once. Rows 1 and 2 are A B C and C A B. What is row 3? Type the three letters with no spaces.

Answer: BCA

Reasoning: Each column needs its missing letter: column 1 has A, C so B; column 2 has B, A so C; column 3 has C, B so A.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S45-F1-Q1 · question page 1026

LOGIC.GRID.02 · Word squares from a list

Word list (the ONLY words allowed): CAT, ARE, TEN, ART, EAR, NET, TAN, ATE, RAT, ERA, TEA, ACE, ONE, NAP, APE, PEN, OAR, ORE, ROE, TOE, TON, NOT, ANT. In a word square, the three rows, read left to right, are the same words as the three columns, read top to bottom, in the same order. The first row of a word square is ATE. Give the second and third rows.

Answer: any values that satisfy: w2 == 'tea' and w3 == 'ear' (example: w2 = TEA, w3 = EAR)

Reasoning: Row 2 = column 2 starts with T, and row 3 = column 3 starts with E. TEA and EAR fit: columns ATE, TEA, EAR.

Tempting answers the rule rejects: w2 = TEN, w3 = EAR; w2 = TAN, w3 = ERA

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Search: 1 word square(s) start with ATE. Automatic: the example passes the rule and 2 listed mistakes fail it.

S45-F1-Q2 · question page 1027

LOGIC.GRID.02 · Word squares from a list

Word list (the ONLY words allowed): CAT, ARE, TEN, ART, EAR, NET, TAN, ATE, RAT, ERA, TEA, ACE, ONE, NAP, APE, PEN, OAR, ORE, ROE, TOE, TON, NOT, ANT. In a word square, the three rows, read left to right, are the same words as the three columns, read top to bottom, in the same order. How many words on the list can be the first row of at

least one word square?

Answer: 15

Reasoning: A search of every pair of words for each possible first row finds the squares; count the first rows that work.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S45-F1-Q3 · question page 1028

LOGIC.GRID.02 · Word squares from a list

Word list (the ONLY words allowed): CAT, ARE, TEN, ART, EAR, NET, TAN, ATE, RAT, ERA, TEA, ACE, ONE, NAP, APE, PEN, OAR, ORE, ROE, TOE, TON, NOT, ANT. In a word square, the three rows, read left to right, are the same words as the three columns, read top to bottom, in the same order. A word square can start with the row ONE.

Answer: False

Reasoning: Row 2 must start with N (NET, NAP or NOT) and row 3 with E. The middle letter of row 3 must equal the last letter of row 2 (T or P), but no listed word starts with ET or EP.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S45-F1-Q4 · question page 1029

LOGIC.GRID.02 · Word squares from a list

Rule: each row and each column must contain each of the letters A, B, C and D exactly once. The grid in the figure has six letters filled in. Type the complete top row (four letters, no spaces).

Answer: ADBC

Reasoning: Row 2 is C _ D B, so its gap is A. Column 4 needs A in row

4. Continuing cell by cell gives the only completion: ADBC, CADB, BCAD, DBCA.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S45-F1-Q5 · question page 1030

LOGIC.GRID.02 · Word squares from a list

Rule: a 2 by 3 grid is filled with A and B so that no two cells sharing a side hold the same letter. How many such grids are there?

Answer: A. 2

Reasoning: The top-left letter decides the whole checkerboard pattern: 2 grids.

- ✗ 1: the top-left cell can be either letter
- ✗ 4: once the top-left cell is chosen, every other cell is forced
- ✗ 6: that counts cells, not grids

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S45-F2-Q1 · question page 1031

LOGIC.GRID.02 · Prove a grid is impossible or count it

Rule: each row and each column must contain each of the letters A, B, C and D exactly once. How many ways are there to fill the whole grid if the first row is A B C D?

Answer: 24

Reasoning: Count the second rows that avoid the letter above each cell (9 ways), then the completions of each (search): 24 in all.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S45-F2-Q2 · question page 1032

LOGIC.GRID.02 · Prove a grid is impossible or count it

Rule: a 3 by 3 grid is filled with A and B so that no two cells sharing a side hold the same letter. The top-left cell holds A and the top-right cell holds B.

Answer: False

Reasoning: The top row must alternate A, B, A, so its two ends match. A grid with A and B at the two top corners breaks the rule.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S45-F2-Q3 · question page 1033

LOGIC.GRID.02 · Prove a grid is impossible or count it

Rule: each row and each column must contain each of the letters A, B and C exactly once. Which partly filled grid can NOT be completed?

Answer: B. Row 1 starts A B, and row 2 ends in C

Reasoning: Row 1 is A B _, so its last cell must be C. But column 3 already has C in row 2, so no letter fits.

- ✗ Row 1 is A B C: it can be completed in 2 ways
- ✗ Row 1 starts A, and row 2 starts B: for example A B C, B C A, C A B
- ✗ The diagonal from top left to bottom right is A, A, A: for example A B C, C A B, B C A

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S45-F2-Q4 · question page 1034

LOGIC.GRID.02 · Prove a grid is impossible or count it

Word list (the ONLY words allowed): CAT, ARE, TEN, ART, EAR, NET, TAN, ATE, RAT, ERA, TEA, ACE, ONE, NAP, APE, PEN, OAR, ORE, ROE, TOE, TON, NOT, ANT. In a word square, the three rows, read left to right, are the same words as the three columns, read top to bottom, in the same order. Give the second and third rows of the word square whose first row is ROE.

Answer: any values that satisfy: w2 == 'oar' and w3 == 'era' (example: w2 = OAR, w3 = ERA)

Reasoning: Row 2 starts with O and row 3 with E. OAR and ERA give columns ROE, OAR, ERA.

Tempting answers the rule rejects: w2 = ORE, w3 = ERA; w2 = OAR, w3 = EAR

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Search: 1 word square(s) start with ROE. Automatic: the example passes the rule and 2 listed mistakes fail it.

S45-F2-Q5 · question page 1035

LOGIC.GRID.02 · Prove a grid is impossible or count it

Rule: each row and each column must contain each of the letters A, B and C exactly once. How many ways are there to fill an empty 3 by 3 grid?

Answer: 12

Reasoning: The first row can be any of the 6 orders of A, B, C, and each has 2 completions: 12.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S46-M-Q1 · question page 1037

LOGIC.EVIDENCE.01 · Identify the unstated assumption an argument needs and evaluate competing explanations

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "The school's robot passed every test in the lab, so it will work at the competition." Which assumption does this argument need?

Answer: C. Conditions at the competition are like those in the lab.

Reasoning: Lab success supports competition success only if the two settings are alike.

- ✗ The robot was built by students.: who built it does not connect the lab results to the competition
- ✗ The robot failed some tests last year.: that is about the past and weakens nothing here
- ✗ Other teams also tested their robots.: other teams' testing is irrelevant to this robot

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S46-M-Q2 · question page 1038

LOGIC.EVIDENCE.01 · Identify the unstated assumption an argument needs and evaluate competing explanations

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Mia's plant grew faster after she moved it to the window, so sunlight made it grow faster." Which assumption does this argument need?

Answer: A. Nothing else that affects growth changed when the plant was moved.

Reasoning: If the temperature or watering also changed, those could explain the faster growth instead.

- ✗ Plants need water to grow.: true, but it does not link the move to the faster growth

✗ The window faces south.: the direction is not needed for the conclusion

✗ Mia has other plants indoors.: other plants are not part of the argument

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S46-M-Q3 · question page 1039

LOGIC.EVIDENCE.01 · Identify the unstated assumption an argument needs and evaluate competing explanations

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Every member of the chess club is in grade 10. Jo is in grade 10. So Jo is in the chess club." This argument would become valid if we add the assumption "Everyone in grade 10 is in the chess club."

Answer: True

Reasoning: With that added premise, Jo being in grade 10 forces Jo into the club. The original premise points the other way.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S46-M-Q4 · question page 1040

LOGIC.EVIDENCE.01 · Identify the unstated assumption an argument needs and evaluate competing explanations

The cafeteria sold fewer sandwiches this week than last week. Which explanation is best supported if we also learn that the cafeteria was closed on two days this week?

Answer: B. The cafeteria was open on fewer days to sell sandwiches.

Reasoning: The given fact directly explains fewer sales without any other change.

✗ Students stopped liking sandwiches.: nothing given says tastes changed

✗ The sandwich price went up.: nothing given mentions prices

✗ Fewer students attend the school this year.: the comparison is week to week, and enrollment was not mentioned

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S46-M-Q5 · question page 1041

LOGIC.EVIDENCE.01 · Identify the unstated assumption an argument needs and evaluate competing explanations

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "This bridge is safe because it has stood for 80 years." The argument needs the assumption that the bridge will be used the same way as it has been.

Answer: True

Reasoning: If much heavier loads start crossing, 80 years of standing would say little about safety under the new loads.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S46-B1-Q1 · question page 1042

LOGIC.EVIDENCE.01 · Find the gap between premise and conclusion

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Sam is taller than Ana, so Sam is taller than Raj." Which assumption fills the gap?

Answer: D. Ana is at least as tall as Raj.

Reasoning: Sam > Ana and Ana >= Raj give Sam > Raj.

✗ Raj is taller than Ana.: that would make the conclusion less certain, not more

✗ Sam is the tallest in his class.: Raj may not be in Sam's class
✗ Ana and Raj are friends.: friendship says nothing about height

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S46-B1-Q2 · question page 1043

LOGIC.EVIDENCE.01 · Find the gap between premise and conclusion

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. Every true statement about the topic of an argument is an assumption that the argument needs.

Answer: False

Reasoning: A true fact that does not bridge the gap between premise and conclusion is not needed by the argument.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S46-B1-Q3 · question page 1044

LOGIC.EVIDENCE.01 · Find the gap between premise and conclusion

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "The park was crowded on Saturday, so it will be crowded this Saturday." What is the gap?

Answer: C. From one past Saturday to a future Saturday

Reasoning: The argument needs this Saturday to be like the last one.

✗ From the park to the city: the argument never mentions a city
✗ From crowded to empty: the conclusion also says crowded
✗ From Saturday to Sunday: both days in the argument are Saturdays

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S46-B1-Q4 · question page 1045

LOGIC.EVIDENCE.01 · Find the gap between premise and conclusion

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. If an assumption an argument needs turns out to be false, the conclusion no longer follows from the premises.

Answer: True

Reasoning: That is what it means for the argument to need it.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S46-B1-Q5 · question page 1046

LOGIC.EVIDENCE.01 · Find the gap between premise and conclusion

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "All squares are rectangles, so all rectangles are squares" needs no assumption to be valid.

Answer: False

Reasoning: It reverses the premise. It would need the extra premise "all rectangles are squares", which is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S46-F1-Q1 · question page 1047

LOGIC.EVIDENCE.01 · Weaken and strengthen explanations

Claim: "Students who eat breakfast score higher on tests, so eating breakfast raises test scores." Which fact would most WEAKEN the claim?

Answer: D. Students who eat breakfast also sleep more, and sleep is

linked to higher scores.

Reasoning: Another factor (sleep) could explain both, so breakfast might not be the cause.

- ✗ Many students eat breakfast.: how common breakfast is does not affect the link
- ✗ Test scores rose last year.: a general rise does not touch the breakfast link
- ✗ Breakfast foods are cheap.: price does not touch the link between breakfast and scores

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S46-F1-Q2 · question page 1048

LOGIC.EVIDENCE.01 · Weaken and strengthen explanations

Claim: "The new crosswalk signal reduced accidents at Elm Street." Which fact would most STRENGTHEN the claim?

Answer: C. Accidents did not drop at similar streets that did not get the signal.

Reasoning: If accidents fell only where the signal was added, the signal is the likely cause.

- ✗ The signal cost a lot of money.: cost says nothing about the effect
- ✗ Elm Street is near a school.: location alone does not show the signal worked
- ✗ Drivers like the new signal.: liking it does not show accidents fell because of it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S46-F1-Q3 · question page 1049

LOGIC.EVIDENCE.01 · Weaken and strengthen explanations

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "No reptiles have fur. This animal has fur. So this animal is not a reptile." This argument needs an unstated assumption to be valid.

Answer: False

Reasoning: It is already valid: if it were a reptile, it would have no fur, but it has fur.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S46-F1-Q4 · question page 1050

LOGIC.EVIDENCE.01 · Weaken and strengthen explanations

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Our team lost every game we played in the rain. It will rain on Friday. So we will lose Friday's game." Which assumption does the argument need?

Answer: B. The rain caused the losses, rather than something else about those games.

Reasoning: The past pattern predicts Friday only if rain is the reason for the losses.

- ✗ Our team has played in the rain before.: that is already stated in the premise
- ✗ Friday's game is at home.: location is not part of the argument
- ✗ It rains more in the spring.: that does not connect rain to losing

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S46-F1-Q5 · question page 1051

LOGIC.EVIDENCE.01 · Weaken and strengthen explanations

Two explanations both fit the facts. Then no new fact can help decide between them.

Answer: False

Reasoning: A new fact that one explanation predicts and the other does not supports the first over the second.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S46-F2-Q1 · question page 1052

LOGIC.EVIDENCE.01 · Judge explanations with care

A town's ice-cream sales and its number of sunburns rise and fall together during the year. Which explanation is best?

Answer: A. Hot, sunny weather increases both.

Reasoning: A common cause (weather) explains why the two move together.

- ✗ Eating ice cream causes sunburn.: nothing suggests ice cream affects skin
- ✗ Sunburns make people buy ice cream.: the timing fits weather better than this
- ✗ The two numbers match by chance every year.: a yearly pattern is unlikely to be chance alone

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S46-F2-Q2 · question page 1053

LOGIC.EVIDENCE.01 · Judge explanations with care

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Ming read the book, so Ming knows how it ends." Which assumption is needed?

Answer: B. Ming read the book all the way to the end.

Reasoning: Reading part of the book would not reveal the ending.

- ✗ The book has a surprise ending.: the type of ending does not matter
- ✗ Ming liked the book.: liking it is not needed to know the ending
- ✗ The book is long.: length is not needed for the conclusion

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S46-F2-Q3 · question page 1054

LOGIC.EVIDENCE.01 · Judge explanations with care

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Every prize winner practiced daily. Kai practiced daily. So Kai won a prize." Adding "Only prize winners practiced daily" makes the argument valid.

Answer: True

Reasoning: Then anyone who practiced daily is a prize winner, and Kai practiced daily.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S46-F2-Q4 · question page 1055

LOGIC.EVIDENCE.01 · Judge explanations with care

A survey of the chess club finds that 90% of members prefer quiet study rooms. A student concludes that 90% of the whole school prefers quiet study rooms. What assumption does this need?

Answer: C. Chess club members have the same preferences as the whole school.

Reasoning: The conclusion jumps from one group to the school; it needs the group to represent the school.

- ✗ The chess club has more than 10 members.: size alone does not make the club like the school
- ✗ Quiet rooms are better for studying.: that does not link the club to the school
- ✗ The survey asked about study rooms.: that is already part of the information

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S46-F2-Q5 · question page 1056

LOGIC.EVIDENCE.01 · Judge explanations with care

If a needed assumption is shown to be false, the conclusion of the argument must be false.

Answer: False

Reasoning: The argument fails, but the conclusion could still be true for other reasons.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S47-M-Q1 · question page 1058

LOGIC.EVIDENCE.02 · Decide whether the given information determines the answer, and what extra fact would

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) Ana, Ben and Cal ran a race with no ties. Who finished first? Clue (1): Ana finished ahead of Ben. Clue (2): Cal finished last.

Answer: A. Neither clue alone is enough, but both together are.

Reasoning: Clue (1) allows A B C, A C B and C A B (first could be Ana or Cal). Clue (2) allows A B C and B A C. Together only A B C: Ana.

- ✗ Clue (1) alone is enough, but clue (2) alone is not.: check whether clue (1) alone leaves exactly one answer
- ✗ Clue (2) alone is enough, but clue (1) alone is not.: check whether clue (2) alone leaves exactly one answer
- ✗ Each clue alone is enough.: one of the clues alone leaves more than one answer

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S47-M-Q2 · question page 1059

LOGIC.EVIDENCE.02 · Decide whether the given information determines the answer, and what extra fact would

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) x is a whole number from 1 to 20. What is x? Clue (1): x is a multiple of 7 and greater than 10. Clue (2): x is even.

Answer: D. Clue (1) alone is enough, but clue (2) alone is not.

Reasoning: Clue (1) leaves only 14. Clue (2) leaves 10 numbers.

- ✗ Clue (2) alone is enough, but clue (1) alone is not.: check whether clue (2) alone leaves exactly one answer
- ✗ Neither clue alone is enough, but both together are.: check each clue alone before combining them
- ✗ Each clue alone is enough.: one of the clues alone leaves more than one answer

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S47-M-Q3 · question page 1060

LOGIC.EVIDENCE.02 · Decide whether the given information determines the answer, and what extra fact would

Two whole numbers, each from 1 to 9, add up to 17. Knowing only this is enough to know which two numbers they are.

Answer: True

Reasoning: Only 8 + 9 = 17 (in either order).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S47-M-Q4 · question page 1061

LOGIC.EVIDENCE.02 · Decide whether the given information determines the answer, and what extra fact would

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) A rectangle has whole-number side lengths. What is its area? Clue (1): its perimeter is 12. Clue (2): one side is twice as long as the other.

Answer: B. Neither clue alone is enough, but both together are.

Reasoning: Clue (1) allows 1 by 5, 2 by 4, 3 by 3 (areas 5, 8, 9). Clue (2) allows 1 by 2, 2 by 4, and more. Together: 2 by 4, area 8.

- ✗ Clue (1) alone is enough, but clue (2) alone is not.: check whether clue (1) alone leaves exactly one answer
- ✗ Clue (2) alone is enough, but clue (1) alone is not.: check whether clue (2) alone leaves exactly one answer
- ✗ Each clue alone is enough.: one of the clues alone leaves more than one answer

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S47-M-Q5 · question page 1062

LOGIC.EVIDENCE.02 · Decide whether the given information determines the answer, and what extra fact would

x is a whole number from 1 to 50. You know that x is a multiple of 6 and also a multiple of 4. How many values of x are still possible?

Answer: 4

Reasoning: Multiples of both 6 and 4 are multiples of 12: 12, 24, 36, 48. Four values remain, so the clues are not enough to find x.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S47-B1-Q1 · question page 1063

LOGIC.EVIDENCE.02 · Strategy: list what each clue leaves

n is a whole number from 1 to 10, and n is a multiple of 4. How many values of n are possible?

Answer: 2

Reasoning: 4 and 8.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S47-B1-Q2 · question page 1064

LOGIC.EVIDENCE.02 · Strategy: list what each clue leaves

n is a whole number from 1 to 10. Knowing that n is prime and even is enough to find n.

Answer: True

Reasoning: 2 is the only even prime.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S47-B1-Q3 · question page 1065

LOGIC.EVIDENCE.02 · Strategy: list what each clue leaves

Two whole numbers add to 10. Knowing only this is enough to find the larger number.

Answer: False

Reasoning: 1 + 9, 2 + 8, 3 + 7, 4 + 6 and 5 + 5 all work.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S47-B1-Q4 · question page 1066

LOGIC.EVIDENCE.02 · Strategy: list what each clue leaves

A clue is "enough" to answer a question when...

Answer: A. exactly one answer is left after using it.

Reasoning: Enough means the answer is determined.

- ✗ at least one answer is left after using it.: two or more answers left means not enough
- ✗ it rules out at least one answer.: it may still leave several answers
- ✗ it is true.: a true clue can still leave several answers

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S47-B1-Q5 · question page 1067

LOGIC.EVIDENCE.02 · Strategy: list what each clue leaves

n is a whole number from 1 to 20. Clue: n is a square number and n is odd. How many values of n are left?

Answer: 2

Reasoning: 1 and 9.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S47-F1-Q1 · question page 1068

LOGIC.EVIDENCE.02 · Which clue is enough?

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) Four friends sit in a row of four seats numbered 1 to 4: Dee, Eli, Fay and Gus. Who sits in seat 1? Clue (1): Dee sits in seat 4. Clue (2): Eli sits directly left of Fay, and Gus sits directly left of Eli.

Answer: C. Neither clue alone is enough, but both together are.

Reasoning: Clue (2) forces G E F together in that order, so the order is G E F D or D G E F: not enough. Clue (1) alone leaves 6 orders. Together: G E F D, so Gus.

- ✗ Clue (1) alone is enough, but clue (2) alone is not.: check whether clue (1) alone leaves exactly one answer
- ✗ Clue (2) alone is enough, but clue (1) alone is not.: check whether clue (2) alone leaves exactly one answer
- ✗ Each clue alone is enough.: one of the clues alone leaves more than one answer

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S47-F1-Q2 · question page 1069

LOGIC.EVIDENCE.02 · Which clue is enough?

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) Pat is older than Quinn. How old is Quinn, in whole years? Clue (1): their ages add to 30. Clue (2): Pat is 4 years older than Quinn.

Answer: D. Neither clue alone is enough, but both together are.

Reasoning: Clue (1) allows many pairs (16, 14), (20, 10), ... Clue (2) allows many pairs. Together: P + Q = 30 and P - Q = 4, so Q = 13.

- ✗ Clue (1) alone is enough, but clue (2) alone is not.: check whether clue (1) alone leaves exactly one answer
- ✗ Clue (2) alone is enough, but clue (1) alone is not.: check whether clue (2) alone leaves exactly one answer
- ✗ Each clue alone is enough.: one of the clues alone leaves more than one answer

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S47-F1-Q3 · question page 1070

LOGIC.EVIDENCE.02 · Which clue is enough?

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) n is a whole number from 1 to 30. What is n? Clue (1): n leaves remainder 1 when divided by 7. Clue (2): n is a perfect square.

Answer: C. Neither clue alone is enough, but both together are.

Reasoning: Clue (1) allows 1, 8, 15, 22, 29. Clue (2) allows 1, 4, 9, 16,

25. Only 1 is in both lists.

- ✗ Clue (1) alone is enough, but clue (2) alone is not.: check whether clue (1) alone leaves exactly one answer
- ✗ Clue (2) alone is enough, but clue (1) alone is not.: check whether clue (2) alone leaves exactly one answer
- ✗ Each clue alone is enough.: one of the clues alone leaves more than one answer

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S47-F1-Q4 · question page 1071

LOGIC.EVIDENCE.02 · Which clue is enough?

A bag holds only red and blue marbles, 12 in all. Knowing that there are 4 more red marbles than blue ones is enough to find the number of red marbles.

Answer: True

Reasoning: r + b = 12 and r - b = 4 give r = 8.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S47-F1-Q5 · question page 1072

LOGIC.EVIDENCE.02 · Which clue is enough?

x is a whole number from 1 to 20 and x is a multiple of 3. Give one extra clue as a single number: a value m from 2 to 10 such that knowing "x is also a multiple of m" leaves exactly one possible x.

Answer: any values that satisfy: m == 4 or m == 5 (example: m = 4)

Reasoning: The multiples of 3 up to 20 are 3, 6, 9, 12, 15, 18. "Also a multiple of 4" leaves only 12, and "also a multiple of 5" leaves only 15.

Other values of m leave none or several.

Tempting answers the rule rejects: m = 2; m = 6; m = 9; m = 7

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked every m from 2 to 10: exactly one x remains for m = [4, 5]. Automatic: the example passes the rule and 4 listed mistakes fail it.

S47-F2-Q1 · question page 1073

LOGIC.EVIDENCE.02 · Design the missing clue

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) n is a whole number from 1 to 20. What is n? Clue (1): n + 5 = 12. Clue (2): 2n = 14.

Answer: D. Each clue alone is enough.

Reasoning: Each clue alone gives n = 7.

- ✗ Clue (1) alone is enough, but clue (2) alone is not.: check whether clue (1) alone leaves exactly one answer
- ✗ Clue (2) alone is enough, but clue (1) alone is not.: check whether clue (2) alone leaves exactly one answer
- ✗ Neither clue alone is enough, but both together are.: check each clue alone before combining them

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S47-F2-Q2 · question page 1074

LOGIC.EVIDENCE.02 · Design the missing clue

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) n is a whole number from 1 to 20. What is n? Clue (1): n is odd. Clue (2): n is a

multiple of 13.

Answer: A. Clue (2) alone is enough, but clue (1) alone is not.

Reasoning: Clue (2) alone leaves only 13 (26 is too big). Clue (1) leaves 10 odd numbers.

- ✗ Clue (1) alone is enough, but clue (2) alone is not.: check whether clue (1) alone leaves exactly one answer
- ✗ Neither clue alone is enough, but both together are.: check each clue alone before combining them
- ✗ Each clue alone is enough.: one of the clues alone leaves more than one answer

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S47-F2-Q3 · question page 1075

LOGIC.EVIDENCE.02 · Design the missing clue

Two whole numbers from 1 to 9 multiply to 12. Give a sum s such that also knowing "the two numbers add to s" tells you exactly which two numbers they are.

Answer: any values that satisfy: s == 7 or s == 8 (example: s = 7)

Reasoning: The pairs with product 12 are 2 and 6 (sum 8) and 3 and 4 (sum 7). Either sum picks out one pair.

Tempting answers the rule rejects: s = 13; s = 6; s = 12

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked every sum from 2 to 18: the pair is determined for s = [7, 8]. Automatic: the example passes the rule and 3 listed mistakes fail it.

S47-F2-Q4 · question page 1076

LOGIC.EVIDENCE.02 · Design the missing clue

Two whole numbers, each from 1 to 9, multiply to 36. Knowing only this is enough to know which two numbers they are.

Answer: False

Reasoning: Both 4 x 9 and 6 x 6 equal 36.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S47-F2-Q5 · question page 1077

LOGIC.EVIDENCE.02 · Design the missing clue

Three different whole numbers from 1 to 9 add up to 23. How many different sets of three numbers are possible?

Answer: 1

Reasoning: The largest possible total is 9 + 8 + 7 = 24. To get 23, lower one number by 1 without repeating: only 9 + 8 + 6. So the total alone is enough to find the set.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S48-M-Q1 · question page 1079

LOGIC.SPATIAL.06 · Reason about seven-segment digits: segments used, rotations and segment moves

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. How many segments are lit when the display shows 2026?

Answer: 22

Reasoning: 2 uses 5, 0 uses 6, 2 uses 5, 6 uses 6: 22.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S48-M-Q2 · question page 1080

LOGIC.SPATIAL.06 · Reason about seven-segment digits: segments used, rotations and segment moves

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Which digit uses exactly 3 lit segments?

Answer: B. 7

Reasoning: 7 lights the top bar and the two right-hand segments.

- ✗ 1: 1 uses 2 segments
- ✗ 4: 4 uses 4 segments
- ✗ 2: 2 uses 5 segments

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S48-M-Q3 · question page 1081

LOGIC.SPATIAL.06 · Reason about seven-segment digits: segments used, rotations and segment moves

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. When the display showing 69 is turned upside down, it shows 69 again.

Answer: True

Reasoning: A half turn turns 6 into 9 and 9 into 6, and also reverses their order: 69 becomes 69.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S48-M-Q4 · question page 1082

LOGIC.SPATIAL.06 · Reason about seven-segment digits: segments used, rotations and segment moves

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Give a digit that looks exactly the same after the display is turned upside down.

Answer: any values that satisfy: d == 0 or d == 2 or d == 5 or d == 8 (example: d = 8)

Reasoning: 0, 2, 5 and 8 are unchanged by a half turn. 6 and 9 swap, and 1, 3, 4, 7 do not become digits.

Tempting answers the rule rejects: d = 6; d = 1; d = 9; d = 3

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked by turning every digit: unchanged digits ['0', '2', '5', '8']. Automatic: the example passes the rule and 4 listed mistakes fail it.

S48-M-Q5 · question page 1083

LOGIC.SPATIAL.06 · Reason about seven-segment digits: segments used, rotations and segment moves

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Exactly one lit segment of the digit 2 is moved to an unlit position. Which digit can result?

Answer: D. 3

Reasoning: Move the lower-left segment to the lower-right position: 2 becomes 3.

- ✗ 5: turning 2 into 5 needs two segments moved
- ✗ 8: 8 has more segments than 2
- ✗ 7: 7 has fewer segments than 2

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S48-B1-Q1 · question page 1084

LOGIC.SPATIAL.06 · Read the segments of each digit

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. How many segments does the digit 4 light?

Answer: 4

Reasoning: Upper left, middle, and both right-hand segments: 4.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S48-B1-Q2 · question page 1085

LOGIC.SPATIAL.06 · Read the segments of each digit

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. The digit 8 lights all seven segments.

Answer: True

Reasoning: 8 uses every segment.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S48-B1-Q3 · question page 1086

LOGIC.SPATIAL.06 · Read the segments of each digit

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Which digit lights the fewest segments?

Answer: C. 1

Reasoning: 1 lights only 2 segments.

- ✗ 7: 7 lights 3
- ✗ 4: 4 lights 4
- ✗ 0: 0 lights 6

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S48-B1-Q4 · question page 1087

LOGIC.SPATIAL.06 · Read the segments of each digit

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. The digits 0 and 6 light the same number of segments.

Answer: True

Reasoning: Each lights 6 segments (in different positions).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S48-B1-Q5 · question page 1088

LOGIC.SPATIAL.06 · Read the segments of each digit

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. The digit 3 lights the middle segment and both left-hand segments.

Answer: False

Reasoning: 3 lights the top, middle, bottom and both right-hand segments; no left-hand ones.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S48-F1-Q1 · question page 1089

LOGIC.SPATIAL.06 · Turned displays and segment counts

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. The display shows 1958. How many segments are lit?

Answer: 20

Reasoning: 2 + 6 + 5 + 7 = 20.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S48-F1-Q2 · question page 1090

LOGIC.SPATIAL.06 · Turned displays and segment counts

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its

lit segments are exactly those of a digit in the reference figure. The display shows 625. What does it show after being turned upside down?

Answer: A. 529

Reasoning: Reverse the order and turn each digit: 5 stays 5, 2 stays 2, 6 becomes 9: 529.

- ✗ 526: the 6 also turns into a 9
- ✗ 925: the order of the digits is also reversed
- ✗ It does not show a number: 6, 2 and 5 all turn into digits

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S48-F1-Q3 · question page 1091

LOGIC.SPATIAL.06 · Turned displays and segment counts

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. The display showing 1001 still shows a number after it is turned upside down.

Answer: False

Reasoning: The 1 lights the two right-hand segments; turned, they become the two left-hand segments, which is not a digit in the reference.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S48-F1-Q4 · question page 1092

LOGIC.SPATIAL.06 · Turned displays and segment counts

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. How many two-digit numbers

(10 to 99) still show a number after the display is turned upside down?

Answer: 30

Reasoning: Both digits must be from 0, 2, 5, 6, 8, 9; the first cannot be 0: 5 x 6 = 30.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S48-F1-Q5 · question page 1093

LOGIC.SPATIAL.06 · Turned displays and segment counts

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Give a digit d such that adding exactly one lit segment to d (moving nothing) makes the digit 8.

Answer: any values that satisfy: d == 0 or d == 6 or d == 9 (example: d = 0)

Reasoning: 0, 6 and 9 each light 6 segments, and each is 8 with one segment missing.

Tempting answers the rule rejects: d = 5; d = 3; d = 8

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked: digits one segment short of 8: ['0', '6', '9']. Automatic: the example passes the rule and 3 listed mistakes fail it.

S48-F2-Q1 · question page 1094

LOGIC.SPATIAL.06 · Segment moves and best displays

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Exactly one lit segment of the

digit 3 is moved to an unlit position. Give a digit that can result.

Answer: any values that satisfy: d == 2 or d == 5 (example: d = 5)

Reasoning: Move the upper-right segment to the upper left to make 5, or the lower-right segment to the lower left to make 2.

Tempting answers the rule rejects: d = 9; d = 8; d = 3

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked every one-segment move: 3 can become ['2', '5']. Automatic: the example passes the rule and 3 listed mistakes fail it.

S48-F2-Q2 · question page 1095

LOGIC.SPATIAL.06 · Segment moves and best displays

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. You may light at most 12 segments in total. What is the largest whole number you can show? (Any number of digits; the first digit is not 0.)

Answer: 111111

Reasoning: More digits beat bigger digits, and 1 is the cheapest digit (2 segments). Six 1s use 12 segments: 111111. Swapping a 1 for 7 costs one more segment, which is not available.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S48-F2-Q3 · question page 1096

LOGIC.SPATIAL.06 · Segment moves and best displays

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its

lit segments are exactly those of a digit in the reference figure. Every digit can be turned into some other digit by moving exactly one lit segment.

Answer: False

Reasoning: 1, 4, 7 and 8 cannot: no single move turns them into another digit.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S48-F2-Q4 · question page 1097

LOGIC.SPATIAL.06 · Segment moves and best displays

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. A clock shows hours 1 to 12 and minutes 00 to 59 (for example 9:58 uses the digits 9, 5, 8). Which time lights the most segments?

Answer: B. 10:08

Reasoning: 10:08 lights 2 + 6 + 6 + 7 = 21. A search of every time confirms 21 is the most, and 10:08 is one of the times that reach it.

- ✗ 12:59: 12:59 lights 2 + 5 + 5 + 6 = 18
- ✗ 9:58: 9:58 lights 6 + 5 + 7 = 18
- ✗ 11:11: 11:11 lights only 8

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S48-F2-Q5 · question page 1098

LOGIC.SPATIAL.06 · Segment moves and best displays

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its

lit segments are exactly those of a digit in the reference figure. How many digits light more than 5 segments?

Answer: 4

Reasoning: 0, 6, 9 light 6 and 8 lights 7: four digits.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S49-M-Q1 · question page 1100

LOGIC.FALLACY.01 · Recognize common informal fallacies in self-contained arguments

Definitions: Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. Hasty generalization: drawing a broad conclusion from too few cases. Circular reasoning: using the conclusion, in other words, as a reason for itself. Appeal to popularity: claiming something is true because many people believe it. Dana argues that the library should stay open late

Answer: D. Attacking the person

Reasoning: Raj rejects Dana's claim because of something about Dana, not because of her reasons.

- ✗ Hasty generalization: Raj does not generalize from a few cases
- ✗ Circular reasoning: Raj does not use his conclusion as a reason
- ✗ Appeal to popularity: Raj does not mention what many people believe

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S49-M-Q2 · question page 1101

LOGIC.FALLACY.01 · Recognize common informal fallacies in self-contained arguments

Definitions: False choice: presenting only two options when others are possible. Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. Straw man: attacking a distorted version of someone's argument instead of the real one. After, so

because: concluding that A caused B only because B happened after A. "Either we cancel the field trip, or every student wil

Answer: A. False choice

Reasoning: Other options exist, such as more chaperones or a planned route.

- ✗ Slippery slope: no chain of steps is described
- ✗ Straw man: no one's argument is being distorted
- ✗ After, so because: no claim about one event following another is made

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S49-M-Q3 · question page 1102

LOGIC.FALLACY.01 · Recognize common informal fallacies in self-contained arguments

Definitions: Hasty generalization: drawing a broad conclusion from too few cases. Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. False choice: presenting only two options when others are possible. Circular reasoning: using the conclusion, in other words, as a reason for itself. "I met two rude people from that town, so everyone from that to

Answer: C. Hasty generalization

Reasoning: Two cases are far too few to support a claim about everyone in a town.

- ✗ Attacking the person: no one's argument is being rejected
- ✗ False choice: no two options are presented
- ✗ Circular reasoning: the conclusion is not used as a reason

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S49-M-Q4 · question page 1103

LOGIC.FALLACY.01 · Recognize common informal fallacies in self-contained arguments

Definitions: Circular reasoning: using the conclusion, in other words, as a reason for itself. "This app is the best because it is better than all the others" is an example of circular reasoning.

Answer: True

Reasoning: "Better than all the others" says the same thing as "the best", so the reason is just the conclusion restated.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S49-M-Q5 · question page 1104

LOGIC.FALLACY.01 · Recognize common informal fallacies in self-contained arguments

Definitions: After, so because: concluding that A caused B only because B happened after A. Hasty generalization: drawing a broad conclusion from too few cases. Appeal to popularity: claiming something is true because many people believe it. Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. "After I started wearing my lucky socks, our team won three

Answer: B. After, so because

Reasoning: The wins came after the socks, but nothing shows the socks caused them.

- ✗ Hasty generalization: it does not generalize to a group
- ✗ Appeal to popularity: it does not mention what many people believe
- ✗ Slippery slope: it does not predict a chain of extreme results

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S49-B1-Q1 · question page 1105

LOGIC.FALLACY.01 · Strategy: name what the argument does instead of giving reasons

Definitions: Appeal to popularity: claiming something is true because many people believe it. Straw man: attacking a distorted version of someone's argument instead of the real one. False choice: presenting only two options when others are possible. Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. Which definition describes "Millions of people watch

Answer: B. Appeal to popularity

Reasoning: The only reason given is how many people watch it.

- ✗ Straw man: no one's argument is distorted
- ✗ False choice: no two options are presented
- ✗ Slippery slope: no chain of results is predicted

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S49-B1-Q2 · question page 1106

LOGIC.FALLACY.01 · Strategy: name what the argument does instead of giving reasons

Definitions: Hasty generalization: drawing a broad conclusion from too few cases. Concluding that all dogs are friendly after meeting three friendly dogs is a hasty generalization.

Answer: True

Reasoning: Three dogs are too few cases for a claim about all dogs.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S49-B1-Q3 · question page 1107

LOGIC.FALLACY.01 · Strategy: name what the argument does instead of giving reasons

Definitions: Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. Saying "The bridge design is unsafe because the beams are

too thin for the load" is attacking the person.

Answer: False

Reasoning: The reason is about the design itself, not about a person.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S49-B1-Q4 · question page 1108

LOGIC.FALLACY.01 · Strategy: name what the argument does instead of giving reasons

An argument's reason is about who made a claim rather than about the claim. Which question best helps you notice this?

Answer: A. Is the reason about the claim, or about something else?

Reasoning: Checking what the reason is about exposes attacks on the person.

- ✗ Is the conclusion surprising?: surprising conclusions can be well supported
- ✗ Is the argument long?: length does not reveal what the reasons are about
- ✗ Do I agree with the conclusion?: agreeing does not show whether the reasons are good

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S49-B1-Q5 · question page 1109

LOGIC.FALLACY.01 · Strategy: name what the argument does instead of giving reasons

Definitions: Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. "If we allow phones at lunch, then students will use them in class, then stop learning, then fail school" is a slippery slope.

Answer: True

Reasoning: A chain of extreme results is claimed without support for each step.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S49-F1-Q1 · question page 1110

LOGIC.FALLACY.01 · Straw men, slopes and popularity

Definitions: Straw man: attacking a distorted version of someone's argument instead of the real one. Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. False choice: presenting only two options when others are possible. Hasty generalization: drawing a broad conclusion from too few cases. Ava says the school should add one vegetarian meal option

Answer: C. Straw man

Reasoning: Ben attacks a much stronger claim (a full ban) than the one Ava made.

- ✗ Attacking the person: Ben attacks a claim, not Ava's character
- ✗ False choice: Ben does not offer only two options
- ✗ Hasty generalization: Ben does not generalize from a few cases

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S49-F1-Q2 · question page 1111

LOGIC.FALLACY.01 · Straw men, slopes and popularity

Definitions: Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. False choice: presenting only two options when others are possible. After, so because: concluding that A caused B only because B happened after A. Circular reasoning: using the conclusion, in other words, as a reason for itself. "If we let one student retake the quiz, soon everyone will r

Answer: D. Slippery slope

Reasoning: It predicts an extreme chain of results from one step, without support.

- ✗ False choice: it does not present exactly two options
- ✗ After, so because: it does not claim a past event caused another
- ✗ Circular reasoning: the conclusion is not used as a reason

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S49-F1-Q3 · question page 1112

LOGIC.FALLACY.01 · Straw men, slopes and popularity

Definitions: Appeal to popularity: claiming something is true because many people believe it. "Most dentists in a large survey recommend brushing twice a day, so brushing twice a day is likely a good idea" is an appeal to popularity.

Answer: False

Reasoning: The people cited are experts on teeth, and their recommendation is evidence about the topic. An appeal to popularity relies only on how many people believe something.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S49-F1-Q4 · question page 1113

LOGIC.FALLACY.01 · Straw men, slopes and popularity

Definitions: Circular reasoning: using the conclusion, in other words, as a reason for itself. Appeal to popularity: claiming something is true because many people believe it. Straw man: attacking a distorted version of someone's argument instead of the real one. Hasty generalization: drawing a broad conclusion from too few cases. "The rule must be fair because it is the rule, and rules are fair."

Answer: C. Circular reasoning

Reasoning: Being a rule is offered as proof of fairness only because rules are assumed to be fair: the conclusion supports itself.

- ✗ Appeal to popularity: it does not mention what many people believe
- ✗ Straw man: it does not distort anyone's argument
- ✗ Hasty generalization: it does not reason from a few examples

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S49-F1-Q5 · question page 1114

LOGIC.FALLACY.01 · Straw men, slopes and popularity

Definitions: After, so because: concluding that A caused B only because B happened after A. Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. False choice: presenting only two options when others are possible. "The new principal started in Septemb

Answer: B. After, so because

Reasoning: The timing alone is used as proof that one caused the other.

- ✗ Attacking the person: no argument by the principal is being rejected
- ✗ Slippery slope: no chain of future results is predicted
- ✗ False choice: no two options are presented

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S49-F2-Q1 · question page 1115

LOGIC.FALLACY.01 · Is it really a fallacy?

Definitions: Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. Which reply is NOT attacking the person?

Answer: A. "Your numbers are from 2010, and the prices have changed since then."

Reasoning: The correct reply challenges the evidence itself.

- ✗ "You are too young to understand prices.": it dismisses the claim because of the speaker's age
- ✗ "You only say that because you work for the store.": it dismisses the claim because of the speaker's job, not the reasons
- ✗ "Nobody listens to you anyway.": it is about the speaker, not the claim

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S49-F2-Q2 · question page 1116

LOGIC.FALLACY.01 · Is it really a fallacy?

Definitions: False choice: presenting only two options when others are possible. "A whole number is either even or odd" is a false choice.

Answer: False

Reasoning: Every whole number really is even or odd, so the two options are complete.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S49-F2-Q3 · question page 1117

LOGIC.FALLACY.01 · Is it really a fallacy?

Definitions: Hasty generalization: drawing a broad conclusion from too few cases. After, so because: concluding that A caused B only because B happened after A. Straw man: attacking a distorted version of someone's argument instead of the real one. Appeal to popularity: claiming something is true because many people believe it. Which argument avoids all four fallacies defined above?

Answer: D. A random sample of 500 voters favored the plan, so most voters probably favor it.

Reasoning: A large random sample is good evidence about the whole group.

- ✗ Our team won after the coach changed shoes, so the shoes helped.: after, so because
- ✗ Everyone online says the plan is good, so it is good.: appeal to popularity
- ✗ My two cousins liked the plan, so most voters like it.: hasty generalization from two people

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S49-F2-Q4 · question page 1118

LOGIC.FALLACY.01 · Is it really a fallacy?

Definitions: Straw man: attacking a distorted version of someone's argument instead of the real one. Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. Appeal to popularity: claiming something is true because many people believe it. Circular reasoning: using the conclusion, in other words, as a reason for itself. Mo says homework should be shorter on

Answer: B. Straw man

Reasoning: Lee replaces Mo's modest claim with an extreme one and attacks that.

- ✗ Slippery slope: Lee does not predict a chain of steps from Mo's idea
- ✗ Appeal to popularity: Lee does not mention what many people think
- ✗ Circular reasoning: Lee does not use the conclusion as a reason

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S49-F2-Q5 · question page 1119

LOGIC.FALLACY.01 · Is it really a fallacy?

Definitions: Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. An argument that predicts a chain of results is always a slippery slope, even when each step is supported by evidence.

Answer: False

Reasoning: A slippery slope is a fallacy only when the steps are claimed without support. A well-supported chain of causes can be good reasoning.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S50-M-Q1 · question page 1121

LOGIC.EVIDENCE.03 · Judge whether evidence supports a claim: samples, comparisons and correlation

To learn how often people in a town exercise, a student surveys 100 people leaving a gym. What is the main problem with this evidence?

Answer: A. People leaving a gym probably exercise more than the town as a whole.

Reasoning: The sample is not representative of the town.

- ✗ 100 people is always too few for a survey.: size is not the main issue; who was asked is
- ✗ The survey should have asked about diet too.: that changes the question, not the evidence
- ✗ Surveys can never give useful evidence.: well-chosen samples can be very useful

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S50-M-Q2 · question page 1122

LOGIC.EVIDENCE.03 · Judge whether evidence supports a claim: samples, comparisons and correlation

School A: 45 of its 60 students passed a test. School B: 52 of its 80 students passed. What percent of School A's students passed?

Answer: 75

Reasoning: 45 / 60 = 0.75 = 75%. (School B: 52 / 80 = 65%.) Comparing percents, not counts, is fair when the schools differ in size.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S50-M-Q3 · question page 1123

LOGIC.EVIDENCE.03 · Judge whether evidence supports a claim: samples, comparisons and correlation

In many towns, more ice-cream sales come with more swimming accidents. This shows that eating ice cream causes swimming accidents.

Answer: False

Reasoning: Both rise in hot weather. A correlation alone does not show that one causes the other.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S50-M-Q4 · question page 1124

LOGIC.EVIDENCE.03 · Judge whether evidence supports a claim: samples, comparisons and correlation

Which study design gives the best evidence that a tutoring program raises test scores?

Answer: C. Randomly assign students to get tutoring or not, then compare the two groups' scores.

Reasoning: Random assignment makes the groups alike except for the tutoring.

- ✗ Ask tutored students whether they feel they improved.: feelings are not test scores, and there is no comparison group
- ✗ Compare tutored students with students who chose not to sign up.: students who sign up may differ in other ways
- ✗ Measure the scores of tutored students only.: with no comparison group, the effect is unknown

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S50-M-Q5 · question page 1125

LOGIC.EVIDENCE.03 · Judge whether evidence supports a claim: samples, comparisons and correlation

A claim says that more than half of a school's students prefer a later start time. In a random sample of 40 students, n said they prefer it. Give a value of n that would support the claim.

Answer: any values that satisfy: isInteger(n) and n > 20 and n <= 40 (example: n = 28)

Reasoning: More than half of 40 is more than 20, so n from 21 to 40 supports the claim.

Tempting answers the rule rejects: n = 20; n = 12; n = 41

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 20 valid answers.

S50-B1-Q1 · question page 1126

LOGIC.EVIDENCE.03 · Compare rates, not counts

18 of 24 students in a class walk to school. What percent is that?

Answer: 75

Reasoning: 18 / 24 = 0.75 = 75%.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S50-B1-Q2 · question page 1127

LOGIC.EVIDENCE.03 · Compare rates, not counts

Club X has 30 members and 12 play chess. Club Y has 10 members and 6 play chess. A larger share of Club Y plays chess.

Answer: True

Reasoning: X: 12 / 30 = 40%. Y: 6 / 10 = 60%.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S50-B1-Q3 · question page 1128

LOGIC.EVIDENCE.03 · Compare rates, not counts

Town P had 50 car accidents and Town Q had 30. What else do you need before saying Town P's roads are more dangerous?

Answer: D. How many people or cars each town has

Reasoning: Accidents per car or per person is the fair comparison.

- ✗ The names of the drivers: names do not affect the rate
- ✗ Nothing; 50 is more than 30: a larger town can have more accidents and still be safer

✗ The colors of the cars: color is not relevant to the comparison

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S50-B1-Q4 · question page 1129

LOGIC.EVIDENCE.03 · Compare rates, not counts

A survey of 200 people finds that 30 own a bike. What percent own a bike?

Answer: 15

Reasoning: 30 / 200 = 15%.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S50-B1-Q5 · question page 1130

LOGIC.EVIDENCE.03 · Compare rates, not counts

If Group A has more members with a trait than Group B, then a larger percent of Group A has the trait.

Answer: False

Reasoning: Group A might be much larger: 30 of 100 is less than 20 of 40.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S50-F1-Q1 · question page 1131

LOGIC.EVIDENCE.03 · Samples and comparisons

A website asks visitors to vote on whether the town needs a new park. Of 300 votes, 80% say yes. Why is this weak evidence about the whole town?

Answer: D. People who choose to vote online may care more about parks than others.

Reasoning: A self-selected sample can differ from the population.

- ✗ 300 votes is more than enough.: that does not address who chose to vote
- ✗ 80% is too high to be believed.: a high percent can be accurate
- ✗ Websites cannot count votes.: counting is not the problem

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S50-F1-Q2 · question page 1132

LOGIC.EVIDENCE.03 · Samples and comparisons

Class A: 21 of 28 students finished the project on time. Class B: 24 of 30. What is the higher of the two on-time percents?

Answer: 80

Reasoning: A: 75%. B: 80%. The higher is 80%, even though the counts are close.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S50-F1-Q3 · question page 1133

LOGIC.EVIDENCE.03 · Samples and comparisons

A study finds that students who own more books tend to have higher reading scores. This alone proves that buying books raises reading scores.

Answer: False

Reasoning: Families who buy many books may also read together more.

The correlation alone does not prove cause.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S50-F1-Q4 · question page 1134

LOGIC.EVIDENCE.03 · Samples and comparisons

To test whether a new fertilizer helps tomato plants grow, which design is best?

Answer: A. Give fertilizer to a randomly chosen half of the plants and compare them with the other half.

Reasoning: Random assignment and a comparison group isolate the effect of the fertilizer.

- ✗ Give fertilizer to all plants and measure them.: there is no comparison group
- ✗ Use the fertilizer only on the healthiest plants.: healthier plants would grow more anyway
- ✗ Ask a gardener whether the plants look bigger.: opinion is not a measurement

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S50-F1-Q5 · question page 1135

LOGIC.EVIDENCE.03 · Samples and comparisons

A coin is flipped 50 times to test whether it is fair. Give a number of heads h from 0 to 50 that would be strong evidence the coin favors heads. Use this test: h is at least 35.

Answer: any values that satisfy: isInteger(h) and h >= 35 and h <= 50 (example: h = 40)

Reasoning: The stated test counts 35 or more heads as strong evidence.

(A fair coin rarely gives that many.)

Tempting answers the rule rejects: h = 26; h = 30; h = 51

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 16 valid answers.

S50-F2-Q1 · question page 1136

LOGIC.EVIDENCE.03 · Comparisons that reverse

Two clinics treat a condition. Clinic A treats 100 mild cases (90 recover) and 20 severe cases (8 recover). Clinic B treats 20 mild cases (19 recover) and 100 severe cases (50 recover). What percent of all Clinic A patients recover, to the nearest whole percent?

Answer: 82

Reasoning: (90 + 8) / 120 = 98 / 120, about 81.7%, which rounds to 82%.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S50-F2-Q2 · question page 1137

LOGIC.EVIDENCE.03 · Comparisons that reverse

In the clinics above, Clinic A has the higher overall recovery rate (about 82% to 58%). Which statement is true?

Answer: B. Clinic B has the higher recovery rate for mild cases AND for severe cases.

Reasoning: A treats mostly mild cases, which recover more often, so its combined rate is higher even though B does better within each group.

- ✗ Clinic A is better for both mild and severe cases.: B: 95% mild, 50% severe; A: 90% mild, 40% severe

✗ The two clinics are equally good for mild cases.: 90% and 95% are not equal

✗ Clinic A must be the better clinic.: A looks better only because it treats more mild cases

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S50-F2-Q3 · question page 1138

LOGIC.EVIDENCE.03 · Comparisons that reverse

In the clinics above, comparing only the overall recovery rates gives a fair comparison of the clinics.

Answer: False

Reasoning: The clinics treat very different mixes of cases. Comparing within each group is fairer.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S50-F2-Q4 · question page 1139

LOGIC.EVIDENCE.03 · Comparisons that reverse

A test for a rare condition is positive for 95 of every 100 people who have it and for 5 of every 100 people who do not. In a town of 10,000 people, 100 have the condition. About how many people in the town test positive in all?

Answer: 590

Reasoning: Have it: 95 of 100 test positive. Do not have it: 5% of 9,900 = 495. Total 590.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S50-F2-Q5 · question page 1140

LOGIC.EVIDENCE.03 · Comparisons that reverse

In the town above, 590 people test positive and 95 of them have the condition. What fraction of the positive tests are correct?

Answer: C. About 16%

Reasoning: 95 / 590 is about 0.161, or 16%. Most positives come from the much larger group without the condition.

- ✗ About 95%: 95% is the rate among people who have the condition, not among positive tests
- ✗ About 5%: 5% is the false-positive rate among people without it
- ✗ About 50%: 95 is far less than half of 590

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S51-M-Q1 · question page 1142

LOGIC.OPTIMIZE.01 · Find the best value under stated constraints and show nothing better is possible

Two whole numbers add up to 20. What is the largest possible product?

Answer: 100

Reasoning: 10 x 10 = 100. Moving one number up by k and the other down by k gives (10 + k)(10 - k) = 100 - k squared, which is smaller.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S51-M-Q2 · question page 1143

LOGIC.OPTIMIZE.01 · Find the best value under stated constraints and show nothing better is possible

A rectangular pen uses exactly 24 meters of fence, and its sides are whole numbers of meters. What is the largest possible area, in square meters?

Answer: 36

Reasoning: The sides add to 12. Listing 1 x 11, 2 x 10, ..., 6 x 6 gives areas 11, 20, 27, 32, 35, 36. The 6 by 6 square is largest.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S51-M-Q3 · question page 1144

LOGIC.OPTIMIZE.01 · Find the best value under stated constraints and show nothing better is possible

A 12-hour digital clock shows times from 1:00 to 12:59. Which time has the largest product of its digits?

Answer: B. 9:59

Reasoning: 9 x 5 x 9 = 405. The hour digit is at most 9 and the minute digits at most 5 and 9, so nothing can beat 405, and only 9:59 reaches it.

- ✗ 12:59: 1 x 2 x 5 x 9 = 90
- ✗ 8:59: 8 x 5 x 9 = 360
- ✗ 11:59: 1 x 1 x 5 x 9 = 45

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S51-M-Q4 · question page 1145

LOGIC.OPTIMIZE.01 · Find the best value under stated constraints and show nothing better is possible

Using coins worth 1, 5, 10 and 25 cents, the fewest coins that make exactly 30 cents is 2.

Answer: True

Reasoning: 25 + 5 uses 2 coins, and no single coin is worth 30 cents.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S51-M-Q5 · question page 1146

LOGIC.OPTIMIZE.01 · Find the best value under stated constraints and show nothing better is possible

Whole numbers x and y are each 0 or more, with x + 2y at most 10 and x at most 6. Make x + 3y as large as possible. Give x and y.

Answer: any values that satisfy: x == 0 and y == 5 (example: x = 0, y = 5)

Reasoning: x + 3y = (x + 2y) + y is at most 10 + y, and y is at most 5, so x + 3y is at most 15. Only x = 0, y = 5 reaches 15.

Tempting answers the rule rejects: x = 6, y = 2; x = 2, y = 4; x = 1, y = 5

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked all 30 allowed pairs: the best value is 15, reached only at (0, 5). Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 1 valid answers.

S51-B1-Q1 · question page 1147

LOGIC.OPTIMIZE.01 · Strategy: find a good answer, then prove nothing beats it

Two whole numbers add to 14. What is the largest possible product?

Answer: 49

Reasoning: 7 x 7 = 49.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S51-B1-Q2 · question page 1148

LOGIC.OPTIMIZE.01 · Strategy: find a good answer, then prove nothing beats it

To show that a value is the largest possible, it is enough to find one example that reaches it.

Answer: False

Reasoning: You also need a reason no other answer can be larger.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S51-B1-Q3 · question page 1149

LOGIC.OPTIMIZE.01 · Strategy: find a good answer, then prove nothing beats it

Which numbers from the list 4, 9, 2, 7, 5 give the largest sum if you may choose any two?

Answer: C. 9 and 7

Reasoning: The two largest numbers give the largest sum: 16.

- ✗ 9 and 5: 9 + 5 = 14, less than 16
- ✗ 4 and 9: 4 + 9 = 13
- ✗ 7 and 5: 7 + 5 = 12

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S51-B1-Q4 · question page 1150

LOGIC.OPTIMIZE.01 · Strategy: find a good answer, then prove nothing beats it

Using coins worth 1, 5 and 10 cents, what is the fewest coins that make 27 cents?

Answer: 5

Reasoning: 10 + 10 + 5 + 1 + 1: 5 coins.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S51-B1-Q5 · question page 1151

LOGIC.OPTIMIZE.01 · Strategy: find a good answer, then prove nothing beats it

A rectangle with whole-number sides and perimeter 20 can have area 25.

Answer: True

Reasoning: A 5 by 5 square.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S51-F1-Q1 · question page 1152

LOGIC.OPTIMIZE.01 · When the obvious choice is not best

Coins are worth 1, 4 and 6 cents. What is the fewest coins that make exactly 8 cents?

Answer: 2

Reasoning: Taking the biggest coin first gives 6 + 1 + 1 (3 coins), but 4 + 4 uses only 2.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S51-F1-Q2 · question page 1153

LOGIC.OPTIMIZE.01 · When the obvious choice is not best

Whole numbers add up to 10 (you choose how many numbers). What is the largest possible product?

Answer: 36

Reasoning: Use 3s and 2s: 3 + 3 + 2 + 2 gives 36, and 3 + 3 + 4 also gives 36. Splitting any part 5 or more into smaller parts raises the product, and 1s never help.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S51-F1-Q3 · question page 1154

LOGIC.OPTIMIZE.01 · When the obvious choice is not best

Coins are worth 1, 4 and 6 cents. Always taking the largest coin that fits gives the fewest coins for every amount.

Answer: False

Reasoning: For 8 cents it gives 6 + 1 + 1, but 4 + 4 is better.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S51-F1-Q4 · question page 1155

LOGIC.OPTIMIZE.01 · When the obvious choice is not best

A backpack can hold at most 10 kilograms. Items: tent 6 kg, value 8; stove 4 kg, value 5; food 5 kg, value 6; lamp 1 kg, value 3. Which choice has the greatest total value?

Answer: A. Food, stove, lamp

Reasoning: Food, stove and lamp weigh exactly 10 kg and are worth 14. Taking the most valuable item (the tent) first leaves room for at most 4

more kg, and the best that adds is the stove: 13.

- ✗ Tent, stove: 10 kg with value 13, less than 14
- ✗ Tent, lamp: 7 kg with value 11
- ✗ Food, lamp: 6 kg with value 9

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S51-F1-Q5 · question page 1156

LOGIC.OPTIMIZE.01 · When the obvious choice is not best

Two whole numbers are each at least 1 and their product is 36. Make their sum as small as possible. Give the two numbers.

Answer: any values that satisfy: a == 6 and b == 6 (example: a = 6, b = 6)

Reasoning: The pairs are 1 x 36, 2 x 18, 3 x 12, 4 x 9, 6 x 6 with sums 37, 20, 15, 13, 12. 6 and 6 is best.

Tempting answers the rule rejects: a = 4, b = 9; a = 1, b = 36; a = 3, b = 12

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S51-F2-Q1 · question page 1157

LOGIC.OPTIMIZE.01 · Prove the best value

Whole numbers add up to 20 (you choose how many numbers). What is the largest possible product?

Answer: 1458

Reasoning: Any part 5 or more can be split into 2 and the rest to raise

the product, and 4 = 2 + 2 gives the same product, so use only 2s and 3s. Three 2s (product 8) are worse than two 3s (product 9), so use at most two 2s: 20 = 3 x 6 + 2, product 729 x 2 = 1458.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S51-F2-Q2 · question page 1158

LOGIC.OPTIMIZE.01 · Prove the best value

Coins are worth 1, 7 and 10 cents. What is the fewest coins that make exactly 14 cents?

Answer: 2

Reasoning: Taking the largest coin first gives 10 + 1 + 1 + 1 + 1 (5 coins), but 7 + 7 uses 2, and no single coin is worth 14.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S51-F2-Q3 · question page 1159

LOGIC.OPTIMIZE.01 · Prove the best value

Among rectangles with whole-number sides and area 36, the 6 by 6 square has the smallest perimeter.

Answer: True

Reasoning: Perimeters: 1 x 36: 74, 2 x 18: 40, 3 x 12: 30, 4 x 9: 26, 6 x 6: 24.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S51-F2-Q4 · question page 1160

LOGIC.OPTIMIZE.01 · Prove the best value

Whole numbers a, b and c are each at least 1 and add up to 12. Make the product a x b x c as large as possible. Give a, b and c.

Answer: any values that satisfy: a == 4 and b == 4 and c == 4 (example: a = 4, b = 4, c = 4)

Reasoning: If two of the numbers differ by 2 or more, moving 1 from the larger to the smaller raises the product. So all three are equal: 4, 4, 4, product 64.

Tempting answers the rule rejects: a = 3, b = 4, c = 5; a = 2, b = 5, c = 5; a = 6, b = 6, c = 0

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked all triples: the largest product is 64, reached only at (4, 4, 4). Automatic: the example passes the rule and 3 listed mistakes fail it.

S51-F2-Q5 · question page 1161

LOGIC.OPTIMIZE.01 · Prove the best value

Use each of the digits 1, 2, 3 and 4 once to make two 2-digit numbers. Which pair has the largest product?

Answer: D. 41 x 32

Reasoning: The tens digits should be 4 and 3. Then 41 x 32 = 1312 beats 42 x 31 = 1302: the extra unit goes with the smaller tens digit.

- ✗ 42 x 31: 42 x 31 = 1302, less than 1312
- ✗ 43 x 21: 43 x 21 = 903
- ✗ 31 x 24: 31 x 24 = 744

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S52-M-Q1 · question page 1163

LOGIC.PATTERN.01 · Identify the rule in a stated family that fits a sequence, and continue it

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). What is the next term of 3, 7, 11, 15, ...?

Answer: 19

Reasoning: Rule (1): add 4 each time. 15 + 4 = 19.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S52-M-Q2 · question page 1164

LOGIC.PATTERN.01 · Identify the rule in a stated family that fits a sequence, and continue it

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Which rule fits 2, 6, 18, 54, ...?

Answer: A. Multiply by 3 each time

Reasoning: Each term is 3 times the one before.

- ✗ Add 4 each time: 2 + 4 = 6, but 6 + 4 = 10, not 18
- ✗ Add 4, then 12, then 36: those amounts do not grow by the same number each time
- ✗ Multiply by 2 each time: 2 x 2 = 4, not 6

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S52-M-Q3 · question page 1165

LOGIC.PATTERN.01 · Identify the rule in a stated family that fits a sequence, and continue it

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). What is the next term of 1, 2, 4, 7, 11, ...?

Answer: 16

Reasoning: Rule (3): the amounts added are 1, 2, 3, 4, so next add 5: 16.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S52-M-Q4 · question page 1166

LOGIC.PATTERN.01 · Identify the rule in a stated family that fits a sequence, and continue it

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). The sequence 5, 10, 20, 35, ... follows rule (3).

Answer: True

Reasoning: The amounts added are 5, 10, 15, which grow by 5 each time.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S52-M-Q5 · question page 1167

LOGIC.PATTERN.01 · Identify the rule in a stated family that fits a sequence, and continue it

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Letters are numbered by their place in the alphabet: A = 1, B = 2, ..., Z = 26. The letters C, F, I, L follow rule (1) by their numbers. Give the next letter.

Answer: any values that satisfy: x == 'o' (example: x = O)

Reasoning: C, F, I, L are 3, 6, 9, 12: add 3. Next is 15, which is O.

Tempting answers the rule rejects: x = M; x = N; x = P

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked: next number 15 is letter O. Automatic: the example passes the rule and 3 listed mistakes fail it.

S52-B1-Q1 · question page 1168

LOGIC.PATTERN.01 · Strategy: write the differences

Write the differences of 2, 5, 9, 14. What is the last difference?

Answer: 5

Reasoning: The differences are 3, 4, 5.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S52-B1-Q2 · question page 1169

LOGIC.PATTERN.01 · Strategy: write the differences

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). The sequence 10, 7, 4, 1, ... follows rule (1).

Answer: True

Reasoning: It adds -3 each time.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S52-B1-Q3 · question page 1170

LOGIC.PATTERN.01 · Strategy: write the differences

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Which rule fits 1, 4, 16, 64, ...?

Answer: B. Rule (2)

Reasoning: Each term is 4 times the one before.

- ✗ Rule (1): the differences 3, 12, 48 are not equal
- ✗ Rule (3): the differences 3, 12, 48 do not grow by the same amount
- ✗ None of the rules: multiplying by 4 each time works

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S52-B1-Q4 · question page 1171

LOGIC.PATTERN.01 · Strategy: write the differences

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). What is the next term of 50, 45, 40, 35, ...?

Answer: 30

Reasoning: Add -5: 30.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S52-B1-Q5 · question page 1172

LOGIC.PATTERN.01 · Strategy: write the differences

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). The sequence 3, 6, 12, 24, ... follows rule (1).

Answer: False

Reasoning: The differences are 3, 6, 12, which are not equal. It follows rule (2): multiply by 2.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S52-F1-Q1 · question page 1173

LOGIC.PATTERN.01 · Rules for letters and far terms

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Letters are numbered by their place in the alphabet: A = 1, B = 2, ..., Z = 26. The letters B, D, G, K follow rule (3) by their numbers. Give the next letter.

Answer: any values that satisfy: x == 'p' (example: x = P)

Reasoning: B, D, G, K are 2, 4, 7, 11; the amounts added are 2, 3, 4, so add 5: 16, which is P.

Tempting answers the rule rejects: x = O; x = N; x = Q

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked: next number 16 is letter P. Automatic: the example passes the rule and 3 listed mistakes fail it.

S52-F1-Q2 · question page 1174

LOGIC.PATTERN.01 · Rules for letters and far terms

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow

by the same number each time (for example +1, +2, +3, ...). What is the 10th term of 4, 7, 10, 13, ...?

Answer: 31

Reasoning: Rule (1), add 3. The 10th term is 4 + 9 x 3 = 31.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S52-F1-Q3 · question page 1175

LOGIC.PATTERN.01 · Rules for letters and far terms

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Knowing only the first two terms, 3 and 6, is enough to know the third term.

Answer: False

Reasoning: Rule (1) gives 9 (add 3) and rule (2) gives 12 (multiply by 2), so more terms are needed.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S52-F1-Q4 · question page 1176

LOGIC.PATTERN.01 · Rules for letters and far terms

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). The first three terms are 1, 2, 4. Which statement is true?

Answer: C. Rule (2) and rule (3) both fit, giving 8 or 7 next.

Reasoning: Three terms cannot tell rule (2) from rule (3) here.

✗ Only rule (2) fits, so the next term is 8.: rule (3) also fits: add 1, then 2, then 3

✗ Only rule (3) fits, so the next term is 7.: rule (2) also fits: multiply by 2

✗ Rule (1) fits, so the next term is 6.: the differences 1 and 2 are not equal

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S52-F1-Q5 · question page 1177

LOGIC.PATTERN.01 · Rules for letters and far terms

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). What is the 8th term of 2, 6, 18, 54, ...?

Answer: 4374

Reasoning: Rule (2), multiply by 3. The 8th term is 2 x 3 to the 7th power = 4374.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S52-F2-Q1 · question page 1178

LOGIC.PATTERN.01 · Work with the rule as a formula

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). The sequence is 5, 9, 13, 17, ... Give the 25th term.

Answer: any values that satisfy: t == 101 (example: t = 101)

Reasoning: Rule (1), add 4: the nth term is 5 + 4(n - 1). For n = 25: 5 + 96 = 101.

Tempting answers the rule rejects: t = 105; t = 97; t = 100

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S52-F2-Q2 · question page 1179

LOGIC.PATTERN.01 · Work with the rule as a formula

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). In the sequence 3, 6, 12, 24, ..., which term is the first one greater than 1000? Give its position (1 for the first term).

Answer: 10

Reasoning: Terms: 3, 6, 12, 24, 48, 96, 192, 384, 768, 1536. The 10th term, 1536, is the first above 1000.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S52-F2-Q3 · question page 1180

LOGIC.PATTERN.01 · Work with the rule as a formula

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). What is the 10th term of 1, 3, 6, 10, ...?

Answer: 55

Reasoning: Rule (3): add 2, 3, 4, ... The nth term is 1 + 2 + ... + n, so the 10th is 55.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S52-F2-Q4 · question page 1181

LOGIC.PATTERN.01 · Work with the rule as a formula

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). A sequence that follows rule (1) with a positive amount added will eventually pass every number.

Answer: True

Reasoning: Adding the same positive amount again and again grows without limit.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S52-F2-Q5 · question page 1182

LOGIC.PATTERN.01 · Work with the rule as a formula

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Letters are numbered by their place in the alphabet: A = 1, B = 2, ..., Z = 26. Which letter sequence follows rule (2) by the letters' numbers?

Answer: D. A, B, D, H

Reasoning: A, B, D, H are 1, 2, 4, 8: multiply by 2.

- ✗ A, C, E, G: that adds 2 each time: rule (1)
- ✗ A, B, D, G: that adds 1, 2, 3: rule (3)
- ✗ B, C, E, H: that adds 1, 2, 3: rule (3)

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S53-M-Q1 · question page 1184

LOGIC.WORD.03 · Find anagrams and letter-bank words within a supplied word list and letter rules

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same

Answer: D. MELON

Reasoning: MELON uses M, E, L, O, N once each, like LEMON.

- ✗ SOLEMN: SOLEMN has an extra S
- ✗ LANE: LANE has different letters
- ✗ TONE: TONE has different letters

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S53-M-Q2 · question page 1185

LOGIC.WORD.03 · Find anagrams and letter-bank words within a supplied word list and letter rules

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same

Answer: any values that satisfy: w == 'tears' or w == 'rates' (example: w = TEARS)

Reasoning: TEARS and RATES use A, E, R, S, T.

Tempting answers the rule rejects: w = STARE; w = STAR; w = TASTE; w = ASTER

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked: anagrams of STARE on the list are ['TEARS', 'RATES']. Automatic: the example passes the rule and 4 listed mistakes fail it.

S53-M-Q3 · question page 1186

LOGIC.WORD.03 · Find anagrams and letter-bank words within a supplied word list and letter rules

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same

Answer: True

Reasoning: Each uses E, I, L, N, S, T once.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S53-M-Q4 · question page 1187

LOGIC.WORD.03 · Find anagrams and letter-bank words within a supplied word list and letter rules

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. How many words on the list can be

Answer: 15

Reasoning: Check each word against P, L, A, N, E, T: PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL: 15.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S53-M-Q5 · question page 1188

LOGIC.WORD.03 · Find anagrams and letter-bank words within a supplied word list and letter rules

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. Read the letters STOPAN from righ

Answer: B. POTS

Reasoning: Backward, STOPAN reads NAPOTS, which contains POTS.

- ✗ STOP: STOP appears forward, not in the backward reading NAPOTS
- ✗ SPOT: the letters S, P, O, T are not neighbors in that order in NAPOTS
- ✗ PAN: PAN is not on the list

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S53-B1-Q1 · question page 1189

LOGIC.WORD.03 · Strategy: sort the letters

Write the letters of NOTE in alphabetical order (no spaces).

Answer: ENOT

Reasoning: E, N, O, T.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S53-B1-Q2 · question page 1190

LOGIC.WORD.03 · Strategy: sort the letters

An anagram uses exactly the same letters, each as many times, in a different order. STONE and NOTES are anagrams.

Answer: True

Reasoning: Both sort to E, N, O, S, T.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S53-B1-Q3 · question page 1191

LOGIC.WORD.03 · Strategy: sort the letters

An anagram uses exactly the same letters, each as many times, in a different order. TASTE and STATE are anagrams.

Answer: True

Reasoning: Both sort to A, E, S, T, T.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S53-B1-Q4 · question page 1192

LOGIC.WORD.03 · Strategy: sort the letters

An anagram uses exactly the same letters, each as many times, in a different order. Which word is NOT an anagram of SPOT?

Answer: C. SPOTS

Reasoning: SPOTS has an extra S.

- ✗ STOP: STOP uses S, P, O, T
- ✗ TOPS: TOPS uses S, P, O, T
- ✗ POTS: POTS uses S, P, O, T

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S53-B1-Q5 · question page 1193

LOGIC.WORD.03 · Strategy: sort the letters

An anagram uses exactly the same letters, each as many times, in a different order. BREAD and BEARD are anagrams.

Answer: True

Reasoning: Both sort to A, B, D, E, R.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S53-F1-Q1 · question page 1194

LOGIC.WORD.03 · Letter banks and joined parts

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. Join one part from the first grou

Answer: any values that satisfy: w == 'planet' or w == 'heart' or w == 'stone' (example: w = PLANET)

Reasoning: PL + ANET = PLANET, HE + ART = HEART and ST + ONE = STONE are all on the list.

Tempting answers the rule rejects: w = START; w = PLONE; w = STANET

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked all 9 joins: listed words ['PLANET', 'STONE', 'HEART']. Automatic: the example passes the rule and 3 listed mistakes fail it.

S53-F1-Q2 · question page 1195

LOGIC.WORD.03 · Letter banks and joined parts

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same

Answer: 2

Reasoning: HEART and HATER.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S53-F1-Q3 · question page 1196

LOGIC.WORD.03 · Letter banks and joined parts

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. You

may use the letters S, T, O,

Answer: A. All of SPOT, STOP, POTS, TOPS and POST tie at 4 letters

Reasoning: Only 4-letter words from S, T, O, P are on the list, and there are five of them.

- ✗ STOP is the only longest word: SPOT, POTS, TOPS and POST also have 4 letters
- ✗ STONE: STONE needs N and E
- ✗ SPOTS: SPOTS is not on the list

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S53-F1-Q4 · question page 1197

LOGIC.WORD.03 · Letter banks and joined parts

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same

Answer: True

Reasoning: Both sort to G, H, I, N, T.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S53-F1-Q5 · question page 1198

LOGIC.WORD.03 · Letter banks and joined parts

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN,

LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same

Answer: 2

Reasoning: LISTEN, SILENT, ENLIST, TINSEL; SPOT, STOP, POTS, TOPS, POST; STONE, ONSET, NOTES (only 3). So 2 families.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S53-F2-Q1 · question page 1199

LOGIC.WORD.03 · Phrases and hidden words

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same

Answer: C. TASTE PEN

Reasoning: PAST TEEN has A, E, E, N, P, S, T, T. TASTE (A, E, S, T, T) and PEN (E, N, P) use exactly these.

- ✗ PETAL NET: it has an L and no S
- ✗ STAR PEN: it has an R and only one T
- ✗ NEAT POTS: it has an O and only one E

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S53-F2-Q2 · question page 1200

LOGIC.WORD.03 · Phrases and hidden words

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. Every word on the list that is al

Answer: True

Reasoning: The pairs are NET and TEN, SPOT and TOPS, STOP and POTS, RATS and STAR.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S53-F2-Q3 · question page 1201

LOGIC.WORD.03 · Phrases and hidden words

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. Give a word on the list that is a

Answer: any values that satisfy: w == 'net' or w == 'ten' or w == 'spot' or w == 'stop' or w == 'pots' or w == 'tops' or w == 'rats' or w == 'star' (example: w = STOP)

Reasoning: NET, TEN, SPOT, TOPS, STOP, POTS, RATS and STAR all work.

Tempting answers the rule rejects: w = POST; w = STONE; w = LEMON

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked every list word read backward: ['NET', 'TEN', 'SPOT', 'STOP', 'POTS',

'TOPS', 'RATS', 'STAR']. Automatic: the example passes the rule and 3 listed mistakes fail it.

S53-F2-Q4 · question page 1202

LOGIC.WORD.03 · Phrases and hidden words

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. How many words on the list can be

Answer: 7

Reasoning: NOTE, TONE, STONE, ONSET, NOTES, NET, TEN: 7.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S53-F2-Q5 · question page 1203

LOGIC.WORD.03 · Phrases and hidden words

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same

Answer: 3

Reasoning: STONE, ONSET and NOTES: 3.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S54-M-Q1 · question page 1205

LOGIC.SEARCH.02 · Plan weighings or questions and find the least number that always identifies the answer

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 9 coins and one is heavier. What is the least number of weighings that always finds the heavy coin?

Answer: 2

Reasoning: Weigh 3 against 3. If one side goes down, the heavy coin is among those 3; if they balance, it is among the other 3. Then weigh 1 against 1 of the 3. One weighing cannot work: it has only 3 outcomes for 9 possibilities.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S54-M-Q2 · question page 1206

LOGIC.SEARCH.02 · Plan weighings or questions and find the least number that always identifies the answer

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 9 coins and one is heavier. Which first weighing lets you always finish in the fewest total weighings?

Answer: A. 3 coins against 3 coins

Reasoning: Splitting 9 into three groups of 3 leaves 3 suspects whatever happens, and 3 needs only one more weighing.

- ✗ 4 coins against 4 coins: if one side goes down, 4 suspects remain, which needs 2 more weighings
- ✗ 1 coin against 1 coin: if they balance, 7 suspects remain, which needs

2 more weighings

✗ 2 coins against 2 coins: if they balance, 5 suspects remain, which needs 2 more weighings

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S54-M-Q3 · question page 1207

LOGIC.SEARCH.02 · Plan weighings or questions and find the least number that always identifies the answer

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. With 3 weighings you can always find the heavier coin among 27 coins.

Answer: True

Reasoning: Each weighing splits the suspects into three groups (left, right, off the scale): 27, then 9, then 3, then 1.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S54-M-Q4 · question page 1208

LOGIC.SEARCH.02 · Plan weighings or questions and find the least number that always identifies the answer

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. What is the largest number of coins n for which 2 weighings always find the one heavier coin?

Answer: any values that satisfy: n == 9 (example: n = 9)

Reasoning: Two weighings have 3 x 3 = 9 outcome patterns, so at most 9 coins; and 9 works by splitting into threes.

Tempting answers the rule rejects: n = 8; n = 10; n = 6

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Search: largest n with need(n) <= 2 is 9. Automatic: the example passes the rule and 3 listed mistakes fail it.

S54-M-Q5 · question page 1209

LOGIC.SEARCH.02 · Plan weighings or questions and find the least number that always identifies the answer

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 10 coins and one is heavier. What is the least number of weighings that always finds the heavy coin?

Answer: 3

Reasoning: Two weighings give only 9 outcome patterns, which is fewer than 10 possibilities, so 2 are not enough. Three are: weigh 3 against 3, leaving at most 4 suspects, then finish in 2 more.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S54-B1-Q1 · question page 1210

LOGIC.SEARCH.02 · Strategy: each weighing has three outcomes

How many different outcomes can one weighing on a balance scale have?

Answer: 3

Reasoning: Left side down, right side down, or balance.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S54-B1-Q2 · question page 1211

LOGIC.SEARCH.02 · Strategy: each weighing has three outcomes

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. With 3 coins and one heavier, one weighing always finds the heavy coin.

Answer: True

Reasoning: Weigh 1 against 1; a balance means the third coin is heavy.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S54-B1-Q3 · question page 1212

LOGIC.SEARCH.02 · Strategy: each weighing has three outcomes

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 5 coins and one is heavier. You weigh 2 against 2 and they balance. What do you know?

Answer: B. The fifth coin is the heavy one.

Reasoning: All four weighed coins are genuine, so the coin left off is heavy.

- ✗ One of the four weighed coins is heavy.: a heavy coin on the scale would tip it
- ✗ Nothing yet.: a balance clears all four weighed coins
- ✗ You must weigh again to know.: only one coin is left as a suspect

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S54-B1-Q4 · question page 1213

LOGIC.SEARCH.02 · Strategy: each weighing has three outcomes

How many outcome patterns do 2 weighings have in all?

Answer: 9

Reasoning: Each of the 3 outcomes of the first weighing can be followed by 3 outcomes of the second: 9.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S54-B1-Q5 · question page 1214

LOGIC.SEARCH.02 · Strategy: each weighing has three outcomes

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. With 4 coins and one heavier, one weighing always finds the heavy coin.

Answer: False

Reasoning: One weighing has 3 outcomes, but there are 4 possibilities.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S54-F1-Q1 · question page 1215

LOGIC.SEARCH.02 · Bigger groups and comparisons

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. You may use 4 weighings. What is the largest number of coins among which you can always find the one heavier coin?

Answer: 81

Reasoning: 4 weighings have 3 x 3 x 3 x 3 = 81 outcome patterns, and

splitting into thirds each time handles 81 coins.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S54-F1-Q2 · question page 1216

LOGIC.SEARCH.02 · Bigger groups and comparisons

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 6 coins and one is heavier. You weigh 2 against 2 and the left side goes down. What should the next weighing be?

Answer: D. One left-side coin against the other left-side coin

Reasoning: Only the 2 left-side coins are suspects. Weigh them against each other: the one that goes down is heavy.

- ✗ One left-side coin against a right-side coin: the right-side coins are already known to be genuine, so this tests only one suspect
- ✗ The two coins left off against each other: those coins are genuine; the heavy coin was on the left
- ✗ All six coins again, 3 against 3: that repeats work and may not finish

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S54-F1-Q3 · question page 1217

LOGIC.SEARCH.02 · Bigger groups and comparisons

You have 3 weights of different sizes and compare them two at a time on a balance. Finding the heaviest always takes at least 2 weighings.

Answer: True

Reasoning: Each weight that is not the heaviest must lose at least one

comparison, and one weighing has only one loser. Two weights must lose, so 2 weighings are needed.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S54-F1-Q4 · question page 1218

LOGIC.SEARCH.02 · Bigger groups and comparisons

You have 8 weights of different sizes and compare them two at a time on a balance. What is the least number of weighings that always finds the heaviest?

Answer: 7

Reasoning: Every weight except the heaviest must lose a comparison, and each weighing has one loser: at least 7. A knockout tournament uses exactly 7.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S54-F1-Q5 · question page 1219

LOGIC.SEARCH.02 · Bigger groups and comparisons

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. Give a number of coins n from 20 to 30 for which 3 weighings always find the one heavier coin.

Answer: any values that satisfy: isInteger(n) and n >= 20 and n <= 27 (example: n = 27)

Reasoning: Three weighings handle up to 3 x 3 x 3 = 27 coins, so any n from 20 to 27 works; 28, 29 and 30 do not.

Tempting answers the rule rejects: n = 28; n = 30; n = 19

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Search: need(n) <= 3 exactly for n up to 27. Automatic: the example passes the rule and 3 listed mistakes fail it.

S54-F2-Q1 · question page 1220

LOGIC.SEARCH.02 · Least weighings, proved

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 7 coins and one is LIGHTER (not heavier). What is the least number of weighings that always finds it?

Answer: 2

Reasoning: The same splitting works for a lighter coin: weigh 3 against 3 (or 2 against 2), leaving at most 3 suspects; then one more weighing. One weighing cannot tell 7 cases apart.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S54-F2-Q2 · question page 1221

LOGIC.SEARCH.02 · Least weighings, proved

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. Why can 2 weighings NOT always find the heavy coin among 10?

Answer: A. Two weighings give only 9 outcome patterns, but there are 10 possible heavy coins.

Reasoning: Each possibility needs its own outcome pattern, and 10 > 9.

✗ 10 is an even number.: evenness is not the issue; 8 coins need only 2 weighings

✗ You cannot put 5 coins on each side.: you can, and it still would not be enough

✗ The first weighing must use all 10 coins.: it need not use every coin

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S54-F2-Q3 · question page 1222

LOGIC.SEARCH.02 · Least weighings, proved

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. With 4 coins and one heavier, 2 weighings always find the heavy coin.

Answer: True

Reasoning: Weigh 1 against 1. If one goes down, done; if they balance, weigh the other two.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S54-F2-Q4 · question page 1223

LOGIC.SEARCH.02 · Least weighings, proved

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. Give a number of coins n from 20 to 90 for which 3 weighings are NOT enough but 4 weighings are enough to always find the one heavier coin.

Answer: any values that satisfy: isInteger(n) and n >= 28 and n <= 81 (example: n = 50)

Reasoning: 3 weighings handle at most 27 coins, and 4 handle at most 81. So n from 28 to 81 works.

Tempting answers the rule rejects: n = 27; n = 82; n = 20

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Search: need(n) == 4 exactly for n from 28 to 81. Automatic: the example passes the rule and 3 listed mistakes fail it.

S54-F2-Q5 · question page 1224

LOGIC.SEARCH.02 · Least weighings, proved

You have 4 weights of different sizes and compare them two at a time on a balance. What is the least number of weighings that always finds both the heaviest and the lightest?

Answer: 4

Reasoning: Compare A with B and C with D. The heaviest is the winner of a second comparison between the two winners, and the lightest is the loser of a comparison between the two losers: 4 weighings. A search of every strategy shows 3 are never enough.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.